

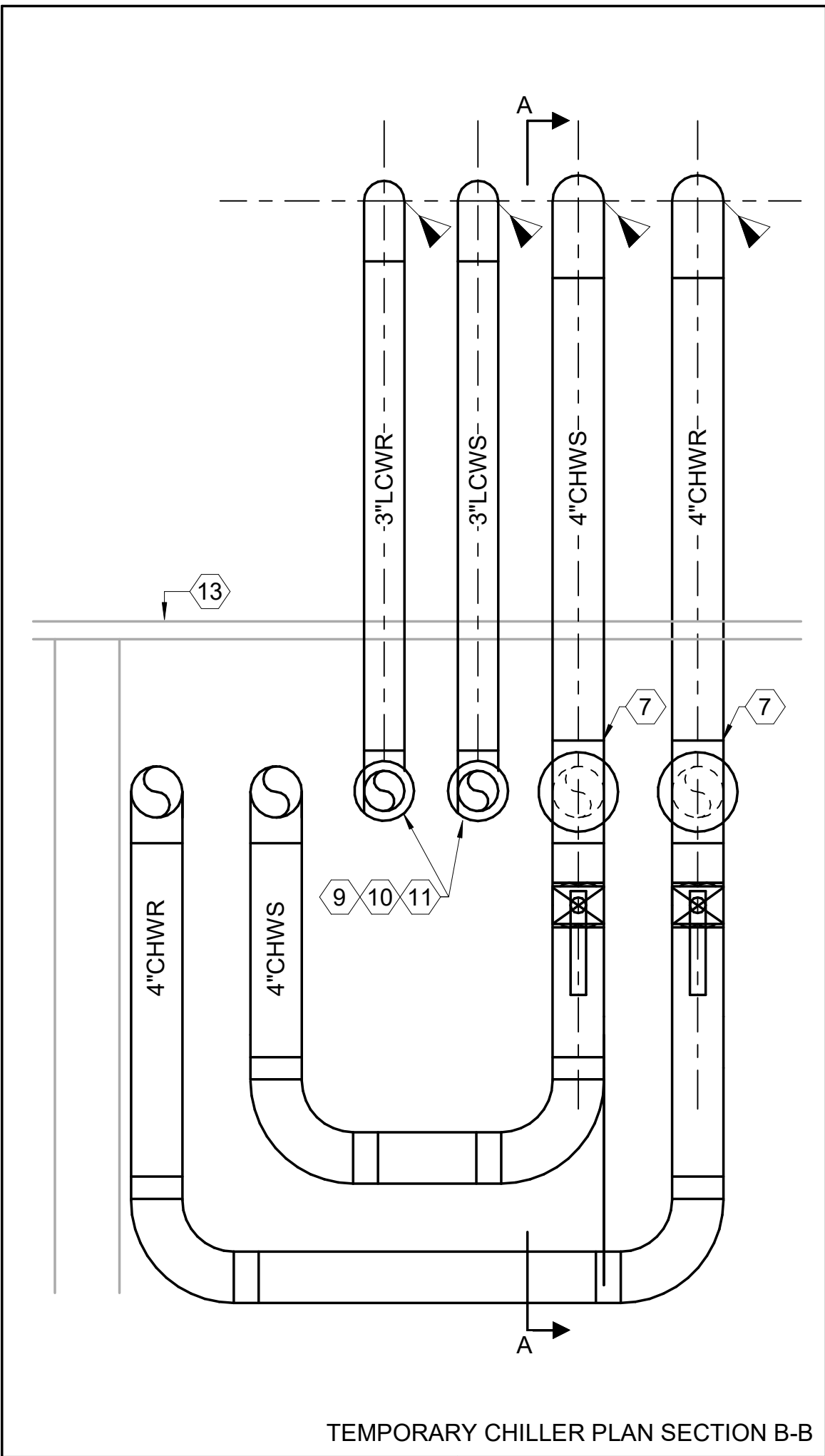
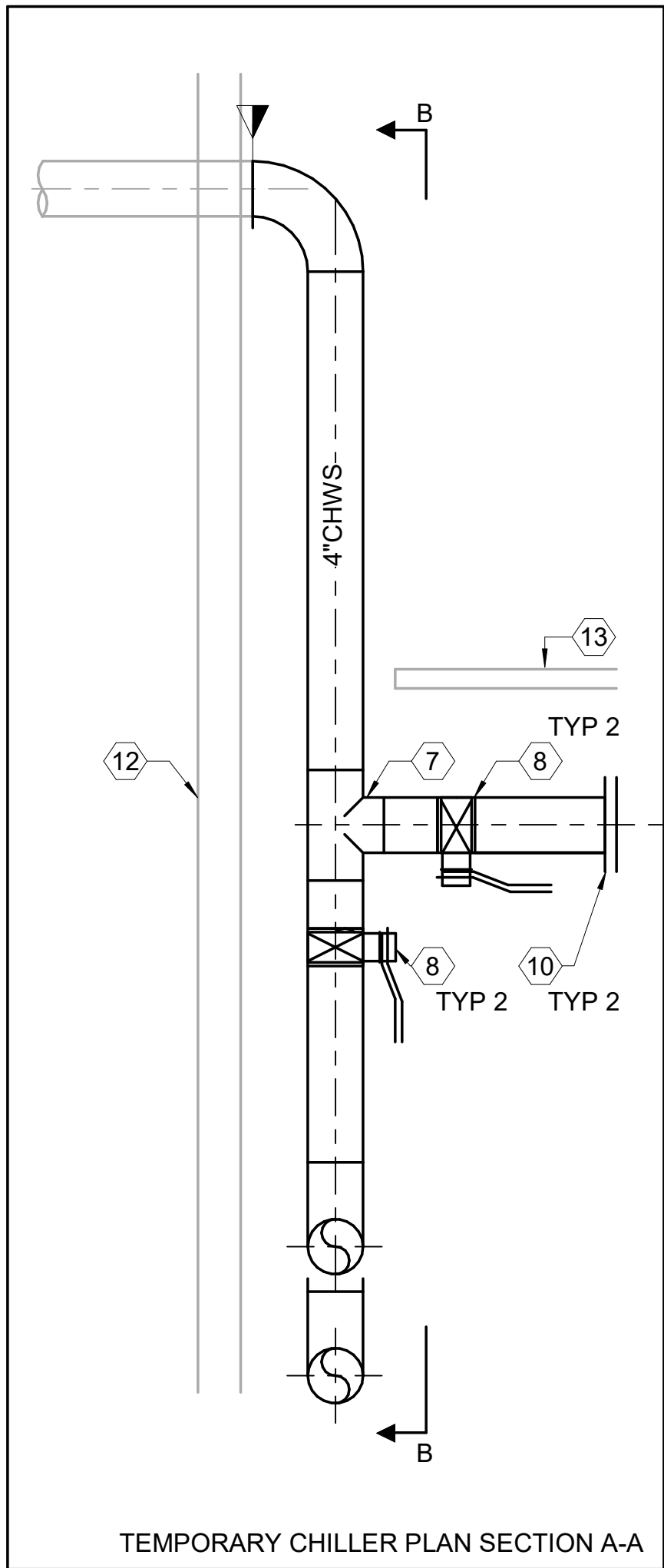
A	DETAIL/SECTION TITLE C1 M-501 DETAIL SCALE: NONE SCALE SHEET NUMBER NUMBER = DETAIL DESIGNATOR LETTER = SECTION DESIGNATOR	GENERAL ABBREVIATIONS <table><tr><td>24X12</td><td>RECT. DUCT DIM. (INCHES)</td><td>IN.W.C.</td><td>INCHES WATER COLUMN</td></tr><tr><td>12Ø</td><td>ROUND DUCT DIM. (INCHES)</td><td>LAT</td><td>LEAVING AIR TEMPERATURE</td></tr><tr><td>AFF</td><td>ABOVE FINISHED FLOOR</td><td>LCW</td><td>LOW-CONDUCTIVITY WATER</td></tr><tr><td>ACU</td><td>AIR CONDITIONING UNIT</td><td>LWT</td><td>LEAVING WATER TEMPERATURE</td></tr><tr><td>ACS</td><td>AIR CONDITIONING SYSTEM</td><td>MA</td><td>MIXED AIR</td></tr><tr><td>ACHPS</td><td>AC HEAT PUMP SYSTEM</td><td>MAX</td><td>MAXIMUM</td></tr><tr><td>AI</td><td>ANALOG INPUT</td><td>MAX.</td><td>MAXIMUM</td></tr><tr><td>ALC</td><td>AUTOMATED LOGIC CONTROLS</td><td>MCC</td><td>MOTOR CONTROL CENTER</td></tr><tr><td>AO</td><td>ANALOG OUTPUT</td><td>MCV</td><td>MOTORIZED CONTROL VALVE</td></tr><tr><td>APD</td><td>AIR PRESSURE DROP</td><td>MECH</td><td>MECHANICAL</td></tr><tr><td>AV</td><td>ANALOG VALUE</td><td>MFR</td><td>MANUFACTURER</td></tr><tr><td>BAS</td><td>BUILDING AUTOMATION SYSTEM</td><td>MHP</td><td>MOTOR HORSEPOWER</td></tr><tr><td>BFP</td><td>BACKFLOW PREVENTER</td><td>MIN</td><td>MINIMUM</td></tr><tr><td>BHP</td><td>BRAKE HORSEPOWER</td><td>MIN.</td><td>MINIMUM</td></tr><tr><td>BI</td><td>BINARY INPUT</td><td>(N)</td><td>NEW</td></tr><tr><td>BO</td><td>BINARY OUTPUT</td><td>NC</td><td>NORMALLY CLOSED</td></tr><tr><td>BOD</td><td>BOTTOM OF DUCT</td><td>NEC</td><td>NATIONAL ELECTRICAL CODE</td></tr><tr><td>BOP</td><td>BOTTOM OF PIPE</td><td>NO</td><td>NORMALLY OPEN</td></tr><tr><td>BV</td><td>BINARY VALUE</td><td>NPSH</td><td>NET POSITIVE SUCTION HEAD</td></tr><tr><td>CCP</td><td>CHILLER CONTROL PANEL</td><td>NPSHR</td><td>NET POSITIVE SUCTION HEAD REQUIRED</td></tr><tr><td>CFM</td><td>CUBIC FEET PER MINUTE</td><td></td><td></td></tr><tr><td>CHW</td><td>CHILLED WATER</td><td>NRTL</td><td>NATIONALLY-RECOGNIZED TESTING LABORATORY</td></tr><tr><td>CL</td><td>CENTERLINE OF PIPE ELEVATION</td><td>NSR</td><td>NON-SPRING-RETURN</td></tr><tr><td>CP</td><td>CONTROL PANEL</td><td>OA</td><td>OUTDOOR AIR</td></tr><tr><td>CPA</td><td>COMPRESSOR, AIR</td><td>OB</td><td>OPPOSED BLADE</td></tr><tr><td>CPD</td><td>COIL PRESSURE DROP (FLUID SIDE)</td><td>PB</td><td>PARALLEL BLADE</td></tr><tr><td>CV</td><td>CONTROL VALVE</td><td>PCHW</td><td>PUMP, CHILLED WATER</td></tr><tr><td>CW</td><td>CITY WATER</td><td>POC</td><td>POINT OF CONNECTION</td></tr><tr><td>(D)</td><td>DEMOLISH (REMOVE AND DISPOSE OF)</td><td>PRV</td><td>PRESSURE REDUCING VALVE</td></tr><tr><td>DB</td><td>DRY BULB</td><td>PS</td><td>POWER SUPPLY</td></tr><tr><td>DCW</td><td>DOMESTIC COLD WATER</td><td>PSIA</td><td>POUNDS, CHILLED WATER ABSOLUTE</td></tr><tr><td>DDC</td><td>DIRECT DIGITAL CONTROL</td><td>PSIG</td><td>POUND PER SQUARE INCH GAUGE</td></tr><tr><td>DDCP</td><td>DIRECT DIGITAL CONTROL PANEL</td><td>RA</td><td>RETURN AIR</td></tr><tr><td>DN</td><td>DOWN</td><td>RCHS</td><td>CHILLER, SCREW</td></tr><tr><td>DP</td><td>DIFFERENTIAL PRESSURE</td><td>RPBFP</td><td>REDUCED PRESSURE BACKFLOW PREVENTER</td></tr><tr><td>DRG</td><td>DIFFUSER, REGISTER, & GRILLE</td><td>RRV</td><td>REFRIGERANT RELIEF VENT</td></tr><tr><td>(E)</td><td>EXISTING-TO-REMAIN</td><td>SCH</td><td>SCHEDULE</td></tr><tr><td>EA</td><td>EXHAUST AIR</td><td>SA</td><td>SUPPLY AIR</td></tr><tr><td>EAT</td><td>ENTERING AIR TEMPERATURE</td><td>SD</td><td>SMOKE DETECTOR</td></tr><tr><td>ELEC</td><td>ELECTRICAL</td><td>SDE</td><td>SEPARATOR, AIR</td></tr><tr><td>EMT</td><td>ELECTRICAL METAL TUBING</td><td>SF</td><td>SUPPLY FAN</td></tr><tr><td>ESP</td><td>EXTERNAL STATIC PRESSURE</td><td>SP</td><td>STATIC PRESSURE</td></tr><tr><td>ETR</td><td>EXISTING-TO-REMAIN</td><td>SPEC</td><td>SPECIFICATION</td></tr><tr><td>EWT</td><td>ENTERING WATER TEMPERATURE</td><td>SR</td><td>SPRING-RETURN</td></tr><tr><td>EXIST</td><td>EXISTING</td><td>(T)</td><td>TEMPORARY (DURING CONSTRUCTION ONLY)</td></tr><tr><td>FACP</td><td>FIRE-ALARM CONTROL PANEL</td><td>TA</td><td>TRANSFER AIR</td></tr><tr><td>FA</td><td>FIRE ALARM CONTROL SYSTEM LOOP</td><td>TCA</td><td>COMPRESSED AIR TANK</td></tr><tr><td>FC</td><td>FAIL-CLOSED</td><td>TCHW</td><td>CHILLED WATER TANK, BUFFER</td></tr><tr><td>FE</td><td>FAN, EXHAUST</td><td>TCH</td><td>CHEMICAL TANK</td></tr><tr><td>FFE</td><td>FILTERED-FAN, EXHAUST</td><td>TCHWX</td><td>EXPANSION TANK, CHILLED WATER</td></tr><tr><td>FHE</td><td>HOOD EXHAUST FAN</td><td>TCP</td><td>TEMPERATURE CONTROL PANEL</td></tr><tr><td>FIP</td><td>FAIL-IN-PLACE</td><td>TEMP</td><td>TEMPERATURE</td></tr><tr><td>FM</td><td>FLOW METER</td><td>VAV</td><td>VARIABLE AIR VOLUME</td></tr><tr><td>FO</td><td>FAIL-OPEN</td><td>VFD</td><td>VARIABLE FREQUENCY DRIVE</td></tr><tr><td>FPI</td><td>FINS PER INCH</td><td>W.C.</td><td>WATER COLUMN</td></tr><tr><td>FR</td><td>FAN, RETURN</td><td>W.G.</td><td>WATER GAUGE</td></tr><tr><td>FS</td><td>FLOW SWITCH</td><td>WB</td><td>WET BULB</td></tr><tr><td>FSL</td><td>FIRE SPRINKLER LINE</td><td>XMFR</td><td>TRANSFORMER</td></tr><tr><td>FT</td><td>FEET</td><td></td><td></td></tr><tr><td>FT.</td><td>FEET</td><td></td><td></td></tr><tr><td>HP</td><td>HORSEPOWER</td><td></td><td></td></tr><tr><td>I/O</td><td>HARD-WIRED INPUT/OUTPUT</td><td></td><td></td></tr><tr><td>ISO</td><td>ISOLATION</td><td></td><td></td></tr><tr><td>IN.</td><td>INCH</td><td></td><td></td></tr></table>		24X12	RECT. 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SEE DRAWINGS FOR SPECIFIC SYMBOLS AND ABBREVIATIONS. - FURNISH AND INSTALL ALL MATERIALS, VALVES, HANGERS, EQUIPMENT, ETC. AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE MECHANICAL SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED, AS REQUIRED BY CODE, AND AS REQUIRED BY STANDARDS. - INSTALL ALL MECHANICAL EQUIPMENT AND APPURTENANCES IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS, CONTRACT DOCUMENTS, STANDARDS, AND APPLICABLE CODES AND REGULATIONS. - COORDINATE CONSTRUCTION OF ALL MECHANICAL WORK WITH OTHER TRADES SHOWN IN CONTRACT DOCUMENT DRAWINGS. - MAINTAIN A MINIMUM OF 6'-8" CLEARANCE TO UNDERSIDE OF PIPES, DUCTWORKS, CONDUITS, ETC. THROUGHOUT ACCESS ROUTES. - LOCATE ALL MECHANICAL EQUIPMENT FOR UNOBSTRUCTED ACCESS TO UNIT ACCESS PANELS, CONTROLS, AND VALVING. MAINTAIN 4'-0" CLEAR DIRECTLY IN FRONT OF EQUIPMENT ELECTRICAL PANELS. - ALL ELEVATIONS ARE ABOVE FINISHED FLOOR TO BOTTOM OF DUCT, DUCT INSULATION, PIPE, OR PIPE INSULATION UNLESS NOTED OTHERWISE. - DUCT DIMENSIONS INDICATED REFER TO SHEET METAL DIMENSIONS. AIR OPENING SIZE AND SHEET METAL SIZE MUST BE THE SAME. - INSTALL WALL-MOUNTED TEMPERATURE CONTROLS AT 48" ABOVE FINISHED FLOOR UNLESS OTHERWISE INDICATED. COORDINATE INSTALLATION WITH OTHER WALL-MOUNTED DEVICES TO INSTALL ALL DEVICES AT SAME ELEVATION. COORDINATE WITH ARCHITECTURAL WALL FINISHES. - ALL EXISTING HVAC COMPONENTS SHOWN ON DRAWINGS ARE SCHEMATIC AND ARE BASED ON SITE SURVEYS AND AVAILABLE AS-BUILT DRAWINGS AND MAY NOT REFLECT EXACT EXISTING CONDITIONS. SUBCONTRACTOR MUST FIELD VERIFY EXISTING CONDITIONS ON JOB SITE. SUBCONTRACTOR MUST REROUTE NEW WORK TO ACCOMMODATE EXISTING CONDITIONS. - DESIGN DRAWINGS SHOW THE SUGGESTED EQUIPMENT LAYOUT. VARIATIONS IN EQUIPMENT LOCATIONS ARE ACCEPTABLE TO ACCOMMODATE EXISTING CONDITIONS PROVIDED THAT THE DESIGN INTENT IS MET, PROPER CLEARANCE AND ACCESS IS PROVIDED AROUND THE EQUIPMENT, AND SIGNIFICANT DEVIATIONS FROM THE DESIGN DRAWINGS HAVE BEEN APPROVED BY THE ENGINEER - ALL NEW PENETRATIONS THROUGH EXISTING WALLS AND FLOORS MUST BE NEATLY CORE-DRILLED TO MINIMIZE OVERSIZED AND SPALLING. - SUBCONTRACTOR MUST INSTALL NEW DUCTWORK CLEAR OF FIRE SPRINKLER HEADS, NEW AND EXISTING, AS REQUIRED BY NFPA 13 STANDARDS. - ALL NEW MECHANICAL SYSTEM INSTALLATIONS MUST NOT INTERFERE WITH EXISTING EQUIPMENT, ACCESS TO CONTROL AND ACCESS FOR MAINTENANCE. - WHEN DUCTWORK AND PIPING INSULATION SHOWS EVIDENCE OF "WET" INSULATION IMMEDIATELY AFTER INSTALLATION, THE DAMAGED INSULATION MUST BE REMOVED, THE PIPE AND/OR DUCT SURFACES MUST BE CLEANED AND DRIED, AND NEW INSULATION MUST BE INSTALLED. - MAKE TEMPORARY ALTERATIONS AND CONNECTIONS AS REQUIRED TO EXECUTE WORK SO THAT ALL SERVICES IN THE BUILDING REMAIN OPERATIONAL WITH MINIMUM POSSIBLE INTERRUPTION. TEMPORARY SHUTDOWNS MUST BE SEGREGATED AND MUST BE OF THE SHORTEST DURATION FEASIBLE. COORDINATE SHUTDOWN SCHEDULE. ALL SERVICES MUST BE KEPT IN CONTINUOUS OPERATION UNLESS PERMISSION IS OTHERWISE GRANTED - ALL NEW MECHANICAL EQUIPMENT WHETHER PURCHASED, MANUFACTURED ONSITE, RENTED, OR BORROWED, THAT MAY PRESENT A SIGNIFICANT RISK TO PERSONS OR PROPERTY SHALL BE "LISTED" BY A NATIONALLY-RECOGNIZED TESTING LABORATORY (NRTL), APPROVED FOR USE BY AN NRTL FIELD EVALUATION ENGINEER, OR APPROVED FOR USE BY AN AUTHORITY HAVING JURISDICTION (AHJ) FIELD REPRESENTATIVE AND LABELED PRIOR TO USE. - ALL FIREWALL PENETRATIONS MUST BE MADE WITH A UL LISTED AND APPROVED FIREWALL PENETRATION SYSTEM/ASSEMBLY. - PRIOR TO ANY DEMOLITION ACTIVITIES, SUBCONTRACTOR MUST COORDINATE TO REVIEW FACILITY ES&H DOCUMENTS, EQUIPMENT TO BE REMOVED, MATERIALS TO BE REMOVED OR DISTURBED, AND IDENTIFY A PLAN PRIOR TO COMMENCING THE WORK. ITEMS TO REVIEW INCLUDE BUT ARE NOT LIMITED TO THE FOLLOWING: NEW WORK IN DISTURBING EXISTING WALLS, FLOORS, ROOFS, OR CEILINGS; ALL CUTTING OR DRILLING; ALL DEMOLITION; REMOVAL AND DISPOSAL. - SURFACE-MOUNTED WIRING/CONDUIT IN ANY ROOM OTHER THAN A MECHANICAL OR ELECTRICAL ROOM IS PROHIBITED UNLESS OTHERWISE SPECIFIED OR APPROVED BY ENGINEER. - CONTROLS SUBCONTRACTOR MUST FURNISH AND INSTALL CONTROLLER ENCLOSURES. REFER TO STRUCTURAL AND ELECTRICAL DRAWINGS FOR CONTROL PANEL INSTALLATION DETAILS. - WHERE ITEMS ARE REMOVED FROM WALLS, SUBCONTRACTOR MUST PATCH AND PAINT WALLS AS NECESSARY TO MATCH SURROUNDING FINISH UNLESS THE AREA IS SMALL AND CAN BE COVERED BY A SMALL WALL COVER PLATE (I.E. THERMOSTATS). FOR LARGER ITEMS (I.E. CONTROL PANELS), WALL MUST BE PATCHED AND PAINTED AS NECESSARY TO MATCH SURROUNDING FINISH. - WORKING IN PROJECT AREAS MAY REQUIRE AN ESCORT. IT IS THE RESPONSIBILITY OF THE MECHANICAL SUBCONTRACTOR TO DETERMINE THESE PROTOCOLS PRIOR TO BID AND INCLUDE ANY COSTS ASSOCIATED WITH COMPLYING WITH THESE PROTOCOLS IN THEIR BID. - DELEGATED DESIGN: FIRE SUPPRESSION SYSTEM NEW WORK INCLUDING FIRE SPRINKLER PIPING, HEADS, VALVES, HANGERS, SEISMIC RESTRAINT BRACKETS, DRAINS, ESCUTCHEONS, WALL PLATES AND TEST FACILITIES NECESSARY FOR A FULLY FUNCTIONING WET-PIPE AUTOMATIC SPRINKLER SYSTEM SHALL BE PROVIDED BY THE DELEGATED DESIGN SUBCONTRACTOR. WORK SHOWN ON DRAWINGS IS ESTIMATE OF FIRE SUPPRESSION SCOPE OF WORK. PROVIDE A COMPLETE DESIGN FOR AFFECTED AREAS; DEVIATION FROM DESIGN DRAWING SPRINKLER SYSTEM LAYOUT IS PERMITTED AS REQUIRED TO PROVIDE A COMPLETE AND FUNCTIONING FIRE SUPPRESSION SYSTEM. DEVIATIONS TO SPRINKLER MAIN AND SPRINKLER HEADS SHALL BE COORDINATED WITH ELECTRICAL, MECHANICAL, STRUCTURAL, AND ARCHITECTURAL CONSTRUCTION DOCUMENTS PRIOR TO SUBMISSION. DELEGATED DESIGN DRAWINGS SHALL BE SUBMITTED TO FIRE PROTECTION ENGINEERING AND BURNS & MCDONNELL CONSTRUCTION MANAGER FOR APPROVAL BEFORE PROCEEDING WITH AFFECTED WORK. REFER TO 21 00 00. - DEMOLITION AND NEW CONSTRUCTION OF FIRE SPRINKLER PIPING MUST BE PERFORMED BY A C16 LICENSED SUBCONTRACTOR. NOTIFY AND SUBMIT PERMIT REQUEST A MINIMUM OF TWO WEEKS BEFORE FIRE SPRINKLER DEMOLITION ACTIVITIES. PERFORM LOCAL IN-SERVICE PRESSURE TESTING ON NEW FIRE SPRINKLER PIPING AND SPRINKLER HEADS. - CLEAN AND FLUSH ALL NEW PIPING SYSTEMS IN ACCORDANCE WITH SPEC. 220111.13. - INSTALL AUTOMATIC AIR VENTS IN HYDRONIC PIPING SYSTEMS AT LOCATIONS INDICATED, AT ALL HIGH POINTS, AND ELSEWHERE AS NECESSARY TO ELIMINATE TRAPPED AIR. EXTEND VENT FROM AUTOMATIC AIR VENT TO NEAREST FLOOR DRAIN OR SOAK PIT. - SUBCONTRACTOR MUST SAMPLE CHILLED WATER FOR CHEMICAL TREATMENT ANALYSIS PRIOR TO CONSTRUCTION START. FORWARD REPORT AND ANY RECOMMENDATIONS FOR RESPONSE. - EQUIPMENT WITH 50 POUNDS OR MORE OF REFERIGERANT MUST BE TRACKED, INVENTORIED, AND MANAGED. COORDINATE DEMOLITION AND INSTALLATION REQUIREMENTS WITH CONTRUCTION MANAGER
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Sheet Title

MECHANICAL
ABBREVIATIONS, LEGEND,
SYMBOLS AND GENERAL
NOTES

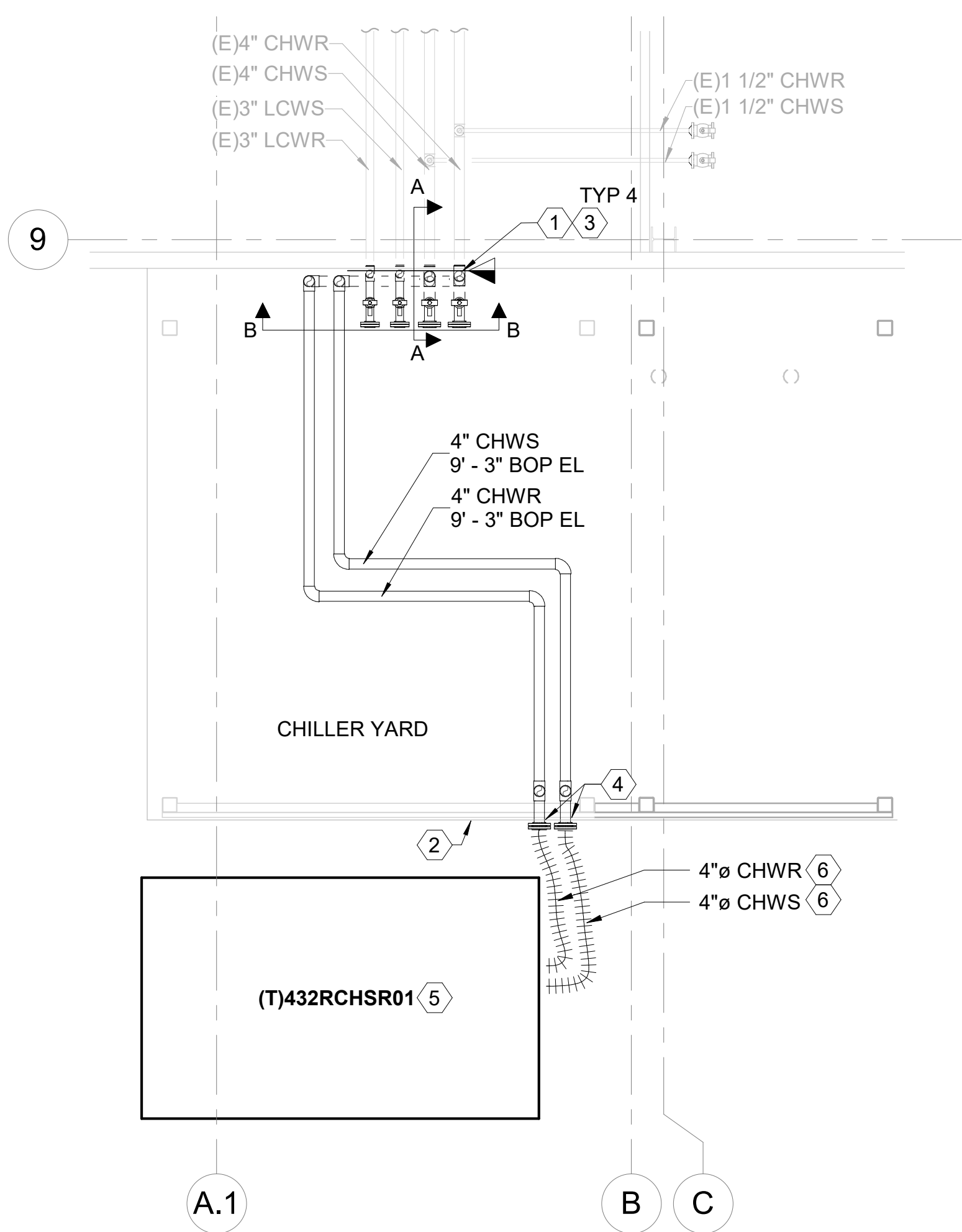
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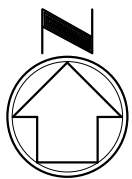
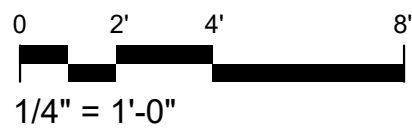
D1 LCW AND CHW PIPING SECTION VIEWS

M-002 SCALE: NO SCALE SHOWN



D5 TEMPORARY CHILLER PLAN

M-002 SCALE: 1/4" = 1'-0"



GENERAL NOTES

1. REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
2. TEMPORARY CHILLER WORK MUST BE SEQUENCED (RECOMMENDED BY KEYED NOTE ORDER) AND COMPLETED SUCH THAT BUILDING SERVICE IS NOT INTERRUPTED FOR MORE THAN 48 HOURS.
3. REMOVE AND DISPOSE OF ALL EXISTING EXTERIOR CHW AND LCW PIPING, FITTINGS AND APPURTENANCES SERVING EXISTING CHILLER AS REQUIRED TO PERFORM NEW WORK.
4. EXISTING CHILLED WATER MAKEUP WATER CONNECTION, PRV, BACKFLOW PREVENTER, AND EXPANSION TANK MUST REMAIN CONNECTED TO CHILLED WATER SYSTEM WHILE TEMPORARY CHILLER IS IN OPERATION.
5. SUBCONTRACTOR MUST SAMPLE CHILLED WATER FOR CHEMICAL TREATMENT ANALYSIS PRIOR TO CONSTRUCTION START. FORWARD REPORT AND ANY RECOMMENDATIONS FOR RESPONSE.
6. PROVIDE LOCKOUT-TAGOUT PLAN FOR APPROVAL PRIOR TO DEMOLITION START.

KEYED NOTES

- 1 EXISTING LCW AND CHW PIPING PENETRATIONS THROUGH EXTERIOR WALL TO REMAIN.
- 2 EXISTING CHILLER SHELTER WALL TO REMAIN.
- 3 INSTALL SELECT CHWS, CHWR, LCWS, & LCWR PIPING AND LINE-SIZE ISOLATION VALVES TO EXTENTS INDICATED IN D1 / M-002 (SEE KEYED NOTES 7 THROUGH 13) WITHIN FIRST SYSTEM SHUTDOWN AT BEGINNING OF CONSTRUCTION PROJECT. ALL WORK REQUIRING CHILLED WATER SYSTEM SHUTDOWN MUST BE ACCOMPLISHED IN THIS OUTAGE. REFER TO M-413 FOR ADDITIONAL WORK REQUIRED DURING FIRST SHUTDOWN. WORK SHALL BE ACCOMPLISHED WITHOUT DRAINING FULL SYSTEM. FREEZING PIPE OR USING HOT TAP PIPE PLUG INSERTION SOLUTION IS ACCEPTABLE.
- 4 EXTEND NEW CHWS AND CHWR PIPING THROUGH EXISTING CHILLER SHELTER WALL TO TEMPORARY CHILLER CONNECTIONS WITH BLIND FLANGED TERMINATIONS AT 12" ABOVE GRADE. APPLY SEALANT AROUND NEW PIPE PENETRATIONS.
- 5 FURNISH, INSTALL, AND STARTUP TEMPORARY 80 TON (NOMINAL) AIR-COOLED CHILLER (WITH INTEGRAL PUMP PACKAGE). RETURN RENTAL CHILLER ONLY AFTER OWNER AND ENGINEER ACCEPTANCE OF NEW WATER-COOLED CHILLER AND CHILLED WATER PUMPS.
- 6 FURNISH AND INSTALL FLEXIBLE 4" CHWS AND CHWR HOSES TO TEMPORARY CHILLER.

KEYED NOTES CONTINUED; DETAIL D1 ONLY

- 7 FURNISH AND INSTALL (2) NEW 4"ø PIPING TEES FOR CHWS AND CHWR CONNECTIONS TO TEMPORARY CHILLER AND NEW CHILLER.
- 8 FURNISH AND INSTALL NEW 4"ø CHW ISOLATION BUTTERFLY VALVE.
- 9 FURNISH AND INSTALL NEW 3"ø LCW ISOLATION BUTTERFLY VALVE.
- 10 FURNISH AND INSTALL TEMPORARY BLIND FLANGE CONNECTION.
- 11 FOR PERMANENT LCW CONNECTIONS TO CHILLER.
- 12 BUILDING EXTERIOR.
- 13 EXISTING STEEL CHILLER YARD STRUCTURE.

KEY PLAN

Sheet Title

MECHANICAL TEMPORARY CHILLER PLAN

Dwg. No.

PLM2021-0432-0002D

Sht. No.

M-002

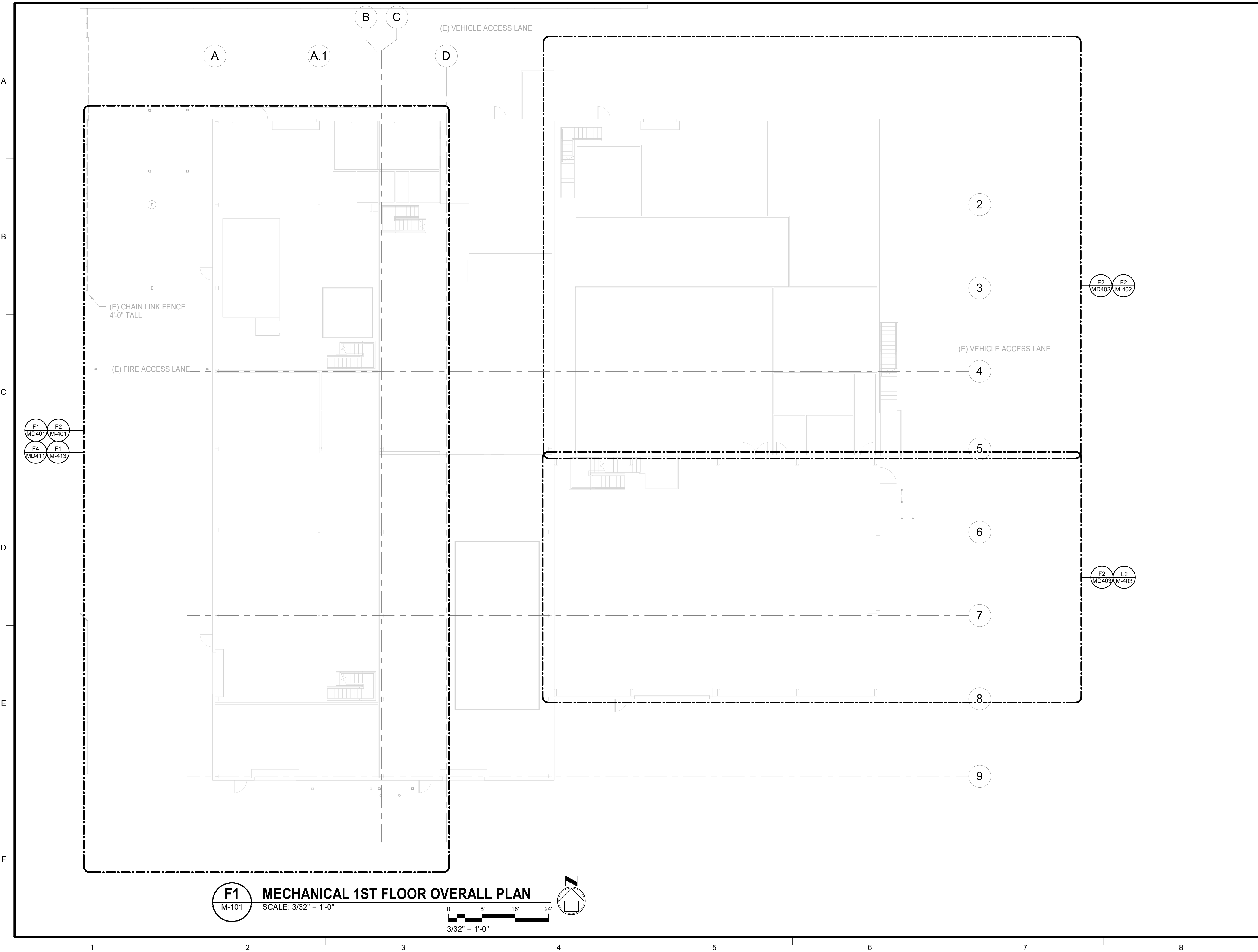
HYDRONIC PIPING MATERIAL SCHEDULE			
PIPING SYSTEM	SIZE	MATERIAL SPECIFICATION	SHUTOFF VALVE
CHILLED WATER SUPPLY & RETURN	2" AND SMALLER	COPPER, TYPE L, ASTM B88, SOLDERED JOINTS	BALL VALVE, BRONZE ALLOY BODY, THREE PIECE FULL PORT, SOLDER ENDS
	2-1/2" - 3"	COPPER, TYPE L, ASTM B88, BRAZED JOINTS	BUTTERFLY VALVE, CARBON STEEL BODY, CLASS 150 FLANGE OR LUG STYLE
	4" AND LARGER	CARBON STEEL, SCHEDULE 40, ASTM A53 GRADE B, WELDED JOINTS	
LOW CONDUCTIVITY WATER SUPPLY & RETURN	2" AND SMALLER	COPPER, TYPE L, ASTM B88, SOLDERED JOINTS	BALL VALVE, BRONZE ALLOY BODY, THREE PIECE FULL PORT, SOLDER ENDS
	2-1/2" - 3"	COPPER, TYPE L, ASTM B88, BRAZED JOINTS	BUTTERFLY VALVE, CARBON STEEL BODY, CLASS 150 FLANGE OR LUG STYLE
	4" AND LARGER	CARBON STEEL, SCHEDULE 40, ASTM A53 GRADE B, WELDED JOINTS	
HEATING HOT WATER SUPPLY & RETURN	2" AND SMALLER	COPPER, TYPE L, ASTM B88, SOLDERED JOINTS	BALL VALVE, BRONZE ALLOY BODY, THREE PIECE FULL PORT, SOLDER ENDS
	2-1/2" - 3"	COPPER, TYPE L, ASTM B88, BRAZED JOINTS	BUTTERFLY VALVE, CARBON STEEL BODY, CLASS 150 FLANGE OR LUG STYLE

TEMPORARY CHILLER SCHEDULE (RCHSR)

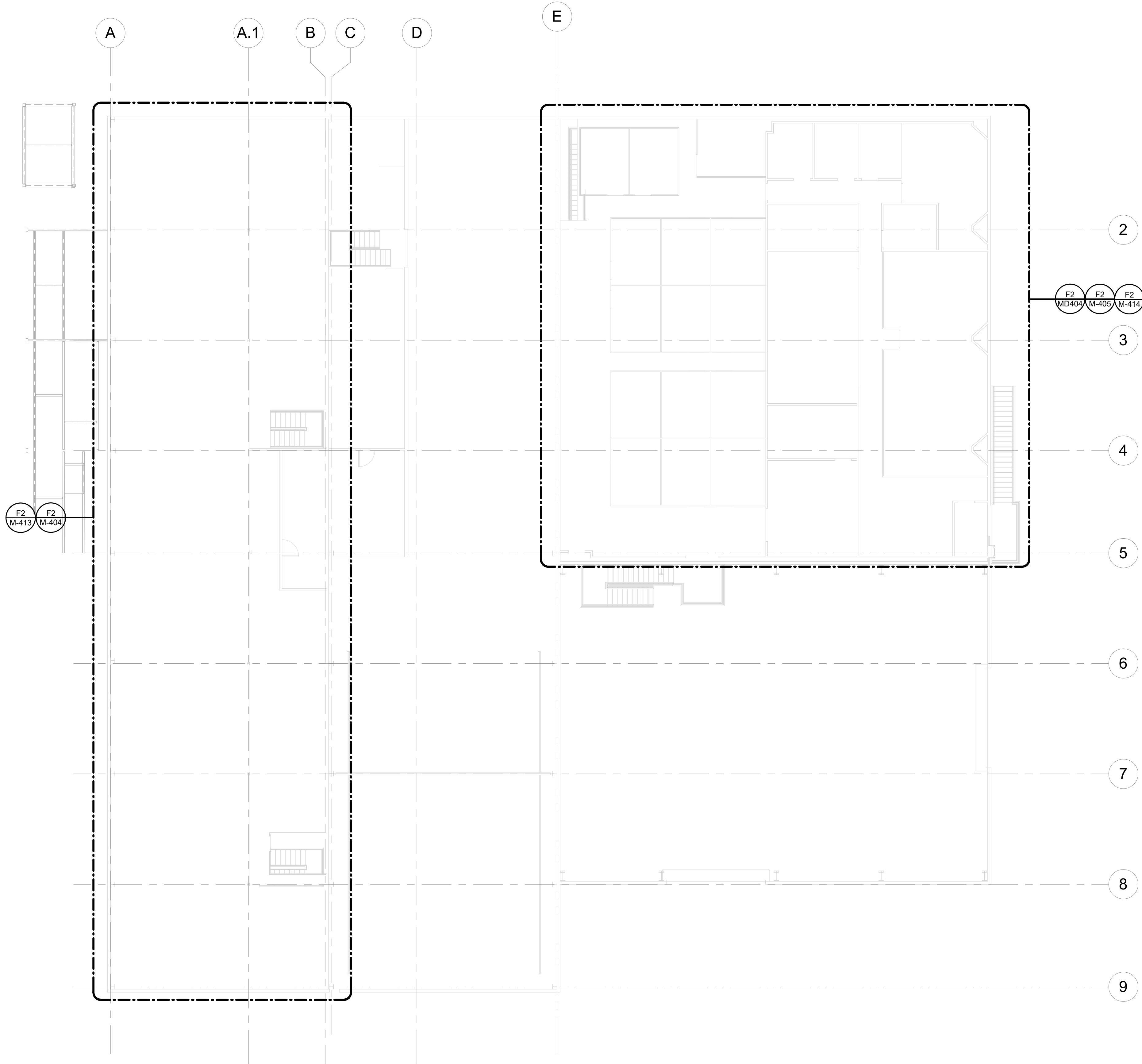
TAG NO.	NOMINAL COOLING CAPACITY (TONS)	EVAPORATOR				COMPRESSOR			ELECTRICAL				INSTALLED WEIGHT (LBS.)	BASIS OF DESIGN		NOTES
		FLUID TYPE	DESIGN FLOW (GPM)	EWT (DEG. F.)	LWT (DEG. F.)	TYPE	QUANTITY / CIRCUITS	REF TYPE	V	PH	MCA	MOCP		MFR	MODEL	
(T)432RCHSR01	80	WATER	192	55	45	SCROLL	1	R-134A	460	3	189	200	10300	TRANE	CGAM	ALL

NOTES:

1. UNIT IS TEMPORARY AND SHOULD BE USED ONLY TO PROVIDE CHILLED WATER BEFORE INSTALLATION AND STARTUP OF NEW CHILLER.
2. PROVIDE UNIT WITH INTEGRAL CIRCUIT BREAKER FOR OVERCURRENT PROTECTION AND DISCONNECTING MEANS.
3. PROVIDE UNIT WITH INTEGRAL PUMP PACKAGE.
4. PROVIDE UNIT WITH TRAILER AND TEMPORARY POWER CABLE AND HOSES OF SUFFICIENT LENGTH FOR INSTALLATION.



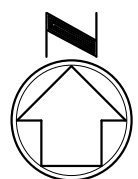
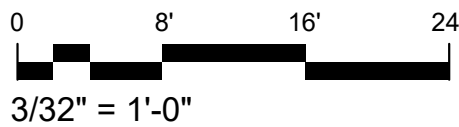
KEY PLAN	
Sheet Title	
MECHANICAL 1ST FLOOR OVERALL PLAN	
Dwg. No.	
PLM2021-0432-0003D	
Sht. No.	
M-101	



KEY PLAN

F2 MECHANICAL 2ND FLOOR OVERALL PLAN

M-102 SCALE: 3/32" = 1'-0"



Sheet Title

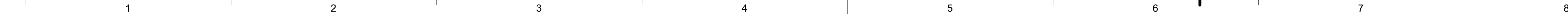
MECHANICAL 2ND FLOOR
OVERALL PLAN

Dwg. No.

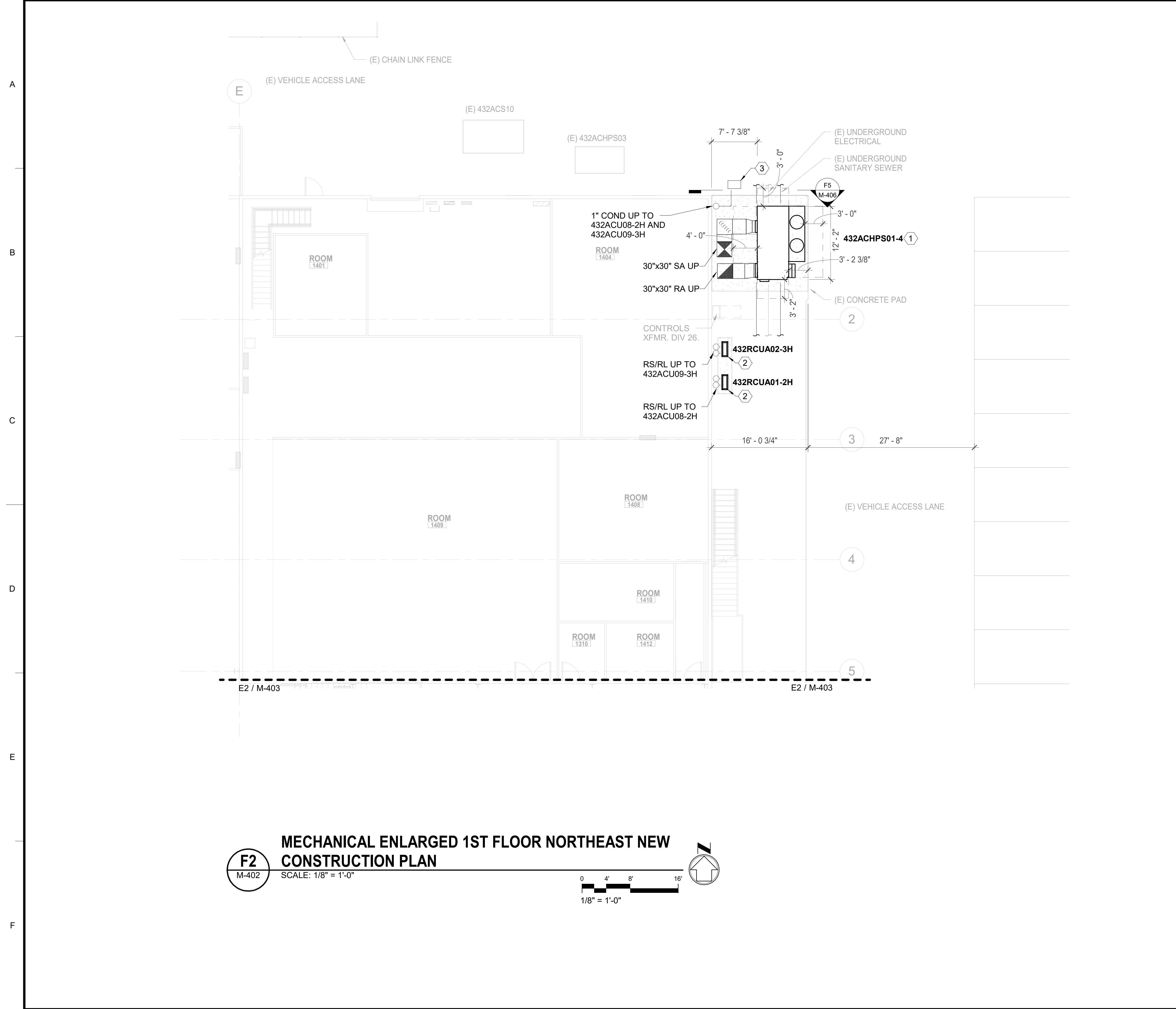
PLM2021-0432-0004D

Sht. No.

M-102



M-401



- GENERAL NOTES**
1. REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
 2. COMPONENTS SHOWN HALF-TONE ON THIS DRAWING ARE ITEMS TO REMAIN; BOLD COMPONENTS ARE NEW.
 3. DASHED REGIONS AROUND EQUIPMENT INDICATE CLEARANCE REQUIRED FOR EQUIPMENT ACCESS AND MAINTENANCE.
 4. REFER TO STRUCTURAL DRAWINGS FOR DUCT SUPPORT AND EQUIPMENT ANCHORAGE DETAILS.

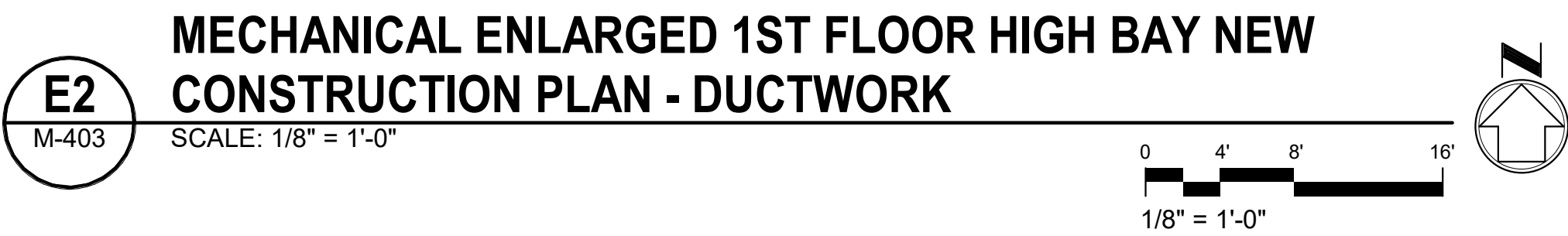
- KEYED NOTES**
- 1 FURNISH AND INSTALL NEW PACKAGED HEAT PUMP UNIT AT GRADE ON EXISTING CONCRETE PAD. REFER TO STRUCTURAL B1 / S-514 FOR UNIT ANCHORAGE TO EXISTING CONCRETE PAD AND B1 / S-515 FOR INSTALLATION DETAILS.
 - 2 FURNISH AND INSTALL NEW SPLIT SYSTEM HEAT PUMP CONDENSING UNIT AT GRADE WITH NEW CONCRETE HOUSEKEEPING PAD. REFER TO B1 / S-514 FOR UNIT ANCHORAGE. INSTALL TO MAINTAIN MFR'S RECOMMENDED UNIT CLEARANCES. REFER TO DETAIL D1 ON M-504. REFER TO M-414 FOR CONTINUATION OF REFRIGERANT PIPING TO INDOOR UNITS.
 - 3 FURNISH AND INSTALL NEW UNDERGROUND CONDENSATE SOAK PIT IN THIS LOCATION, REFER TO DETAIL D6 / M-505. EXTEND 432ACHPS01 CONDENSATE TO THIS LOCATION, REFER TO DETAIL D3 / M-505.

KEY PLAN

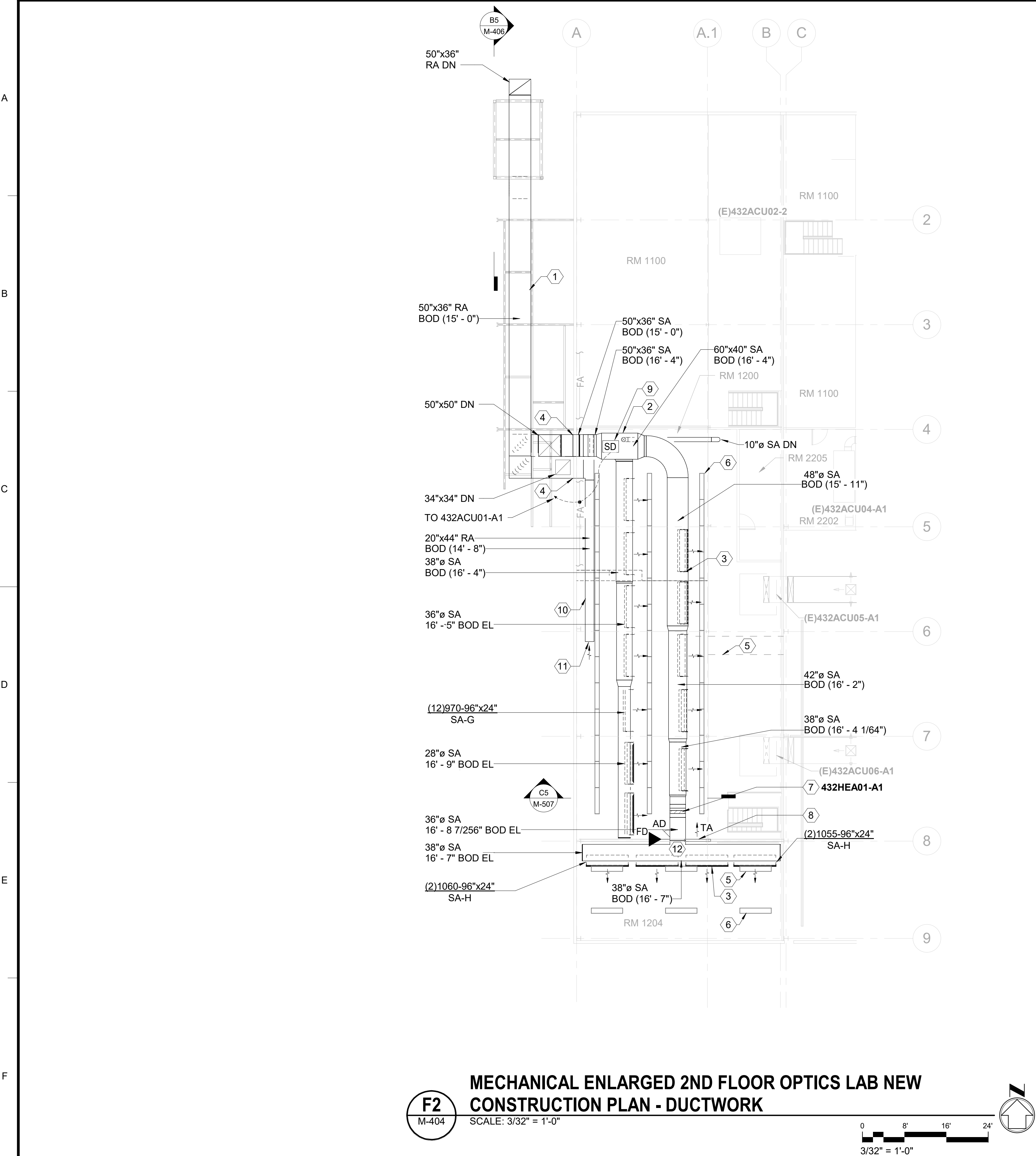
Sheet Title
MECHANICAL ENLARGED 1ST FLOOR NORTHEAST NEW CONSTRUCTION PLAN

Dwg. No.
PLM2021-0432-0013D

Sht. No.
M-402



Sht. No. M-403



- KEYED NOTES**
- 11 PROVIDE RETURN AIR OPENING WITH WIRE MESH BIRDSCREEN.
 - 12 REFER TO M-502 FOR FIRE DAMPER REQUIREMENTS.

- GENERAL NOTES**
- REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
 - COMPONENTS SHOWN HALF-TONE ON THIS DRAWING ARE ITEMS TO REMAIN; BOLD COMPONENTS ARE NEW.
 - REFER TO STRUCTURAL DRAWINGS FOR DUCT SUPPORT AND EQUIPMENT ANCHORAGE DETAILS.
 - COORDINATE INSTALLATION OF NEW DUCTWORK THROUGH OPTICS LAB AREA WITH EXISTING UTILITIES AND OBSTACLES, REROUTE DUCTWORK AS NECESSARY TO AVOID EXISTING UTILITIES AND OBSTACLES.
 - BOTTOM OF DUCT ELEVATIONS ON THIS DRAWING ARE RELATIVE TO 1ST FLOOR LAB FLOOR.
 - INSTALLATION OF NEW INTERIOR DUCTWORK AND AIR TERMINALS WILL REQUIRE A DETAILED WORK PLAN DEVELOPED WITH INPUT FROM CONSTRUCTION MANAGER, SUBCONTRACTORS, AND FACILITY END-USERS. SUBCONTRACTOR MUST REVIEW EXISTING CONDITIONS AND DISCUSS ACCEPTABLE MEANS AND METHODS FOR NEW INSTALLATION WITH CONSTRUCTION MANAGER PRIOR TO BID. USAGE CONTINUITY OF END-USER LAB EQUIPMENT IN PROJECT AREA WILL REQUIRE COMPLETE INSTALLATION OF NEW COMPONENTS OVER A LONG WEEKEND, INCLUDE COSTS TO ACCOMPLISH THIS IN BID. CONSTRUCTION MANAGER TO COORDINATE TEMPORARY LAB EQUIPMENT REMOVAL
 - INSTALL MANUAL VOLUME DAMPER AT EACH SUPPLY AIR DIFFUSER, IN BRANCH DUCTWORK IMMEDIATELY UPSTREAM OF DIFFUSER CONNECTION, AND ELSEWHERE AS INDICATED.

- KEYED NOTES**
- 1 SECURE NEW RETURN AIR DUCTWORK TO NEW DUCT SUPPORT FRAMES. ROUTE WITH SUPPLY DUCT, RETURN STACKED ON TOP OF SUPPLY. REFER TO S-403.
 - 2 ALL INDOOR SUPPLY DUCT CONNECTED TO 432ACU01-A1 MUST BE DOUBLE-WALL INSULATED DUCT.
 - 3 REFER TO DIFFUSER DETAIL C5 ON M-507, TYPICAL OF ALL NEW DIFFUSERS IN LAB AREAS.
 - 4 INSTALL SUPPLY AND RETURN AIR DUCTS TIGHT TOGETHER THROUGH WALL OPENING AND INSULATE AROUND SET OF DUCTS WITH NO GAP IN BETWEEN DUCTS. SEAL WALL OPENING AROUND NEW DUCTWORK. REFER TO A-501 FOR WALL SEAL REQUIREMENTS.
 - 5 DASHED LINE INDICATES LOCATION OF EXISTING HOIST LIFTING AREA AND PATH OF TRAVEL. ALL NEW MEP COMPONENTS MUST NOT INTERRUPT HOIST LIFTING AREA OR PATH OF TRAVEL.
 - 6 EXISTING LIGHT FIXTURE TO REMAIN AS IS, TYPICAL.
 - 7 FURNISH AND INSTALL NEW DUCT-MOUNTED ELECTRIC RESISTANCE HEATING COIL.
 - 8 EXISTING 60"x48" TRANSFER AIR GRILLE TO REMAIN AS IS.
 - 9 EXISTING, RE-USED, SD FURNISHED INSTALLED AND WIRED BY CONTROLS SUBCONTRACTOR. REFER TO MD401. WIRE DEVICE TO EXISTING ADDRESSABLE FIRE ALARM CONTROL LOOP AND WIRE FIRE ALARM CONTROL LOOP TO 432ACU01-A1 SUPPLY FAN MOTORS.
 - 10 INDOOR RETURN AIR DUCT IN OPTICS LAB TO BE UNINSULATED.

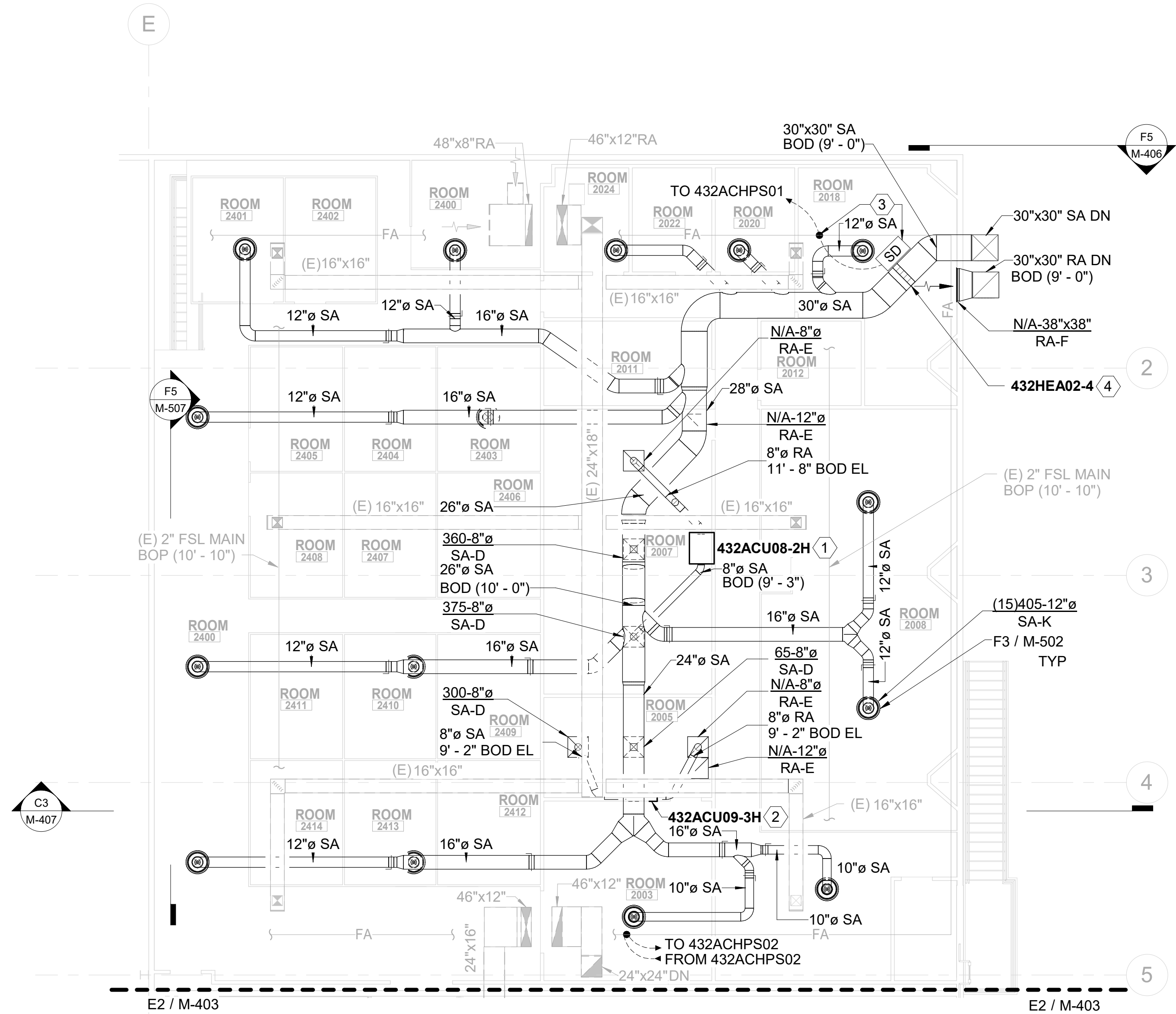
KEY PLAN

Sheet Title
MECHANICAL ENLARGED 2ND FLOOR OPTICS LAB NEW CONSTRUCTION PLAN - DUCTWORK

Dwg. No.
PLM2021-0432-0015D

Sht. No.
M-404

A
B
C
D
E
F



F2
M-405

MECHANICAL ENLARGED 2ND FLOOR OPEN OFFICE NEW CONSTRUCTION PLAN - DUCTWORK

SCALE: 1/8" = 1'-0"

0 4' 8' 16'

1/8" = 1'-0"

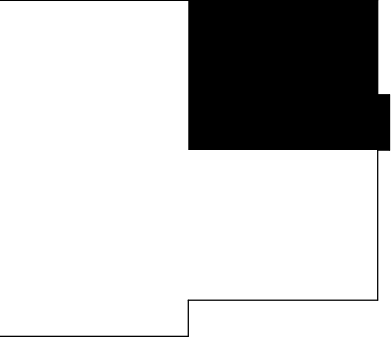
GENERAL NOTES

1. REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
2. COMPONENTS SHOWN HALF-TONE ON THIS DRAWING ARE ITEMS TO REMAIN; BOLD COMPONENTS ARE NEW. REFER TO MD404 FOR DEMOLITION OF EXISTING DUCTWORK AFTER NEW DUCTWORK IS INSTALLED.
3. DASHED REGIONS AROUND EQUIPMENT INDICATE CLEARANCE REQUIRED FOR EQUIPMENT ACCESS AND MAINTENANCE.
4. NEW DUCTWORK MUST BE INSTALLED PRIOR TO DEMOLITION OF EXISTING DUCTWORK TO LIMIT DURATION OF SYSTEM OUTAGE.
5. INSTALL MANUAL VOLUME DAMPER AT EACH SUPPLY AIR DIFFUSER, IN BRANCH DUCTWORK IMMEDIATELY UPSTREAM OF DIFFUSER CONNECTION, AND ELSEWHERE AS INDICATED.
6. REFER TO M-414 FOR SYSTEM THERMOSTATS.
7. EXISTING FIRE SPRINKLER PIPING TO REMAIN AS IS. NEW DUCTWORK MUST BE ROUTED TO AVOID CONFLICTS WITH EXISTING FIRE SPRINKLER PIPING.

KEYED NOTES

- ① FURNISH AND INSTALL NEW SPLIT SYSTEM DX FAN COIL UNIT. INTERLOCK WITH 432RCUA02-2H. INSTALL UNIT ABOVE EXISTING TO REMAIN CEILING, SUSPENDED FROM ROOF STRUCTURE. POSITION UNIT TO BE ACCESSIBLE FOR MAINTENANCE FROM CORRIDOR. REFER TO S-514.
- ② FURNISH AND INSTALL NEW SPLIT SYSTEM DX FAN COIL UNIT. INTERLOCK WITH 432RCUA03-3H. INSTALL UNIT ABOVE EXISTING TO REMAIN CEILING, SUSPENDED FROM ROOF STRUCTURE. POSITION UNIT TO BE ACCESSIBLE FOR MAINTENANCE FROM CORRIDOR. REFER TO S-514.
- ③ EXISTING, RE-USED, SD FURNISHED INSTALLED AND WIRED BY CONTROLS SUBCONTRACTOR. REFER TO MD404. WIRE DEVICE TO EXISTING ADDRESSABLE FIRE ALARM CONTROL LOOP AND WIRE FIRE ALARM CONTROL LOOP TO 432ACHPS01 SUPPLY FAN MOTORS.
- ④ FURNISH AND INSTALL NEW DUCT-MOUNTED ELECTRIC RESISTANCE HEATING COIL.

KEY PLAN



Sheet Title

MECHANICAL ENLARGED 2ND FLOOR OPEN OFFICE NEW CONSTRUCTION PLAN - DUCTWORK

Dwg. No.

PLM2021-0432-0016D

Sht. No.

M-405

A

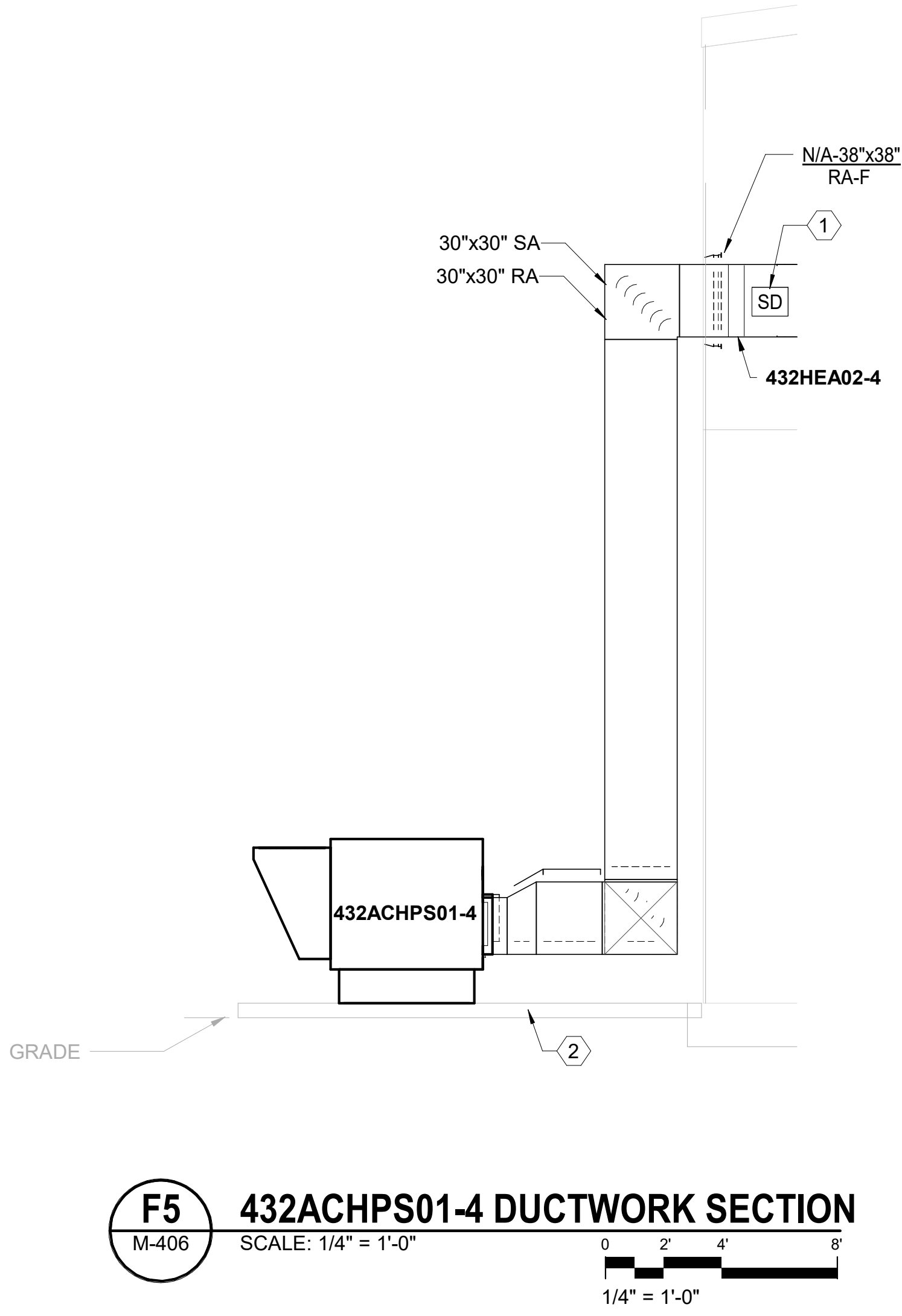
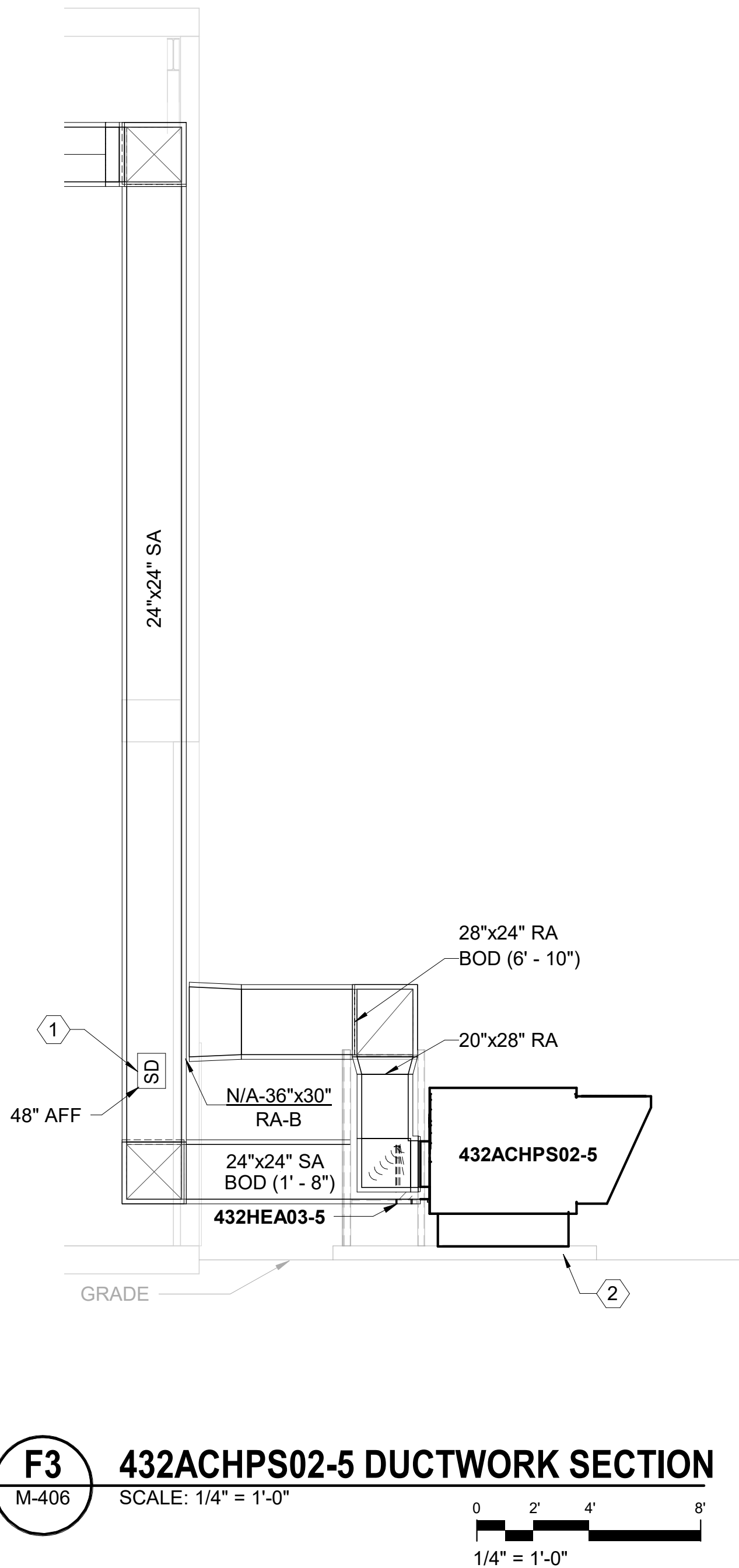
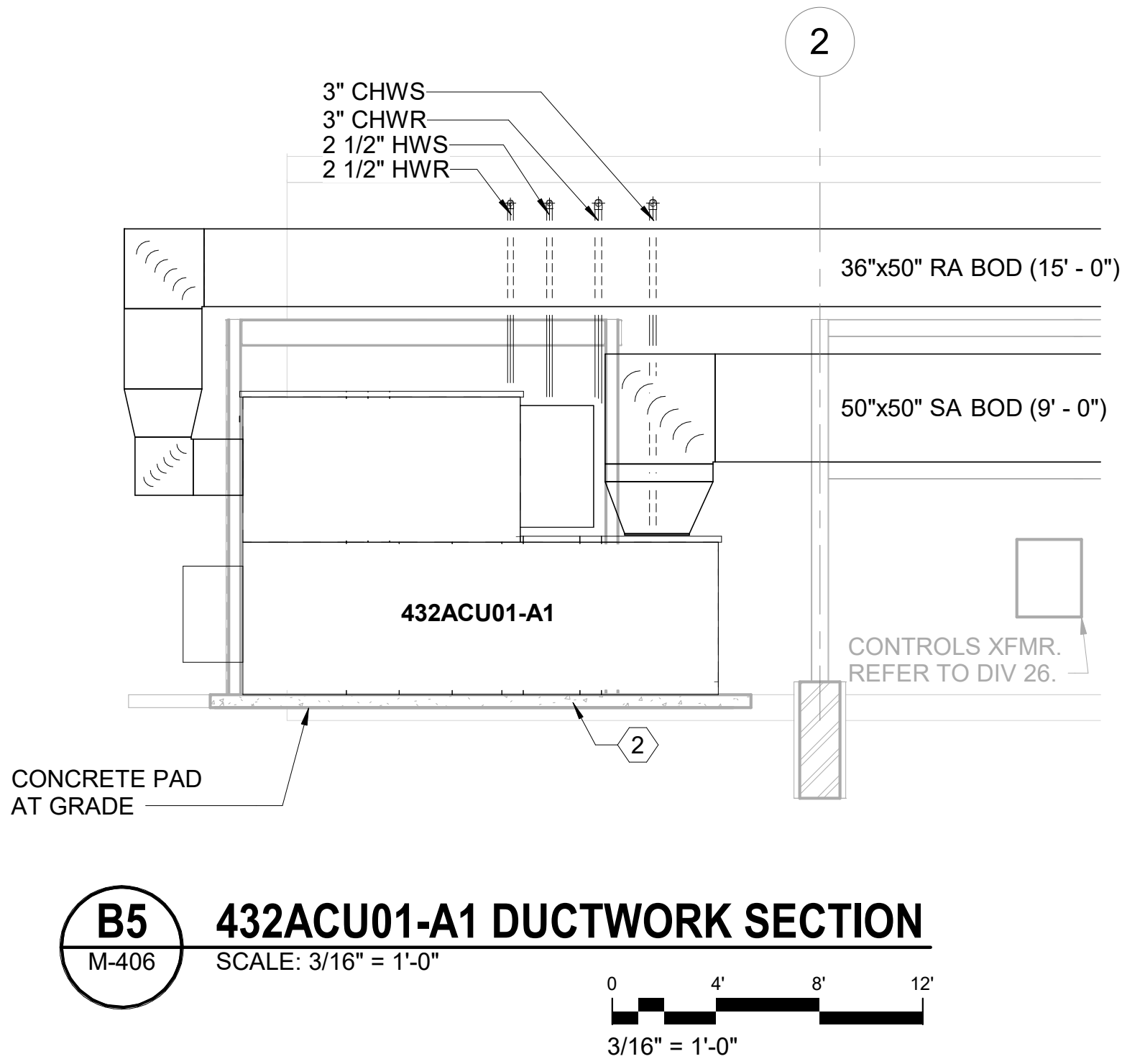
B

C

D

E

F



GENERAL NOTES

1. REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
2. DASHED REGIONS AROUND EQUIPMENT INDICATE CLEARANCE REQUIRED FOR EQUIPMENT ACCESS AND MAINTENANCE.
3. REFER TO STRUCTURAL DRAWINGS FOR DUCT SUPPORT AND EQUIPMENT ANCHORAGE DETAILS.

KEYED NOTES

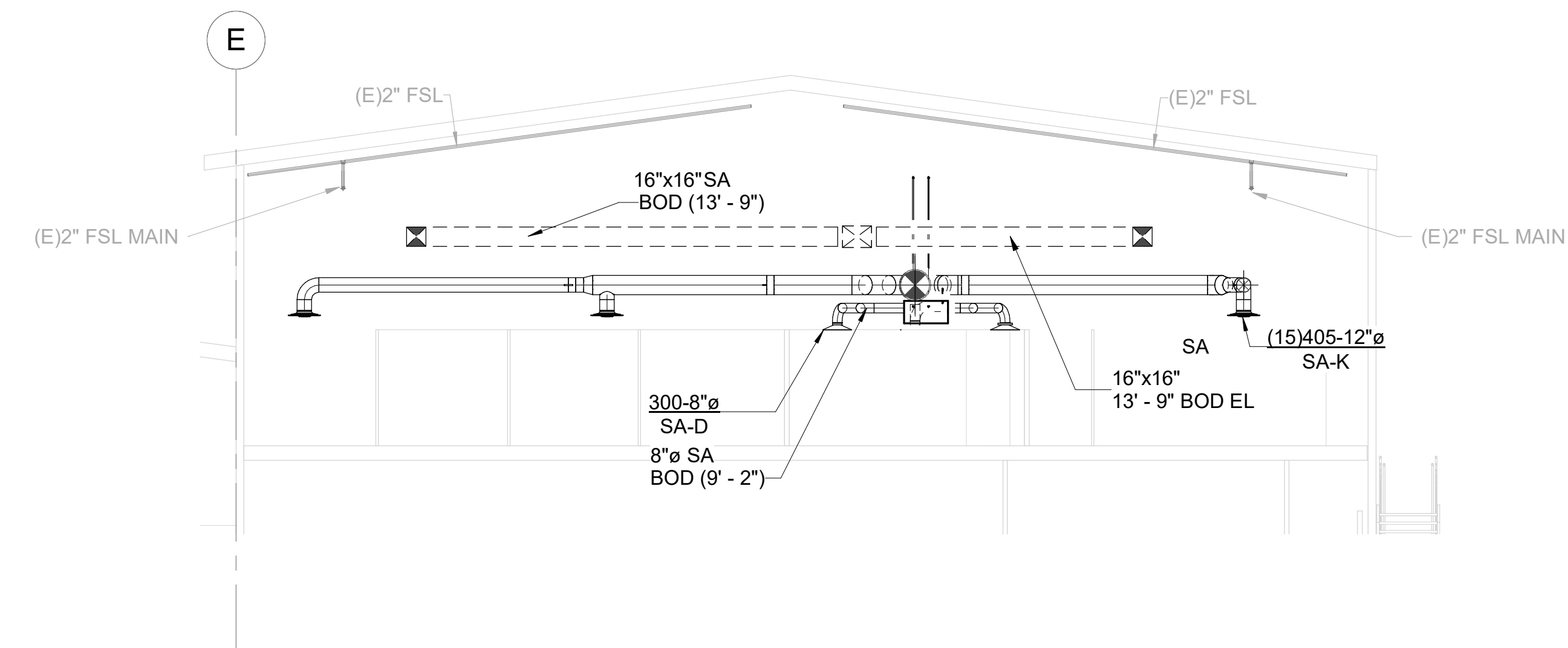
- ① EXISTING, RE-USED SMOKE DETECTOR DEVICE. INSTALLED AND WIRED BY CONTROLS SUBCONTRACTOR
- ② CONCRETE HOUSEKEEPING PAD. REFER TO S-402 FOR UNIT ANCHORAGE.

Sheet Title

**MECHANICAL ENLARGED
EQUIPMENT ELEVATIONS -
DUCTWORK**

Dwg. No.
PLM2021-0432-0017D

Sht. No.
M-406

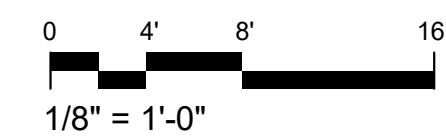


**MECHANICAL ENLARGED 2ND FLOOR NORTHEAST NEW
CONSTRUCTION SECTION (OPEN OFFICE)**

C3

M-407

SCALE: 1/8" = 1'-0"



GENERAL NOTES

1. REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
2. COMPONENTS SHOWN HALF-TONE ON THIS DRAWING ARE ITEMS EXISTING TO REMAIN; BOLD COMPONENTS ARE NEW; DASHED COMPONENTS ARE EXISTING TO BE DEMOLISHED.
3. DASHED REGIONS AROUND EQUIPMENT INDICATE CLEARANCE REQUIRED FOR EQUIPMENT ACCESS AND MAINTENANCE.
4. NEW DUCTWORK SHALL BE INSTALLED PRIOR TO DEMOLITION OF EXISTING DUCTWORK TO LIMIT DURATION OF SYSTEM OUTAGE. REFER TO M-407.
5. EXISTING FIRE SPRINKLER PIPING TO REMAIN AS IS. NEW DUCTWORK MUST BE ROUTED TO AVOID CONFLICTS WITH EXISTING FIRE SPRINKLER PIPING.

Sheet Title

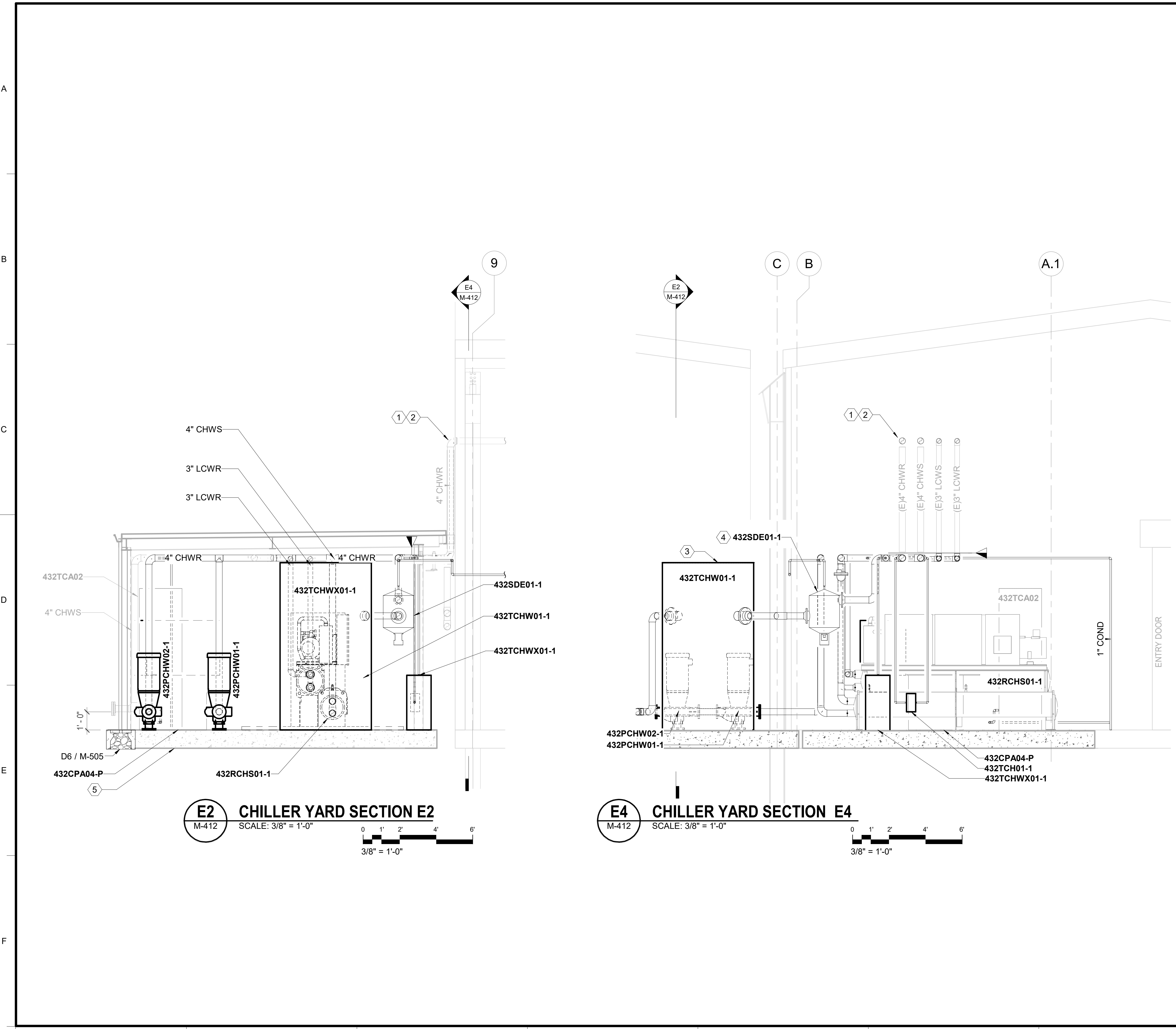
MECHANICAL ENLARGED ELEVATIONS

Dwg. No.

PLM2021-0432-0018D

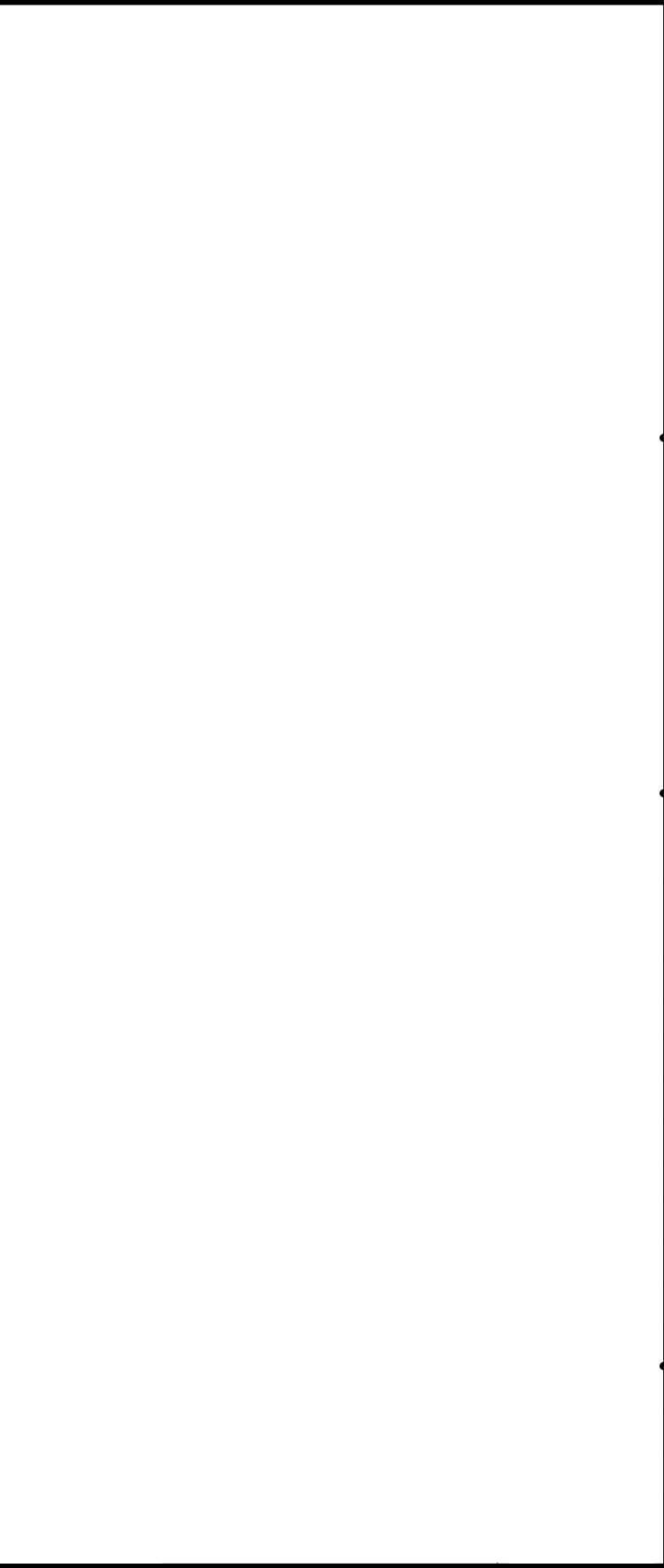
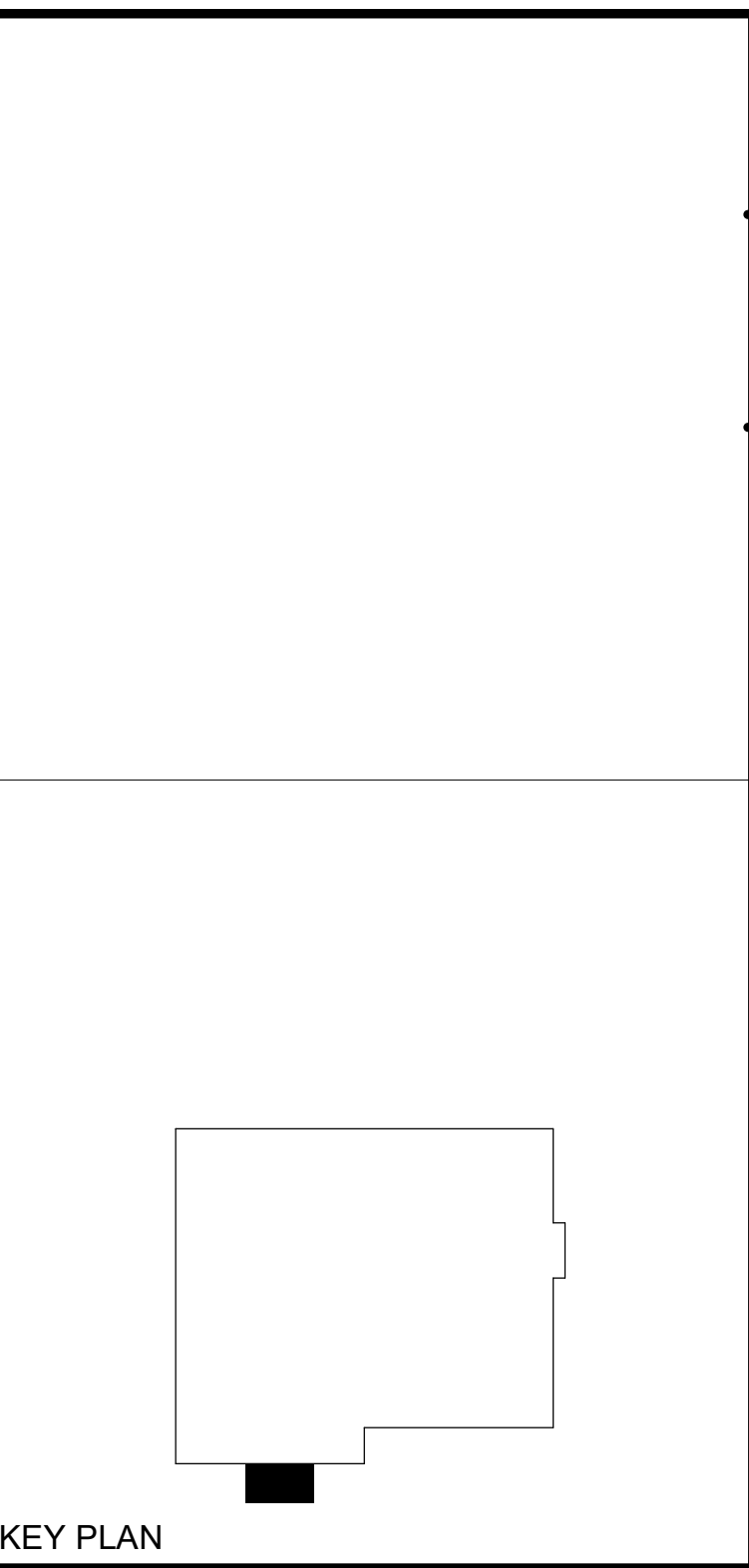
Sht. No.

M-407

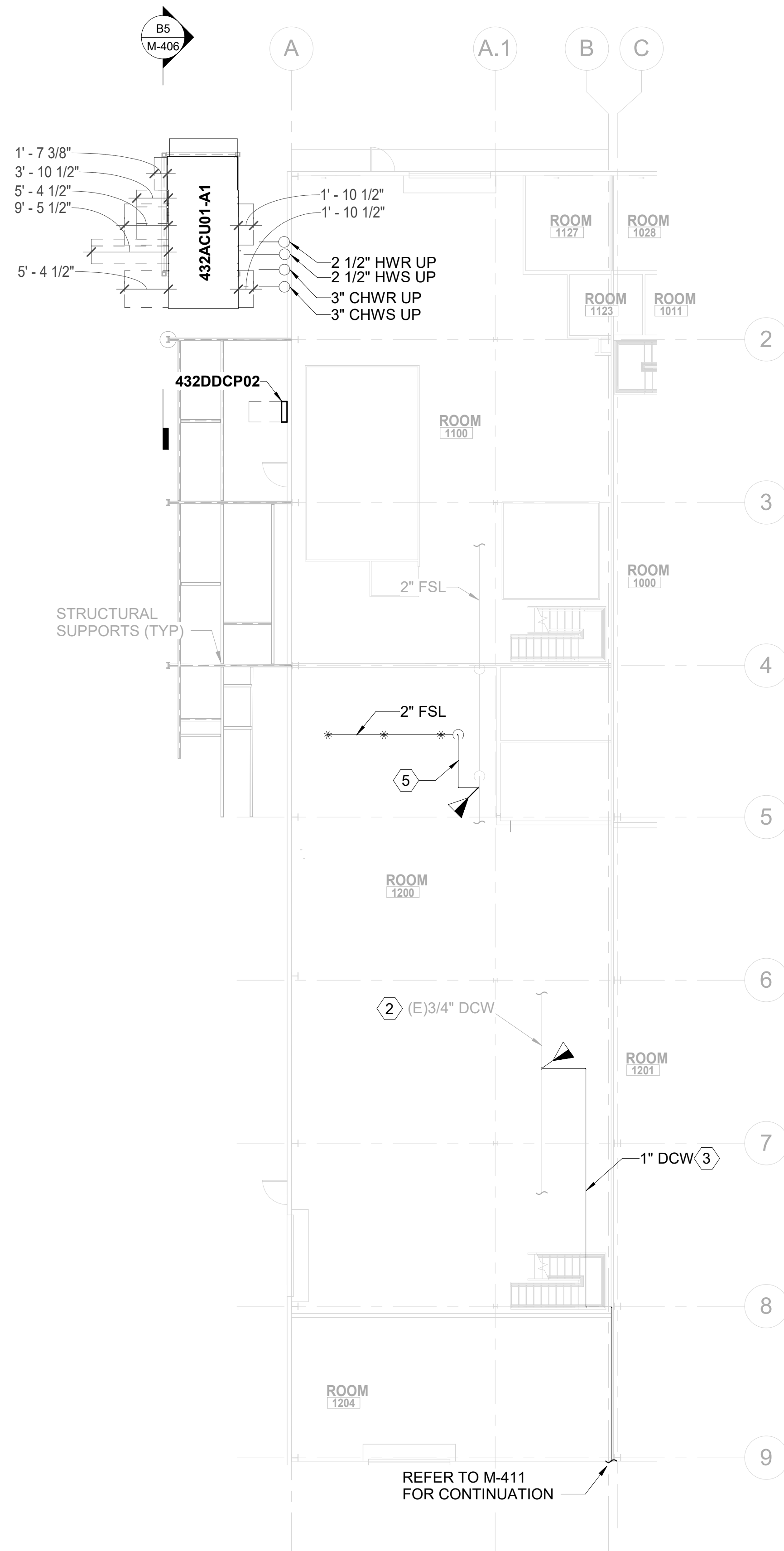


- GENERAL NOTES**
1. REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
 2. CROSS-REFERENCE THIS DRAWING WITH M-411 PLAN VIEW TO UNDERSTAND 3D ARRANGEMENT OF ELEMENTS WITHIN THIS AREA.

- KEYED NOTES**
1. INSTALL SELECT PIPING AND LINE-SIZE ISOLATION VALVES WITHIN FIRST SYSTEM SHUTDOWN AT BEGINNING OF CONSTRUCTION PROJECT. REFER TO M-002.
 2. EXISTING LCW AND CHW PIPING PENETRATIONS THROUGH EXTERIOR WALL TO REMAIN.
 3. FURNISH AND INSTALL BUFFER TANK 432TCHW01-1 WITH AUTOMATIC AIR VENT AND FULL PORT NORMALLY OPEN QUARTER TURN ISOLATION BALL VALVE. ROUTE CONDENSATE TO DRAIN.
 4. FURNISH AND INSTALL AIR AND DIRT SEPARATOR WITH FULL PORT NORMALLY OPEN QUARTER TURN ISOLATION BALL VALVE ON MAKE-UP WATER CONNECTION AND QUARTER TURN BLOW-DOWN BALL VALVE ON DRAIN CONNECTION WITH FLEX TUBING TERMINATING AT 0'-6" AFF. FURNISH AND INSTALL SEPARATE LABEL FOR DRAIN VALVE CONNECTION WITH TEXT: "DIRT CLEANOUT" ON PIPING.
 5. CONCRETE PAD. REFER TO S-401.



Sheet Title
MECHANICAL ENLARGED CHILLER YARD SECTIONS - PIPING
Dwg. No. PLM2021-0432-0020D
Sht. No. M-412



F1

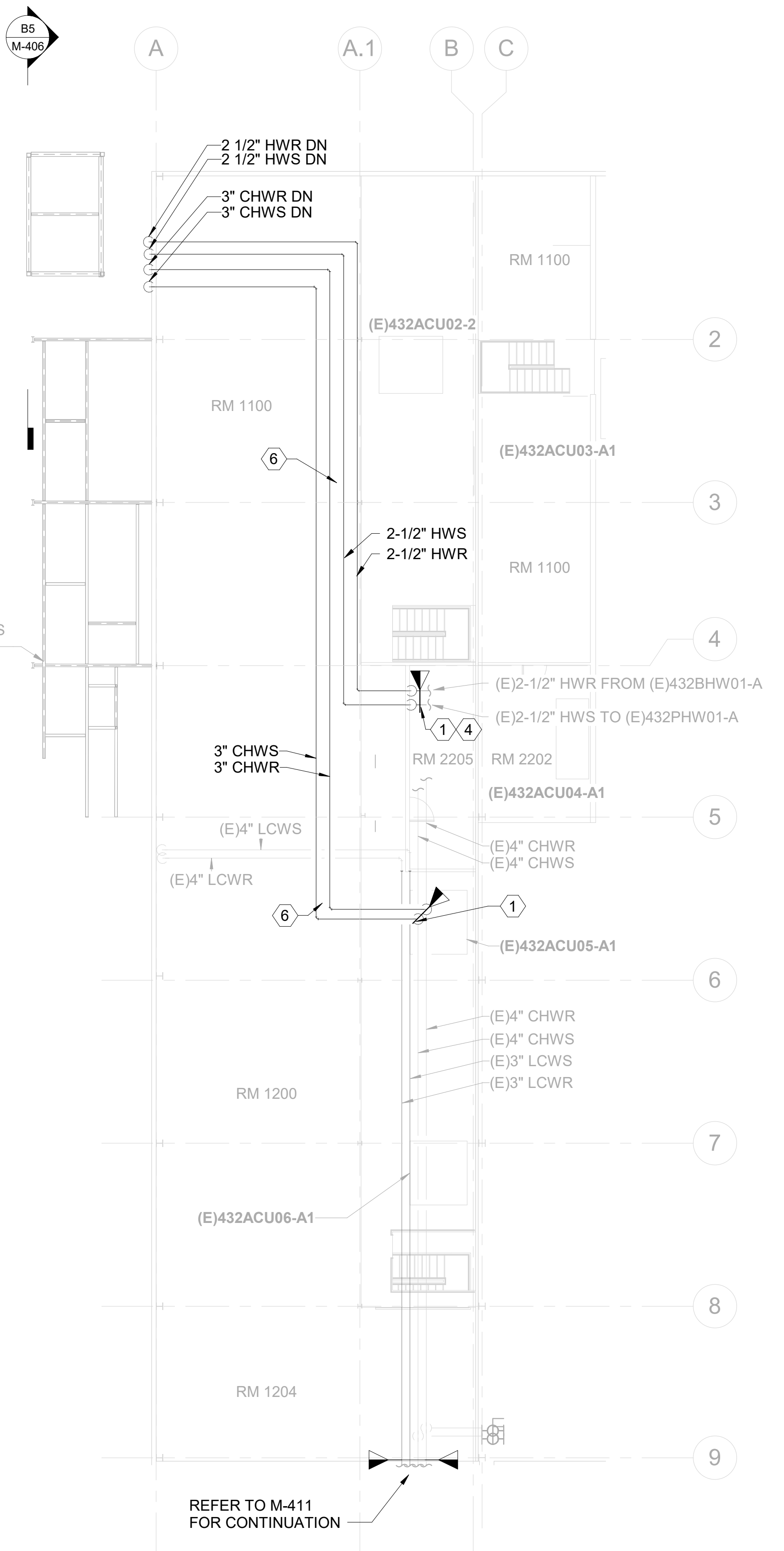
M-413

MECHANICAL ENLARGED 1ST FLOOR OPTICS LAB NEW
CONSTRUCTION PLAN - PIPING

SCALE: 3/32" = 1'-0"

081624

3/32" = 1'-0"



F2

M-413

MECHANICAL ENLARGED 2ND FLOOR OPTICS LAB NEW
CONSTRUCTION PLAN - PIPING

SCALE: 3/32" = 1'-0"

081624

3/32" = 1'-0"

- GENERAL NOTES
1.

REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
2.

COMPONENTS SHOWN HALF-TONE ON THIS DRAWING ARE ITEMS TO REMAIN; BOLD COMPONENTS ARE NEW.
3.

DASHED REGIONS AROUND EQUIPMENT INDICATE CLEARANCE REQUIRED FOR EQUIPMENT ACCESS AND MAINTENANCE.
4.

REFER TO M-401 FOR HVAC EQUIPMENT INSTALLATION.
5.

INSTALLATION OF NEW INTERIOR PIPING WILL REQUIRE A DETAILED WORK PLAN DEVELOPED WITH INPUT FROM CONSTRUCTION MANAGER, SUBCONTRACTORS, AND FACILITY END-USERS. SUBCONTRACTOR MUST REVIEW EXISTING CONDITIONS AND DISCUSS ACCEPTABLE MEANS AND METHODS FOR NEW INSTALLATION WITH CONSTRUCTION MANAGER PRIOR TO BID. USAGE CONTINUITY OF END-USER LAB EQUIPMENT IN PROJECT AREA WILL REQUIRE COMPLETE INSTALLATION OF NEW COMPONENTS OVER A LONG WEEKEND, INCLUDE COSTS TO ACCOMPLISH THIS IN BID. CONSTRUCTION MANAGER TO COORDINATE TEMPORARY LAB EQUIPMENT REMOVAL
6.

CLEAN AND FLUSH ALL NEW PIPING SYSTEMS IN ACCORDANCE WITH SPEC. 220111.13.
7.

INSTALL AUTOMATIC AIR VENTS AT ALL HIGH POINTS, AND ELSEWHERE AS NECESSARY TO ELIMINATE TRAPPED AIR. EXTEND VENT FROM AUTOMATIC AIR VENT TO NEAREST FLOOR DRAIN OR SOAK PIT.

- KEYED NOTES
- 1

INSTALL NEW BRANCH LINE-SIZE ISOLATION VALVES WITHIN FIRST SYSTEM SHUTDOWN AT BEGINNING OF CONSTRUCTION PROJECT TO FACILITATE INSTALLATION OF REMAINING NEW PIPING WITHOUT AN ADDITIONAL SHUTDOWN. WORK SHALL BE ACCOMPLISHED WITHOUT DRAINING FULL SYSTEM. FREEZING PIPE OR USING HOT TAP PIPE PLUG INSERTION SOLUTION IS ACCEPTABLE.
- 2

EXISTING CW INSTALLED UNDER MEZZANINE.
- 3

EXTEND NEW CW UNDER EXISTING STAIR LANDING AND THROUGH INTERIOR AND EXTERIOR WALL AS INDICATED TO TIE-IN POINT SHOWN ON M-411.
- 4

REFER TO M-802 FOR HEATING WATER TIE-IN POINTS WITHIN BOILER ROOM. RE-USE EXISTING WALL OPENINGS, ENLARGE EXISTING OPENINGS AS NECESSARY.
- 5

FURNISH AND INSTALL NEW FIRE SPRINKLER PIPING FROM MAIN TIE-IN. THREE SPRINKLER HEADS IN LOCATION IDENTICAL TO EXISTING SYSTEM, AND ALL APPURTENANCES. SIZE TO MATCH EXISTING SYSTEM BRANCH PIPING. REFER TO MD411 FOR EXISTING SPRINKLER SYSTEM.
- 6

ROUTE NEW PIPING WITHIN 18" OF BUILDING ROOF STRUCTURE.

KEY PLAN

Sheet Title

MECHANICAL ENLARGED 1ST AND 2ND FLOOR OPTICS LAB
NEW CONSTRUCTION PLANS
- PIPING

Dwg. No.

PLM2021-0432-0021D

Sht. No.

M-413



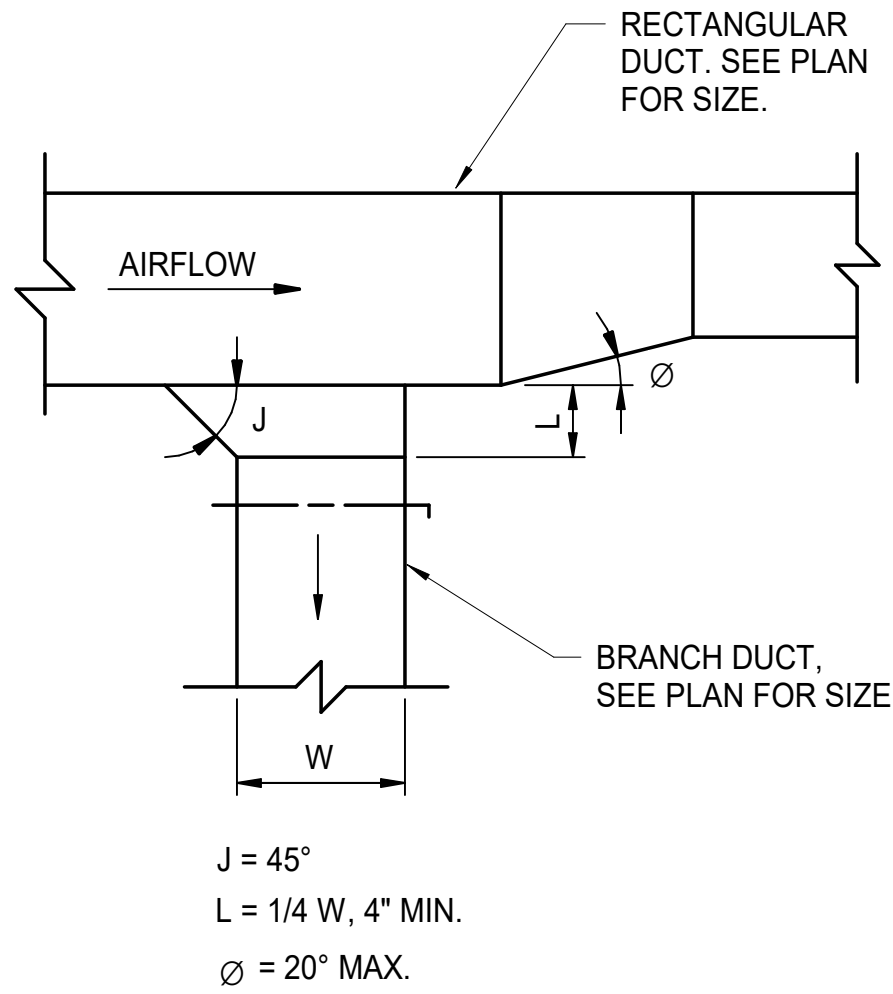
1. REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
2. COMPONENTS SHOWN HALF-TONE ON THIS DRAWING ARE ITEMS TO REMAIN; BOLD COMPONENTS ARE NEW.
3. GRAVITY PIPING MUST BE SLOPED IN DIRECTION OF FLOW AT 1/4" PER FOOT.

- 1 REFER TO M-402 FOR CONTINUATION.
- 2 ROUTE HEADERED CONDENSATE DRAIN TO NEW CONDENSATE SOAK PIT AT GRADE. SLOPE ALL HORIZONTAL CONDENSATE PIPING DOWNWARD IN THE DIRECTION OF FLOW WITH 1% SLOPE. INSTALL HORIZONTAL PIPING MAXIMUM 24" ABOVE CONDENSATE PUMPS.
- 3 REFER TO DETAIL D1 ON M-504.

MECHANICAL ENLARGED
2ND FLOOR OPEN OFFICE
NEW CONSTRUCTION PLAN -
PIPING

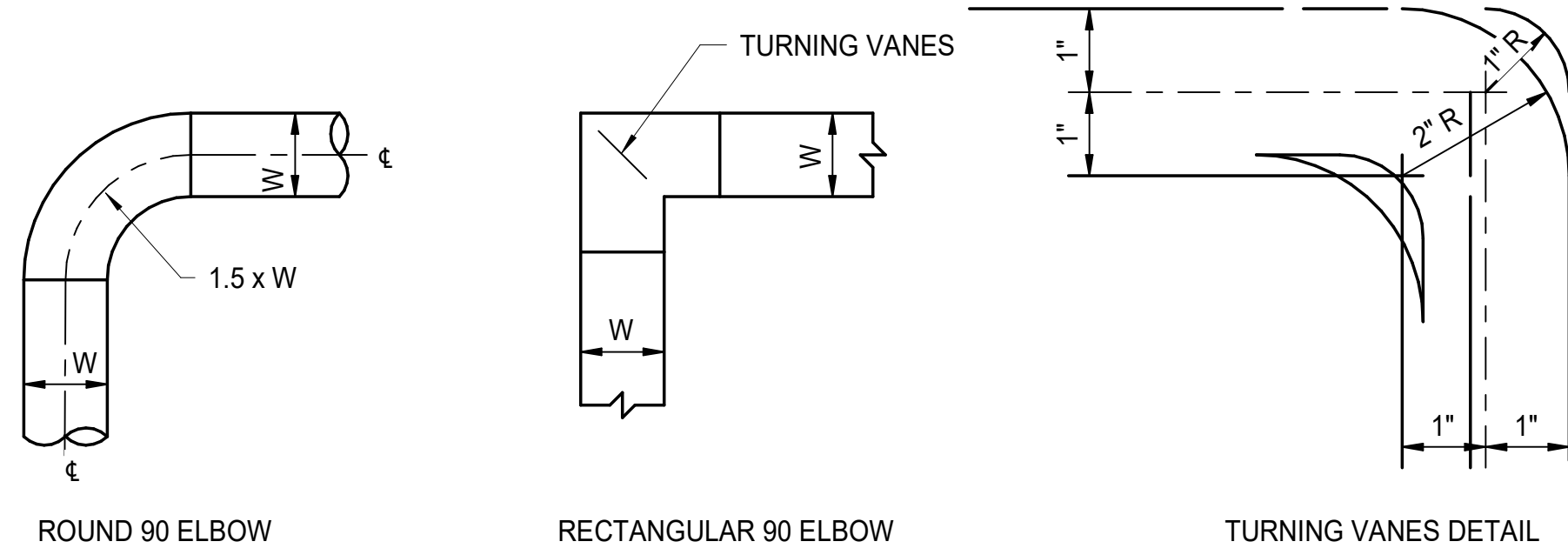
PLM2021-0432-0022D

M-414

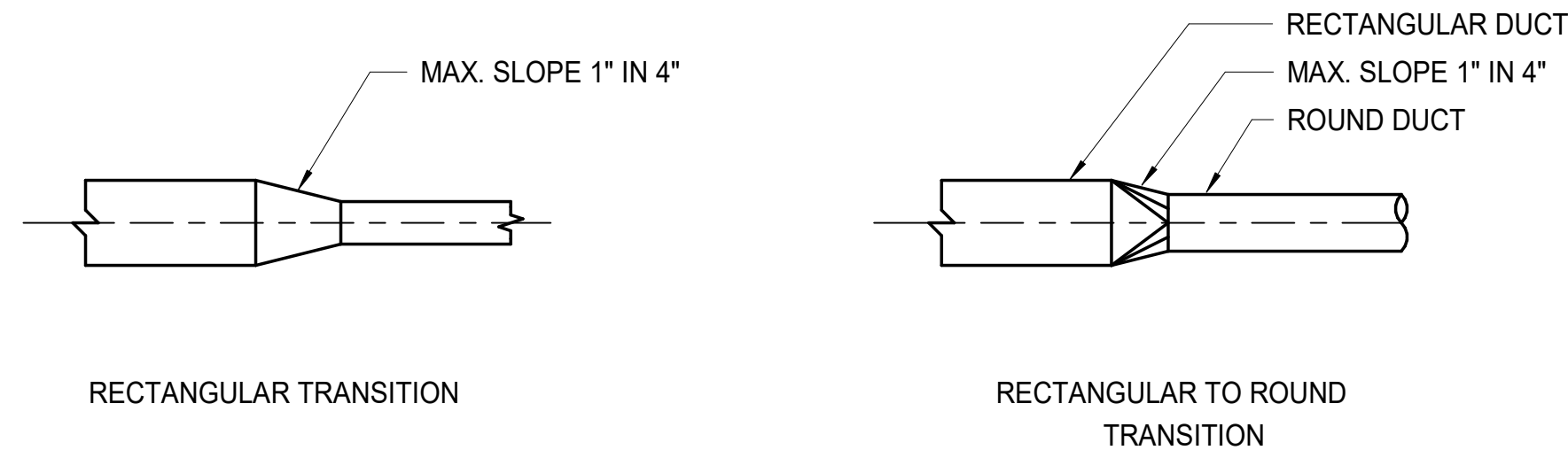


- NOTES:**
1. TAKEOFF ON BOTH SIDES OR SINGLE SIDE AS REQUIRED BY PLANS.
 2. SEE SMACNA MANUAL FOR TAP-IN DETAILS.
 3. WHEN FITTING HAS ONLY ONE SIDE SLOPED FOR A SINGLE SIDE TAKE-OFF, THE TAP-IN MUST BE INSTALLED IN SLOPED SIDE.
 4. CLINCH LOCK CONNECTION TO DUCT MUST HAVE CORNER SEALS. (SEE SMACNA MANUAL).
 5. EXTRACTORS, SCOOPS, DEFLECTORS OR DAMPERS THAT PROTRUDE INTO THE MAIN DUCT MUST NOT BE USED. BALANCING DAMPERS MUST BE LOCATED TO PREVENT PROTRUSION INTO THE MAIN DUCT AND TO FURNISH AND INSTALL STABLE AIR FLOW AND MINIMAL NOISE WHEN ADJUSTED.

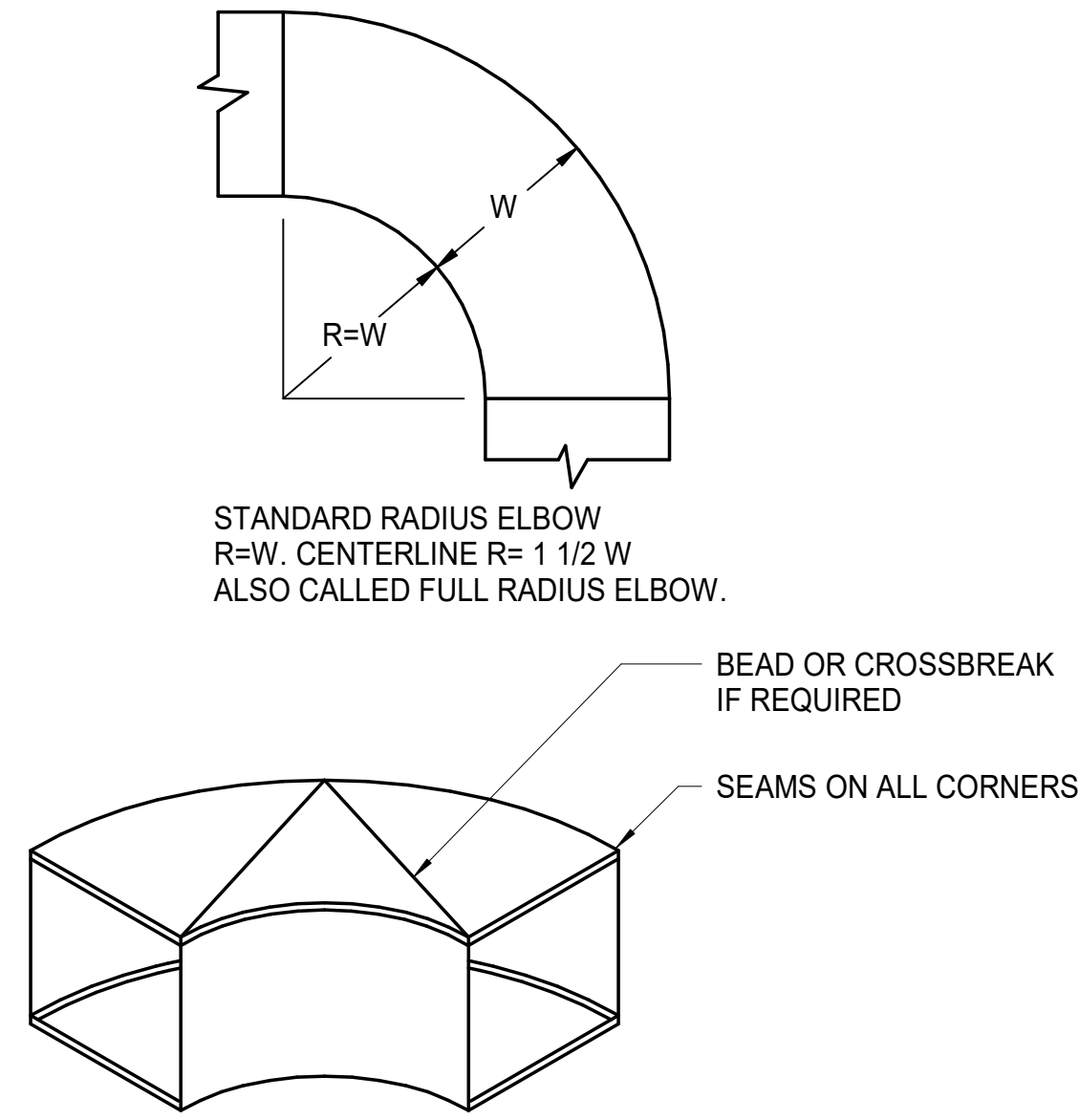
C1 BRANCH DUCT TAKE-OFF
M-501 SCALE: NO SCALE SHOWN



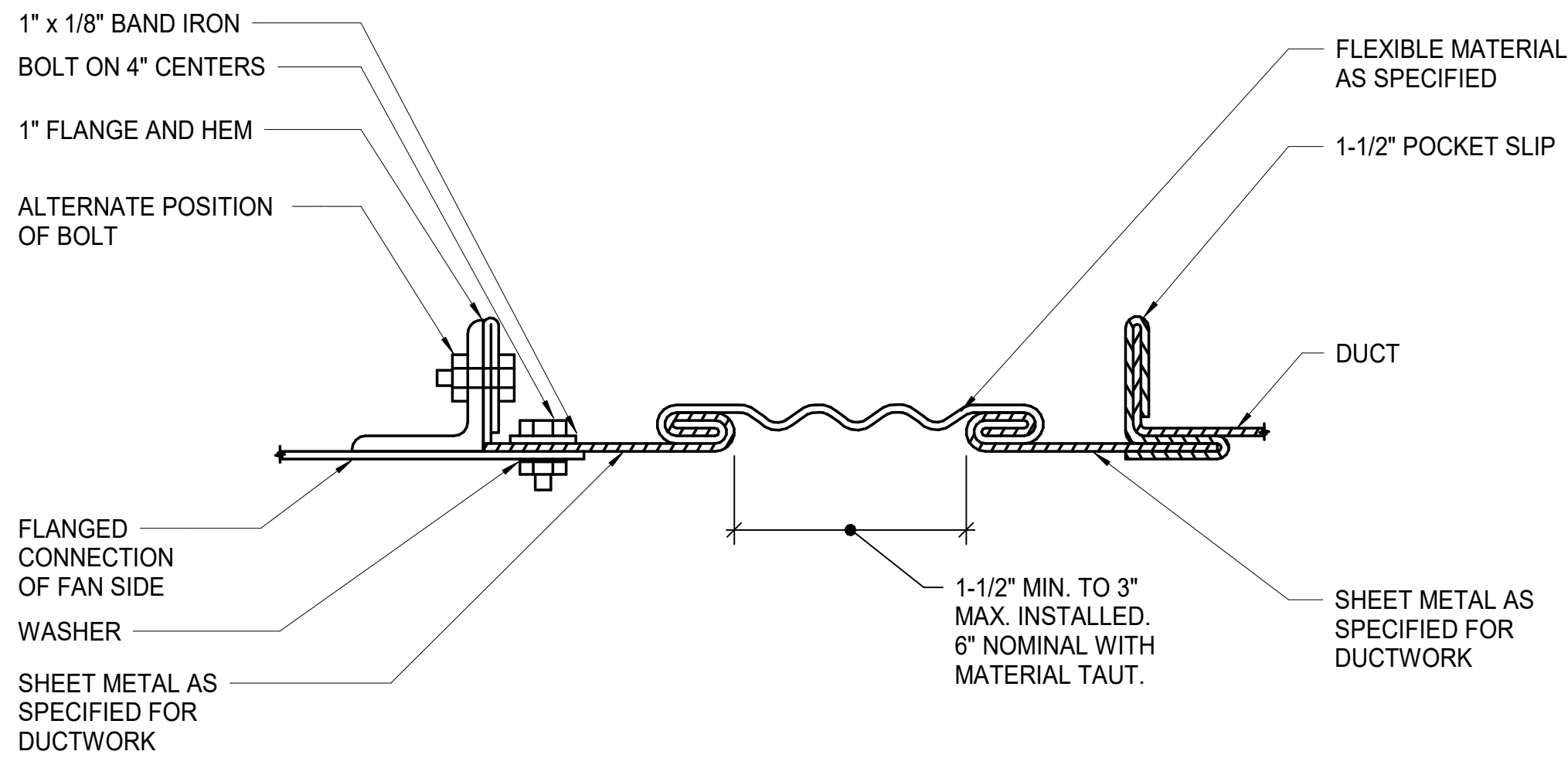
- NOTE:**
1. USE THIS DESIGN WHERE SQUARE 90° ELBOWS ARE SHOWN ON DRAWINGS OR IF SPACE DOES NOT PERMIT ROUND 90° ELBOWS.



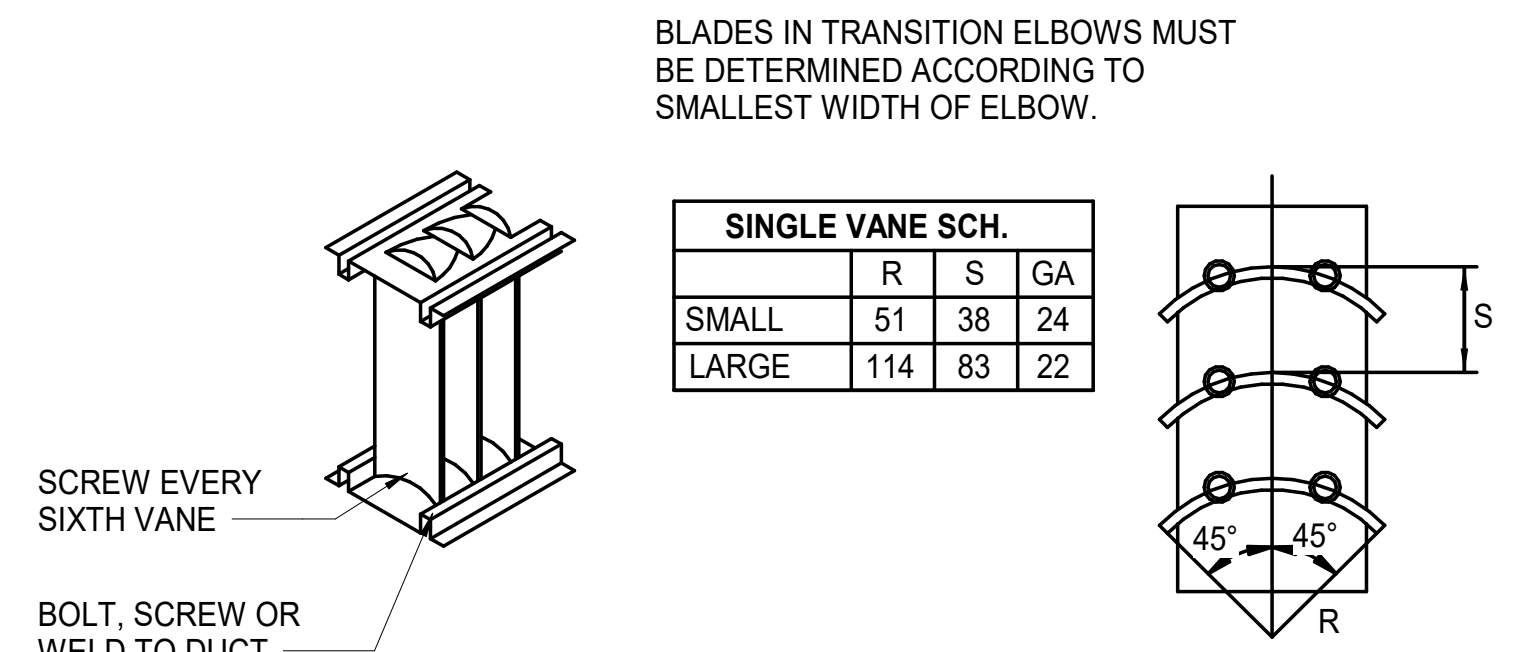
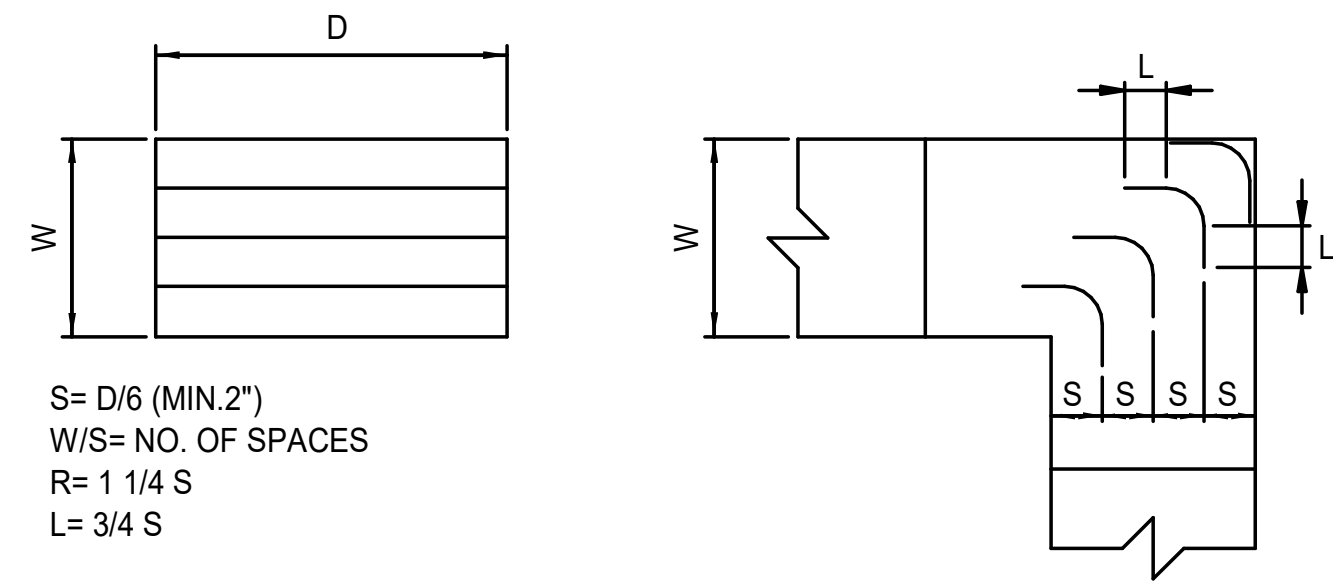
C3 LOW VELOCITY DUCT LAYOUT
M-501 SCALE: NO SCALE SHOWN



C7 STANDARD RADIUS ELBOW AIR DUCT
M-501 SCALE: NO SCALE SHOWN



F3 RECTANGULAR FLEXIBLE CONNECTION
M-501 SCALE: NO SCALE SHOWN

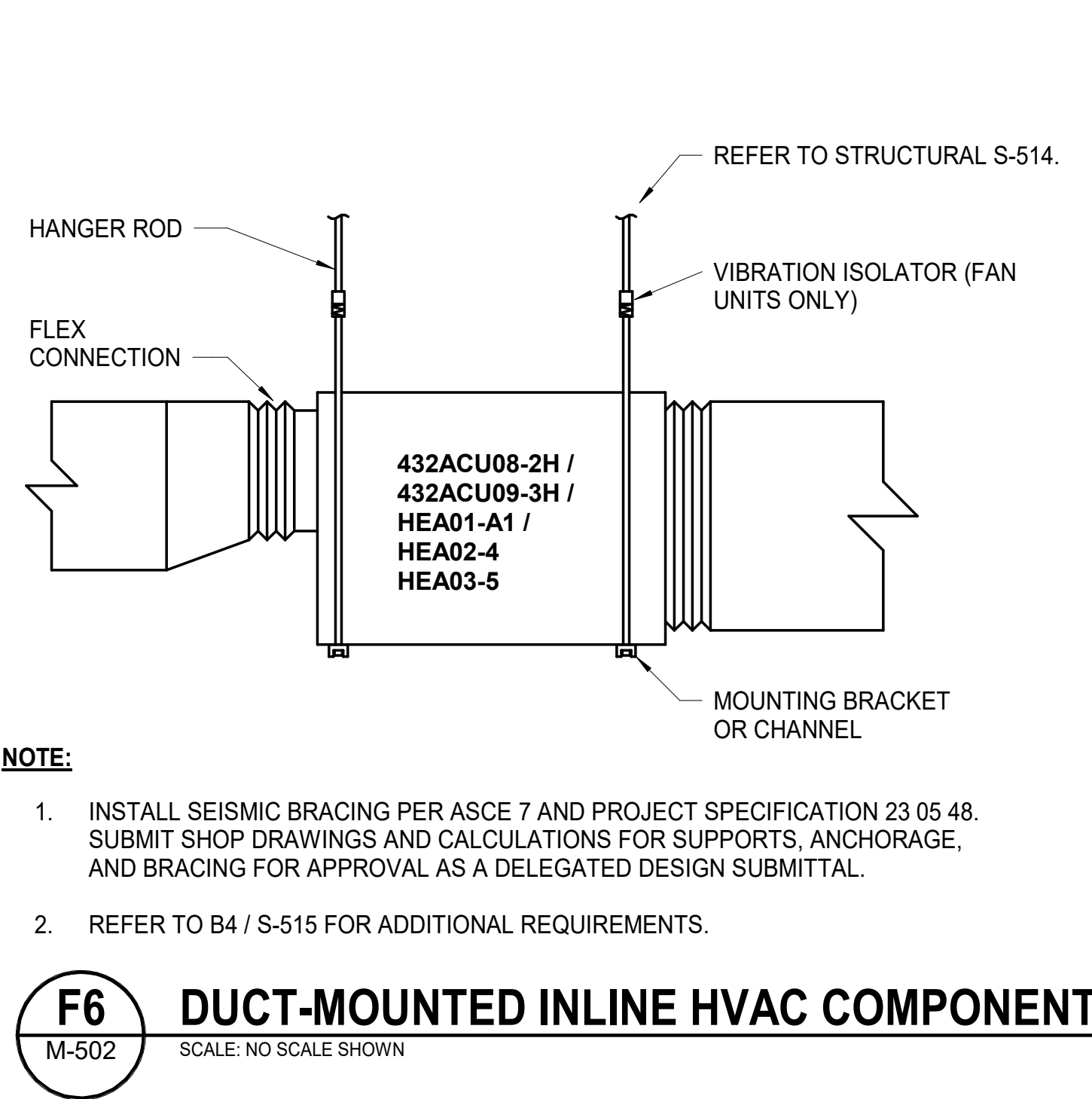
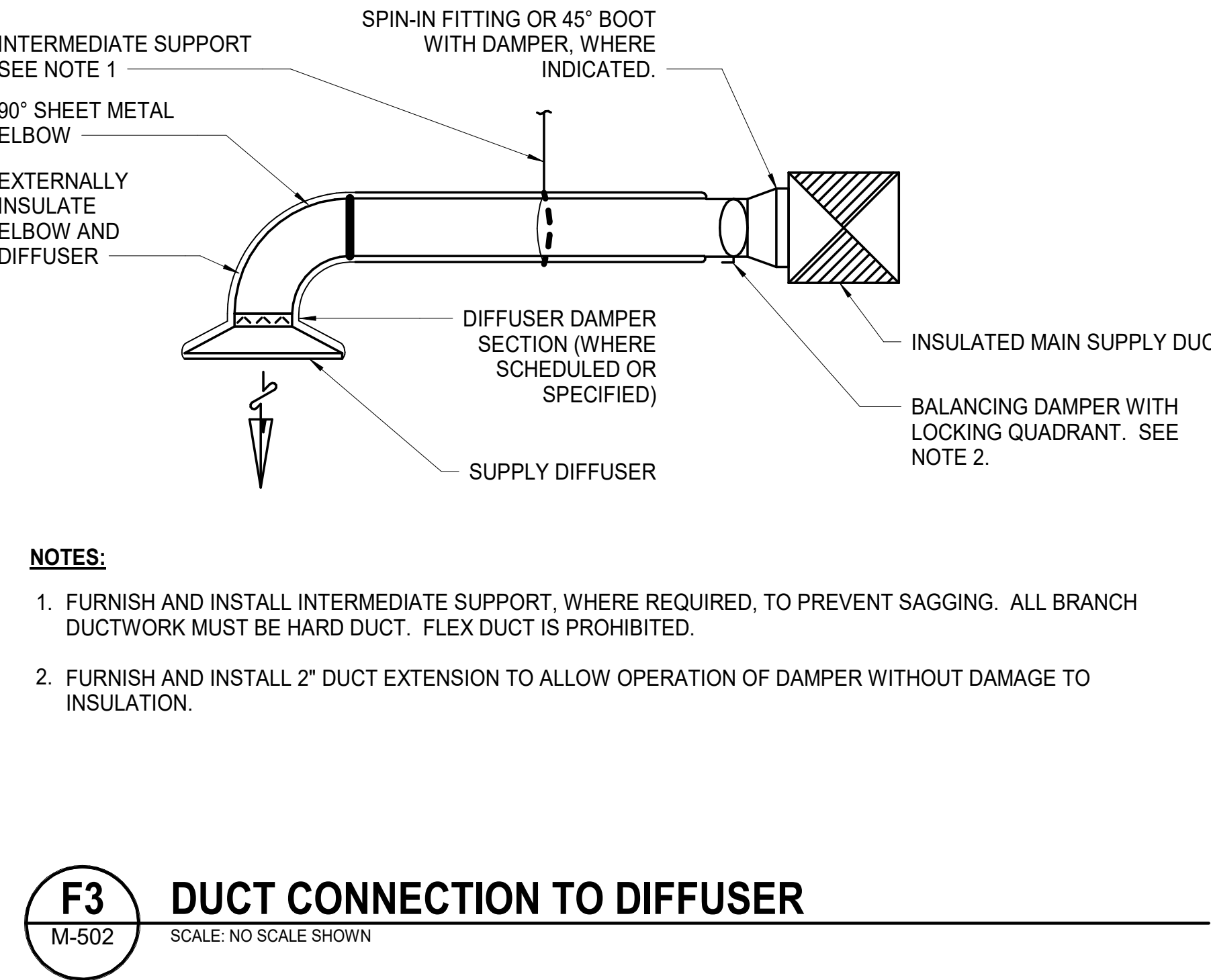
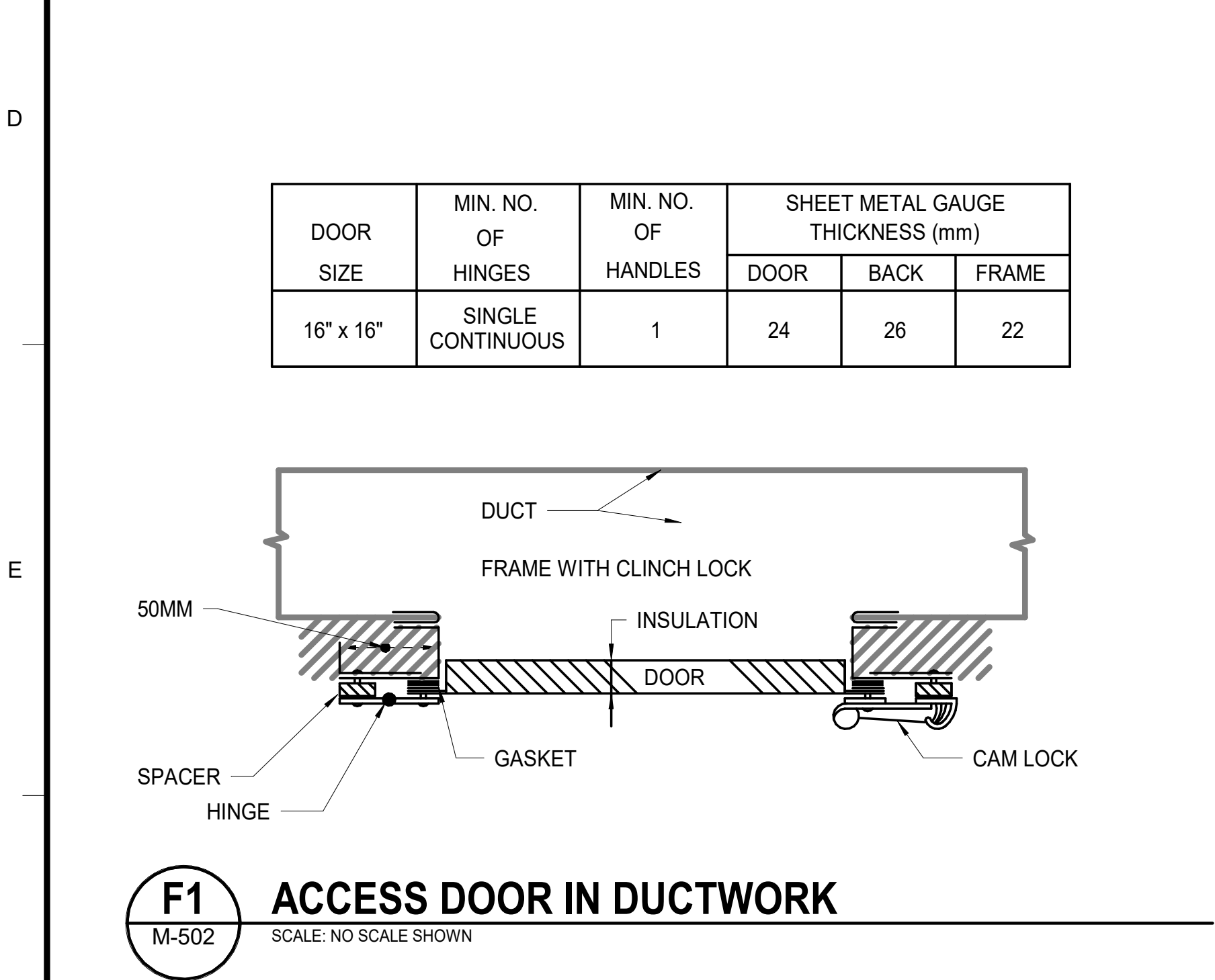
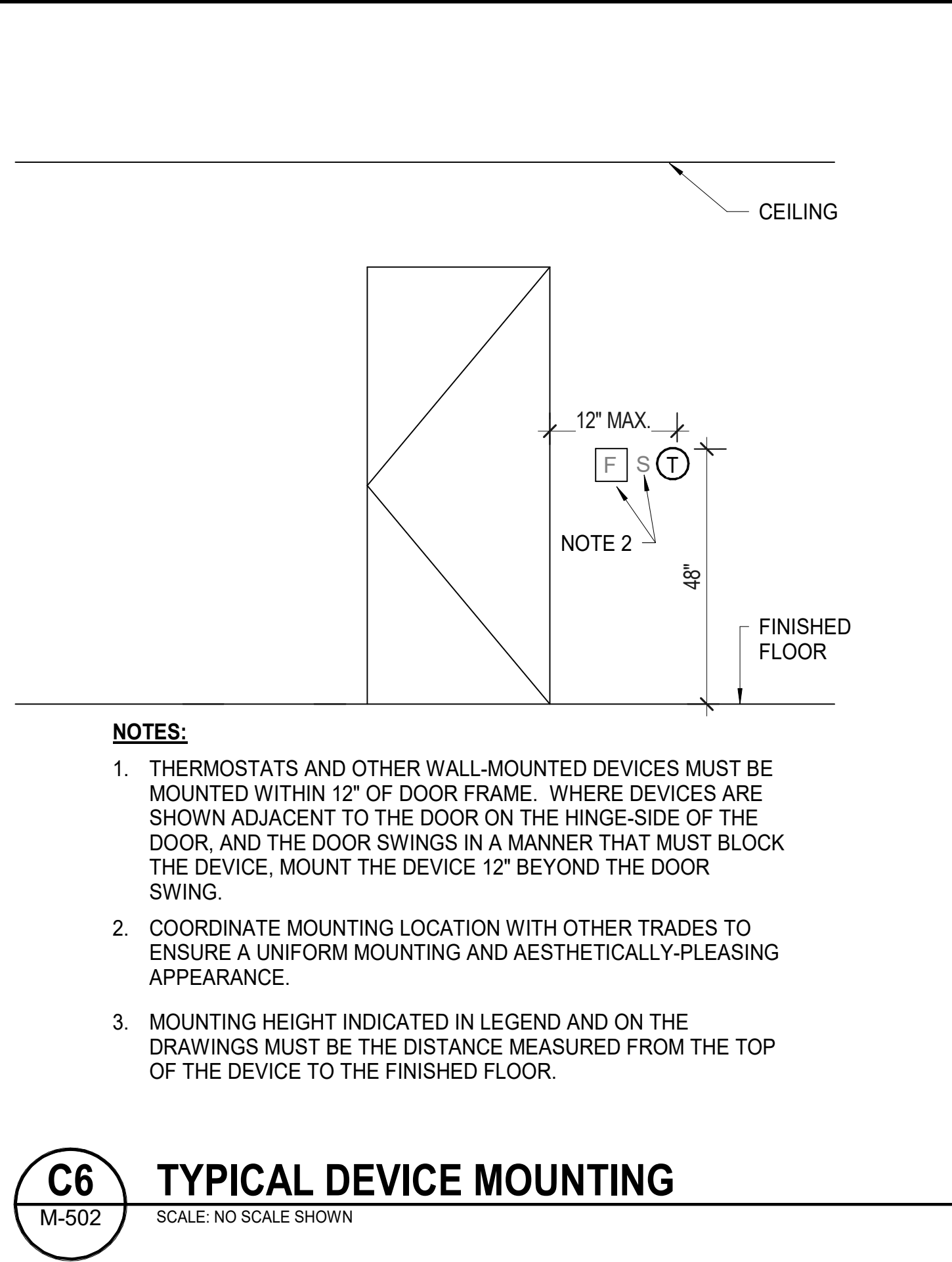
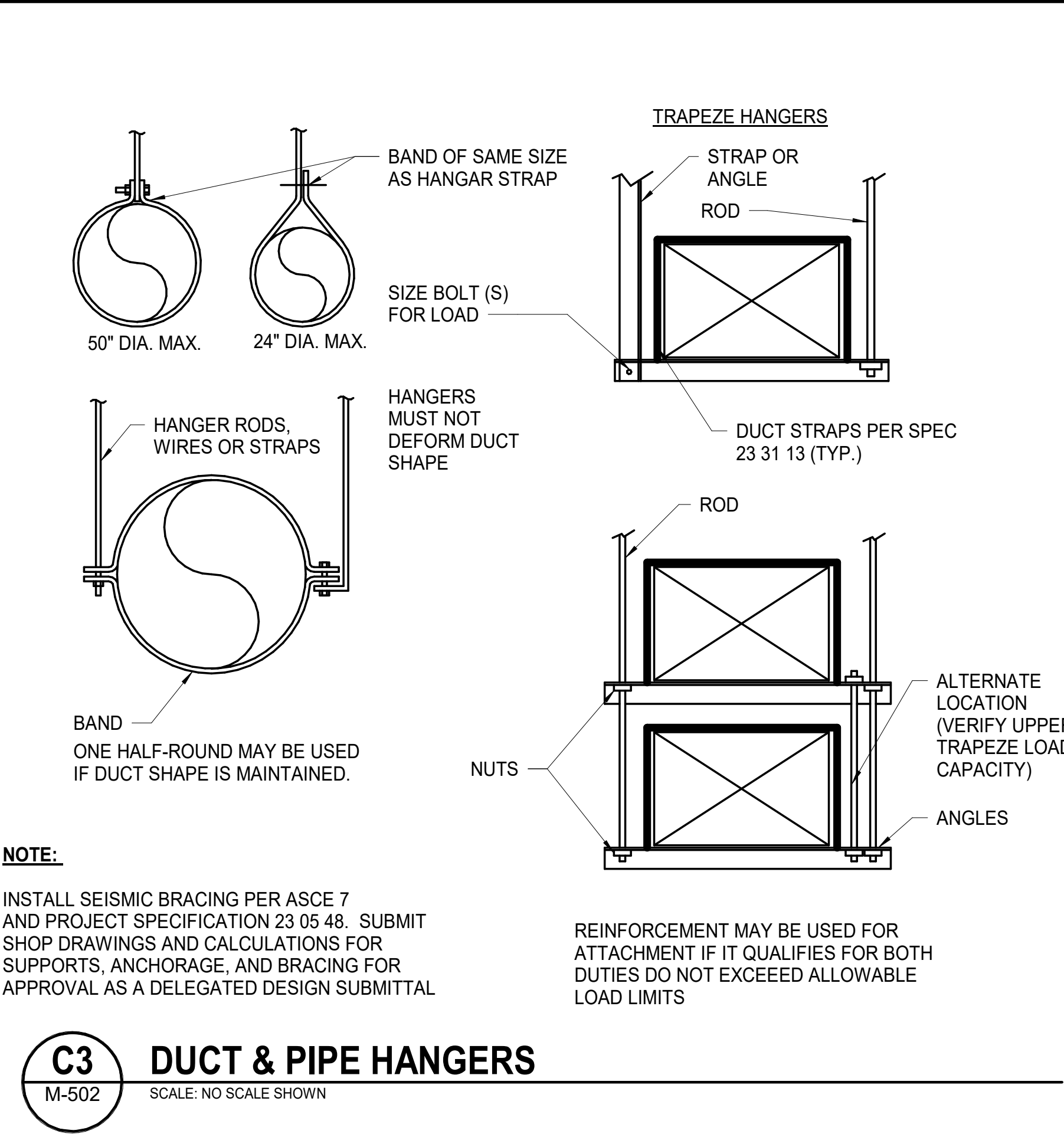
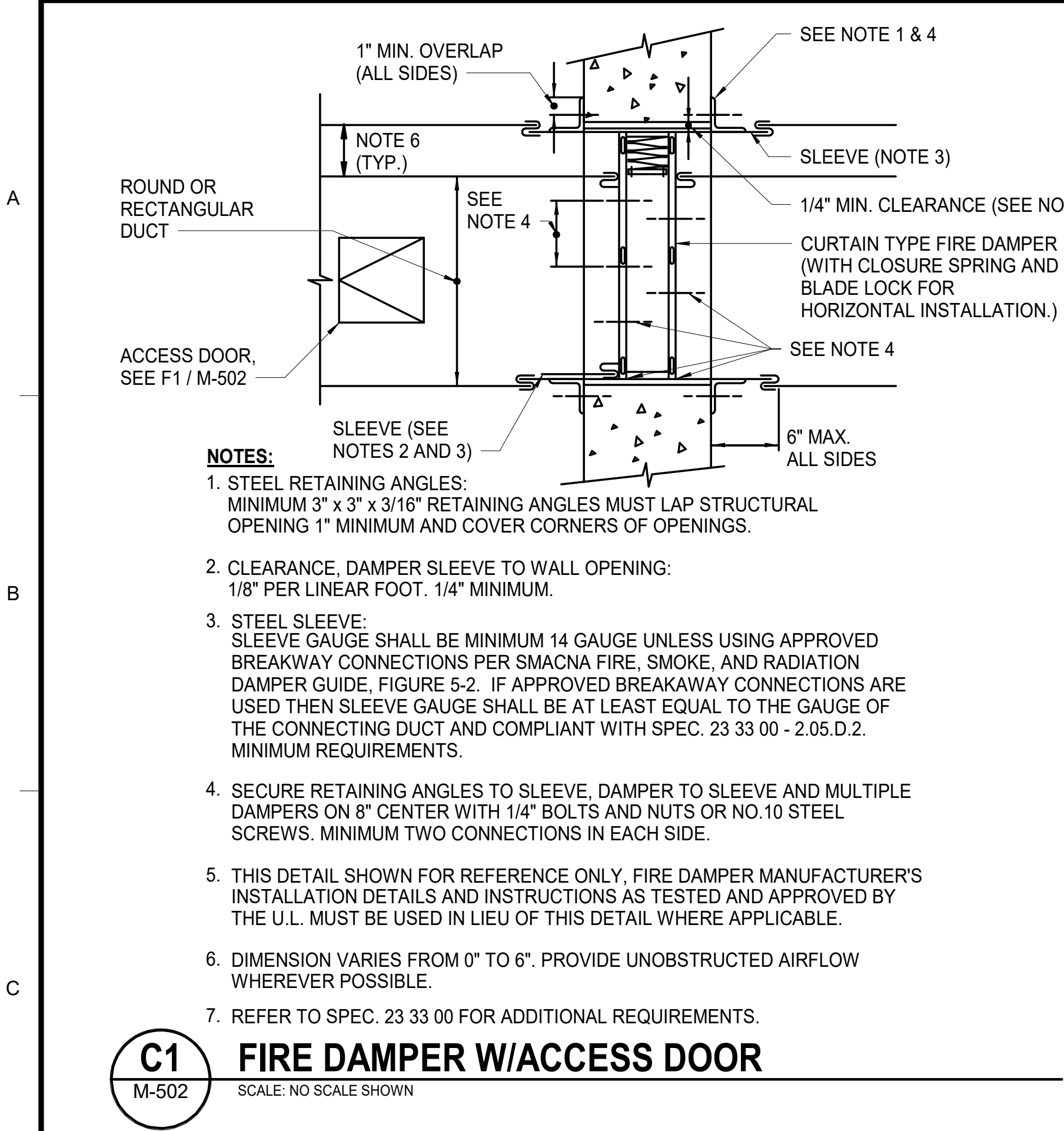


F6 RECTANGULAR DUCT ELBOW TURNING VANES
M-501 SCALE: NO SCALE SHOWN

Sheet Title
MECHANICAL DETAILS

Dwg. No.
PLM2021-0432-0023D

Sht. No.
M-501



Sheet Title
MECHANICAL DETAILS

Dwg. No.
PLM2021-0432-0024D

Sht. No.
M-502

A

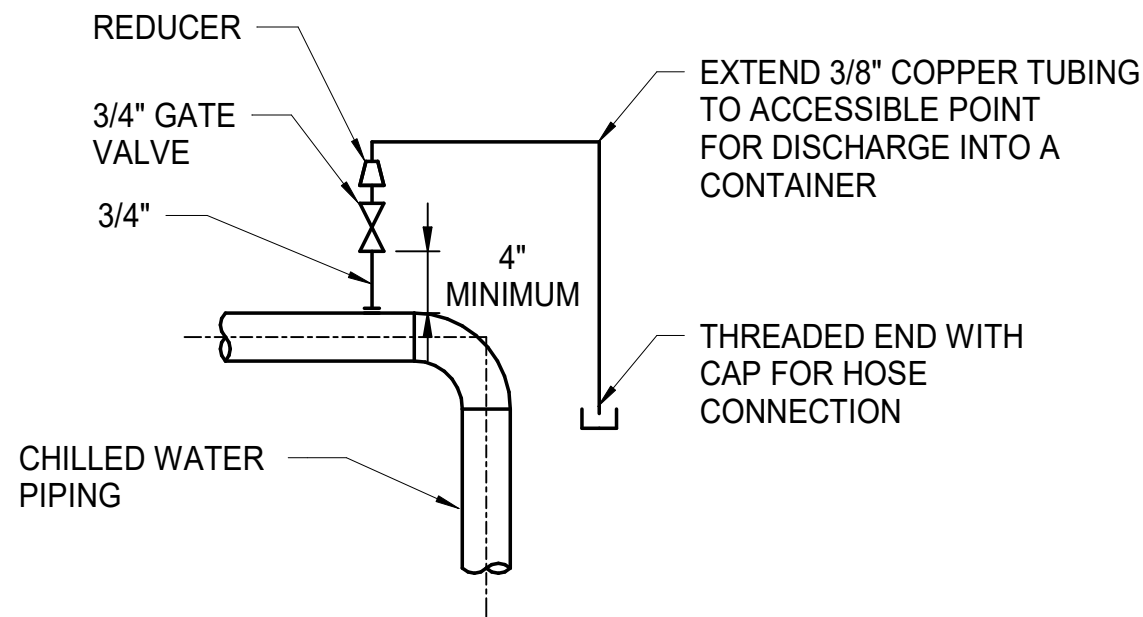
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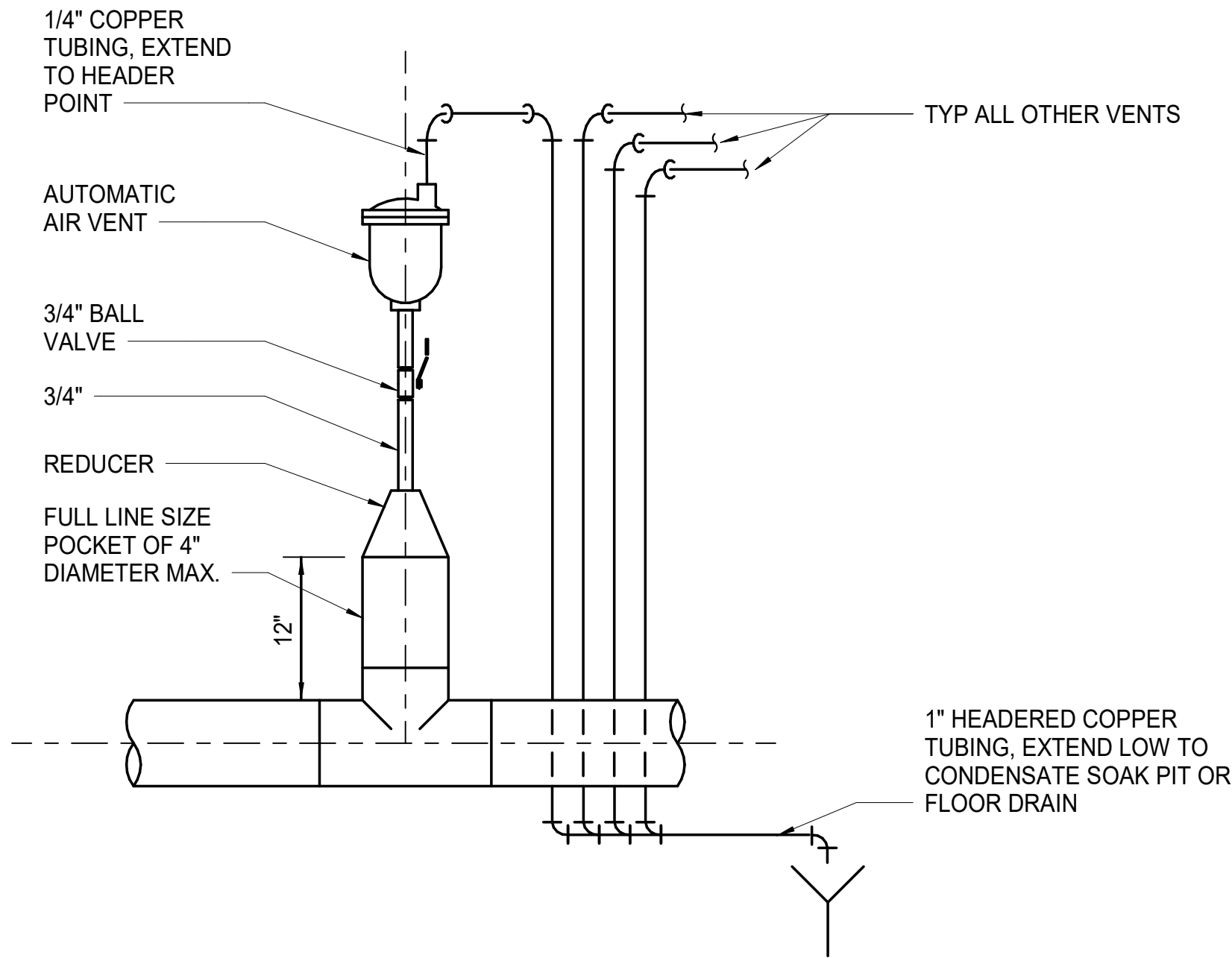
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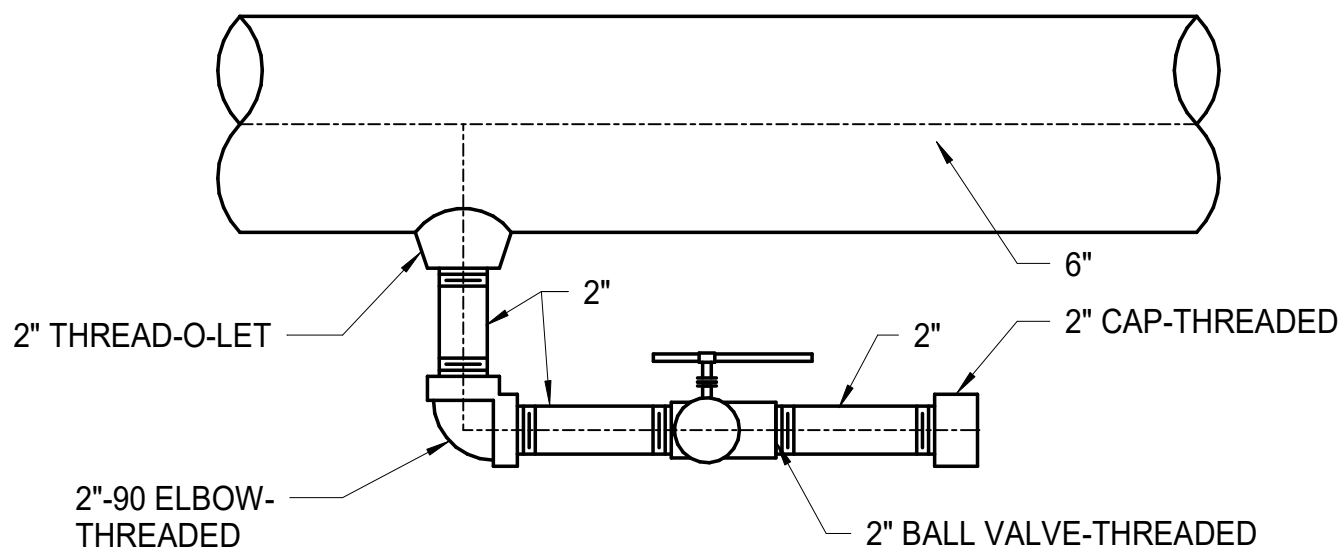
C3 MANUAL AIR VENT
M-503 SCALE: NO SCALE SHOWN



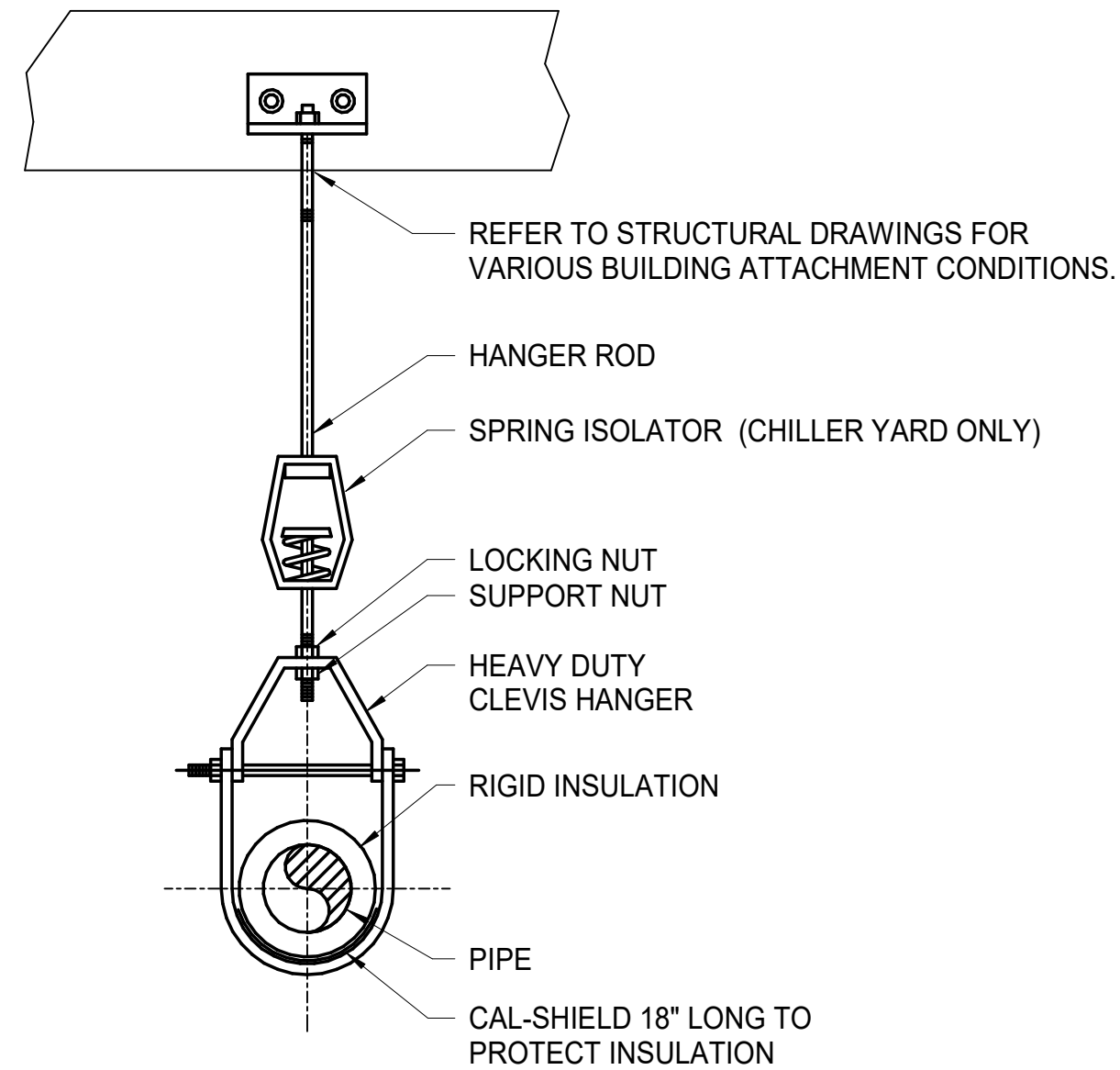
NOTE:

1. ALL AV 1/4" VENT TUBING IN SAME AREA MUST HEADER INTO 1" LINE AT FLOOR. HEADERED LINE MUST ROUTE TO (N) SOAK PIT OR (E) FLOOR DRAIN AS COMBINED DRAIN LINE.

C6 AUTOMATIC AIR VENT
M-503 SCALE: NO SCALE SHOWN



F3 LINE DRAIN AT CHILLER
M-503 SCALE: NO SCALE SHOWN



NOTES:

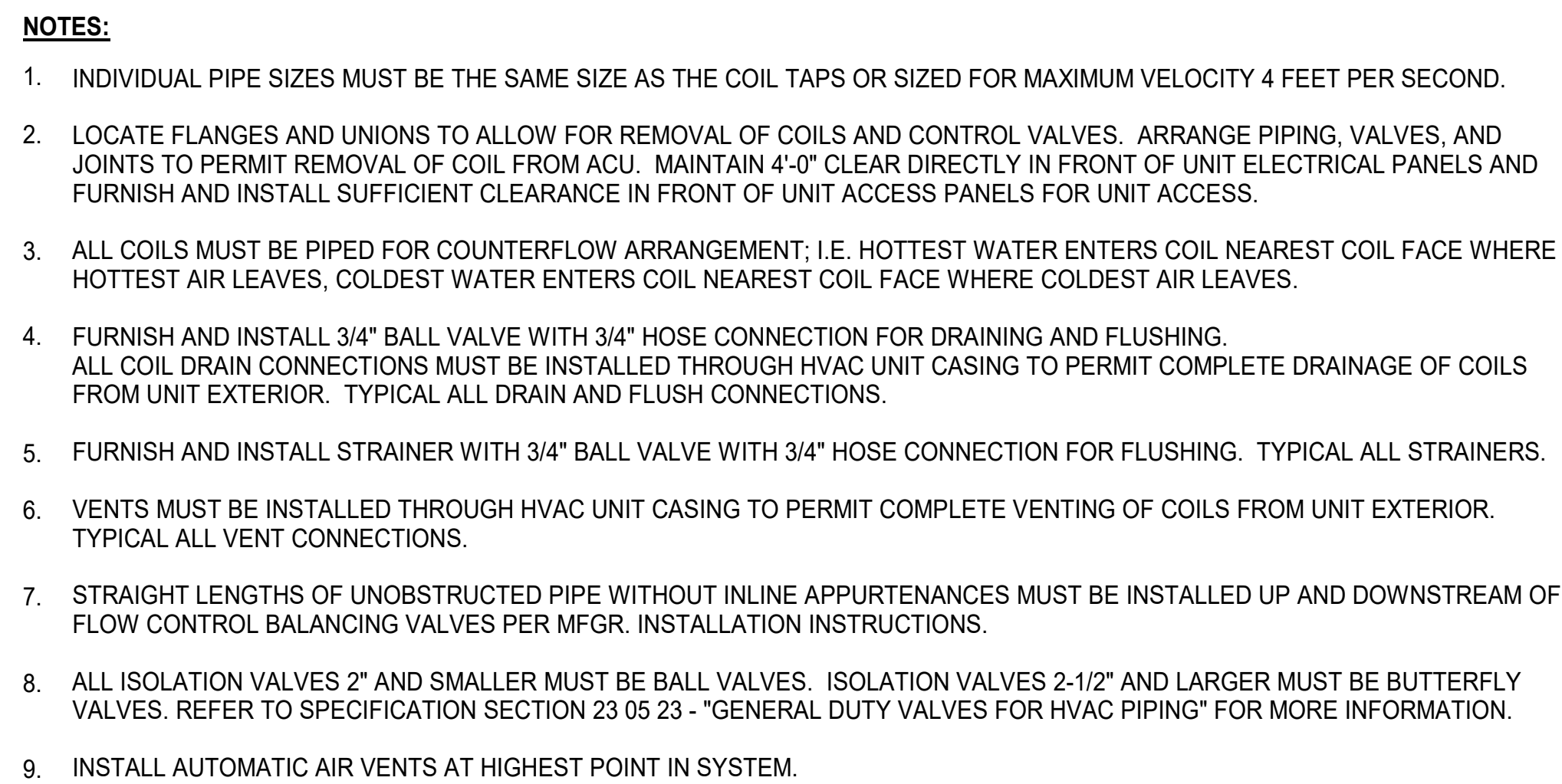
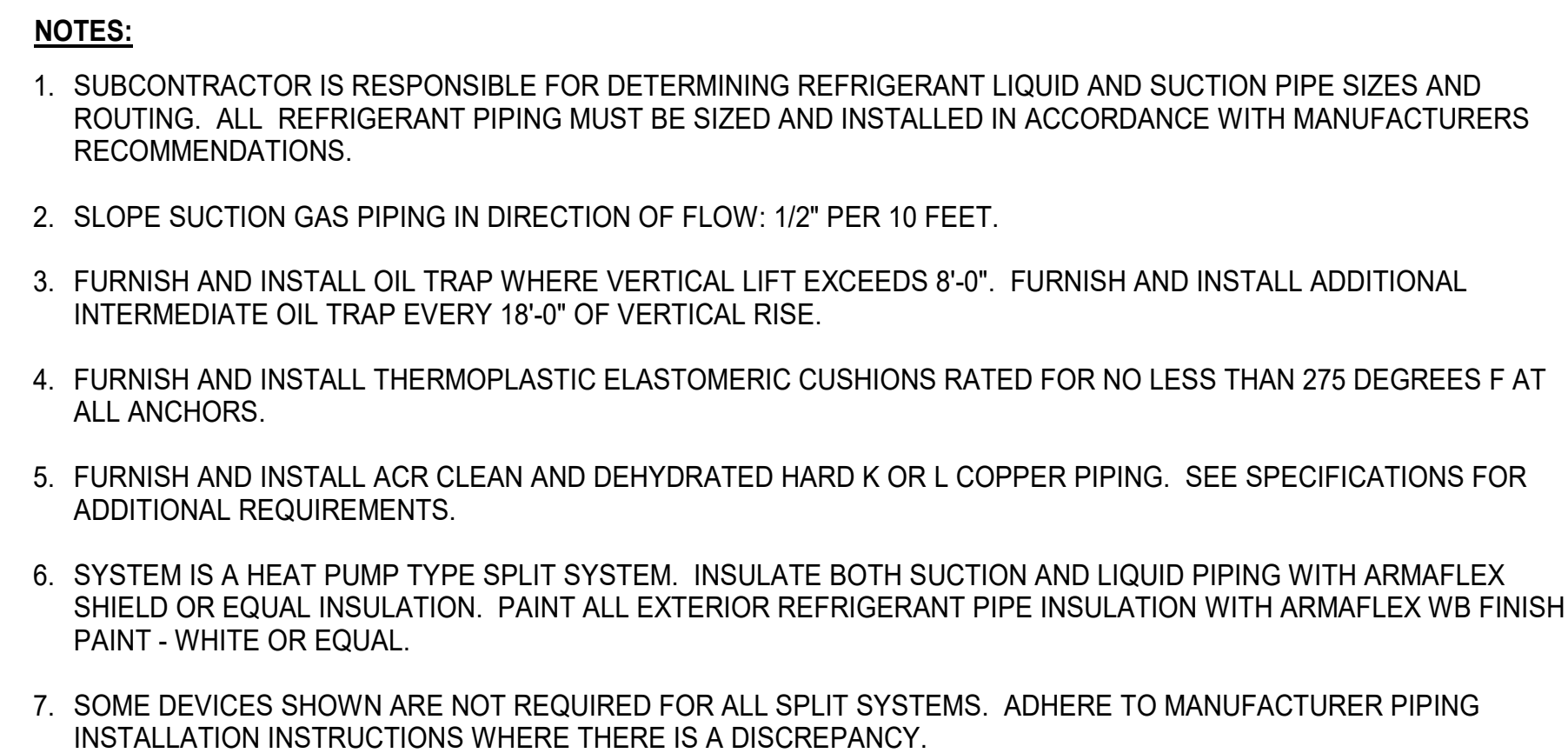
1. ALL CHILLED WATER PIPE HANGERS IN CHILLER YARD MUST INCLUDE SPRING-TYPE VIBRATION ISOLATION ASSEMBLY REFER TO 23 05 48; PIPE HANGERS FOR ALL OTHER SYSTEMS DO NOT REQUIRE VIBRATION ISOLATION.
2. REFER TO SPEC. SECTION 23 05 48 FOR PIPE SWAY BRACING.
3. INSTALL SEISMIC BRACING PER ASCE 7 AND PROJECT SPECIFICATION 23 05 48. SUBMIT SHOP DRAWINGS AND CALCULATIONS FOR SUPPORTS, ANCHORAGE, AND BRACING FOR APPROVAL AS A DELEGATED DESIGN SUBMITTAL.

F6 PIPE HANGER-CLEVIS TYPE
M-503 SCALE: NO SCALE SHOWN

Sheet Title
MECHANICAL DETAILS

Dwg. No.
PLM2021-0432-0025D

Sht. No.
M-503



D5 ACU SINGLE HYDRONIC COIL, 3-WAY CONTROL VALVE
M-504 SCALE: NO SCALE SHOWN

Sheet Title	MECHANICAL DETAILS
Dwg. No.	PLM2021-0432-0026D
Sht. No.	M-504

A

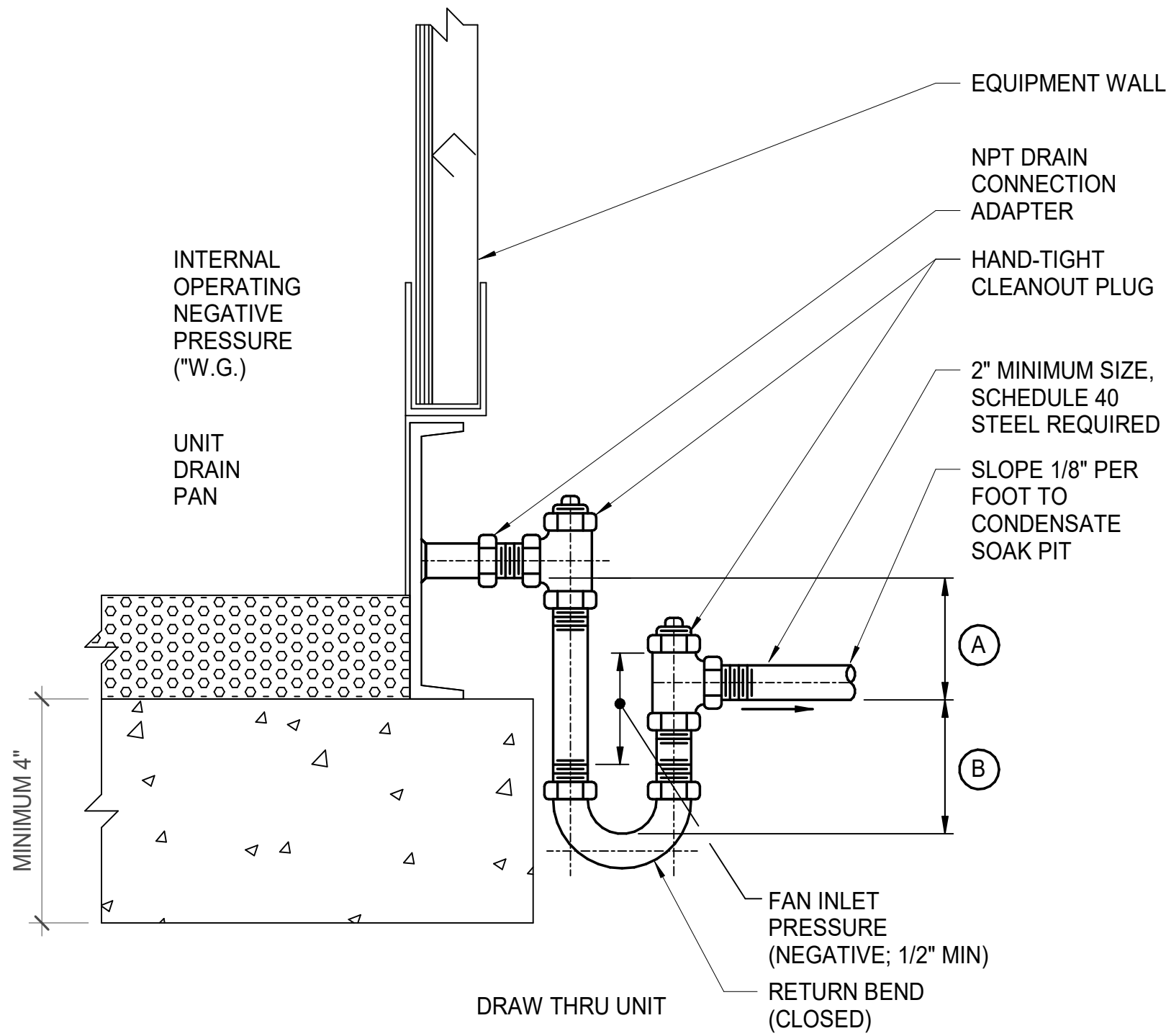
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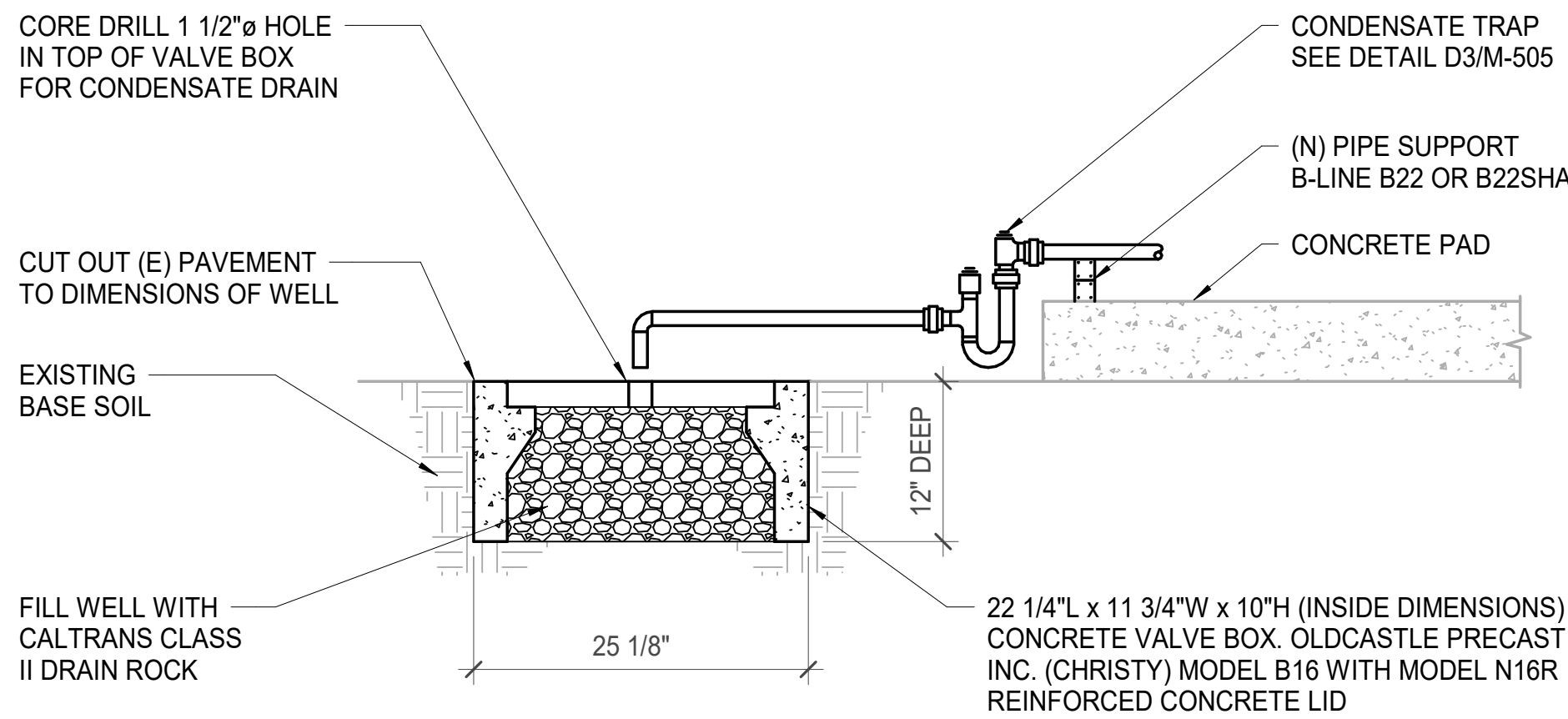
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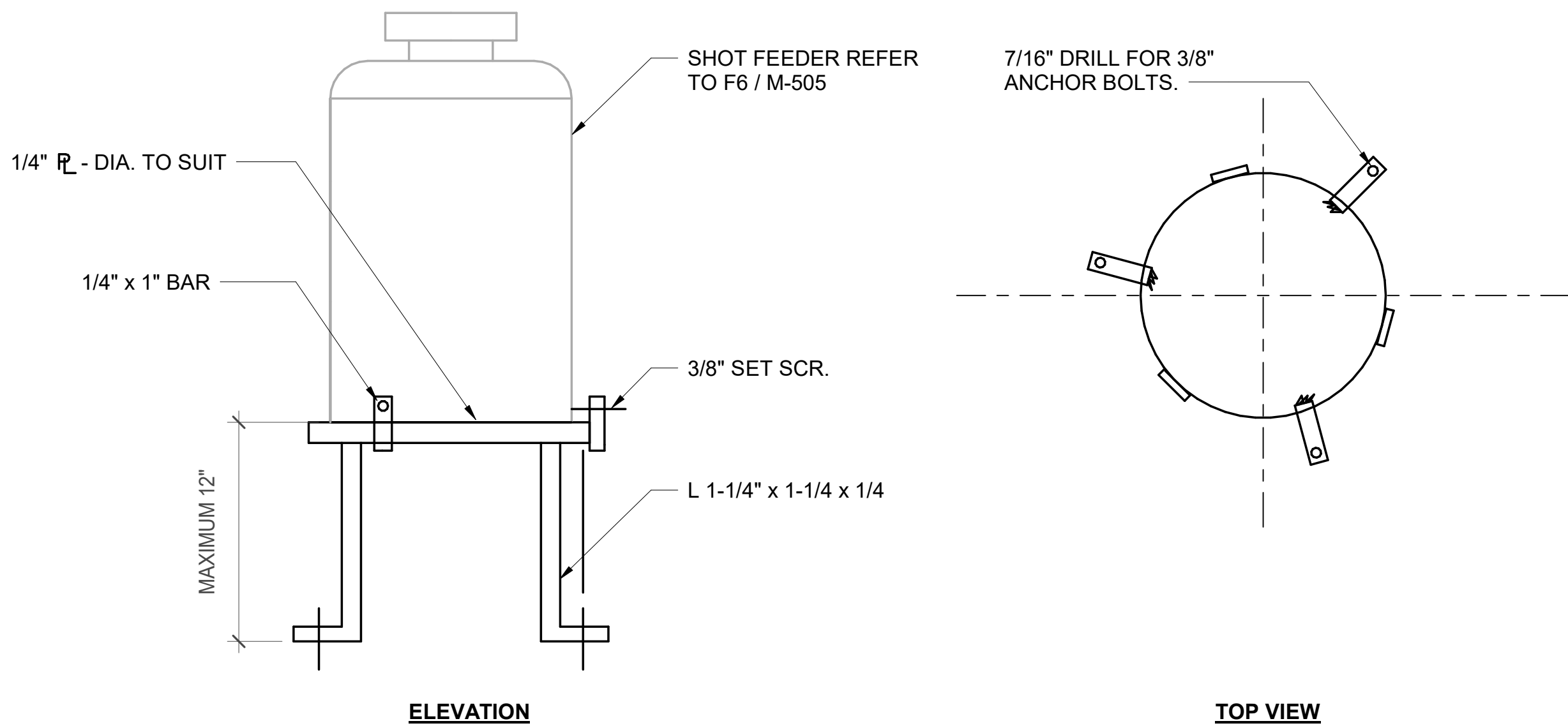
NOTES:

1. (A) DIMENSION EQUAL TO MAXIMUM FAN INLET PRESSURE (INCH WC) + 1/2" (MIN)
2. (B) DIMENSION EQUAL TO (A) /2 (MIN)

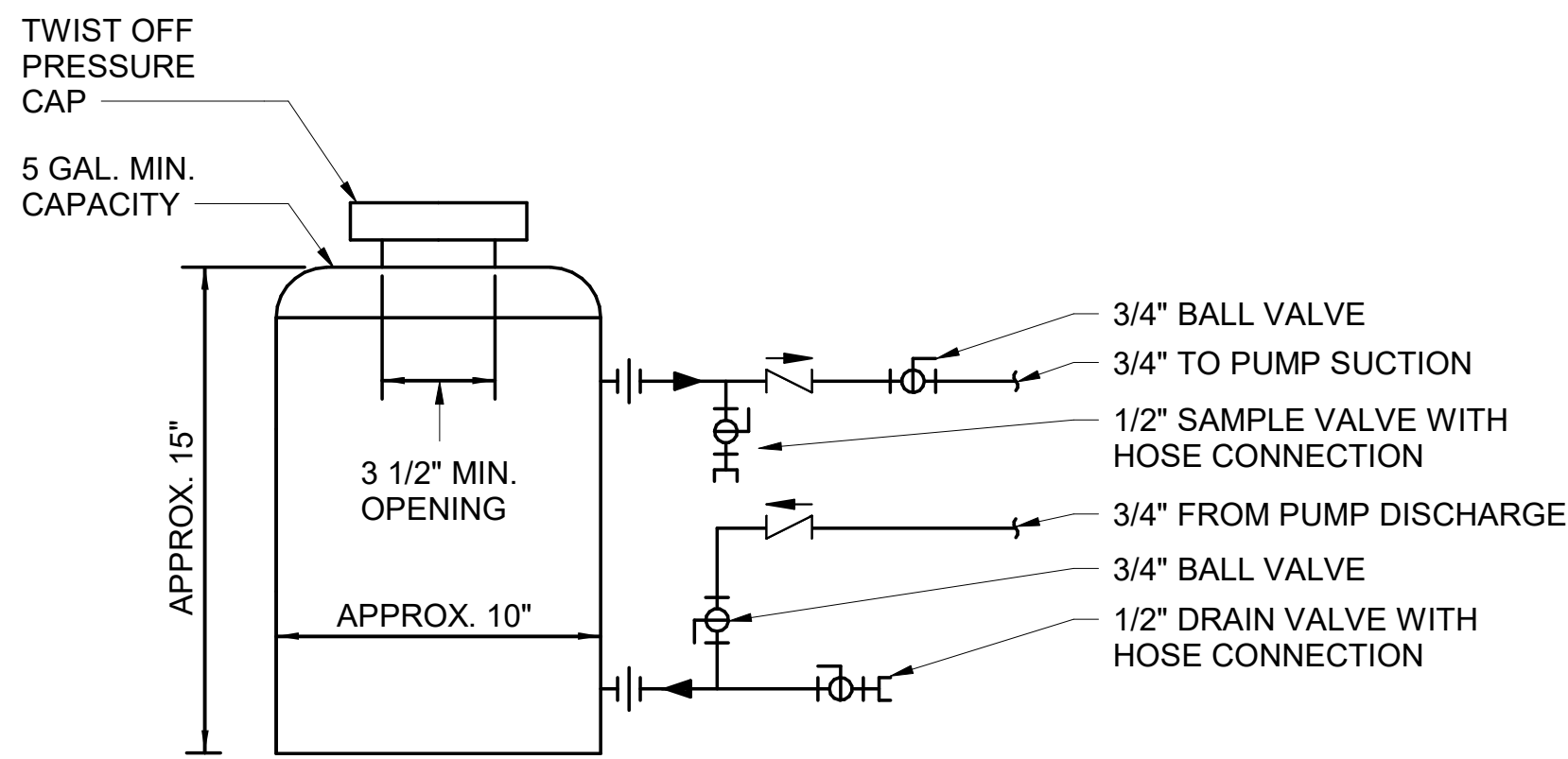
D3 EQUIPMENT CONDENSATE TRAP DRAIN
M-505 SCALE: NO SCALE SHOWN



D6 CONDENSATE SOAK PIT
M-505 SCALE: NO SCALE SHOWN



F3 SHOT FEEDER SUPPORT
M-505 SCALE: NO SCALE SHOWN

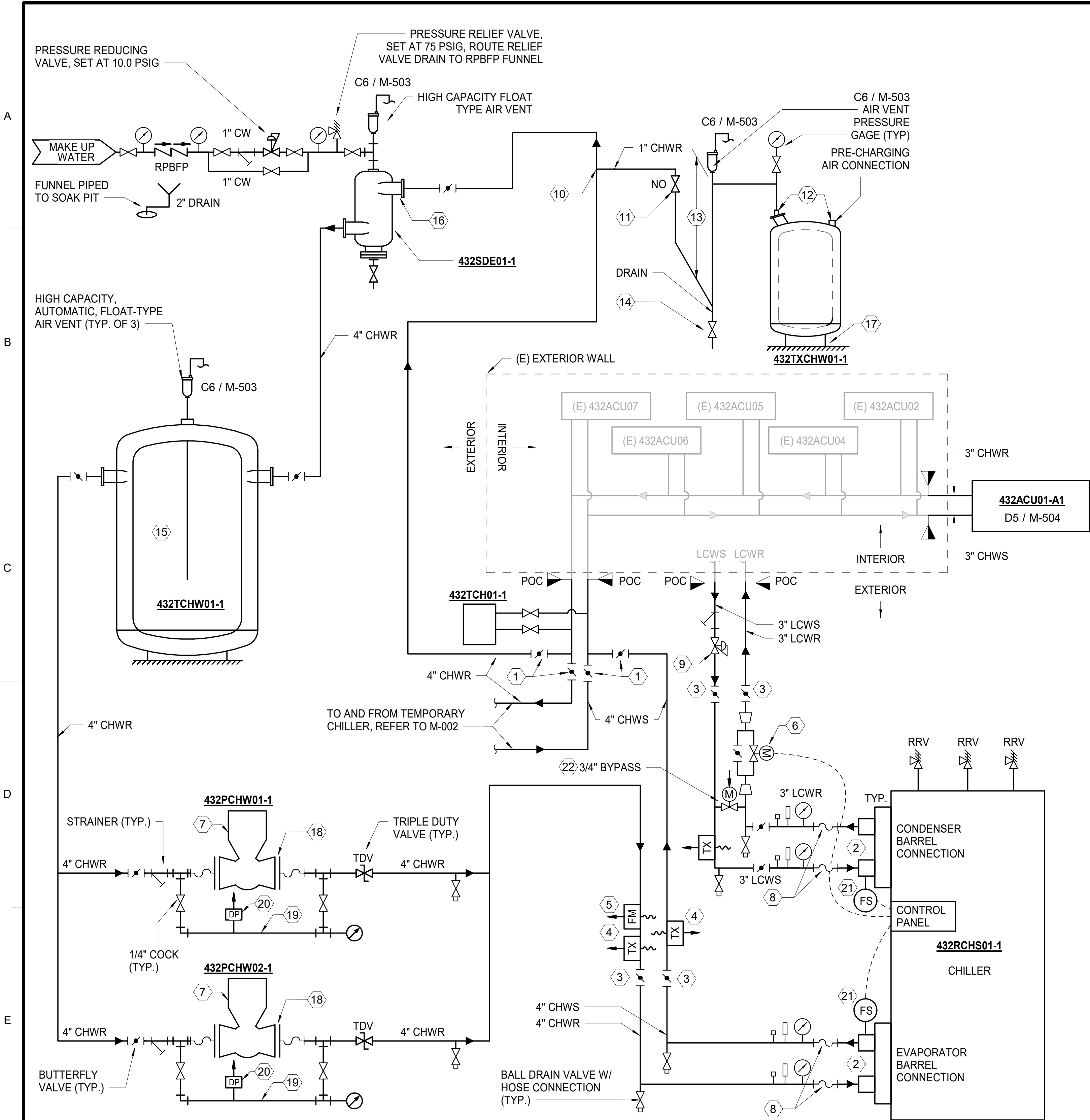


F6 SHOT FEEDER
M-505 SCALE: NO SCALE SHOWN

Sheet Title
MECHANICAL DETAILS

Dwg. No.
PLM2021-0432-0027D

Sht. No.
M-505



NOTES:

1. REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, SYMBOL LEGEND, AND GENERAL NOTES.
2. REFER TO M-701 FOR CONTROLS ABBREVIATIONS, SYMBOL LEGEND, AND GENERAL NOTES.
3. CLEAN AND FLUSH ALL NEW PIPING SYSTEMS IN ACCORDANCE WITH SPEC. 220111.13.
4. INSTALL AUTOMATIC AIR VENTS AT LOCATIONS INDICATED, AT ALL HIGH POINTS, AND ELSEWHERE AS NECESSARY TO ELIMINATE TRAPPED AIR. EXTEND VENT FROM AUTOMATIC AIR VENT TO NEAREST FLOOR DRAIN OR SOAK PIT.

KEYED NOTES:

1. INSTALL SELECT PIPING AND LINE-SIZE ISOLATION VALVES WITHIN FIRST SYSTEM SHUTDOWN AT BEGINNING OF CONSTRUCTION PROJECT. REFER TO M-002.
2. TYPICAL: CONNECTION TO THE CHILLER INLET AND OUTLET FOR CONDENSER AND EVAPORATOR IS 6" NPS GROOVED PIPE CONNECTION. ALL OTHER FITTINGS MUST BE WELDED, OR FLANGED AS SHOWN.
3. EVAPORATOR AND CONDENSER SPOOL PIECES MUST BE FABRICATED SO THAT WHEN REMOVED, THERE IS A CLEAR PATH FOR THE CONDENSER AND EVAPORATOR TUBE BUNDLES TO BE REMOVED FOR MAINTENANCE. SPOOL PIECES MUST MATE TO THE THREADED LUG STYLE ISOLATION VALVE FLANGE CONNECTIONS.
4. CHILLED WATER SUPPLY AND RETURN TEMPERATURE SENSORS AND ALL INTERCONNECTING WIRING AND CONDUIT MUST BE FURNISHED AND INSTALLED BY THE SUBCONTRACTOR. POINTS ARE AVAILABLE FOR VIEWING IN THE ALC DDC, BUT WILL NOT BE USED FOR CONTROL OF THE CHILLER.
5. CHILLED WATER FLOW METER, ONICON F-1200, AND ALL INTERCONNECTING WIRING AND CONDUIT MUST BE FURNISHED AND INSTALLED BY THE SUBCONTRACTOR. POINT IS AVAILABLE FOR VIEWING IN THE ALC DDC, BUT WILL NOT BE USED FOR CONTROL OF THE CHILLER.
6. HEAD PRESSURE CONTROL VALVE: ELECTRONIC REFRIGERANT PRESSURE ACTUATED LCW CONTROL VALVE PLUS ACTUATOR (FURNISHED BY CHILLER VENDOR) MUST BE INSTALLED BY SUBCONTRACTOR AND COMMISSIONED BY SUBCONTRACTOR AND CHILLER MANUFACTURER. VALVE MUST BE ELECTRONICALLY ACTUATED BY CHILLER CONTROLLER IN RESPONSE TO REFRIGERANT PRESSURE. FOLLOW VALVE MANUFACTURER'S INSTRUCTIONS FOR VALVE INSTALLATION. ISOLATION VALVES MUST BE FURNISHED AND INSTALLED ON EITHER SIDE OF THE CONTROL VALVE TO FACILITATE MAINTENANCE OR REPLACEMENT OF THE VALVE. FURNISH AND INSTALL WIRED CONNECTION, WIRING AND CONDUIT, FROM CHILLER CONTROL PANEL TO THE CONTROL VALVE.
7. INSTALL PUMP DEAD LEVEL. PUMP MUST NOT TOUCH OR REST ON ANY PART OF THE BUILDING STRUCTURE. INSTALL PUMP SO THAT IT CAN BE REMOVED WITHOUT DISMANTLING OR FORCING ADJACENT PIPE. BEFORE START-UP, LUBRICATE PUMP IN ACCORDANCE WITH MANUFACTURERS INSTRUCTIONS. (DO NOT RUN PUMP DRY). PARALLEL PUMPS ARE 100% REDUNDANT. SUPPORT PUMPS PER S-514.
8. FURNISH AND INSTALL STAINLESS STEEL BRAIDED FLANGED FLEX CONNECTORS ONTO CHW AND LCW PIPING CONNECTION POINTS. FLEX CONNECTORS MUST BE INSTALLED IN A HORIZONTAL ORIENTATION.
9. FURNISH AND INSTALL PRV TO REDUCE LCW WATER PRESSURE FROM 90 PSIG TO 40 PSIG.
10. CONNECT TO SIDE OF MAIN. AVOID TOP OR BOTTOM CONNECTION TO PREVENT AIR OR DEBRIS FROM ENTERING PIPE TO TANK.
11. SHUT-OFF VALVE (LOCKED NORMALLY OPEN)
12. TANK CONNECTION LOCATIONS MAY VARY DEPENDING ON TANK SIZE AND MODEL.
13. FURNISH AND INSTALL 12" MIN. DROP FOR ANTI-THERMOSYPHON LOOP TO PREVENT HEATING OF TANK.
14. DRAIN VALVE REQUIRED TO PROPERLY PRECHARGE TANK.
15. TANK CONNECTION LOCATIONS MAY VARY DEPENDING ON TANK SIZE AND MODEL.
16. SUPPORT AIR SEPARATOR PER S-514.
17. SUPPORT EXPANSION TANK PER S-514.
18. UNION TYPE PIPE CONNECTION PIPE SO PUMP CAN BE REMOVED FOR SERVICE.
19. 1/4" TYPE L COPPER TUBING FOR GAUGE CONNECTIONS.
20. DP TRANSMITTER SHALL BE WIRED TO ALC SYSTEM FOR USE IN MONITORING THE STATUS OF CHILLED WATER PUMP. INSTALL DEVICE ON UNISTRUT STAND FOR VFD PROVIDED BY ELECTRICAL.
21. IFM EFFECTOR FLOW SWITCH FURNISHED BY CHILLER VENDOR. SUBCONTRACTORS MUST INSTALL AND WIRE DEVICE, INCLUDING PIPE SPOOL PIECE AS NECESSARY, IF DEVICE IS SHIPPED LOOSE BY CHILLER VENDOR FOR FIELD INSTALLATION.
22. NORMALLY-CLOSED 2-WAY ELECTRONIC LCW BYPASS CONTROL VALVE FOR LCW FREEZE PROTECTION.

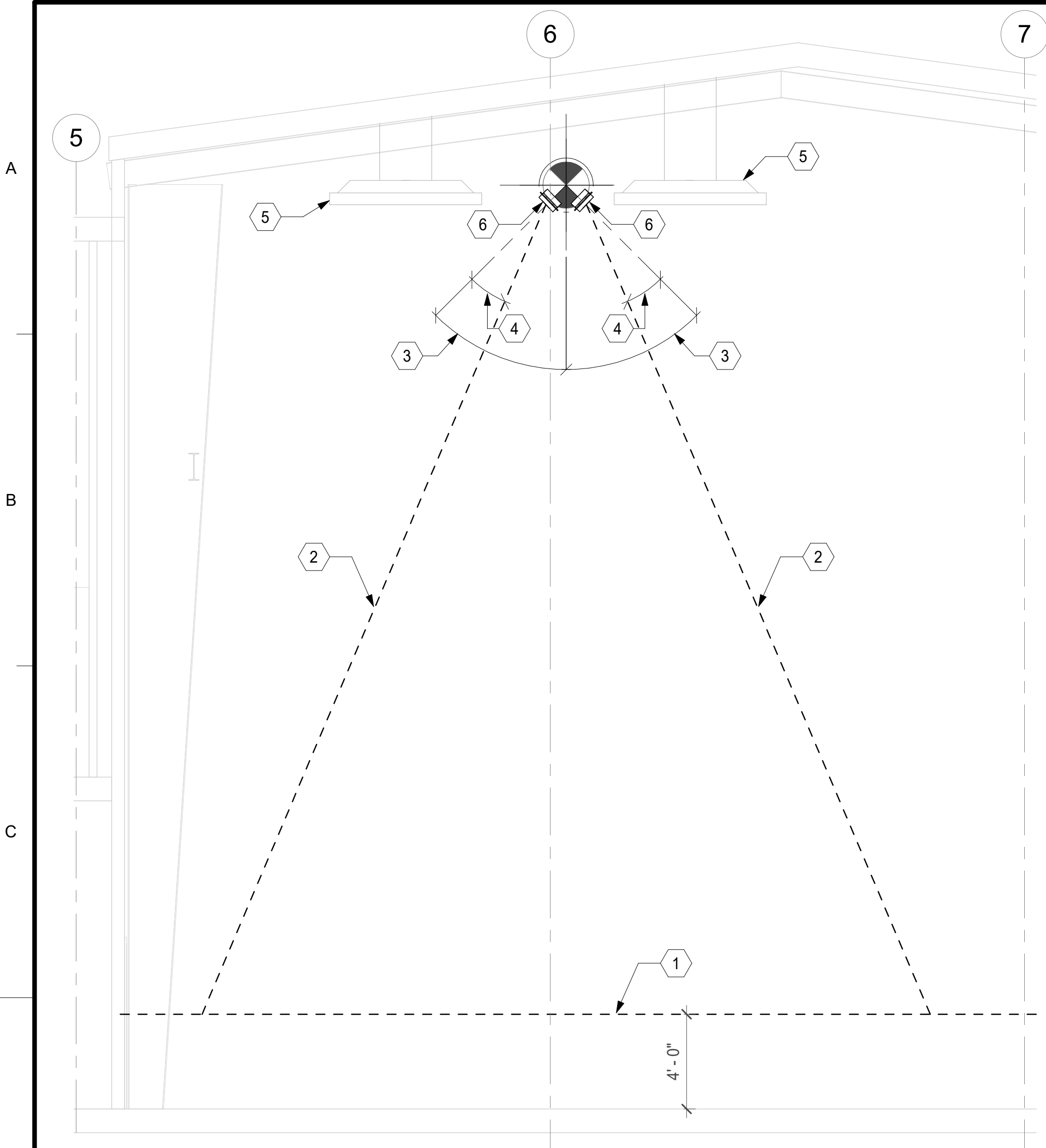
F1 CHILLER YARD DETAILS AND CHILLER CONTROLS APPURTENANCES

M-506 SCALE: NO SCALE SHOWN

Sheet Title
MECHANICAL DETAILS

Dwg. No.
PLM2021-0432-0028D

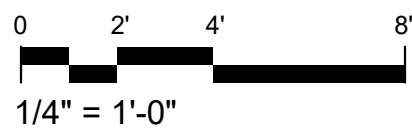
Sht. No.
M-506



- NOTES:**
- 1 WORK PLANE (48" AFF)
 - 2 CENTER OF AIRFLOW PATH
 - 3 INSTALLATION ANGLE FROM PLUMB. REFER TO DRG SCHEDULE.
 - 4 DIFFUSER DEFLECTION ANGLE. REFER TO DRG SCHEDULE.
 - 5 EXISTING LIGHTING TO REMAIN.
 - 6 MOUNT DIFFUSER ON DUCTWORK AT ANGLE SCHEDULED. CONSTRUCT DUCT TAP TO ACCOMMODATE DUCT-MOUNTED VOLUME DAMPER AND TO ACCOMMODATE INSTALLATION OF BASIS OF DESIGN DIFFUSER. SUPPORT DIFFUSER TAKEOFFS WITH DEDICATED STRAP HANGER SUPPORTS AS NECESSARY TO PREVENT SAG.

E1 TYPICAL DIFFUSER INSTALLATION - HIGH BAY

M-507 SCALE: 1/4" = 1'-0"

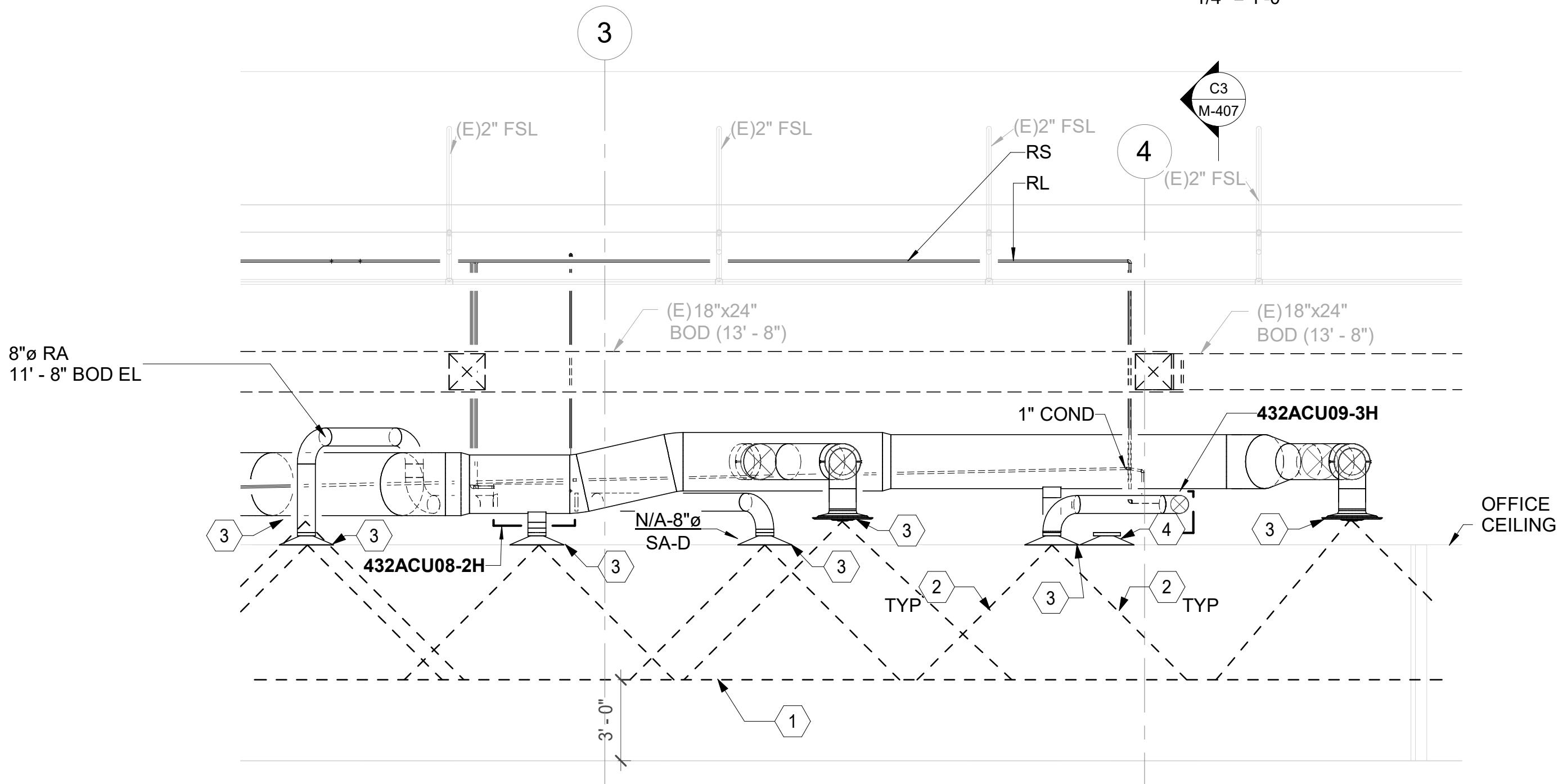
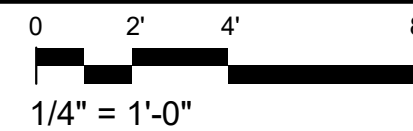


NOTES:

- 1 WORK PLANE (36" AFF)
- 2 CENTER OF AIRFLOW PATH
- 3 INSTALLATION ANGLE FROM PLUMB. REFER TO DRG SCHEDULE.
- 4 DIFFUSER DEFLECTION ANGLE. REFER TO DRG SCHEDULE.
- 5 EXISTING LIGHTING TO REMAIN.
- 6 MOUNT DIFFUSER ON DUCTWORK AT ANGLE SCHEDULED. CONSTRUCT DUCT TAP TO ACCOMMODATE DUCT-MOUNTED VOLUME DAMPER AND TO ACCOMMODATE INSTALLATION OF BASIS OF DESIGN DIFFUSER. USE LOW-LOSS 45DEG.TAPS. SUPPORT DIFFUSER TAKEOFFS WITH DEDICATED STRAP HANGER SUPPORTS AS NECESSARY TO PREVENT SAG.
- 7 BASIS OF DESIGN DIFFUSERS ARE SELECTED TO PRODUCE MAXIMUM VELOCITY OF 50FPM AT WORK PLANE WITH BLADE DEFLECTION SCHEDULED. REFER TO DRG SCHEDULE.

C5 TYPICAL DIFFUSER INSTALLATION - OPTICS LAB

M-507 SCALE: 1/4" = 1'-0"

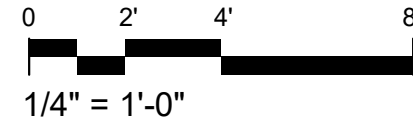


NOTES:

- 1 WORK PLANE (36" AFF)
- 2 CENTER OF AIRFLOW PATHS (4 WAY DRG)
- 3 MOUNT DIFFUSER ON DUCTWORK AT ANGLE SCHEDULED. DIFFUSERS MUST BE SIZED FOR THROWS FROM DUCT TO WORK PLANE AT NO GREATER THAN 100FPM. REFER TO DRG SCHEDULE.
- 4 HARD LID CEILING FOR ROOMS 2005 AND 2007 EXCLUSIVELY.

F5 TYPICAL DIFFUSER INSTALLATION - OFFICE

M-507 SCALE: 1/4" = 1'-0"



Sheet Title
MECHANICAL DETAILS

Dwg. No.
PLM2021-0432-0029D

Sht. No.
M-507

A

WATER COOLED CHILLER SCHEDULE (RCHS)																											
TAG NO.	NOMINAL COOLING CAPACITY (TONS)	NON-STANDARD PART-LOAD VALUE (INPLV)	EVAPORATOR							CONDENSER						COMPRESSOR				ELECTRICAL				INSTALLED WEIGHT (LBS.)	BASIS OF DESIGN		NOTES
			FLUID TYPE	DESIGN FLOW (GPM)	MIN FLOW (GPM)	EWI (DEG. F.)	LWT (DEG. F.)	MAX. WATER PRESSURE DROP (FT. H2O)	FOULING FACTOR	FLUID TYPE	DESIGN LCW FLOWRATE (GPM)	EWI (DEG. F.)	LWT (DEG. F.)	MAX.WPD (FT. H2O)	FOULING FACTOR	TYPE	QUANTITY / CIRCUITS	REF TYPE	MAX. CHARGE (LBS.)	V	PH	MCA	MOCP		MFR	MODEL	
432RCHS01-1	90	0.5816	WATER	220	92	53	43	19.1	0.0001	LCW	291	85	95	24.5	0.005000	SCREW	2	R-134A	194	460	3	108	150	5800	TRANE	RTWD090	ALL
NOTES: 1. OPERATING FLUID IS WATER. 2. PROVIDE WITH INTEGRAL CONTROLS PACKAGE AND CONTROL PANEL WITH BACnet/MSTP INTERFACE TO ALC DDC SYSTEM. 3. PROVIDE WITH LOCKABLE DISCONNECT AND SINGLE POINT POWER CONNECTION BY DIV 26, 65K SCC. 4. PROVIDE WITH INTEGRAL EVAPORATOR AND CONDENSER FLOW SWITCHES, FACTORY-INSTALLED. 5. PROVIDE WITH (1) 2-WAY BUTTERFLY-TYPE (LUG-STYLE) LCW CONTROL VALVES PLUS VALVE ACTUATORS FOR FIELD-INSTALLATION. 6. OIL PUMP STARTER SHALL BE BY MANUFACTURER AND DISCONNECT SHALL BE BY DIVISION 26. 7. PROVIDE NRTL 'LISTED' EQUIPMENT INCLUDING 'LISTED' ELECTRICAL PANEL.																											

B

VERTICAL SPLIT-CASE HVAC PUMP SCHEDULE																							
TAG NO.	TYPE	LOCATION	SERVICE	MAXIMUM FLUID TEMP. (°F)	MINIMUM FLUID TEMP. (°F)	CAPACITY (GPM)	SPEED CONTROL TYPE	HEAD (FT. W.C.)	SUCTION SIZE (IN.)	DISCHARGE SIZE (IN.)	IMPELLER SIZE (IN.)	PUMP EFFICIENCY (%)	MOTOR				STARTER / DISCONNECT PROVIDED BY	ELECTRICAL		APPROX WEIGHT (LBS)	BASIS OF DESIGN		NOTES
													RPM	HP	BHP	TYPE		VOLTS	PHASE		MFR	MODEL	
432PCHW01-1	SPLIT CASE VERTICAL INLINE CENTRIFUGAL	CHILLER YARD	CHILLED WATER	105	43	220	VFD	130	2.5	2.5	6.25	73.6	3600	15	9.64	TEFC	DIV 26	460	3	425	BELL & GOSSETT	2.5x2.5x7	ALL
432PCHW02-1	SPLIT CASE VERTICAL INLINE CENTRIFUGAL	CHILLER YARD	CHILLED WATER	105	43	220	VFD	130	2.5	2.5	6.25	73.6	3600	15	9.64	TEFC	DIV 26	460	3	425	BELL & GOSSETT	2.5x2.5x7	ALL
NOTES: 1. PUMPS ARE EACH SIZED FOR 100% DESIGN CHILLED WATER FLOWRATE. 2. OPERATING FLUID IS WATER. 3. DISCONNECT AND VFD SHALL BE FURNISHED AND INSTALLED BY DIVISION 26. DIV. 23 TO TEST AND ADJUST VFD AND PUMP TO MEET PROJECT DESIGN CONDITIONS. 4. MOUNT IN LINE WITH PIPING. REFER TO STRUCTURAL S-514 FOR SUPPORTS AND ANCHORAGE. 5. NPSHR SHALL BE A MAXIMUM OF 10 FT. WC. 6. PUMP SHALL BE INSTALLED TO MFR'S RECOMMENDATIONS AND ORIENTATION. 7. INSULATE PUMP PER SPEC. SECTION 20 07 00.																							

C

TANK SCHEDULE																
TAG NO.	TYPE	ORIENTATION	LOCATION	SYSTEM	MIN. TANK VOLUME (GAL)	MIN. TANK ACCEPTANCE VOLUME (GAL)	RELIEF VALVE SETPOINT (PSI)	DESIGN TEMP. (°F)	MIN. TEMP (°F)	COLD FILL PRESSURE (PSI)	MAX. WORKING PRESS. (PSI)	COLD WATER FILL SIZE (IN)	BASIS OF DESIGN		APPROX OPERATING WEIGHT (LBS)	NOTES
													MFR	MODEL		
432TCHWX01-1	BLADDER EXPANSION TANK	VERITCAL	CHILLER YARD	CHILLED WATER	16.63	8.7	79	105	45	18	75	1	BELL & GOSSETT	B-85LA	95	1,2,3,4
432TCHW01-1	BUFFER TANK	VERITCAL	CHILLER YARD	CHILLED WATER	1050	1050	N/A	105	45	18	150	1	TACO	BTH	2260	1,2,3,5,6
NOTES:																
1. TANK SHALL BE PROVIDED WITH FLANGED CONNECTIONS.																
2. REFER TO STRUCTURAL S-514 FOR SUPPORTS AND ANCHORAGE.																
3. TANK SHALL BE ASME-RATED.																
4. PRE-CHARGE TANK TO 14 PSIG.																
5. INSULATE TANK PER SPEC. SECTION 20 07 00.																
6. PROVIDE WITH HIGH FLANGE CONNECTIONS.																

D

AIR SEPARATOR SCHEDULE											
TAG NO.	LOCATION	SYSTEM	DESIGN FLOWRATE (GPM)	MAX ALLOWABLE PRESSURE DROP (FT W.G.)	CONNECTION TYPE	CONNECTION SIZE (IN)	MAX PRESSURE (PSIG)	BASIS OF DESIGN		WEIGHT (LBS)	NOTES
								MFR	MODEL		
432SDE01-1	CHILLER YARD	CHW	220	0.82	FLANGE	4	125	BELL & GOSSETT	SRS 4FB	250	ALL
NOTES: 1. OPERATING FLUID IS WATER. 2. AIR SEPARATOR SHALL BE THE TANGENTIAL TYPE WITHOUT STRAINER. 3. PROVIDE WITH MANUAL BLOWDOWN VALVE TO FACILIATE ROUTINE CLEANING. 4. INSULATE AIR SEPARATOR PER SPEC. SECTION 20 07 00.											

E

F

Sheet Title

MECHANICAL SCHEDULES - WATERSIDE

Dwg. No.
PLM2021-0432-0030D

Sht. No.
M-601

A

AIR CONDITIONING UNIT SCHEDULE (1/2)																														
TAG	SUPPLY FAN			COOLING COIL													HEATING COIL										FILTER			
	DESIGN AIRFLOW - COOLING MODE (CFM)	DESIGN AIRFLOW - HEATING MODE (CFM)	DESIGN MIN. OUTDOOR AIRLFOW (CFM)	EAT DB (°F)	EAT WB (°F)	LAT DB (°F)	LAT WB (°F)	MAX. GPM	EWT (°F)	LWT (°F)	NET TOTAL CAPACITY (MBH)	SENSIBLE CAPACITY (MBH)	APD (IN. W.G.)	WPD (FT. W.G.)	ROWS	FPI	EER	HEATING CAPCITY (MBH)	EAT (°F)	MAX LAT (°F)	EWT (°F)	LWT (°F)	FLOW (GPM)	MAX. WPD (FT. WC)	ROWS	FPI	MERV		APD (IN. W.G.)	
																											PRE	FINAL		
432ACU01-A1	16,400	16,400	980	80	67	55	54.42	127.0	45	55	637.45	451.95	0.535	16.33	4	11	N/A	241.5	64.8	78.7	140	117	21	10	1	7	8	15	2	
432ACHPS01-4	6,500	4,400	1,260	80	67	57.2	55.1	N/A	N/A	N/A	218.4	157.9	0.44	N/A	6	14	8.4	118.5	56.2	95	N/A	N/A	N/A	N/A	N/A	6	14	8	15	0.65
432ACHPS02-5	5,000	3,600	450	80	67	58.2	56.4	N/A	N/A	N/A	187.1	127.2	0.3	N/A	6	14	8.4	112	63.5	95	N/A	N/A	N/A	N/A	N/A	6	14	8	15	0.65

B

AIR CONDITIONING UNIT SCHEDULE (2/2)																							
TAG	SUPPLY FAN				RELIEF FAN					CONDENSER COIL		ELECTRICAL								APPROX. UNIT WEIGHT (LBS)	BASIS OF DESIGN		NOTES
	TOTAL STATIC (IN W.G.)	EXTERNAL STATIC (IN W.C.)	MHP / QTY	TOTAL BHP	MAX. AIRFLOW (CFM)	MHP / QTY	TOTAL BHP	TOTAL STATIC (IN. W.C.)	EXTERNAL STATIC (IN. W.C.)	ROWS	FPI	CIRCUIT 1				CIRCUIT 2							
												V	PH	MCA	MOCP	V	PH	MCA	MOCP				
432ACU01-A1	4.76	3.25	10 / 4	19.29	16400	3 / 4	8.01	1.80	0.5	N/A	N/A	460	3	20.4	25	460	3	59.5	70	9892	TRANE	CLIMATE CHANGER	2, 3, 4, 6, 8, 9,10,11
432ACHPS01-4	4.58	3.25	7.5 / 1	5.7	6000	3 / 1	2.6	0.75	0.5	3	12	460	3	54.4	60	N/A	N/A	N/A	N/A	3331	TRANE	HORIZON	1, 3, 4, 5, 7, 8, 9,10,11
432ACHPS02-5	4.10	3.25	7.5 / 1	5.7	5500	5 / 1	2.75	0.75	0.5	3	12	460	3	52.4	60	N/A	N/A	N/A	N/A	3384	TRANE	HORIZON	1, 3, 4, 5, 7, 8, 9,10,11
NOTES:																							
1. PROVIDE WITH MANUFACTURER'S DISCONNECT, VFD'S WITH BACnet/MSTP CONNECTION, AND SINGLE POINT POWER CONNECTION.																							
2. PROVIDE WITH MANUFACTURER'S DISCONNECT, VFD'S WITH BACnet/MSTP CONNECTION, AND DUAL POINT POWER CONNECTION. POWER EXHAUST ECONOMIZER REQUIRES SEPARATE CIRCUIT.																							
3. PROVIDE AHU WITH MANUFACTURER'S INTEGRAL 0-100% MODULATING POWER EXHAUST ECONOMIZER.																							
4. EXTERNAL STATIC PRESSURE DROP VALUES IN SCHEDULE INCLUDES 2 IN. W.G. ALLOWANCE PER UNIT FOR DIRTY FILTERS.																							
5. R-410A IS THE REFRIGERANT.																							
6. WATER IS THE COOLING AND HEATING MODE OPERATING FLUID.																							
7. PROVIDE WITH INTEGRAL CONTROLS PACKAGE AND UNIT CONTROLLER WITH TRANE BCI, INTEGRATE WITH ALC BUILDING AUTOMATION SYSTEM VIA BACnet/MSTP CONNECTION.																							
8. PROVIDE UNIT WITH 2" PLEATED MERV 8 PRE-FILTER. PROVIDE UNIT WITH 12 IN. DEEP SIDE LOAD BAG TYPE MERV 15 FILTER CARTRIDGE.																							
9. PROVIDE UNIT WITH MFR'S BASERAIL (MINIMUM 6 INCHES).																							
10. SMOKE DETECTOR(S) FURNISHED. EXCLUDE FACTORY-INSTALLED SMOKE DETECTOR(S).																							
11. PROVIDE WITH AIRFLOW MEASUREMENT FAN INLET RING FOR EACH SUPPLY AND EACH RELIEF FAN. PROVIDE WITH OUTSIDE AIRFLOW MEASUREMENT DEVICE.																							

C

COIL SCHEDULE										
TAG NO.	TYPE	MAX DESIGN AIRFLOW (CFM)	EAT (°F)	LAT (°F)	AIR PRESSURE DROP (IN W.G.)	ELECTRICAL				NOTES
						POWER (kW)	V	PH	A	
432HEA01-A1	ELECTRIC RESISTIVE HEATING	4230	56	70	0.05	20	460	3	25.1	1,2
432HEA02-4	ELECTRIC RESISTIVE HEATING	1260	40	65	0.05	10	460	3	12.5	1,2
432HEA03-5	ELECTRIC RESISTIVE HEATING	1260	40	65	0.05	10	460	3	12.5	1,2,3
NOTES: 1. PROVIDE UNIT WITH ELECTRICAL DISCONNECT BY DIVISION 26. 2. PROVIDE UNIT WITH SILICON CONTROLLED RECTIFIER (SCR) CONTROLS HARD-WIRED TO ALC. 3. PROVIDE UNIT WITH NEMA 3R ENCLOSURE FOR OUTDOOR INSTALLATION.										

D

CONDENSING UNIT SCHEDULE (RCUA)																
TAG NO.	INTERLOCK WITH	COOLING CAPACITY (BTU/H)	HEATING CAPACITY (BTU/H)	STAGE	COMPRESSOR TYPE	MIN. EFFICIENCY (EER)	RATED OUTDOOR DB (°F)	REFRIGERANT TYPE	UNIT ELECTRICAL			WEIGHT (LBS)	SIZE (H X W X D) (IN)	BASIS OF DESIGN		NOTES
									VOLT	PH.	MCA			MFR	MODEL	
432RCUA01-2H	432ACU08-2H	24000	24000	SINGLE	SCROLL	12.5	95	R-410A	208	1	19	150	38x38x14	TRANE/MITSUBISHI	TPEAD/TRUZ	ALL
432RCUA02-3H	432ACU09-3H	12000	12000	SINGLE	SCROLL	12.5	95	R-410A	208	1	11	140	32x12x24	TRANE/MITSUBISHI	TPEAD/TRUZ	ALL
NOTES:																
1. PROVIDE UNIT WITH MFR'S LOCKABLE DISCONNECT AND SINGLE POINT POWER CONNECTION IN ACCORDANCE WITH DIVISION 26 SPECIFICATIONS.																
2. PROVIDE CONDENSING UNIT WITH CONCRETE HOUSEKEEPING PAD. REFER TO STRUCTURAL S-402 FOR ANCHORAGE.																
3. PROVIDE ALUMINUM JACKETING FOR EXTERIOR REFRIGERANT LINES.																
4. HEAT PUMP UNIT SHALL BE CAPABLE OF OPERATION TO 30°F.																
5. PROVIDE UNIT WITH ELECTRIC RESISTIVE DEFROST CYCLE CAPABILITY.																

E

AIR CONDITIONING UNIT SCHEDULE (ACU)												
TAG NO.	INTERLOCK WITH	SUPPLY FAN						COOLING COIL				NOTES
		FAN TYPE	TOTAL AIRFLOW (CFM)	MOTOR			POWER PROVIDED BY	COOLING COIL TYPE	TOTAL CAPACITY (MBH)	LAT DB (°F)	LAT WB (°F)	
				MCA	VOLTS	PHASE						
432ACU08-2H	432RCUA01-2H	BLOWER	1770	1	208	1	432RCUA01-2H	DIRECT EXPANSION	24	57	57	ALL
432ACU09-3H	432RCUA02-3H	BLOWER	400	1	208	1	432RCUA02-3H	DIRECT EXPANSION	12	57	57	ALL
NOTES:												
1. UNIT POWER PROVIDED BY CONDENSING UNIT ON GRADE AT EXTERIOR OF BUILDING. REFER TO RCUA SCHEDULE.												
2. PROVIDE UNIT WITH DUCTED SUPPLY AND DUCTED RETURN FOR UNIT.												
3. PROVIDE UNIT WITH MFR'S CONDENSATE LIFT PUMP WITH INTERGRAL RESERVOIR TANK AND CHECK VALVE. PROVIDE POWER FOR UNIT BY INTERLOCKED CONDENSING UNIT.												
4. PROVIDE FIELD-INSTALLED CONDENSATE DRAIN PAN AND CONDENSATE OVERFLOW SWITCH.												
5. PROVIDE UNIT WITH MFR'S DISCONNECT.												
6. PROVIDE UNIT WITH MFR'S HARDWIRED TOUCHSCREEN THERMOSTAT INTERFACE WITH BACnet/MSTP COMMUNICATION CAPABILITIES.												

F

Sheet Title
MECHANICAL SCHEDULES - AIRSIDE

Dwg. No.
PLM2021-0432-0031D

Sht. No.
M-602

ABBREVIATIONS & ACRONYMS

2P	TWO-POSITION (CONTROL SIGNAL)	M&C	MONITORING & CONTROL (SOFTWARE)
ADJ	ADJUSTABLE/ADJUSTMENT	MS	MOTOR STARTER
AI	ANALOG INPUT	N/A	NOT APPLICABLE
ALM	ALARM	NC	NORMALLY CLOSED
AO	ANALOG OUTPUT	NCI	NETWORK CONFIGURATION INPUT
BI	BINARY INPUT	NO	NORMALLY OPEN
BLDG	BUILDING	NVI	NETWORK VARIABLE INPUT
BO	BINARY OUTPUT	NVO	NETWORK VARIABLE OUTPUT
BYP	BYPASS	OCC	OCCUPIED
C	COMMAND (MODULATING CONTROL SIGNAL)	ODT	ON DELAY TIMER
CLD	CLOSED	OL	OVERLOAD
CLG	COOLING	OPN	OPEN
COM	COMMON	OVRD	OVERRIDE
COV	CHANGE OF VALUE	P	PRESSURE
CSR	CURRENT SENSING RELAY	PID	PROPORTIONAL INTEGRAL DERIVATIVE (CONTROL)
CT	CURRENT TRANSFORMER/SWITCH	PWR	POWER
D	DAMPER	R	RELAY
DB	DEADBAND	REV	REVERSE (CONTROL ACTION)
DDC	DIRECT DIGITAL CONTROL(LER)	RM	ROOM
DIFF	DIFFERENCE	RQST	REQUEST
DIR	DIRECT (CONTROL ACTION)	RST	RESET
DIS	DISABLE	RT	RATE
DISP	DISPLAY	S	STATUS
EA	EXHAUST AIR	SNVT	STANDARD NETWORK VARIABLE TYPE
ECM	ELECTRICALLY COMMUTATED MOTOR	SO	SHUT-OFF
EF	EXHAUST FAN	SP	SETPOINT
ENA	ENABLE	SS	START/STOP COMMAND
F	FLOW	STAT	THERMOSTAT
FACP	FIRE ALARM CONTROL PANEL	SYS	SYSTEM
FL	FAIL IN LAST POSITION	SCHD	SCHEDULER
FLT	FILTER	T	TEMPERATURE
HL	HIGH LIMIT	TRB	TROUBLE
HTG	HEATING	TSTAT	THERMOSTAT
I/O	INPUT/OUTPUT	UNOCC	UNOCCUPIED
LDP	LOCAL DISPLAY PANEL	VFD	VARIABLE FREQUENCY DRIVE
LL	LOW LIMIT	XFMR	TRANSFORMER
LVG	LEAVING	ZN	ZONE
M	MOTOR or MAIN		

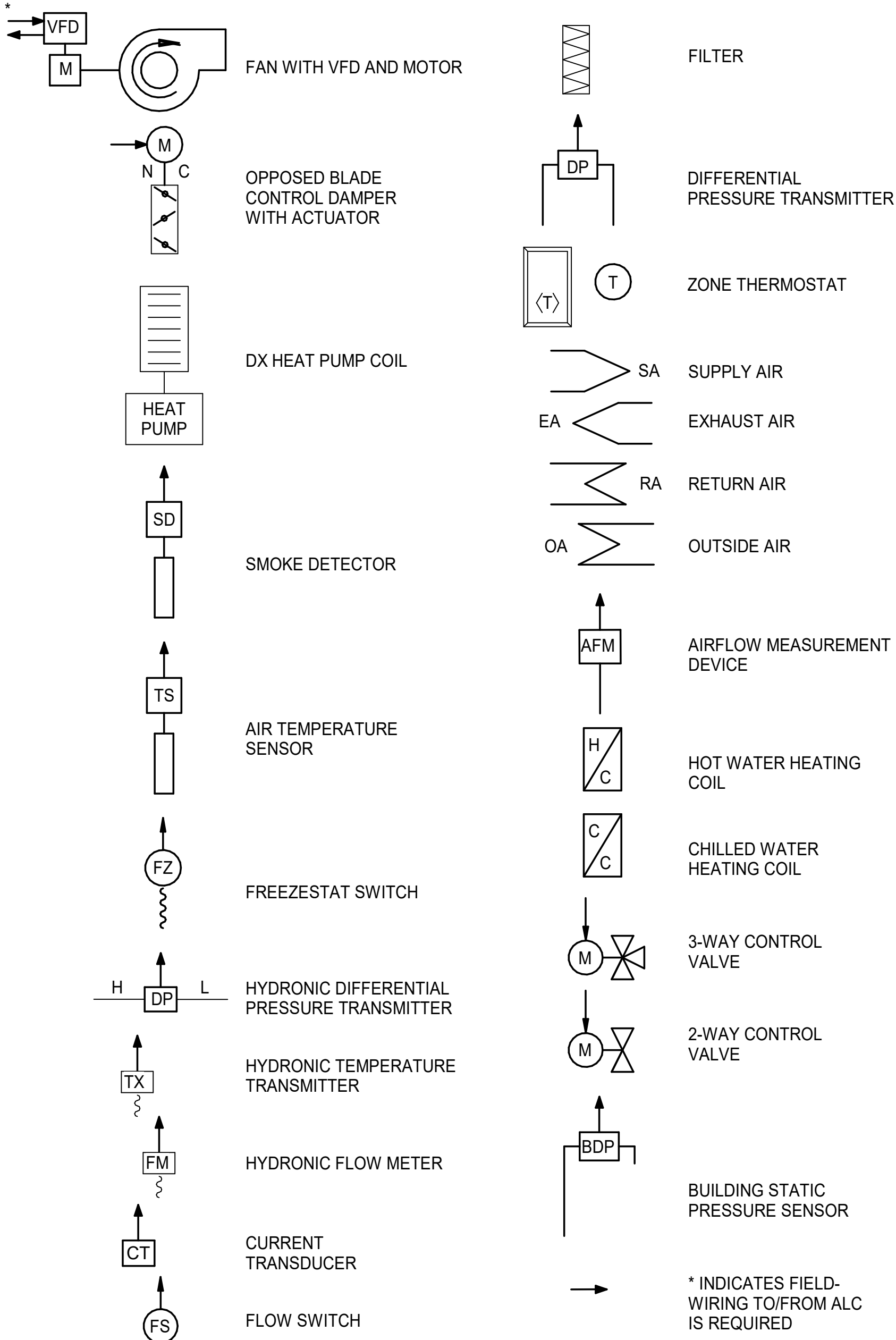
"HVAC CONTROLS" GENERAL NOTES:

1. THE CONTROL DIAGRAMS AND SCHEMATICS ILLUSTRATE THE FUNCTIONAL REQUIREMENTS AND CONTROL RELATIONSHIPS, ALL ACCESSORIES AND DEVICES MAY NOT BE SHOWN IN DETAIL
2. THE SPECIFICATIONS SHOULD BE CONSULTED FOR DETAILED REQUIREMENTS.
3. BUILDING CONTROL SYSTEM MUST BE COMPLETELY OPERATIONAL BEFORE BUILDING COMMISSIONING CAN TAKE PLACE

NOTES:

1. FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS REFER TO DRAWING M-001.
2. REFER TO ELECTRICAL FOR NEW LINE-VOLTAGE POWER EQUIPMENT, WIRING, AND CONDUIT.
3. LOW-VOLTAGE POWER WIRING AND CONDUIT FROM NEW 120V/24V POWER SUPPLIES, CONTROL WIRING AND CONDUIT FROM CONTROLLERS TO END CONTROL DEVICES, AND COMMUNICATION WIRING AND CONDUIT BETWEEN NETWORKED ALC EQUIPMENT MUST BE FURNISHED AND INSTALLED BY CONTROLS SUBCONTRACTOR FOR A COMPLETE AND OPERATING SYSTEM. REFER TO SPEC. SECTION 26 05 33.13 FOR LOW-VOLTAGE POWER, CONTROL, AND COMMUNICATION CONDUIT SPECIFICATIONS. REFER TO SPEC. SECTION 23 09 00 FOR LOW-VOLTAGE POWER, CONTROL, AND COMMUNICATION CONDUCTOR SPECIFICATIONS.
4. IT IS THE RESPONSIBILITY OF THE MECHANICAL SUBCONTRACTOR TO COORDINATE THE LOCATION OF ALL CONTROLS EQUIPMENT AND CONTROL DEVICES WITH THE CONTROLS SUBCONTRACTOR AND THE ELECTRICAL SUBCONTRACTOR AND NOTIFY THE CONTROLS SUBCONTRACTOR AND ELECTRICAL SUBCONTRACTOR OF CONDUIT ROUTING PREFERENCES AND POWER SUPPLY LOCATION PREFERENCES.
5. CONTROLS SUBCONTRACTOR MUST FURNISH AND INSTALL WIRE AND CONDUIT FOR ALL SMOKE DETECTORS: FROM DEVICE TO FIRE ALARM CONTROL SYSTEM ADDRESSABLE LOOP, FROM DEVICE TO ALC, AND FROM FIRE ALARM CONTROL LOOP TO FAN MOTOR.
6. CONTROLS SUBCONTRACTOR MUST FURNISH AND INSTALL WIRE AND CONDUIT REQUIRED TO CONNECT INDIVIDUAL MODULES OF FIELD-ASSEMBLED EQUIPMENT.
7. ALL CONTROL DEVICES EXPOSED TO WEATHER MUST BE INSTALLED IN WEATHERPROOF ENCLOSURES.

HVAC CONTROL SYMBOL LEGEND



CONTROLS CONDUIT SCHEDULE				
CONDUCTOR TYPE	EXPOSED - NOT SUBJECT TO DAMAGE	EXPOSED - SUBJECT TO DAMAGE	CONCEALED	
	MECH/ELEC/FIRE RISER ROOMS	ROOF & ELSEWHERE 0-36" AFF	CEILING PLENUM	WALL PLENUM
LOW-VOLTAGE POWER/CONTROL WIRE	EMT OR RMC	RMC	NO CONDUIT REQUIRED	EMT OR RMC
ETHERNET WIRE/CABLE	EMT OR RMC	RMC	EMT	EMT OR RMC
OTHER COMMUNICATION WIRE/CABLE	EMT OR RMC	RMC	INNERDUCT OR EMT	EMT OR RMC
LINE-VOLTAGE POWER WIRING	PROVIDED BY DIV. 26; REFER TO ELECTRICAL	PROVIDED BY DIV. 26; REFER TO ELECTRICAL	PROVIDED BY DIV. 26; REFER TO ELECTRICAL	PROVIDED BY DIV. 26; REFER TO ELECTRICAL

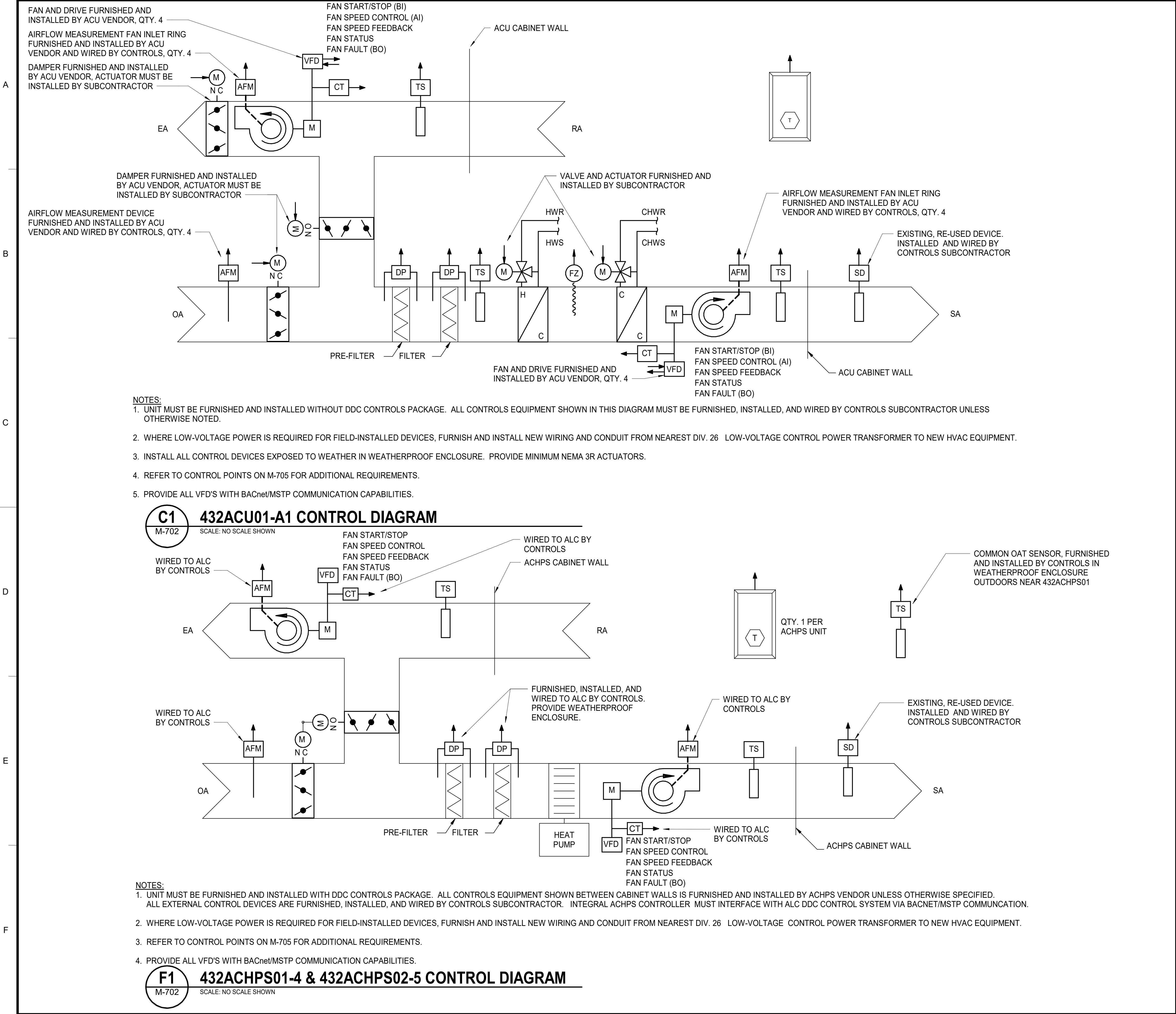
NOTES:

- 1) ALL CONDUCTORS INSTALLED IN A WALL CAVITY OR CEILING PLENUM SHALL BE PLENUM-RATED.
- 2) ALL CONDUCTORS INSTALLED IN CEILING PLENUM SHALL BE TIED OFF TO NEW OR EXISTING SUPPORTS; NOT LAID ON CEILING TILES.
- 3) ALL CONDUIT SHOWN HERE SHALL BE FURNISHED AND INSTALLED BY CONTROLS SUBCONTRACTOR UNLESS OTHERWISE SPECIFIED EITHER IN THIS TABLE OR ON A PLAN.
- 4) REFER TO SPEC. SECTION 26 05 33.13 FOR LOW-VOLTAGE POWER, CONTROL, AND COMMUNICATION CONDUIT SPECIFICATIONS. REFER TO SPEC. SECTION 23 09 00 FOR LOW-VOLTAGE POWER, CONTROL, AND COMMUNICATION CONDUCTOR SPECIFICATIONS.
- 5) SUBCONTRACTOR SHALL SUBMIT REQUEST FOR APPROVAL TO INSTALL EXPOSED CONDUIT IN FINISHED AREAS. CONDUIT REQUIRED FOR EXPOSED, FINISHED AREAS IS RIGID METAL CONDUIT UNLESS APPROVAL IS PROVIDED BY ENGINEER TO USE EMT. SUBCONTRACTOR SHALL PAINT EXPOSED CONDUIT IN FINISHED AREAS TO MATCH COLOR OF ADJACENT SURFACE.
- 6) MAINTAIN 6 INCHES (MIN.) CLEARANCE BETWEEN CONDUIT AND OTHER MECH. PIPING AND TUBING. MAINTAIN 1 INCH (MIN.) CLEARANCE BETWEEN CONDUIT AND DUCTWORK. OFFSETS APPLY TO SURFACE OF INSULATION ON INSULATED PIPING.
- 7) NEW HORIZONTAL CONDUIT RUNS SHALL BE ROUTED OVERHEAD AND AS CLOSE TO STRUCTURE AS POSSIBLE. DO NOT FLOOR-MOUNT NEW CONDUIT.
- 8) OUTER CASING OF ALL NEW CONDUIT PENETRATING FLOOR OR CEILING SHALL BE SEALED.
- 9) FLEX CONDUIT IS ALLOWED AT ALL DEVICE TERMINATIONS.

Sheet Title
MECHANICAL CONTROLS
LEGEND AND GENERAL
NOTES

Dwg. No.
PLM2021-0432-0032D

Sht. No.
M-701



GENERAL NOTES

- REFER TO M-701 FOR CONTROLS ABBREVIATIONS, SYMBOL LEGEND, AND GENERAL NOTES.

Sheet Title
**MECHANICAL AIR HANDLING
UNIT CONTROL DIAGRAMS**

Dwg. No.
PLM2021-0432-0033D

Sht. No.
M-702

A

B

C

D

E

F

POINT NAME	HARDWARE POINTS				SOFTWARE POINTS					
	AI	AO	BI	BO	AV	BV	LOOP	SCHED	TREND	ALARM
432RCHS01-1 (NOTE 1)										
Chiller Start/Stop Command				1				1	1	
Chiller Status			1						1	1
Chilled Water Supply Temp Setpoint Reset					1				1	
Chiller General Alarm			1							1
Chiller CHWS-T > SP + 5deg.F Alarm						1				1
Chiller Run Fault			1							1
Chiller Compressor Status						2			2	2
Chiller Current					1				1	
Chiller Instantaneous Power (kW)					1				1	
Chiller Misc. Monitoring 1					1				1	1
Chiller Misc. Monitoring 2					1				1	1
Condenser Water Head Pressure Control Valve (NOTE 2)										
Condenser Water Head Pressure Control Valve Position					1				1	1
Chilled Water Flow Switch Status (NOTE 2)										
Condenser Water Flow Switch Status (NOTE 2)										
Condenser Water Bypass Control Valve				1				1	1	
Chilled Water Return Temp (NOTE 3)	1								1	1
Condenser Water Supply Temp (NOTE 3)	1								1	1
Chilled Water Flow Meter (NOTE 3)	1								1	1
432PCHW01-1 & 432PCHW02-1 (NOTE 1)										
VFD Pump Start/Stop Command				2				2	2	
VFD Pump Speed Control Command		2						2	2	
VFD Status			2						2	2
VFD Speed/Current	2								2	
VFD Instantaneous Power (kW)					2				2	
VFD General Alarm			2						2	2
Pump Differential Pressure (NOTE 3)	2								2	2
432ACU08-2H / 432RCUA01-2H & 432ACU09-3H / 432RCUA02-3H										
Unit Enable Command						2		2	2	
Room Temp Setpoint					2				2	
Unit General Alarm						2				2
Unit Run Fault						2				2
Unit Current					2				2	
Unit Instantaneous Power (kW)					2				2	
Unit Misc. Monitoring 1					2					2
Unit Misc. Monitoring 2					2					2
Room Temp (NOTE 3)	2								2	2
432HEA01-A1 & 432HEA02-4 & 432HEA03-5										
Heater Enable Command		3							3	
Room Temp Setpoint					3				3	
Room Temp (NOTE 3)	3								3	3
Supply Airflow (NOTE 3)	3								3	3
TOTALS	15	5	7	4	21	9	0	8	50	34

Total Hardware (31)
Total Software (122)
NOTES:
1. NEW ALC DDC UNIT CONTROLLERS SERVING 432RCHS01-1, 432PCHW01-1, & 432PCHW02-1 SHALL BE SIZED AND SELECTED TO SERVE AS CENTRAL CONTROL PANEL (432DDCP01) FOR ALL NEW HVAC EQUIPMENT REQUIRED BY PROJECT AND INCLUDE 10% SPARE CAPACITY FOR EACH CATEGORY OF HARD-WIRED POINT.
2. WIRE CONTROL DEVICE DIRECTLY TO FACTORY CHILLER CONTROL PANEL. IF POINT CAN BE MONITORED OVER ALC, MONITOR POINT. FLOW SWITCHES ARE FACTORY-INSTALLED AND FACTORY-WIRED.
3. EQUIPMENT VENDOR WILL NOT PROVIDE CONTROL DEVICE, SUBCONTRACTOR TO PROVIDE FIELD-INSTALLED, HARD-WIRED CONTROL DEVICE.
4. REFER TO SEQUENCES OF OPERATION FOR ALL DEFAULT SYSTEM SETPOINTS. COORDINATE FINAL SYSTEM SETPOINTS WITH LLNL.
5. CONTROLS SUBCONTRACTOR TO PROVIDE ALL PROGRAMMING REQUIRED TO DELIVER NEW ALARM NOTIFICATIONS OVER NEW ALC SYSTEM TO LLNL EMAILS AND PAGERS. COORDINATE WITH LLNL.
6. CELLS WITH "<__>" SHALL BE DETERMINED BY THE CONTROLS SUBCONTRACTOR. SPACES WHERE NO ENTRY IS REQUIRED CONTAIN "--".

HEAT PUMP SPLIT SYSTEMS
432ACU08-2H / 432RCUA01-2H & 432ACU09-3H / 432RCUA02-3H

- RUN CONDITIONS - CONTINUOUS:
EACH EVAPORATOR AND INTERLOCKED CONDENSING UNIT (SEE EQUIPMENT SCHEDULES FOR INTERLOCKED CONDENSING UNIT) MUST RUN 24/7 TO SATISFY THE FOLLOWING SETPOINTS:
- 432ACU08: A 72°F (ADJ.) OCCUPIED COOLING SETPOINT
A 70°F (ADJ.) OCCUPIED HEATING SETPOINT
A 85°F (ADJ.) UNOCCUPIED COOLING SETPOINT
A 55°F (ADJ.) UNOCCUPIED HEATING SETPOINT
 - 432ACU09: A 72°F (ADJ.) OCCUPIED COOLING SETPOINT
A 70°F (ADJ.) OCCUPIED HEATING SETPOINT
A 85°F (ADJ.) UNOCCUPIED COOLING SETPOINT
A 55°F (ADJ.) UNOCCUPIED HEATING SETPOINT
 - 432ACU10: A 72°F (ADJ.) OCCUPIED COOLING SETPOINT
A 70°F (ADJ.) OCCUPIED HEATING SETPOINT
A 85°F (ADJ.) UNOCCUPIED COOLING SETPOINT
A 55°F (ADJ.) UNOCCUPIED HEATING SETPOINT

SPLIT SYSTEMS WILL COMMUNICATE STATUS AND ALARMS TO THE BAS FROM NATIVE UNIT CONTROLLER (REFER TO POINTS LIST). SPLIT SYSTEMS WILL OPERATE ACCORDING TO EXISTING, STAND-ALONE OEM PROGRAMMING.

- ADDITIONAL ALARMS MUST BE PROVIDED AS FOLLOWS:
- HIGH ZONE TEMP: IF THE ZONE TEMPERATURE IS GREATER THAN THE COOLING SETPOINT BY A USER DEFINABLE AMOUNT (ADJ.).

ALARM CONDITION	GRAPHIC	POINT SETTING (WITH UNITS)	POINT RANGE (WITH UNITS)	ALARM TYPE
~	Y	~	ON/OFF	~
CHILLER COMMAND IS ON, BUT CHILLER STATUS IS OFF	Y	~	ON/OFF	CRIT
~	Y	43 DEG. F.	<__>	~
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	CRIT
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
CHILLER COMMAND IS ON, BUT COMPRESSOR STATUS IS OFF	Y	~	ON/OFF	CRIT
~	Y	~	<__>	~
~	Y	~	<__>	~
FOR FUTURE USE AT OWNER'S DISCRETION	Y	~	~	INFO
FOR FUTURE USE AT OWNER'S DISCRETION	Y	~	~	INFO
~	Y	~	0-100%	~
FOR FUTURE USE AT OWNER'S DISCRETION	Y	~	<__>	INFO
~	Y	~	OPEN/CLOSED	~
~	Y	~	OPEN/CLOSED	~
~	Y	~	OPEN/CLOSED	~
CHWR TEMP GREATER THAN 57 DEG F OR LESS THAN 42 DEG F	Y	~	<__>	CRIT
LCWS TEMP GREATER THAN 86 DEG F OR LESS THAN 60 DEG F	Y	~	<__>	INFO
FLOW IS NOT EQUAL TO 220 GPM	Y	~	OPEN/CLOSED	INFO
~	Y	~	ON/OFF	~
~	Y	~	0-100%	~
VFD COMMAND IS ON, BUT VFD STATUS IS OFF	Y	~	ON/OFF	CRIT
~	Y	~	0-100%	~
~	Y	~	<__>	~
FOR FUTURE USE AT OWNER'S DISCRETION	Y	~	~	INFO
VFD COMMAND IS ON, BUT DP IS LESS THAN 5 PSIG	Y	56 PSIG	<__>	CRIT
~	Y	~	ON/OFF	~
~	Y	72 DEG. F.	<__>	~
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
~	Y	~	<__>	~
~	Y	~	<__>	~
FOR FUTURE USE AT OWNER'S DISCRETION	Y	~	~	INFO
FOR FUTURE USE AT OWNER'S DISCRETION	Y	~	~	INFO
HIGH LIMIT ALARM, LOW LIMIT ALARM - REFER TO SEQUENCE	Y	~	<__>	INFO
~	Y	~	ON/OFF	~
~	Y	72 DEG. F.	<__>	~
~	Y	~	<__>	~
HIGH LIMIT ALARM, LOW LIMIT ALARM - REFER TO SEQUENCE	Y	~	<__>	INFO

WATER FLOW METER (CHILLED WATER)

THE ALC CONTROLLER MUST MONITOR THE CHILLED WATER METER FOR CHILLED WATER FLOW (GPM) VIA ANALOG INPUT. PROVIDE CONNECTION FROM WATER FLOW METER TO ALC PANEL.

- ALARM MUST BE GENERATED AS FOLLOWS:
- FLOW IS DIFFERENT THAN DESIGN FLOWRATE (220 GPM) BY MORE THAN 5 GPM (ADJ.)

USAGE HISTORY:
THE ALC CONTROLLER MUST CALCULATE AND RECORD BTU/HR USING FLOW METER GPM VALUE AND TEMPERATURE SENSORS SO AS TO PROVIDE A CHILLED WATER CONSUMPTION HISTORY. USAGE READINGS MUST BE RECORDED ON A DAILY, MONTH-TO-DATE, AND YEAR-TO-DATE BASIS.

432HEA01-A1

WHEN ROOM 1204 TEMPERATURE IS LESS THAN 65 DEG. F. (ADJ.), SCR PROPORTIONAL ELECTRIC DUCT-MOUNTED HEATING COIL SHALL BE ENABLED TO MAINTAIN MINIMUM 65 DEG. F. (ADJ.) ROOM TEMPERATURE.

HEATING COIL SHALL BE DISABLED WHEN ROOM TEMPERATURE IS AT LEAST 65 DEG. F. (ADJ.) FOR 10 MINUTES (ADJ.)

CONTROLLER SHALL MODULATE HEATER LOAD ACCORDING TO THE TEMPERATURE CONTROL SIGNAL. HEATER SHALL PROVIDE MAXIMUM HEATING WHEN WHEN NEEDED WITH NORMAL MINIMUM AIRFLOW (TBD DURING CX), REDUCE HEATING WHEN AIRFLOW IS LOWER THAN NORMAL MINIMUM AIRFLOW (TBD DURING CX), AND STOP HEATING WHEN AIRFLOW IS ZERO.

432HEA02-4 & 432HEA03-5

WHEN ACHPS SUPPLY AIR TEMPERATURE IS LESS THAN 40 DEG. F. (ADJ.), AN ALARM SHALL BE PROVIDED OVER ALC, AND SCR PROPORTIONAL ELECTRIC DUCT MOUNTED HEATING COIL SHALL BE ENABLED TO MAINTAIN MINIMUM 65 DEG. F. (ADJ.) SUPPLY AIR TEMPERATURE.

HEATING COIL SHALL BE DISABLED AFTER 30 MINUTES (ADJ.)

CONTROLLER SHALL MODULATE HEATER LOAD ACCORDING TO THE TEMPERATURE CONTROL SIGNAL. HEATER SHALL PROVIDE MAXIMUM HEATING WHEN WHEN NEEDED WITH NORMAL MINIMUM AIRFLOW (TBD DURING CX), REDUCE HEATING WHEN AIRFLOW IS LOWER THAN NORMAL MINIMUM AIRFLOW (TBD DURING CX), AND STOP HEATING WHEN AIRFLOW IS ZERO.

Sheet Title
MECHANICAL CONTROL
POINTS AND SEQUENCES

Dwg. No.
PLM2021-0432-0035D

Sht. No.
M-704

A

B

C

D

E

F

Total Hardware (68)
Total Software (434)
NOTES:
1. SERVED BY NEW 432DDCP02. CONNECT 432DDCP02 TO 432DDCP01 VIA ARCNET OR BACNET/MSTP.
2. 432ACHPS01 AND 432ACHPS02 UTILIZE INTEGRAL ON-BOARD UNIT CONTROLLERS. CONTROLLERS SHALL BE WIRED TO COMMUNICATE WITH NEW ALC DDC SYSTEM OVER BACNET/MSTP OR MODBUS. BACNET/IP IS PROHIBITED. 432ACHPS04 IS AN EXISTING TO REMAIN UNIT THAT IS TO RECEIVE A NEW BACNET/MSTP NETWORK COMMUNICATION CARD - ONLY LISTED SOFTWARE POINTS THAT ARE AVAILABLE OVER BACNET/MSTP ARE REQUIRED OVER ALC FOR EXISTING TO REMAIN 432ACHPS04.
3. EQUIPMENT VENDOR WILL NOT PROVIDE CONTROL DEVICE. MECHANICAL SUBCONTRACTOR TO PROVIDE FIELD-INSTALLED, HARD-WIRED CONTROL DEVICE.
4. REFER TO M-702. SUBCONTRACTOR SHALL RE-USE EXISTING DUCT SMOKE DETECTOR AT EXISTING HVAC EQUIPMENT AND RE-INSTALL AND RE-WIRE DEVICE FOR USE WITH NEW HVAC EQUIPMENT.
5. REFER TO M-702. INSTALL DEVICE NEAR 432ACHPS01.
6. PROVIDE ALL CONTROL RELAYS NECESSARY TO REPORT SMOKE DETECTOR ALARM STATUS TO NEW ALC DDC SYSTEM.
7. REFER TO SEQUENCES OF OPERATION FOR ALL DEFAULT SYSTEM SETPOINTS. COORDINATE FINAL SYSTEM SETPOINTS WITH LLNL.
8. CONTROLS SUBCONTRACTOR TO PROVIDE ALL PROGRAMMING REQUIRED TO DELIVER NEW ALARM NOTIFICATIONS OVER NEW ALC SYSTEM TO LLNL EMAILS AND PAGERS. COORDINATE WITH LLNL.
9. CELLS WITH "Y", ">" SHALL BE DETERMINED BY THE CONTROLS SUBCONTRACTOR. SPACES WHERE NO ENTRY IS REQUIRED CONTAIN "~".

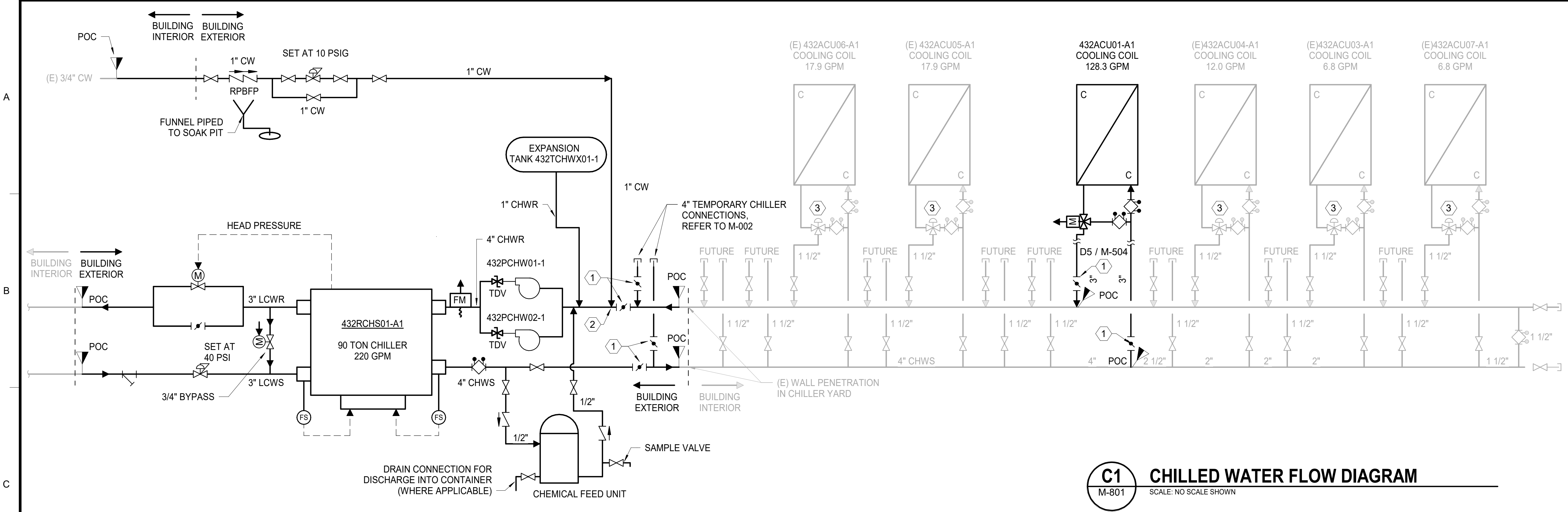
POINT NAME	HARDWARE POINTS				SOFTWARE POINTS					
	AI	AO	BI	BO	AV	BV	LOOP	SCHED	TREND	ALARM
642ACU01-A1 (NOTE 1)										
VFD Supply Fan Start/Stop Command				4				4	4	
VFD Relief Fan Start/Stop Command				4				4	4	
Smoke Detector Supply Fan Shutdown (NOTE 4)				4					4	
VFD Supply Fan Speed Control Command		4						4	4	
VFD Relief Fan Speed Control Command		4						4	4	
Heating Control Valve Command (NOTE 3)		1							1	
Cooling Control Valve Command (NOTE 3)		1							1	
Outside Air Control Damper Command (NOTE 3)		1						1	1	
Return Air Control Damper Command (NOTE 3)			1					1	1	
Relief Air Control Damper Command (NOTE 3)		1						1	1	
Pre-Filter Differential Pressure (NOTE 3)	1									1
Final Filter Differential Pressure (NOTE 3)	1									1
Outside Air Flowrate Measurement	1								1	1
VFD Supply Fan Current Transducer (NOTE 3)	4								4	4
VFD Relief Fan Current Transducer (NOTE 3)	4								4	4
Supply Fan Inlet Air Flowrate Measurement Ring	4								4	4
Relief Fan Inlet Air Flowrate Measurement Ring	4								4	4
Outside Air Damper Position Feedback (NOTE 3)	1								1	1
Return Air Damper Position Feedback (NOTE 3)	1								1	1
Relief Air Damper Position Feedback (NOTE 3)	1								1	1
Room Temp (NOTE 3)	1								1	1
Freezestat Switch Status (NOTE 3)			1						1	1
Freezestat Switch Auto Reset				1					1	
Supply Air Temp (NOTE 3)	1								1	1
Return Air Temp (NOTE 3)	1								1	1
Mixed Air Temp (NOTE 3)	1								1	1
Supply Air Temp Setpoint (NOTE 3)					1					
Mixed Air Temp Setpoint (NOTE 3)					1					
Room Temp Setpoint (NOTE 3)					1				1	
VFD Supply Fan Status					4				4	4
VFD Relief Fan Status					4				4	4
VFD Supply Fan Speed/Current					4				4	
VFD Relief Fan Speed/Current					4				4	
VFD Supply Fan Instantaneous Power (kW)					4				4	
VFD Relief Fan Instantaneous Power (kW)					4				4	
VFD Supply Fan Alarm/Failure			4						4	4
VFD Relief Fan Alarm/Failure			4						4	4
Supply Air Smoke Detector Status (NOTE 6)			1						1	1
432ACHPS01-4 & 432ACHPS02-5 & 432ACHPS04 (NOTE 2)										
Unit Enable Command						3		3	3	
Unit Heat Lockout Command						3			3	
Unit Cool Lockout Command						3			3	
Occupancy Mode Command					3			3	3	
Economizer Enable Command					3				3	
Supply Fan Start/Stop Command						3		3	3	
Relief Fan Start/Stop Command						3		3	3	
Supply Fan Speed Control Command					3			3	3	
Relief Fan Speed Control Command					3				3	
Smoke Detector Unit Shutdown (NOTE 4)				2						
Pre-Filter Differential Pressure (NOTE 3)	2									3
Final Filter Differential Pressure (NOTE 3)	2									3
Outside Air Flowrate Measurement	1								1	1
VFD Supply Fan Current Transducer (NOTE 3)	2								3	3
VFD Relief Fan Current Transducer (NOTE 3)	2								3	3
Supply Fan Inlet Airflow Measurement Ring	1								1	1
Relief Fan Inlet Airflow Measurement Ring	1								1	1
Room Temp (NOTE 3)	3								3	3
Outside Air Temp (NOTE 5)	1								2	
Supply Air Temp					3				3	3
Return Air Temp					3				3	3
Supply Air Temp Setpoint					3				3	
Room Temp Setpoint (NOTE 3)					3				3	
Unit Economizer Minimum Position Command					3			3	3	
Unit Economizer Minimum Position Setpoint					3				3	
Unit Status						3			3	
Unit General Alarm						3			3	3
Unit Run Fault						3			3	3
Unit Current					3				3	
Unit Instantaneous Power (kW)					3				3	
Unit Misc. Monitoring 1					3				3	3
Unit Misc. Monitoring 2					3				3	3
VFD Supply Fan Status					3				3	3
VFD Relief Fan Status					3				3	3
VFD Supply Fan Alarm/Failure			2			1			3	3
VFD Relief Fan Alarm/Failure			2			1			3	3
Supply Air Smoke Detector Status (NOTE 6)			1						1	1
Outside Air Damper Failure						3			3	3
Economizer Failure						3			3	3
MISCELLANEOUS										
Supply Air Smoke Detector to Fire Alarm (NOTE 3)	3									3
Optics Lab Purge Push Button Status			1						1	
TOTALS	44	13	16	15	75	32	0	40	185	102

ALARM CONDITION	GRAPHIC	POINT SETTING (WITH UNITS)	POINT RANGE (WITH UNITS)	ALARM TYPE
~	Y	~	ON/OFF	~
~	Y	~	ON/OFF	~
~	Y	~	ON/OFF	~
~	Y	~	0-100%	~
~	Y	~	0-100%	~
~	Y	~	0-100%	~
~	Y	~	0-100%	~
~	Y	~	0-100%	~
~	Y	~	0-100%	~
~	Y	~	0-100%	~
DIFFERENTIAL PRESSURE EXCEEDS SETPOINT	Y	1.0" W.C.	< >	CRIT
DIFFERENTIAL PRESSURE EXCEEDS SETPOINT	Y	1.0" W.C.	< >	CRIT
LOW LIMIT ALARM CORRESPONDING TO MINIMUM OA	Y	980 CFM	< >	CRIT
HIGH LIMIT ALARM - REFER TO SEQUENCE	Y	~	< >	CRIT
HIGH LIMIT ALARM - REFER TO SEQUENCE	Y	~	< >	CRIT
HIGH LIMIT ALARM, LOW LIMIT ALARM - REFER TO SEQUENCE	Y	16,400 CFM	< >	INFO
HIGH LIMIT ALARM, LOW LIMIT ALARM - REFER TO SEQUENCE	Y	VARIES	< >	INFO
DAMPER POSITION DOES NOT MATCH COMMANDED POSITION	Y	~	0-100%	INFO
DAMPER POSITION DOES NOT MATCH COMMANDED POSITION	Y	~	0-100%	INFO
DAMPER POSITION DOES NOT MATCH COMMANDED POSITION	Y	~	0-100%	INFO
HIGH LIMIT ALARM, LOW LIMIT ALARM - REFER TO SEQUENCE	Y	~	< >	INFO
LOW LIMIT ALARM	Y	35 DEG. F.	OPEN/CLOSED	INFO
~	Y	~	OPEN/CLOSED	~
HIGH LIMIT ALARM, LOW LIMIT ALARM - REFER TO SEQUENCE	Y	~	< >	INFO
HIGH LIMIT ALARM, LOW LIMIT ALARM - REFER TO SEQUENCE	Y	~	< >	INFO
HIGH LIMIT ALARM, LOW LIMIT ALARM - REFER TO SEQUENCE	Y	~	< >	INFO
~	Y	54 DEG. F.	< >	~
~	Y	SEE SEQUENCE	< >	~
~	Y	70 DEG. F.	< >	~
VFD COMMAND IS ON, BUT VFD STATUS IS OFF	Y	~	ON/OFF	CRIT
VFD COMMAND IS ON, BUT VFD STATUS IS OFF	Y	~	ON/OFF	CRIT
~	Y	~	0-100%	~
~	Y	~	0-100%	~
~	Y	~	< >	~
~	Y	~	< >	~
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
~	Y	~	ON/OFF	~
~	Y	~	ON/OFF	~
~	Y	~	ON/OFF	~
~	Y	~	ON/OFF	~
~	Y	~	ON/OFF	~
~	Y	~	ON/OFF	~
~	Y	~	ON/OFF	~
DIFFERENTIAL PRESSURE EXCEEDS SETPOINT	Y	1.0" W.C.	< >	CRIT
DIFFERENTIAL PRESSURE EXCEEDS SETPOINT	Y	1.0" W.C.	< >	CRIT
LOW LIMIT ALARM CORRESPONDING TO MINIMUM OA	Y	REFER TO SCHED.	< >	CRIT
HIGH LIMIT ALARM - REFER TO SEQUENCE	Y	~	< >	CRIT
HIGH LIMIT ALARM - REFER TO SEQUENCE	Y	~	< >	CRIT
HIGH LIMIT ALARM, LOW LIMIT ALARM - REFER TO SEQUENCE	Y	REFER TO SCHED.	< >	INFO
HIGH LIMIT ALARM, LOW LIMIT ALARM - REFER TO SEQUENCE	Y	VARIES	< >	INFO
HIGH LIMIT ALARM, LOW LIMIT ALARM - REFER TO SEQUENCE	Y	~	< >	INFO
~	Y	~	< >	~
HIGH LIMIT ALARM, LOW LIMIT ALARM - REFER TO SEQUENCE	Y	~	< >	INFO
HIGH LIMIT ALARM, LOW LIMIT ALARM - REFER TO SEQUENCE	Y	~	< >	INFO
~	Y	54 DEG. F.	< >	~
~	Y	72 DEG. F.	< >	~
~	Y	~	< >	~
~	Y	~	< >	~
~	Y	~	ON/OFF	~
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
~	~	~	< >	~
~	Y	~	< >	~
FOR FUTURE USE AT OWNER'S DISCRETION	~	~	~	INFO
FOR FUTURE USE AT OWNER'S DISCRETION	~	~	~	INFO
FAN COMMAND IS ON, BUT FAN STATUS IS OFF	Y	~	ON/OFF	CRIT
FAN COMMAND IS ON, BUT FAN STATUS IS OFF	Y	~	ON/OFF	CRIT
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	INFO
STATUS IS "ALARM"	Y	~	NORMAL/ALARM	CRIT
~	Y	~	ON/OFF	~

Sheet Title
MECHANICAL CONTROL POINTS

Dwg. No.
PLM2021-0432-0036D

Sht. No.
M-705



WATER COOLED CHILLED WATER SYSTEM WITH PRIMARY ONLY PUMPING SEQUENCE OF OPERATION

GENERAL:
432PCHW01-A1 and 432PCHW02-A1 ARE 100% REDUNDANT AND MUST OPERATE IN LEAD/STANDBY CONFIGURATION. EQUIPMENT RUN-TIME OR OPERATOR SELECTION MUST DETERMINE ACTIVE LEAD PUMP.

MINIMUM ALLOWABLE CHILLED WATER FLOW THROUGH SYSTEM IS 95 GPM DUE TO MINIMUM FLOW REQUIREMENTS OF 432RCHS01-A1. FOR REFERENCE ONLY, SYSTEM SHALL OPERATE AT CONSTANT VOLUME (220 GPM).

CHILLED WATER PUMPS MUST BE MONITORED AND CONTROLLED BY THE DDC SYSTEM. 432RCHS01-1 CONTROL FUNCTIONS, SAFETY INTERLOCKS, ALARMS, AND MONITORED PARAMETERS MUST BE MADE AVAILABLE TO AND MONITORED BY THE DDC SYSTEM. THE DDC SYSTEM IS AN AUTOMATED LOGIC CONTROLLER.

CHILLER CONTROL:
DDC SYSTEM SHALL MONITOR THE STATUS OF THE CHILLER AND CHILLED WATER PUMPS AND INITIATE ENABLE/DISABLE COMMANDS TO EACH CHILLER AND PUMP. DDC SYSTEM SHALL ENABLE CHILLER AFTER LEAD CHILLED WATER PUMP STATUS PROVES ON.

THE FACTORY CHILLER CONTROL PANEL (CCP) MUST CONTROL CHILLER AND EXTERNAL CHILLER HEAD PRESSURE CONTROL VALVE TO SATISFY CHILLED WATER SUPPLY TEMPERATURE SETPOINT.

DDC SYSTEM MUST DISABLE CHILLER WHEN LEAD CHILLED WATER PUMP STATUS CHANGES TO OFF.

IF CHILLER IS COMMANDED TO RUN AND COMPRESSOR STATUS IS OFF AFTER A 30 SECOND (ADJ.) DELAY OR CHILLED WATER SUPPLY TEMPERATURE IS GREATER THAN CHILLED WATER SUPPLY TEMPERATURE SEPOINT PLUS 5 DEG. F. (ADJ.) FOR 4 MINUTES (ADJ.), AN ALARM MUST BE SENT TO THE DDC SYSTEM. AFTER A 300 SECOND (ADJ.) DELAY, LEAD CHILLER MUST BE COMMANDED OFF.

CHILLER CONTROL FUNCTIONS, MINIMUM RUNTIME, SAFETY INTERLOCKS, AND MAXIMUM STARTS PER HOUR MUST BE AS RECOMMENDED BY THE CHILLER MANUFACTURER DURING EQUIPMENT STARTUP.

CHILLER MUST AUTOMATICALLY DISABLE AFTER OPERATING AT OR BELOW 25% CAPACITY (CONFIRM GREATER THAN EQUIPMENT TURNDOWN) (ADJ.) FOR 15 MINUTES (ADJ.) CORRESPONDING LEAD CHILLED WATER PUMP MUST REMAIN RUNNING DURING LOW DEMAND, STATUS OF PUMP IS ACCORDING TO USER-DEFINED OPERATING SCHEDULE OR OPERATOR MANUAL START/STOP OVER BAS.

CHILLED WATER PUMP CONTROL:
THE LEAD CHILLED WATER PUMP MUST RUN ACCORDING TO A USER-DEFINED OPERATING SCHEDULE OR OPERATOR MANUAL START/STOP OVER BAS.

PUMP VARIABLE SPEED DRIVE MUST MAINTAIN CONSTANT FLOW (220 GPM) THROUGH SYSTEM BY OPERATING AT CONSTANT SPEED. SPEED CORRESPONDING TO CONSTANT FLOW SETPOINT (220 GPM) TO BE DETERMINED DURING TAB.

IF PUMP IS COMMANDED TO RUN AND COMMON DIFFERENTIAL PRESSURE TRANSDUCER DOES NOT INDICATE PROOF OF FLOW AFTER A 30 SECOND (ADJ.) DELAY, AN ALARM MUST BE SENT TO THE DDC SYSTEM. AFTER A 180 SECOND (ADJ.) DELAY, LEAD PUMP MUST BE COMMANDED OFF AND STANDBY PUMP MUST BECOME LEAD PUMP.

CHILLED WATER FREEZE PROTECTION:
WHEN OUTSIDE AIR TEMPERATURE IS LESS THAN 45 DEG. F. (ADJ.), ENABLE BOTH CHILLED WATER PUMPS. RUN BOTH PUMPS AT LOW SPEED (TO BE DETERMINED DURING TAB) UNTIL OUTSIDE AIR TEMPERATURE IS GREATER THAN 45 DEG. F. (ADJ.)

CONDENSER WATER FLOW CONTROL:
THE CONDENSER WATER FLOW CONTROL IS NOT MANAGED BY THE BAS SYSTEM. CHILLER CONTROLLER MUST COMMUNICATE DIRECTLY WITH EXTERNAL CONDENSER WATER CONTROL VALVE THAT IS ACTUATED BASED ON REFRIGERANT PRESSURE IN THE CIRCUIT MEASURED BY THE CHILLER. THIS VALVE MUST AUTOMATICALLY ADJUST CONDENSER WATER FLOW BASED ON REFRIGERANT PRESSURE.

THE CONDENSER SHALL HAVE A FLOW SWITCH WIRED DIRECTLY TO CHILLER. THE CHILLER MUST NOT BE ALLOWED TO START IF FLOW SWITCH DOES NOT INDICATE PROOF OF FLOW THROUGH THE CHILLER.

CONDENSER WATER FREEZE PROTECTION:
WHEN OUTSIDE AIR TEMPERATURE IS LESS THAN 45 DEG. F. (ADJ.), OPEN BYPASS 2-WAY CONTROL VALVE UNTIL OUTSIDE AIR TEMPERATURE IS GREATER THAN 45 DEG. F. (ADJ.)

EVAPORATOR WATER FLOW CONTROL:
THE CHILLED WATER FLOW MUST BE CONSTANT. EXISTING AND NEW 3-WAY ACU COIL CONTROL VALVES MUST MODULATE OPEN AND CLOSED IN RESPONSE TO ACU COMMAND FOR MORE OR LESS CHILLED WATER.

THE EVAPORATOR SHALL HAVE A FLOW SWITCH WIRED DIRECTLY TO CHILLER. THE CHILLER MUST NOT BE ALLOWED TO START IF FLOW SWITCH DOES NOT INDICATE PROOF OF FLOW THROUGH THE CHILLER.

CHILLED WATER PRESSURE/TEMPERATURE CONTROL:
THE CHILLED WATER SUPPLY TEMPERATURE MUST BE CONSTANT AT 43 DEG. F. (ADJ.)

THE MINIMUM SPEED ALLOWED BY THE CHILLED WATER PUMP MUST NOT PROVIDE LESS THAN THE MINIMUM REQUIRED FLOW THROUGH EITHER CHILLER, NOR MUST THE MINIMUM SPEED BE BELOW 30% OF PUMP DESIGN OPERATING SPEED.

MISCELLANEOUS MONITORING:
PUMP STATUS MUST BE MONITORED AT THE VFD AND BY THE DDC SYSTEM. VFD STATUS AND PUMP STATUS VIA DIFFERENTIAL PRESSURE TRANSDUCER IS REQUIRED.

CHILLER MONITORING POINTS SHALL BE PROVIDED FROM THE CCP.

EXISTING AIR HANDLING UNIT CONTROL VALVE:
EXISTING 3-WAY PNEUMATIC CONTROL VALVE FUNCTIONALITY TO REMAIN AS-IS.

NEW AIR HANDLING UNIT CONTROL VALVE: REFER TO M-703.

GENERAL NOTES

- REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
- REFER TO M-701 FOR CONTROLS ABBREVIATIONS, SYMBOL LEGEND, AND GENERAL NOTES.
- COMPONENTS SHOWN HALF-TONE ON THIS DRAWING ARE ITEMS TO REMAIN; BOLD COMPONENTS ARE NEW.
- REFER TO DETAIL F2 ON M-506 AND D5 ON M-504 FOR COMPLETE VALVE AND PIPELINE ACCESSORY DETAILS FOR NEW CHILLED WATER AND MAKEUP WATER SYSTEM COMPONENTS. CONTENT INCLUDED ON THIS DRAWING IS LIMITED TO ONLY THE PIPELINE ACCESSORIES THAT IMPACT SYSTEM FLOW.

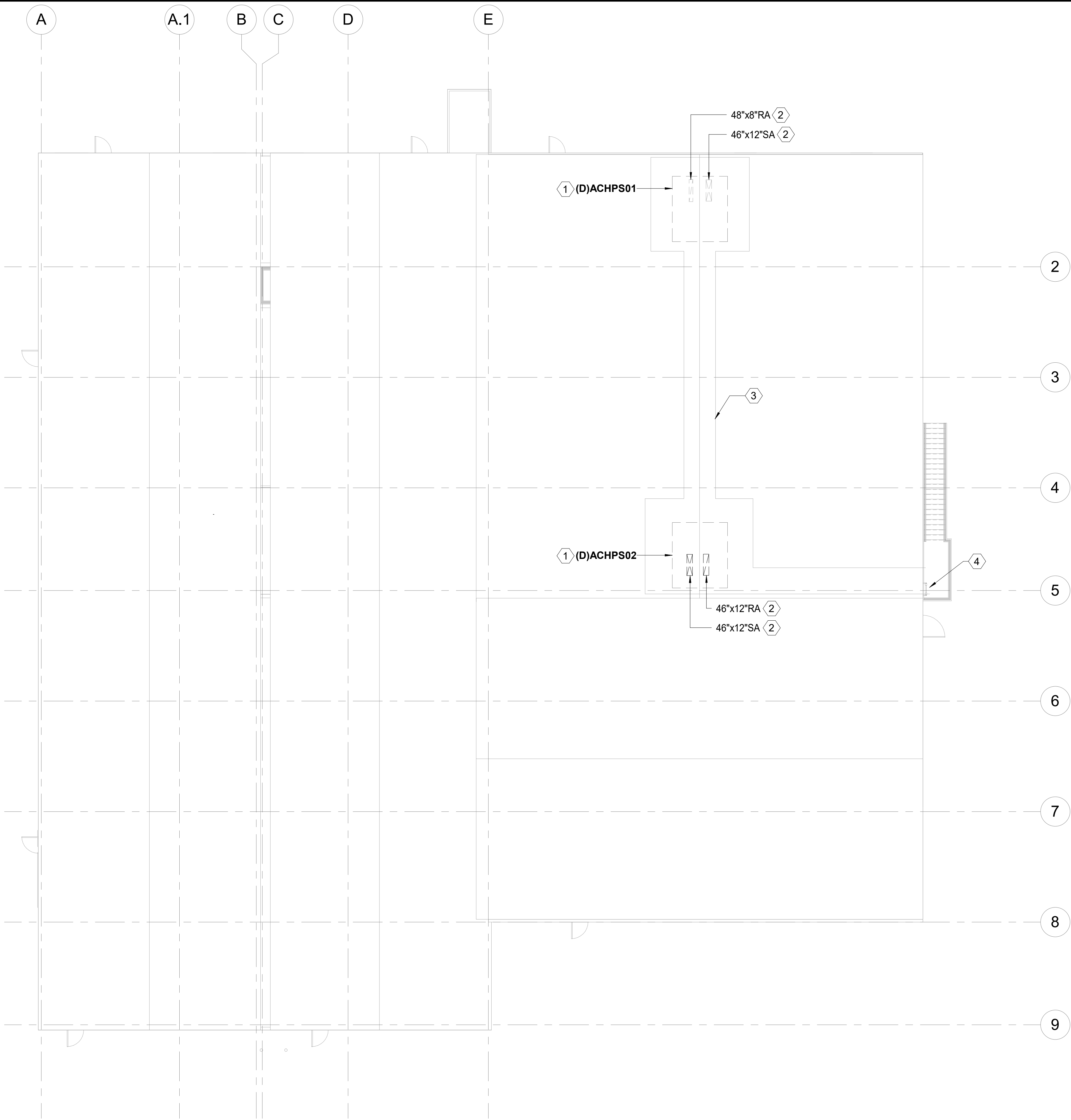
KEYED NOTES

- INSTALL SELECT PIPING AND LINE-SIZE ISOLATION VALVES WITHIN FIRST SYSTEM SHUTDOWN AT BEGINNING OF CONSTRUCTION PROJECT. REFER TO M-002 AND M-413. WORK SHALL BE ACCOMPLISHED WITHOUT DRAINING FULL SYSTEM. FREEZING PIPE OR USING HOT TAP PIPE PLUG INSERTION SOLUTION IS ACCEPTABLE.
- CW MAKEUP WATER CONNECTION TO CHILLED SYSTEM IS MADE AT NEW AIR SEPARATOR 432SDE01-1, NOT SHOWN ON THIS DIAGRAM FOR SIMPLICITY.
- FUNCTIONALITY OF EXISTING PNEUMATIC CONTROL VALVE TO REMAIN AS-IS. EXISTING PNEUMATIC TUBING AND CENTRALIZED CONTROL COMPONENTS SERVING EXISTING VALVE TO REMAIN AS-IS.

Sheet Title
**MECHANICAL CHILLED
WATER FLOW DIAGRAM**

Dwg. No.
PLM2021-0432-0037D

Sht. No.
M-801



F1

MD120

MECHANICAL DEMO ROOF PLAN

SCALE: 3/32" = 1'-0"

08'16'24'

3/32" = 1'-0"

N

GENERAL NOTES

1. REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
2. COMPONENTS SHOWN DASHED ON THIS DRAWING ARE ITEMS TO BE DEMOLISHED; HALF-TONE COMPONENTS ARE EXISTING-TO-REMAIN.
3. PRIOR TO ANY DEMOLITION ACTIVITIES, SUBCONTRACTOR MUST COORDINATE WITH THE STR TO REVIEW FACILITY ES&H DOCUMENTS, EQUIPMENT TO BE REMOVED, MATERIALS TO BE REMOVED OR DISTURBED, AND IDENTIFY A PLAN PRIOR TO COMMENCING THE WORK. ITEMS TO REVIEW INCLUDE BUT ARE NOT LIMITED TO THE FOLLOWING: NEW WORK IN DISTURBING EXISTING WALLS, FLOORS, ROOFS, OR CEILINGS; ALL CUTTING OR DRILLING; ALL DEMOLITION; REMOVAL AND DISPOSAL.
4. CUT, REMOVE, AND LEGALLY DISPOSE OF ALL MECHANICAL ITEMS SELECTED FOR DEMOLITION AND ANY MECHANICAL ITEMS MADE OBSOLETE BY THE NEW WORK, INCLUDING EXISTING EQUIPMENT/PIPE/TUBE SUPPORTS AND HANGERS.

KEYED NOTES

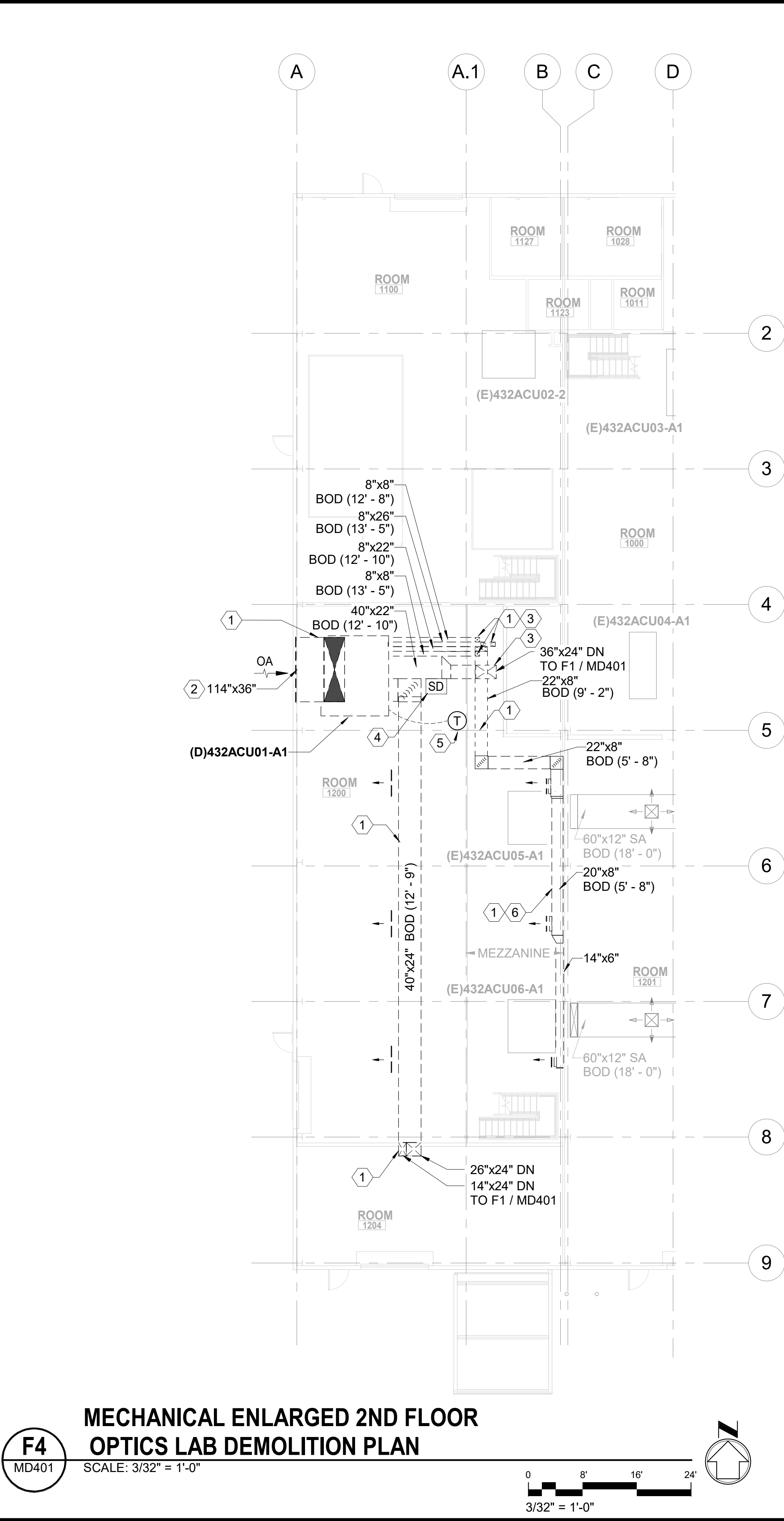
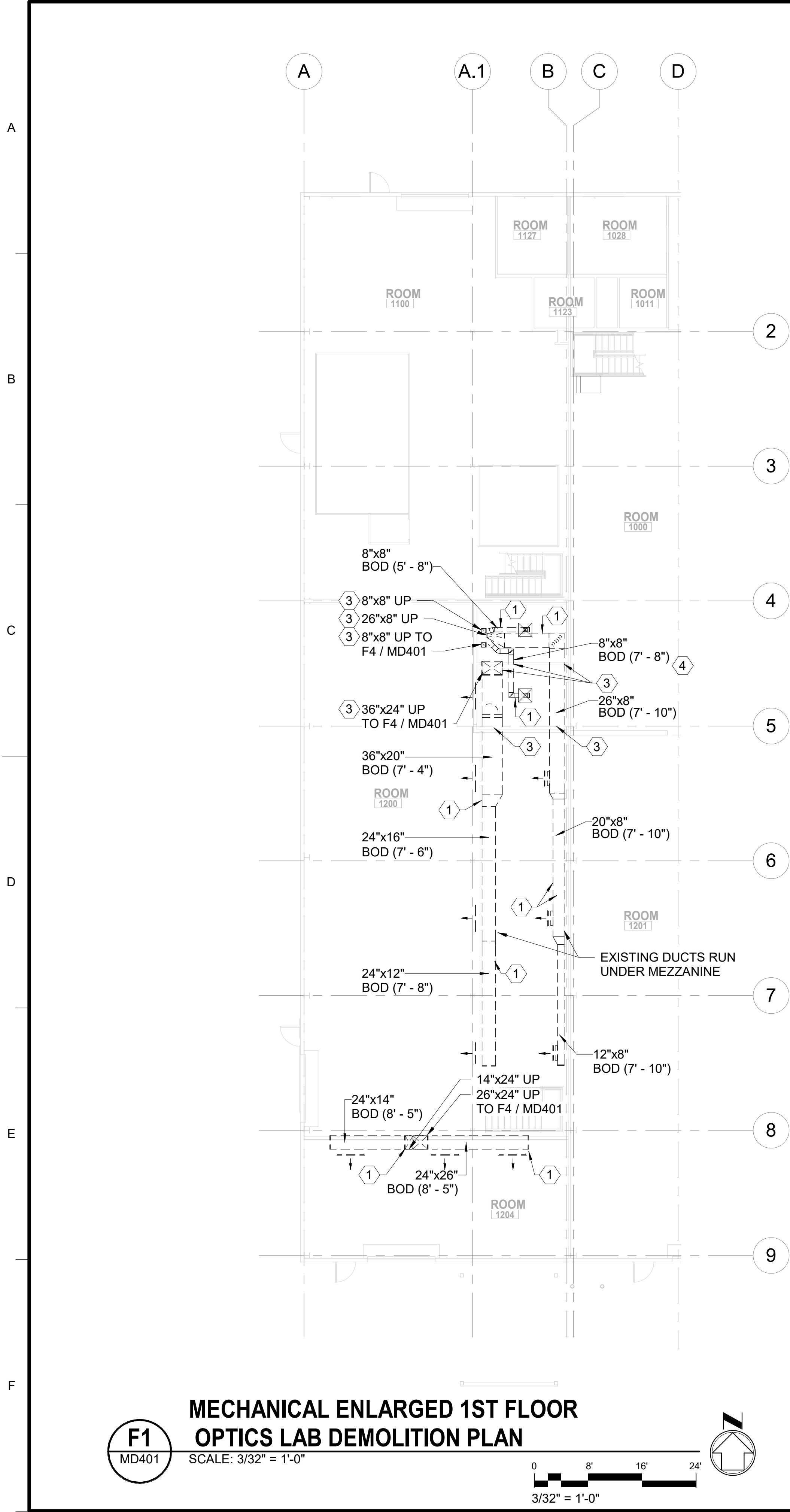
- (1) REMOVE AND DISPOSE OF EXISTING ROOFTOP AIR CONDITIONING UNIT, PIPING, DUCT AND UTILITY CONNECTIONS, SUPPORTS, ANCHORAGE, AND ALL APPURTENANCES. INFILL EXISTING ROOF OPENINGS EXPOSED BY REMOVAL OF ROOFTOP ACHPS UNIT AND SEAL WATER TIGHT PER ARCHITECTURAL AND STRUCTURAL DRAWINGS AND SPECIFICATIONS. REFER TO F3 / A-501 AND S-512.
- (2) REMOVE AND DISPOSE OF EXISTING DUCTWORK THROUGH ROOF TO OFFICE SPACE BELOW. REFER TO MD404 FOR CONTINUATION.
- (3) EXISTING ROOFTOP CATWALK, FLOOR GRATING, AND RAILINGS TO REMAIN.
- (4) EXISTING LADDER TO ROOF AND LADDER SUPPORTS TO REMAIN AS IS.

KEY PLAN

Sheet Title
MECHANICAL DEMOLITION
ROOF PLAN

Dwg. No.
PLM2021-0432-0005D

Sht. No.
MD120



GENERAL NOTES

1. REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
2. COMPONENTS SHOWN DASHED ON THIS DRAWING ARE ITEMS TO BE DEMOLISHED; HALF-TONE COMPONENTS ARE EXISTING-TO-REMAIN.
3. PRIOR TO ANY DEMOLITION ACTIVITIES, SUBCONTRACTOR MUST COORDINATE WITH THE STR TO REVIEW FACILITY ES&H DOCUMENTS, EQUIPMENT TO BE REMOVED, MATERIALS TO BE REMOVED OR DISTURBED, AND IDENTIFY A PLAN PRIOR TO COMMENCING THE WORK. ITEMS TO REVIEW INCLUDE BUT ARE NOT LIMITED TO THE FOLLOWING: NEW WORK IN DISTURBING EXISTING WALLS, FLOORS, ROOFS, OR CEILINGS; ALL CUTTING OR DRILLING; ALL DEMOLITION; REMOVAL AND DISPOSAL.
4. CUT, REMOVE, AND LEGALLY DISPOSE OF ALL MECHANICAL ITEMS SELECTED FOR DEMOLITION AND ANY MECHANICAL ITEMS MADE OBSOLETE BY THE NEW WORK, INCLUDING EXISTING EQUIPMENT/PIPE/TUBE SUPPORTS AND HANGERS.
5. REFER TO STRUCTURAL AND CIVIL DRAWINGS FOR SITE DEMOLITION REQUIRED TO SUPPORT INSTALLATION OF NEW HVAC EQUIPMENT.
6. REMOVAL OF EXISTING SUSPENDED AIR CONDITIONING UNIT FROM ROOM WILL REQUIRE A DETAILED WORK PLAN DEVELOPED WITH INPUT FROM CONSTRUCTION MANAGER, SUBCONTRACTORS, AND FACILITY END-USERS. SUBCONTRACTOR MUST REVIEW EXISTING INSTALLATION AND DISCUSS ACCEPTABLE MEANS AND METHODS FOR REMOVAL WITH CONSTRUCTION MANAGER PRIOR TO BID. USAGE CONTINUITY OF END-USER LAB EQUIPMENT UNDER ACU WILL REQUIRE COMPLETE REMOVAL OF EXISTING ACU AND APPURTENANCES OVER A LONG WEEKEND, INCLUDE COSTS TO ACCOMPLISH THIS IN BID. CONSTRUCTION MANAGER TO COORDINATE TEMPORARY LAB EQUIPMENT REMOVAL

KEYED NOTES

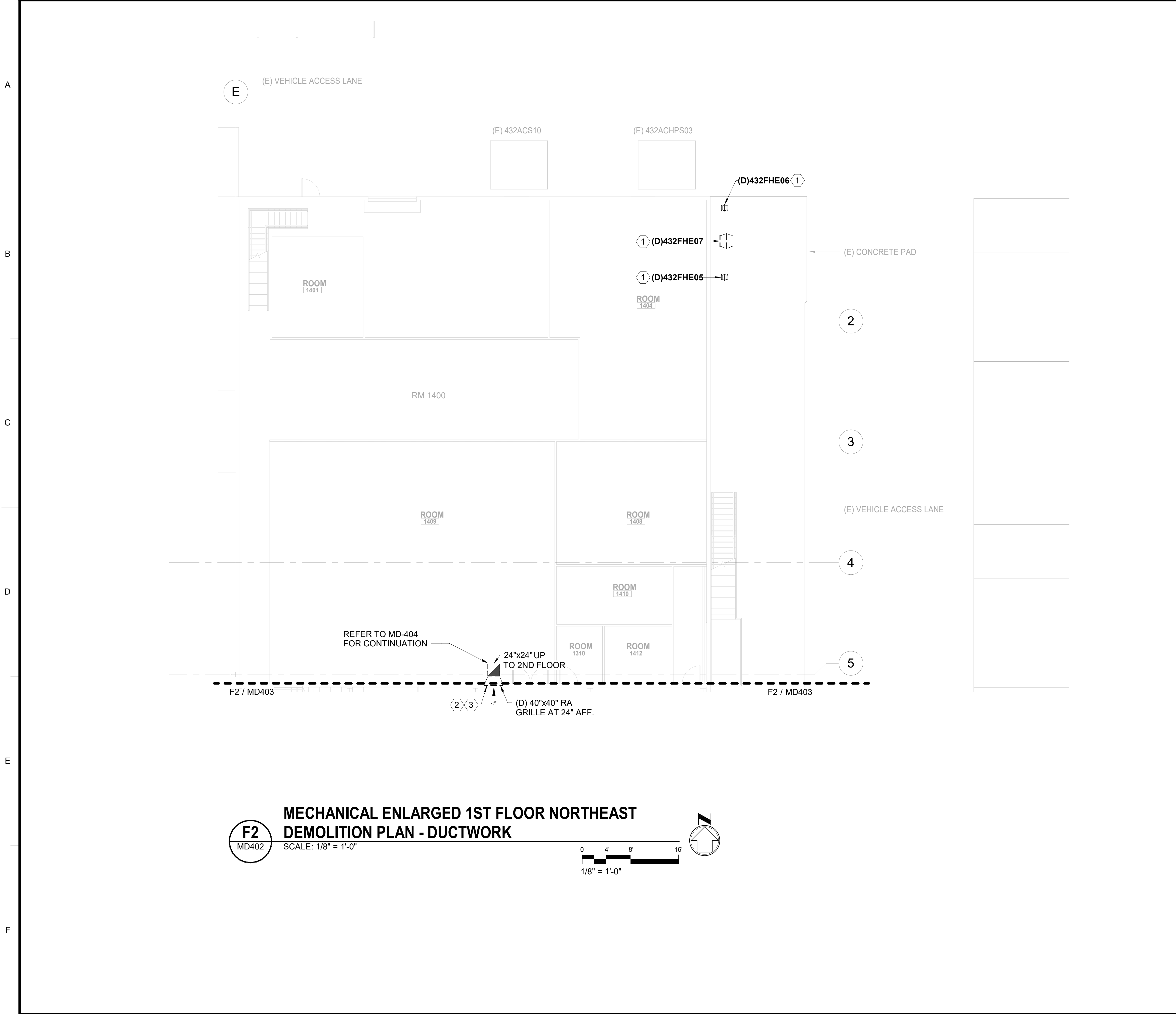
1. REMOVE AND DISPOSE OF EXISTING, SUSPENDED AIR CONDITIONING UNIT, DUCTWORK CONNECTIONS, SUPPORTS, ANCHORAGE, AND ALL APPURTENANCES. REMOVE AND DISPOSE OF ALL SUPPLY, RETURN AND OUTDOOR AIR DUCTWORK INCLUDING: AIR TERMINALS, SUPPORTS, ANCHORAGE, CONROLS, AND APPURTENANCES ASSOCIATED WITH AIR CONDITIONING UNIT REMOVAL.
2. REMOVE AND DISPOSE OF EXISTING 114"x36" WALL-MOUNTED OUTSIDE AIR LOUVER. PREPARE EXISTING WALL OPENING TO RECEIVE NEW SUPPLY AND RETURN DUCTWORK PENETRATIONS PER M-401. SEAL WALL OPENING AROUND NEW DUCTWORK. REFER TO A-501 FOR WALL SEALING REQUIREMENTS.
3. INFILL WALL/CEILING PENETRATION AFTER DUCT REMOVAL. REFER TO A-501.
4. EXISTING DUCT SMOKE DETECTOR. RETAIN AND RE-INSTALL ON REPLACEMENT ACU SYSTEM. REFER TO M-404.
5. REMOVE AND DISPOSE OF EXISTING WALL-MOUNTED THERMOSTAT AND WIRING AND CONDUIT FROM DEVICE TO (D)432ACU01-A1. PATCH AND PAINT WALLS WHERE DEMOLITION RESULTS IN UNFINISHED CONDITION.
6. EXISTING DUCTWORK IS LOCATED BEHIND EXISTING AIR HANDLING UNITS ON MEZZANINE. DETERMINE MEANS AND METHODS FOR REMOVAL PRIOR TO BID.

KEY PLAN

Sheet Title
MECHANICAL ENLARGED 1ST AND 2ND FLOOR OPTICS LAB DEMOLITION PLANS - DUCTWORK

Dwg. No.
PLM2021-0432-0006D

Sht. No.
MD401



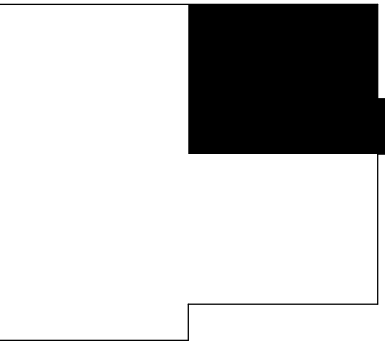
GENERAL NOTES

1. REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
2. COMPONENTS SHOWN DASHED ON THIS DRAWING ARE ITEMS TO BE DEMOLISHED; HALF-TONE COMPONENTS ARE EXISTING-TO-REMAIN.
3. PRIOR TO ANY DEMOLITION ACTIVITIES, SUBCONTRACTOR MUST COORDINATE WITH THE STR TO REVIEW FACILITY ES&H DOCUMENTS, EQUIPMENT TO BE REMOVED, MATERIALS TO BE REMOVED OR DISTURBED, AND IDENTIFY A PLAN PRIOR TO COMMENCING THE WORK. ITEMS TO REVIEW INCLUDE BUT ARE NOT LIMITED TO THE FOLLOWING: NEW WORK IN DISTURBING EXISTING WALLS, FLOORS, ROOFS, OR CEILINGS; ALL CUTTING OR DRILLING; ALL DEMOLITION; REMOVAL AND DISPOSAL.
4. CUT, REMOVE, AND LEGALLY DISPOSE OF ALL MECHANICAL ITEMS SELECTED FOR DEMOLITION AND ANY MECHANICAL ITEMS MADE OBSOLETE BY THE NEW WORK, INCLUDING EXISTING EQUIPMENT/PIPE/TUBE SUPPORTS AND HANGERS.

KEYED NOTES

- 1 REMOVE AND DISPOSE OF EXHAUST FAN, EXTERIOR INTAKE DUCTWORK TO WALL PENETRATION (MINIMUM OF 25FT, 18" DIAMETER), EXTERIOR VERTICAL DISCHARGE DUCTWORK TO ROOF (MINIMUM OF 40FT, 18" DIAMETER), SUPPORTS, ANCHORAGE, AND ALL APPURTENANCES. RETAIN EXISTING CONCRETE PAD.
- 2 REMOVE AND DISPOSE OF EXISTING GRILLE PENETRATION THROUGH WALL, INFILL WALL, PENETRATION AND SEAL PER ARCHITECTURAL DRAWINGS AND SPECIFICATIONS. REFER TO A-501.
- 3 REMOVE AND DISPOSE OF EXISTING RETURN AIR DUCTWORK TO SECOND FLOOR INCLUDING: AIR TERMINALS, SUPPORTS, ANCHORAGE, AND APPURTENANCES.

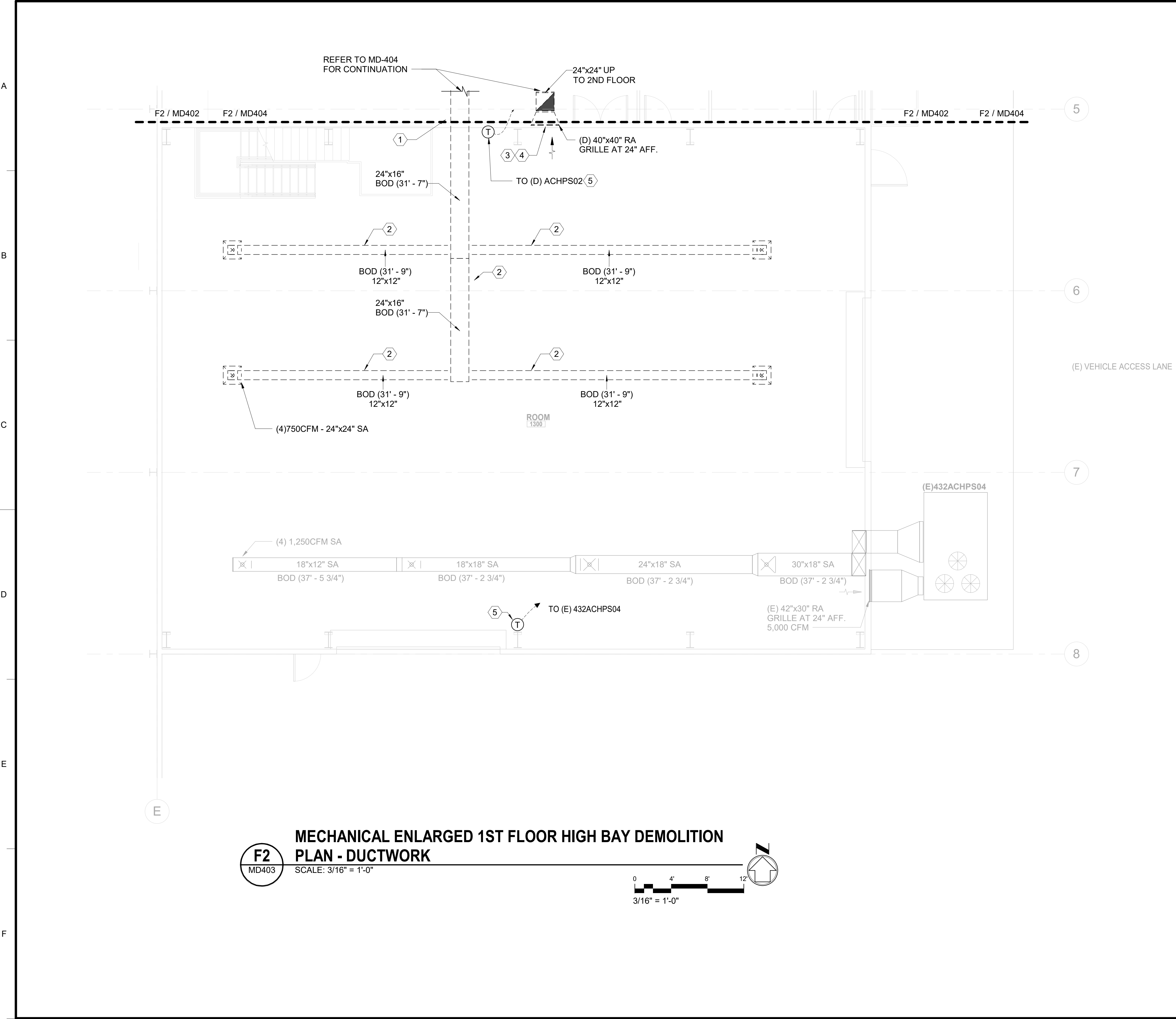
KEY PLAN



Sheet Title
MECHANICAL ENLARGED 1ST FLOOR NORTHEAST DEMOLITION PLAN - DUCTWORK

Dwg. No.
PLM2021-0432-0007D

Sht. No.
MD402



GENERAL NOTES

1. REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
2. COMPONENTS SHOWN DASHED ON THIS DRAWING ARE ITEMS TO BE DEMOLISHED; HALF-TONE COMPONENTS ARE EXISTING-TO-REMAIN.
3. PRIOR TO ANY DEMOLITION ACTIVITIES, SUBCONTRACTOR MUST COORDINATE WITH THE STR TO REVIEW FACILITY ES&H DOCUMENTS, EQUIPMENT TO BE REMOVED, MATERIALS TO BE REMOVED OR DISTURBED, AND IDENTIFY A PLAN PRIOR TO COMMENCING THE WORK. ITEMS TO REVIEW INCLUDE BUT ARE NOT LIMITED TO THE FOLLOWING: NEW WORK IN DISTURBING EXISTING WALLS, FLOORS, ROOFS, OR CEILINGS; ALL CUTTING OR DRILLING; ALL DEMOLITION; REMOVAL AND DISPOSAL.
4. CUT, REMOVE, AND LEGALLY DISPOSE OF ALL MECHANICAL ITEMS SELECTED FOR DEMOLITION AND ANY MECHANICAL ITEMS MADE OBSOLETE BY THE NEW WORK, INCLUDING EXISTING EQUIPMENT/PIPE/TUBE SUPPORTS AND HANGERS.
5. REFER TO STRUCTURAL AND CIVIL DRAWINGS FOR SITE DEMOLITION REQUIRED TO SUPPORT INSTALLATION OF NEW HVAC EQUIPMENT.
6. NEW DUCTWORK MUST BE INSTALLED PRIOR TO DEMOLITION OF EXISTING DUCTWORK TO LIMIT DURATION OF SYSTEM OUTAGE. REFER TO M-403.

KEYED NOTES

- 1 REMOVE AND DISPOSE OF EXISTING DUCTWORK WALL PENETRATION. INFILL WALL PENETRATION AND SEAL PER ARCHITECTURAL DRAWINGS AND SPECIFICATIONS. REFER TO A-501.
- 2 REMOVE AND DISPOSE OF EXISTING SUPPLY AIR DISTRIBUTION DUCTWORK BELOW CEILING INCLUDING: AIR TERMINALS, SUPPORTS, ANCHORAGE, AND APPURTENANCES.
- 3 REMOVE AND DISPOSE OF EXISTING GRILLE PENETRATION THROUGH WALL. INFILL WALL PENETRATION AND SEAL PER ARCHITECTURAL DRAWINGS AND SPECIFICATIONS. REFER TO A-501.
- 4 REMOVE AND DISPOSE OF EXISTING RETURN AIR DUCTWORK TO SECOND FLOOR INCLUDING: AIR TERMINALS, SUPPORTS, ANCHORAGE, AND APPURTENANCES.
- 5 REMOVE AND DISPOSE OF EXISTING WALL-MOUNTED THERMOSTAT. REMOVE AND DISPOSE OF EXISTING THERMOSTAT, WIRING, AND CONDUIT FROM DEVICE TO ASSOCIATED ACHPS UNIT.

KEY PLAN

Sheet Title
MECHANICAL ENLARGED 1ST FLOOR HIGH BAY DEMOLITION PLAN - DUCTWORK

Dwg. No.
PLM2021-0432-0008D

Sht. No.
MD403

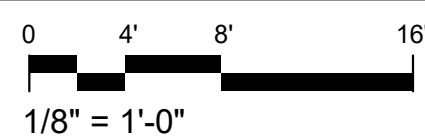


MECHANICAL ENLARGED 2ND FLOOR OPEN OFFICE DEMOLITION PLAN - DUCTWORK

F2

MD404

SCALE: 1/8" = 1'-0"



GENERAL NOTES

1. REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
2. COMPONENTS SHOWN DASHED ON THIS DRAWING ARE ITEMS TO BE DEMOLISHED; HALF-TONE COMPONENTS ARE EXISTING-TO-REMAIN.
3. PRIOR TO ANY DEMOLITION ACTIVITIES, SUBCONTRACTOR MUST COORDINATE WITH THE STR TO REVIEW FACILITY ES&H DOCUMENTS, EQUIPMENT TO BE REMOVED, MATERIALS TO BE REMOVED OR DISTURBED, AND IDENTIFY A PLAN PRIOR TO COMMENCING THE WORK. ITEMS TO REVIEW INCLUDE BUT ARE NOT LIMITED TO THE FOLLOWING: NEW WORK IN DISTURBING EXISTING WALLS, FLOORS, ROOFS, OR CEILINGS; ALL CUTTING OR DRILLING; ALL DEMOLITION; REMOVAL AND DISPOSAL.
4. CUT, REMOVE, AND LEGALLY DISPOSE OF ALL MECHANICAL ITEMS SELECTED FOR DEMOLITION AND ANY MECHANICAL ITEMS MADE OBSOLETE BY THE NEW WORK, INCLUDING EXISTING EQUIPMENT/PIPE/TUBE SUPPORTS AND HANGERS.
5. NEW DUCTWORK MUST BE INSTALLED PRIOR TO DEMOLITION OF EXISTING DUCTWORK TO LIMIT DURATION OF SYSTEM OUTAGE. REFER TO M-402, M-405, AND M-407.



KEYED NOTES

- 1 REMOVE AND DISPOSE OF EXISTING DUCTWORK THROUGH ROOF. INFILL EXISTING ROOF OPENINGS EXPOSED BY REMOVAL OF ROOFTOP ACHPS UNIT AND SEAL WATER TIGHT PER ARCHITECTURAL AND STRUCTURAL DRAWINGS AND SPECIFICATIONS. REFER TO F3 / A-501 AND S-512.
- 2 REMOVE AND DISPOSE OF EXISTING DISTRIBUTION DUCTWORK BELOW CEILING INCLUDING: AIR TERMINALS, SUPPORTS, ANCHORAGE, AND APPURTENANCES.
- 3 REMOVE AND DISPOSE OF EXISTING RETURN AIR PLENUM BELOW CEILING INCLUDING: AIR TERMINALS, SUPPORTS, ANCHORAGE, AND APPURTENANCES.
- 4 EXISTING DUCT SMOKE DETECTOR. RETAIN AND RE-INSTALL ON REPLACEMENT ACHPS SYSTEM. REFER TO M-403 AND M-405.
- 5 REMOVE AND DISPOSE OF WALL-MOUNTED THERMOSTAT AND WIRING AND CONDUIT FROM DEVICE TO ROOF-MOUNTED (D)ACHPS01. PATCH AND PAINT WALLS WHERE DEMOLITION RESULTS IN UNFINISHED CONDITION.

KEY PLAN

Sheet Title

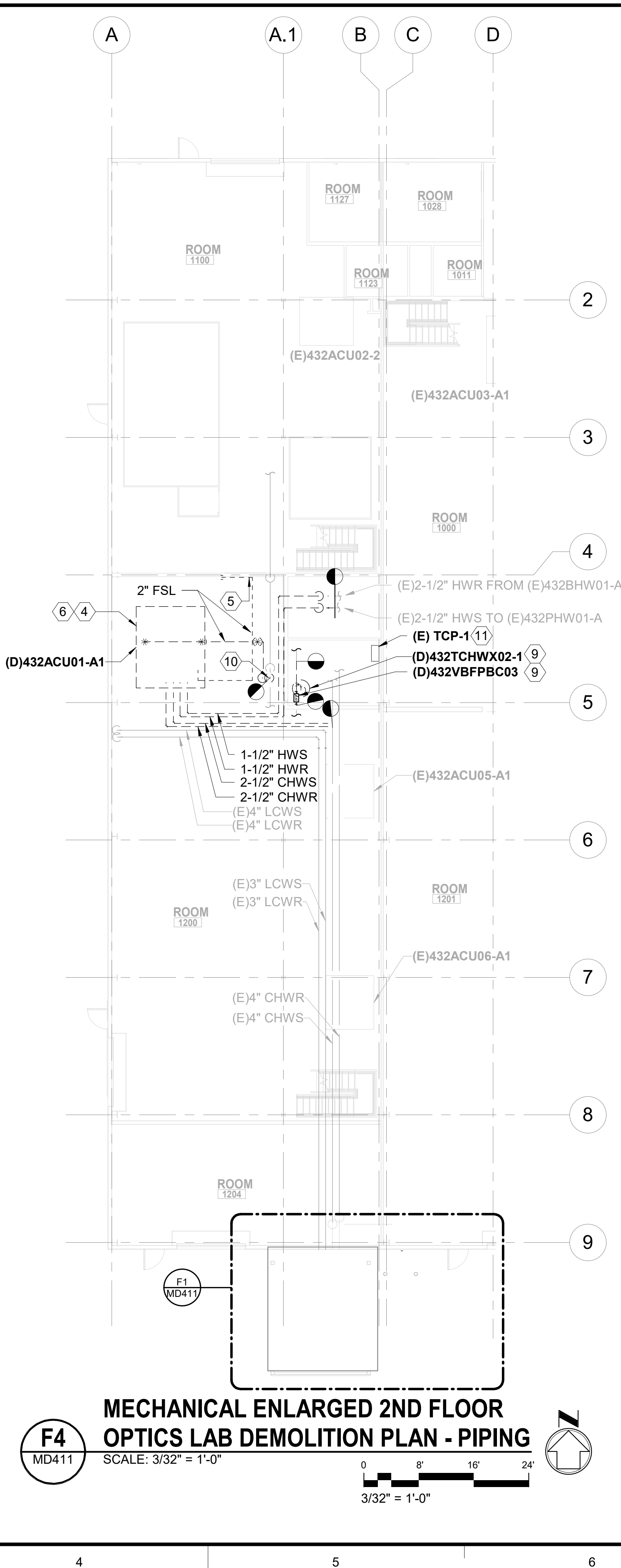
MECHANICAL ENLARGED
2ND FLOOR OPEN OFFICE
DEMOLITION PLAN -
DUCTWORK

Dwg. No.

PLM2021-0432-0009D

Sht. No.

MD404



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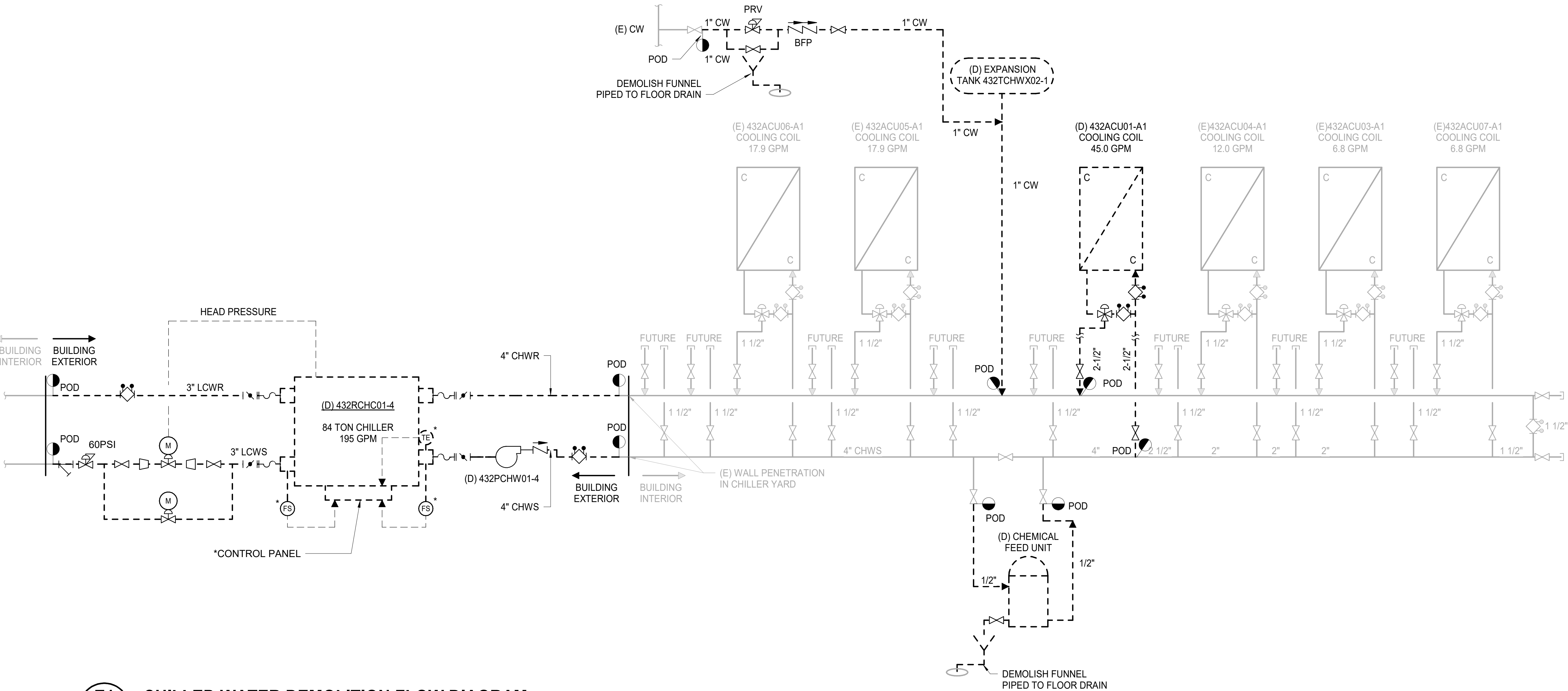
B

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E1
MD801

CHILLED WATER DEMOLITION FLOW DIAGRAM

SCALE: NO SCALE SHOWN

GENERAL NOTES

- REFER TO M-001 FOR MECHANICAL ABBREVIATIONS, LEGEND, SYMBOLS, AND GENERAL NOTES.
- REFER TO M-701 FOR CONTROLS ABBREVIATIONS, SYMBOL LEGEND, AND GENERAL NOTES.
- COMPONENTS SHOWN HALF-TONE ON THIS DRAWING ARE ITEMS TO REMAIN; DASHED COMPONENTS ARE TO BE DEMOLISHED.
- DEMOLITION FLOW DIAGRAM IS SCHEMATIC IN NATURE. SUBCONTRACTOR MUST FIELD VERIFY CONDITIONS.
- CAP EXISTING TO REMAIN OBSOLETE, ABANDONED BRANCH LINES DOWNSTREAM OF EXISTING BRANCH ISOLATION VALVE.
- DEMOLITION WORK SHALL BE ACCOMPLISHED WITHOUT DRAINING FULL SYSTEM. FREEZING PIPE OR USING HOT TAP PIPE PLUG INSERTION SOLUTION IS ACCEPTABLE.
- EXISTING CHILLED WATER MAKEUP WATER CONNECTION, PRV, BACKFLOW PREVENTER, AND EXPANSION TANK MUST REMAIN CONNECTED TO CHILLED WATER SYSTEM WHILE TEMPORARY CHILLER IS IN OPERATION.
- COMMUNICATE VALVE FUNCTION TO CONSTRUCTION MANAGER AND RECEIVE APPROVAL FOR LOCKOUT-TAGOUT PRIOR TO DEMOLITION START.

Sheet Title

**MECHANICAL CHILLED
WATER DEMOLITION FLOW
DIAGRAM**

Dwg. No.

PLM2021-0432-0011D

Sht. No.

MD801

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GENERAL NOTES:

1. STATEMENT OF SPECIAL INSPECTIONS

A. THIS "STATEMENT OF SPECIAL INSPECTIONS" HAS BEEN PREPARED IN ACCORDANCE WITH CBC 2019, SECTION 1704.

B. SUBCONTRACTOR SHALL EMPLOY ONE OR MORE APPROVED AGENCIES TO PROVIDE SPECIAL INSPECTION AND TEST DURING CONSTRUCTION IN ACCORDANCE WITH CBC 2019, CHAPTER 17.

C. EACH SPECIAL INSPECTOR SHALL BE A QUALIFIED PERSON WHO SHALL DEMONSTRATE COMPETENCE, TO THE SATISFACTION OF BURNS AND MCDONNELL FOR INSPECTION OF THE PARTICULAR TYPE OF CONSTRUCTION OR OPERATION REQUIRING SPECIAL INSPECTION.

D. SPECIAL INSPECTION AGENCY SHALL SUBMIT INSPECTION REPORTS DURING CONSTRUCTION FOR VERIFICATION, INCLUDING FINAL REPORTS IN ACCORDANCE WITH CBC 2019, SECTION 1704.2.4.

E. SPECIAL INSPECTOR SHALL USE THE LATEST ISSUE OF THE STRUCTURAL DRAWINGS FOR ALL INSPECTIONS. SHOP FABRICATION DRAWINGS SHALL NOT BE USED FOR SUCH PURPOSES.

F. THE FOLLOWING TABLES IDENTIFY THE MATERIALS, SYSTEMS, AND COMPONENTS FOR WHICH SPECIAL INSPECTION IS REQUIRED.
2. TESTING REQUIREMENTS

A. SUBCONTRACTOR SHALL EMPLOY ONE OR MORE APPROVED AGENCIES TO PROVIDE STRUCTURAL TESTING DURING CONSTRUCTION IN ACCORDANCE WITH CBC 2019, CHAPTER 17.

B. TESTING AGENCY SHALL SUBMIT TEST RESULTS DURING CONSTRUCTION FOR VERIFICATION INCLUDING A FINAL REPORT IN ACCORDANCE WITHCBC 2019, 1704.2.4.

C. THE TABLES BELOW IDENTIFY THE STRUCTURAL TESTS REQUIRED FOR THIS PROJECT.

3. STRUCTURAL OBSERVATIONS:

A. SUBCONTRACTOR SHALL EMPLOY ONE OR MORE APPROVED THIRD-PARTY REGISTERED DESIGN PROFESSIONALS TO PROVIDE STRUCTURAL OBSERVATIONS DURING CONSTRUCTION IN ACCORDANCE WITH CBC 2019, CHAPTER 17.

B. EACH STRUCTURAL OBSERVER SHALL SUBMIT WRITTEN STATEMENTS IDENTIFYING FREQUENCY AND EXTENT OF STRUCTURAL OBSERVATIONS AND ANY REPORTED DEFICIENCIES WHICH, TO THE BEST OF THE STRUCTURAL OBSERVER'S KNOWLEDGE, HAVE NOT BEEN RESOLVED.

3. STRUCTURAL OBSERVATIONS (CONTINUED):

C. O = OBSERVE THESE ITEMS ON A RANDOM BASIS. OPERATIONS NEED NOT TO BE DELAYED PENDING THESE INSPECTIONS.

D. P = PERFORM THESE TASKS FOR EACH JOINT OR MEMBER.

E. QA = INSPECTION THAT THE WORK IS IN COMPLIANCE WITH THE CONSTRUCTION DOCUMENTS

F. QC = INSPECTION THAT THE WORK IS PERFORMED IN ACCORDANCE WITH THE CONSTRUCTION DOCUMENTS

G. ALL COSTS DUE TO SPECIAL INSPECTIONS IN ACCORDANCE WITH CBC 2019, TMS 402 AND AISC 360-2016 SHALL BE THE RESPONSIBILITY OF THE SUBCONTRACTOR. THIS INCLUDES ALL QC AND QA REQUIREMENTS INDICATED ON S-003 & S-004.

H. QUALITY CONTROL: CONTROLS AND INSPECTIONS IMPLEMENTED BY THE FABRICATOR OR ERECTOR, AS APPLICABLE, TO ENSURE THAT THE MATERIAL PROVIDED AND WORK PERFORMED MEET THE REQUIREMENTS OF THE APPROVED CONSTRUCTION DOCUMENTS AND REFERENCED STANDARDS.

I. QUALITY ASSURANCE: MONITORING AND INSPECTION TASKS PERFORMED BY AN AGENCY OR FIRM OTHER THAN THE FABRICATOR OR ERECTOR TO ENSURE THAT THE MATERIAL PROVIDED AND WORK PERFORMED BY THE FABRICATOR AND ERECTOR MEET THE REQUIREMENTS OF THE APPROVED CONSTRUCTION DOCUMENTS AND REFERENCED STANDARDS. QUALITY ASSURANCE INCLUDES THOSE TASKS DESIGNATED "SPECIAL INSPECTION" BY THE APPLICABLE BUILDING CODE.
- | CBC 2019 TABLE 1705.3 REQUIRED VERIFICATION AND INSPECTION OF CONCRETE CONSTRUCTION | | | | | |
|---|---|------------|----------|---|--------------------------------|
| VERIFICATION AND INSPECTION | | CONTINUOUS | PERIODIC | REFERENCED STANDARD | CBC REFERENCE |
| 1. | INSPECT REINFORCEMENT, INCLUDING PRESTRESSING TENDONS, AND VERIFY PLACEMENT. | - | X | ACI 318: 20, 25.2, 25.3, 26.6.1-26.6.3 | 1908.4 |
| 2. | REINFORCING BAR WELDING. | | | AWS D1.4 ACI 318: 26.6.4 | |
| a. | VERIFY WELDABILITY OF REINFORCING BARS OTHER THAN ASTM A706. | - | X | | |
| b. | INSPECT SINGLE-PASS FILLET WELDS, MAXIMUM 5/16". | - | X | | |
| c. | INSPECT ALL OTHER WELDS. | X | - | | |
| 3. | INSPECTION OF ANCHORS CAST IN CONCRETE. | - | X | ACI 318: 17.8.2 | |
| 4. | INSPECTION OF ANCHORS POST-INSTALLED IN HARDENED CONCRETE MEMBERS. | | | | |
| a. | ADHESIVE ANCHORS INSTALLED IN HORIZONTALLY OR UPWARDLY INCLINED ORIENTATIONS TO RESIST SUSTAINED TENSION LOADS. | X | - | ACI 318: 17.8.2.4 | |
| b. | MECHANICAL ANCHORS AND ADHESIVE ANCHORS NOT DEFINED IN 4.a. | - | X | ACI 318: 17.8.2 | |
| 5. | VERIFYING USE OF REQUIRED DESIGN MIX. | - | X | ACI 318: 19, 26.4.3, 26.4.4 | 1904.1, 1904.2, 1908.2, 1908.3 |
| 6. | PRIOR TO CONCRETE PLACEMENT, FABRICATE SPECIMENS FOR STRENGTH TESTS, PERFORM SLUMP AND AIR CONTENT TESTS, AND DETERMINE THE TEMPERATURE OF THE CONCRETE. | X | - | ASTM C 172, ASTM C 31, ACI 318: 26.5, 26.12 | 1908.10 |
| 7. | INSPECT CONCRETE PLACEMENT FOR PROPER APPLICATION TECHNIQUES. | X | - | ACI 318: 26.5 | 1908.6, 1908.7, 1908.8 |
| 8. | VERIFY MAINTENANCE OF SPECIFIED CURING TEMPERATURE AND TECHNIQUES. | - | X | ACI 318: 26.5.3-26.5.5 | 1908.9 |
| 9. | INSPECT PRESTRESSED CONCRETE. | | | ACI 318: 26.10 | |
| a. | APPLICATION OF PRESTRESSING FORCES. | X | - | | |
| b. | GROUTING OF BONDED PRESTRESSING TENDONS. | X | - | | |
| 10. | INSPECT ERECTION OF PRECAST CONCRETE MEMBERS. | - | X | ACI 318: 26.9 | |
| 11. | VERIFY IN-SITU CONCRETE STRENGTH, PRIOR TO STRESSING OF TENDONS IN POST-TENSIONED CONCRETE AND PRIOR TO REMOVAL OF SHORES AND FORMS BEAMS AND STRUCTURAL SLABS. | - | X | ACI 318: 26.11.2 | |
| 12. | INSPECT FORMWORK FOR SHAPE, LOCATION AND DIMENSIONS OF THE CONCRETE MEMBER BEING FORMED. | - | X | ACI 318: 26.11.1.2(b) | |
- FOR SI: 1 INCH = 25.4 MM

a. WHERE APPLICABLE, SEE ALSO SECTION 1705.12, SPECIAL INSPECTIONS FOR SEISMIC RESISTANCE.

b. SPECIFIC REQUIREMENTS FOR SPECIAL INSPECTION SHALL BE INCLUDED IN THE RESEARCH REPORT FOR THE ANCHOR ISSUED BY AN APPROVED SOURCE IN ACCORDANCE WITH 17.8.2 IN ACI 318, OR OTHER QUALIFICATION PROCEDURES. WHERE SPECIFIC REQUIREMENTS ARE NOT PROVIDED, SPECIAL INSPECTION REQUIREMENTS SHALL BE SPECIFIED BY THE REGISTERED DESIGN PROFESSIONAL AND SHALL BE APPROVED BY THE BUILDING OFFICIAL PRIOR TO THE COMMENCEMENT OF THE WORK.
- Sheet Title

STATEMENT OF SPECIAL INSPECTIONS

Dwg. No.

PLS2021-0432-0003D

Sht. No.

S-003

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AISC 360-10 INSPECTIONS FOR WELDING		
TABLE N5.4-1 INSPECTION TASKS PRIOR TO WELDING	QC	QA
WELDING PROCEDURE SPECIFICATION (WPSs) AVAILABLE	P	P
MANUFACTURER CERTIFICATIONS FOR WELDING CONSUMABLES AVAILABLE	P	P
MATERIAL IDENTIFICATION (TYPE/GRADE)	O	O
WELDER IDENTIFICATION SYSTEM (1)	O	O
FIT-UP OF GROOVE WELDS (INCLUDING JOINT GEOMETRY) - JOINT PREPARATION - DIMENSIONS (ALIGNMENT, ROOT OPENING, ROOT FACE, BEVEL) - CLEANLINESS (CONDITION OF STEEL SURFACES) - TACKING (TACK WELD QUALITY AND LOCATION) - BACKING TYPE AND FIT (IF APPLICABLE)	O	O
CONFIGURATION AND FINISH OF ACCESS HOLES	O	O
FIT-UP OF FILLET WELDS - DIMENSIONS (ALIGNMENT, GAPS AT ROOT) - CLEANLINESS (CONDITION OF STEEL SURFACES) - TACKING (TACK WELD QUALITY AND LOCATION)	O	O
CHECK WELDING EQUIPMENT	O	—
TABLE N5.4-2 INSPECTION TASKS DURING WELDING		
USE OF QUALIFIED WELDERS	O	O
CONTROL AND HANDLING OF WELDING CONSUMABLES - PACKAGING - EXPOSURE CONTROL	O	O
NO WELDING OVER CRACKED TACK WELDS	O	O
ENVIRONMENTAL CONDITIONS - WIND SPEED WITHIN LIMITS - PRECIPITATION AND TEMPERATURE	O	O
WPS FOLLOWED - SETTINGS ON WELDING EQUIPMENT - TRAVEL SPEED - SELECTED WELDING MATERIALS - SHIELDING GAS TYPE/FLOW RATE - PREHEAT APPLIED - INTERPASS TEMPERATURE MAINTAINED (MIN/MAX) - PROPER POSITION (F, V, H, OH)	O	O
WELDING TECHNIQUES - INTERPASS AND FINAL CLEANING - EACH PASS WITHIN PROFILE LIMITATIONS - EACH PASS MEETS QUALITY REQUIREMENTS	O	O
TABLE N5.4-3 INSPECTION TASKS AFTER WELDING		
WELDS CLEANED	O	O
SIZE, LENGTH AND LOCATION OF WELDS	P	P
WELDS MEET VISUAL ACCEPTANCE CRITERIA - CRACK PROHIBITION - WELD/BASE-METAL FUSION - CRATER CROSS SECTION - WELD PROFILES - WELD SIZE - UNDERCUT - POROSITY	P	P
ARC STRIKES	P	P
K-AREA (2)	P	P
BACKING REMOVED AND WELD TABS REMOVED (IF REQUIRED)	P	P
REPAIR ACTIVITIES	P	P
DOCUMENT ACCEPTANCE OR REJECTION OF WELDED JOINT OR MEMBER	P	P
(1) THE FABRICATOR OR ERECTOR, AS APPLICABLE, SHALL MAINTAIN A SYSTEM BY WHICH A WELDER WHO HAS WELDED A JOINT OR MEMBER CAN BE IDENTIFIED. STAMPS, IF USED, SHALL BE THE LOW-STRESS TYP. (2) WHEN WELDING OF DOUBLER PLATES, CONTINUITY PLATES OR STIFFENERS HAS BEEN PERFORMED IN THE K-AREA, VISUALLY INSPECT THE WEB K-AREA FOR CRACKS WITHIN 3 IN. (75 MM) OF THE WELD.		

AISC 360-10 FABRICATOR AND ERECTOR QUALITY CONTROL PROGRAM		
FABRICATOR INSPECTION TASKS	QC	QA
DETAILS IN ACCORDANCE WITH SECTION N5	P	-
SHOP CUT AND FINISHED SURFACES IN ACCORDANCE WITH SECTION M2	P	-
SHOP HEATING FOR STRAIGHTENING, CAMBERING AND CURVING IN ACCORDANCE WITH SECTION M2.1	P	-
TOLERANCES FOR SHOP FABRICATION IN ACCORDANCE WITH SECTION 6 OF THE <i>CODE OF STANDARD PRACTICE</i>	P	-
INSPECT THE FABRICATED STEEL TO VERIFY COMPLIANCE WITH DETAILS SHOWN ON SHOP DRAWINGS, SUCH AS PROPER APPLICATION OF JOINT DETAILS AT EACH CONNECTION	P	-
ERECTOR INSPECTION TASKS		
DETAILS IN ACCORDANCE WITH SECTION N5	P	-
STEEL DECK HEADED STEEL STUD ANCHOR PLACEMENT AND ATTACHMENT IN ACCORDANCE WITH SECTION N6	P	-
FIELD CUT SURFACES IN ACCORDANCE WITH SECTION M2.2	P	-
FIELD HEATING FOR STRAIGHTENING IN ACCORDANCE WITH SECTION M2.1	P	-
TOLERANCES FOR FIELD ERECTION IN ACCORDANCE WITH SECTION 7.13 OF THE <i>CODE OF STANDARD PRACTICE</i>	P	-
INSPECT THE ERECTED STEEL FRAME TO VERIFY COMPLIANCE WITH THE DETAILS SHOWN ON THE ERECTION DRAWINGS, SUCH AS BRACES, STIFFENERS, MEMBER LOCATIONS AND PROPER APPLICATION OF JOINT DETAILS AT EACH CONNECTION	P	-
INSPECT PLACEMENT OF ANCHOR RODS AND OTHER EMBEDMENTS SUPPORTING STRUCTURAL STEEL FOR COMPLIANCE WITH CONSTRUCTION DOCUMENTS. AS A MINIMUM, THE DIAMETER, GRADE, TYPE AND LENGTH OF THE ANCHOR ROD OR EMBEDDED ITEM, AND THE EXTENT OR DEPTH OF EMBEDMENT INTO CONCRETE, SHALL BE VERIFIED PRIOR TO PLACEMENT OF CONCRETE	-	P
INSPECT THE FABRICATED STEEL OR ERECTED STEEL FRAME, AS APPROPRIATE, TO VERIFY COMPLIANCE WITH THE DETAILS SHOWN ON CONSTRUCTION DOCUMENTS, SUCH AS BRACES, STIFFENERS, MEMBER LOCATIONS AND PROPER APPLICATION OF JOINT DETAILS AT EACH CONNECTION	-	O
REVIEW THE MATERIAL TEST REPORTS AND CERTIFICATIONS FOR COMPLIANCE WITH THE CONSTRUCTION DOCUMENTS	-	P

GENERAL NOTES

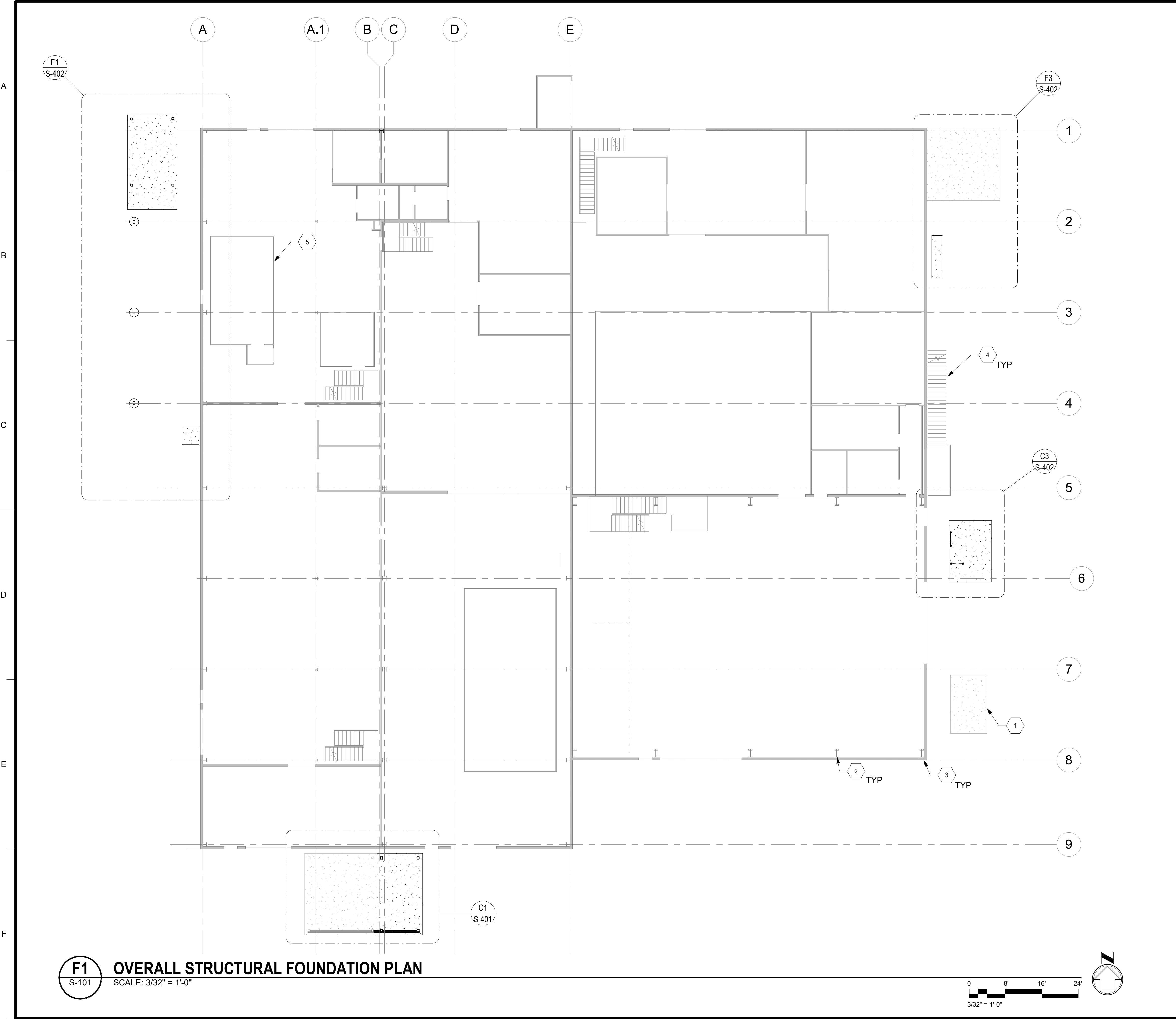
1) SEE SHEET S-003 FOR SPECIAL INSPECTION PROGRAM NOTES.

Sheet Title

STATEMENT OF SPECIAL INSPECTIONS

Dwg. No.
PLS2021-0432-0004D

Sht. No.
S-004



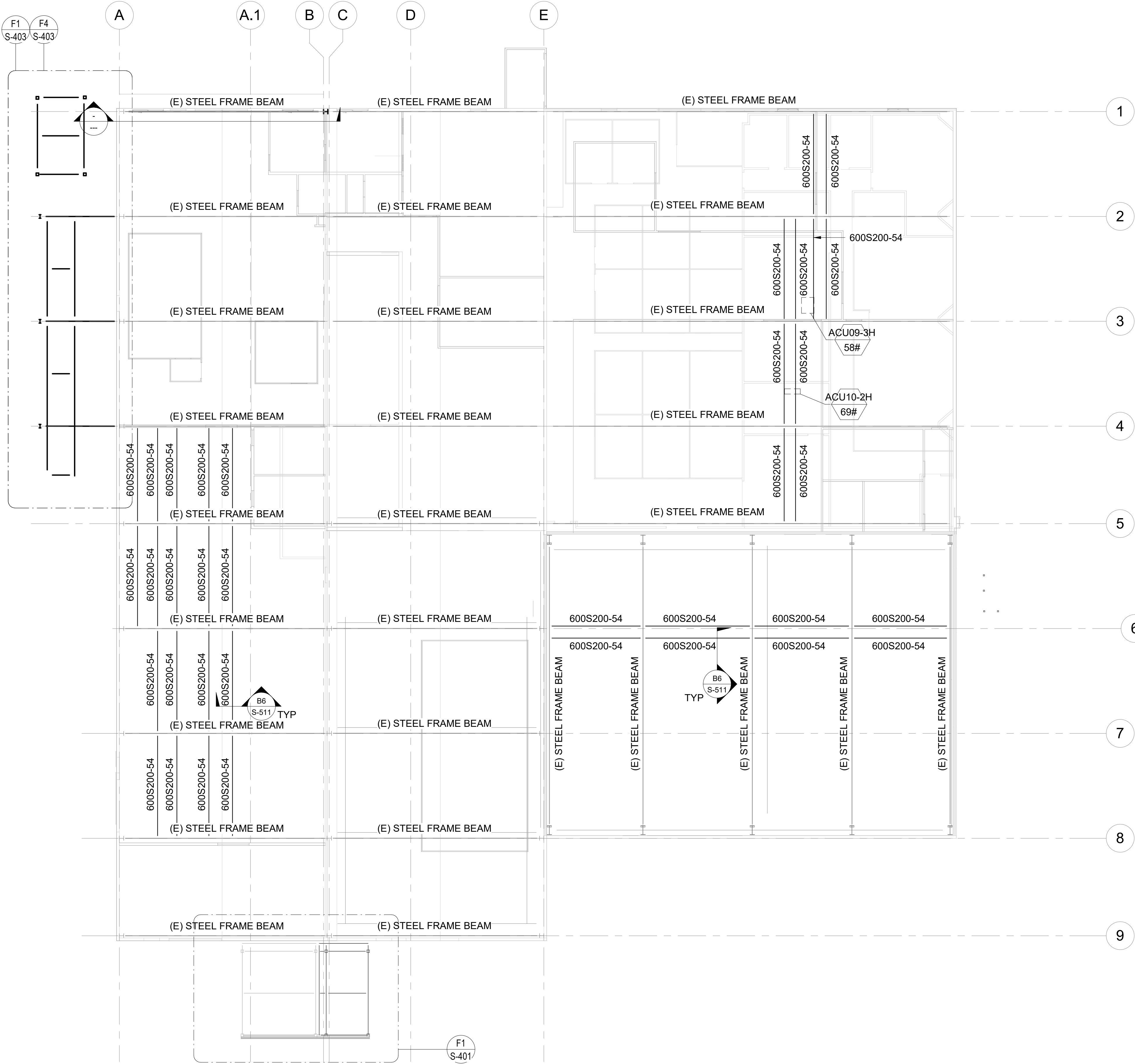
GENERAL NOTES

1. FOR STRUCTURAL GENERAL NOTES, SEE S-001.
2. FOR STRUCTURAL LEGNEDS AND ABBREVIATIONS, SEE S-002.
3. FOR STRUCTURAL SPECIAL INSPECTIONS, SEE S-003 AND S-004.
4. FOR ALL EQUIPMENT, REFER TO EQUIPMENT ANCHORAGE SCHEDULE (B1/S-513).

KEYED NOTES

1. EXISTING CONC PAD AND HEAT PUMP TO REMAIN
2. TAPERED BUILDING COLUMN
3. EXISTING EXTERIOR METAL PANEL WALL
4. EXISTING STAIRS
5. EXISTING CLEAN ROOM

Sheet Title	OVERALL FOUNDATION PLAN
Dwg. No.	PLS2021-0432-0005D
Sht. No.	S-101



F1
S-102

OVERALL ROOF FRAMING PLAN
SCALE: 3/32" = 1'-0"

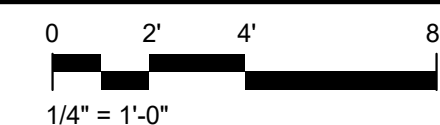
GENERAL NOTES

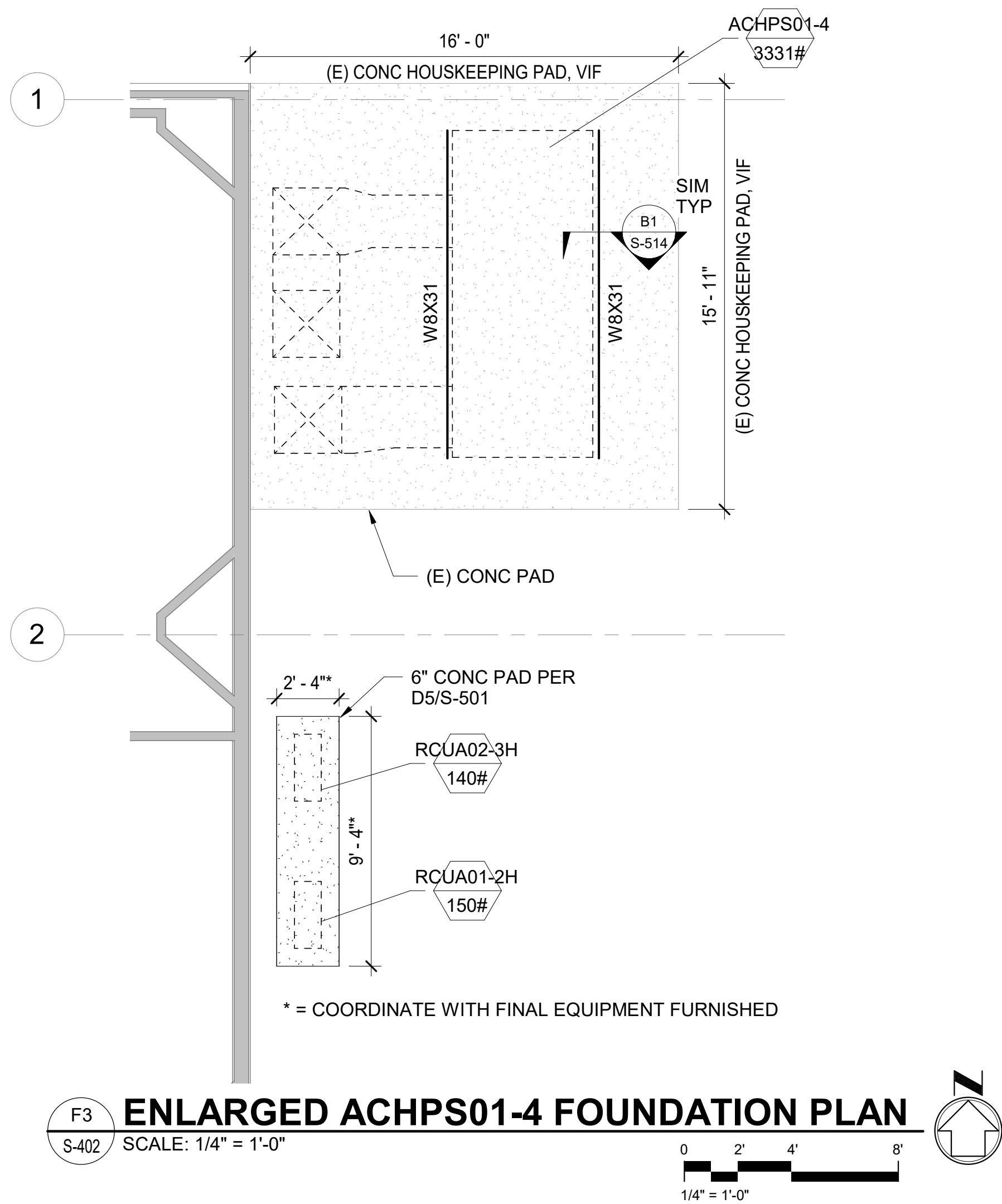
1. FOR STRUCTURAL GENERAL NOTES, SEE S-001.
2. FOR STRUCTURAL LEGNEDS AND ABBREVIATIONS, SEE S-002.
3. FOR STRUCTURAL SPECIAL INSPECTIONS, SEE S-003 AND S-004.

Sheet Title
**OVERAL STRUCTURAL ROOF
PLAN**

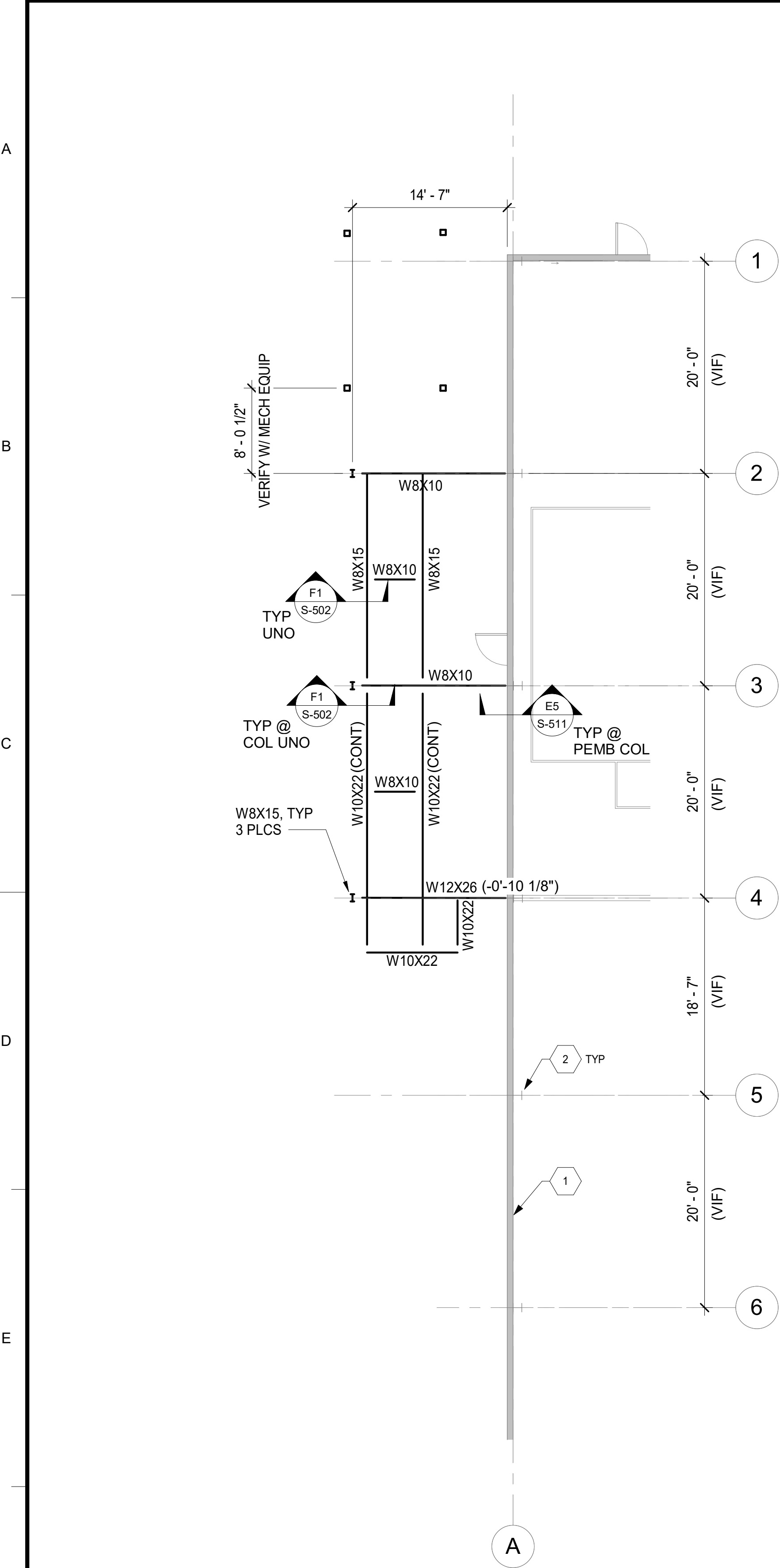
Dwg. No.
PLS2021-0432-0006D

Sht. No.
S-102



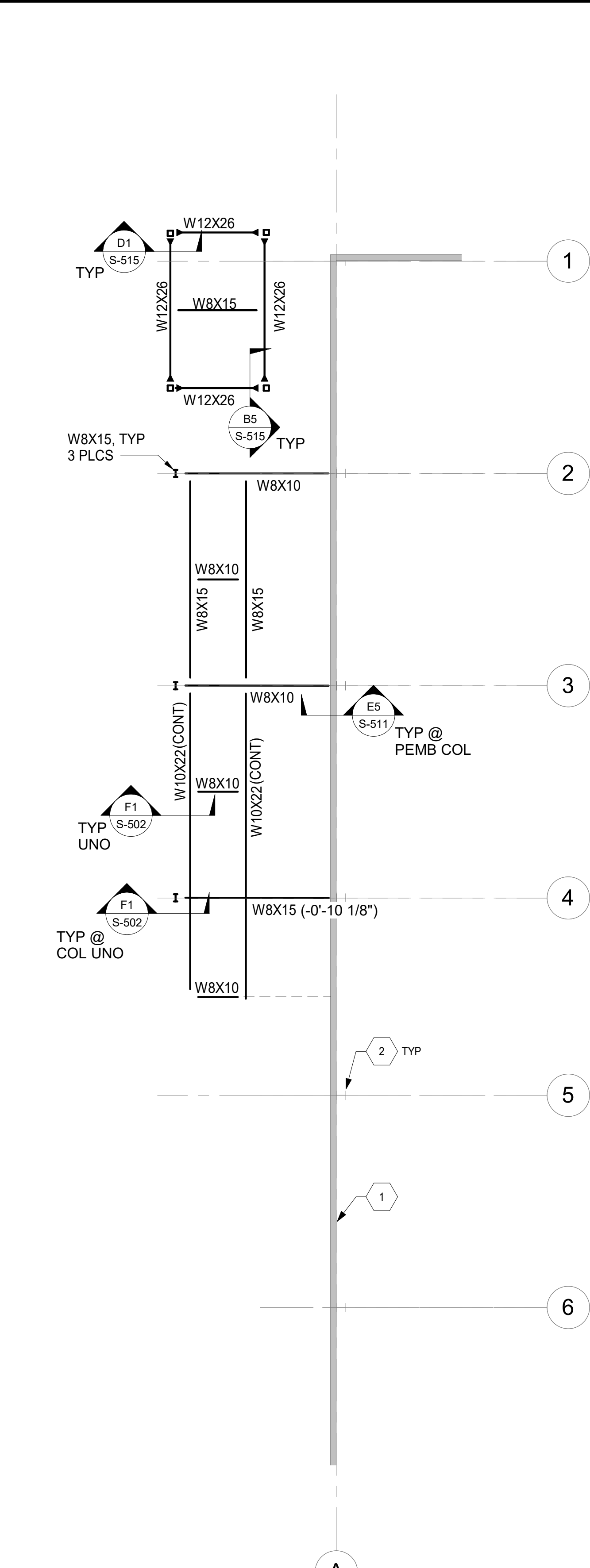
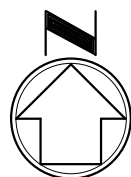


Sht. No. S-402



F1
S-403 LOWER DUCT SUPPORT FRAMING PLAN

SCALE: 1/8" = 1'-0" TOS = 109'-0", UNO



F4
S-403 UPPER DUCT SUPPORT FRAMING PLAN

SCALE: 1/8" = 1'-0" TOS = 115'-0", UNO



GENERAL NOTES

1. FOR STRUCTURAL GENERAL NOTES, SEE S-001.
2. FOR STRUCTURAL LEGNEDS AND ABBREVIATIONS, SEE S-002.
3. FOR STRUCTURAL SPECIAL INSPECTIONS, SEE S-003 AND S-004.
4. REFER TO MECHANICAL FOR ANY DIMENSIONS NOT SHOWN.
5. ALL TOP OF STEEL ELEVATIONS SHALL BE COORDINATED AND CONFIRMED WITH MECHANICAL.

KEYED NOTES

1. EXISTING EXTERIOR BUILDING WALL.
2. EXISTING BUILDING STEEL COLUMN.

LEGEND

- MOMENT CONNECTION
- (-0' - 10 1/8") DESIGNATES TOP OF STEEL BEAM ELEVATION RELATIVE TO TYPICAL TOP OF STEEL ELEVATION NOTED ON PLAN

Sheet Title

ENLARGED DUCT SUPPORT FRAMING PLANS

Dwg. No.

PLS2021-0432-0009D

Sht. No.

S-403

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S-501

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DETAIL 2



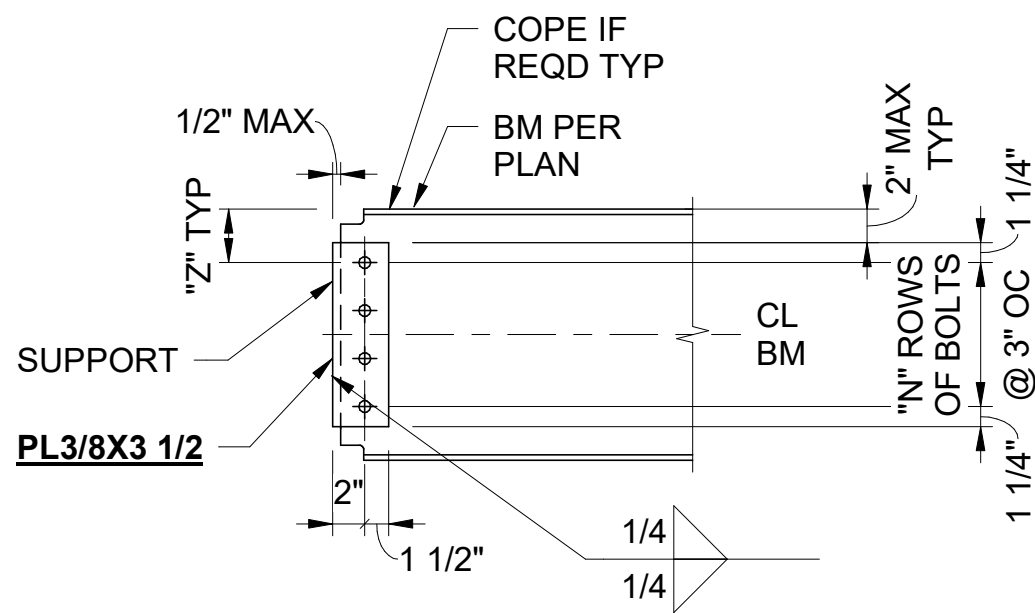
SCALE: 1" = 1'-0"



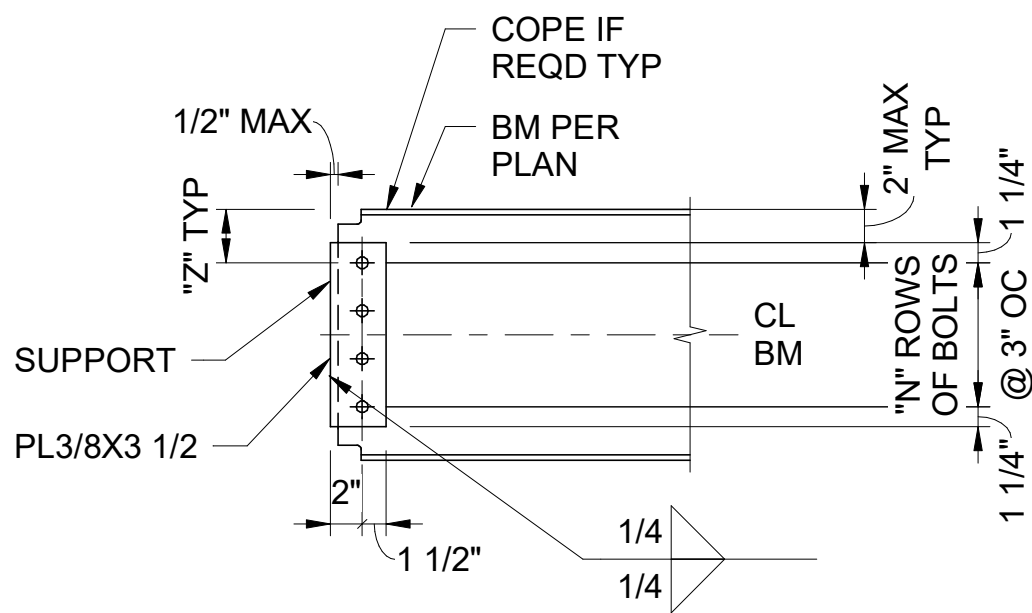
F5
S-501

SCALE: 3/4" = 1'-0"

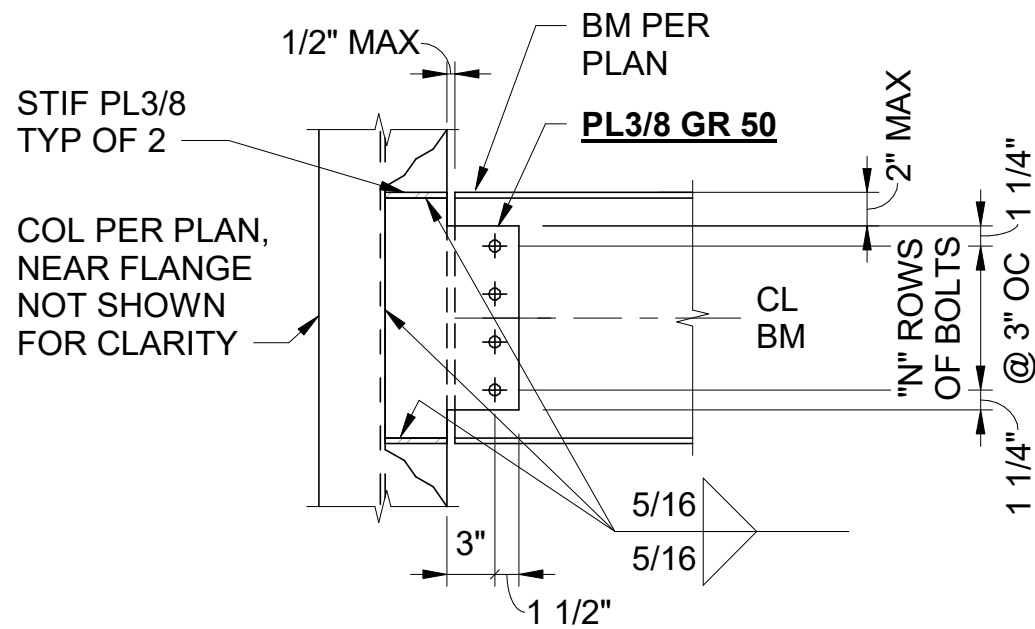
S-501



TYPICAL CONNECTION



TYPICAL CONNECTION

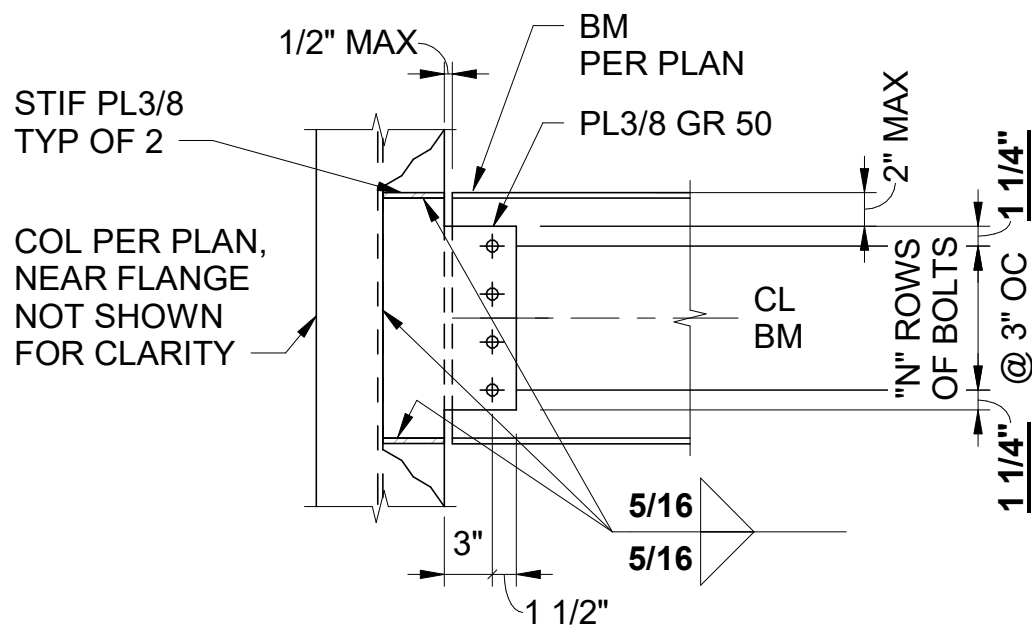


EXTENDED PLATE CONNECTION/STANDARD CONNECTION AT COLUMN WEB

MEMBER SIZE	"N" UNO
W30, W33	7
W27, W24	6
W21	5
W16, W18	4
W12, W14	3
W8, W10	2

- NOTES:**
- STANDARD CONNECTIONS ABOVE SHALL USE 3/4" DIA A325-PT-STD BOLTS.
 - USE WHERE CONNECTION IS NOT OTHERWISE INDICATED ON PLANS, NOTES, ELEVATIONS, SECTIONS, OR DETAILS.
 - "Z": 2" MIN FOR STANDARD HOLES (1 1/4" MIN FOR BEAMS LESS THAN A W12). 2 1/4" MIN FOR OVS AND SSLT HOLES (1 3/8" MIN FOR BEAMS LESS THAN A W12).

STANDARD SINGLE PLATE BEAM CONNECTION



EXTENDED PLATE CONNECTION CONNECTION AT COLUMN WEB

- NOTES:**
- SPECIAL CONNECTION SYMBOL AS NOTED ON PLANS.
 - THE LETTER INDICATES THE CONNECTION DESIGNATION PER THE SPECIAL CONNECTION SCHEDULE
 - "Z": 2" MIN FOR STANDARD HOLES (1 1/4" MIN FOR BEAMS LESS THAN A W12). 2 1/4" MIN FOR OVS AND SSLT HOLES (1 3/8" MIN FOR BEAMS LESS THAN A W12).

SPECIAL SINGLE PLATE BEAM CONNECTION

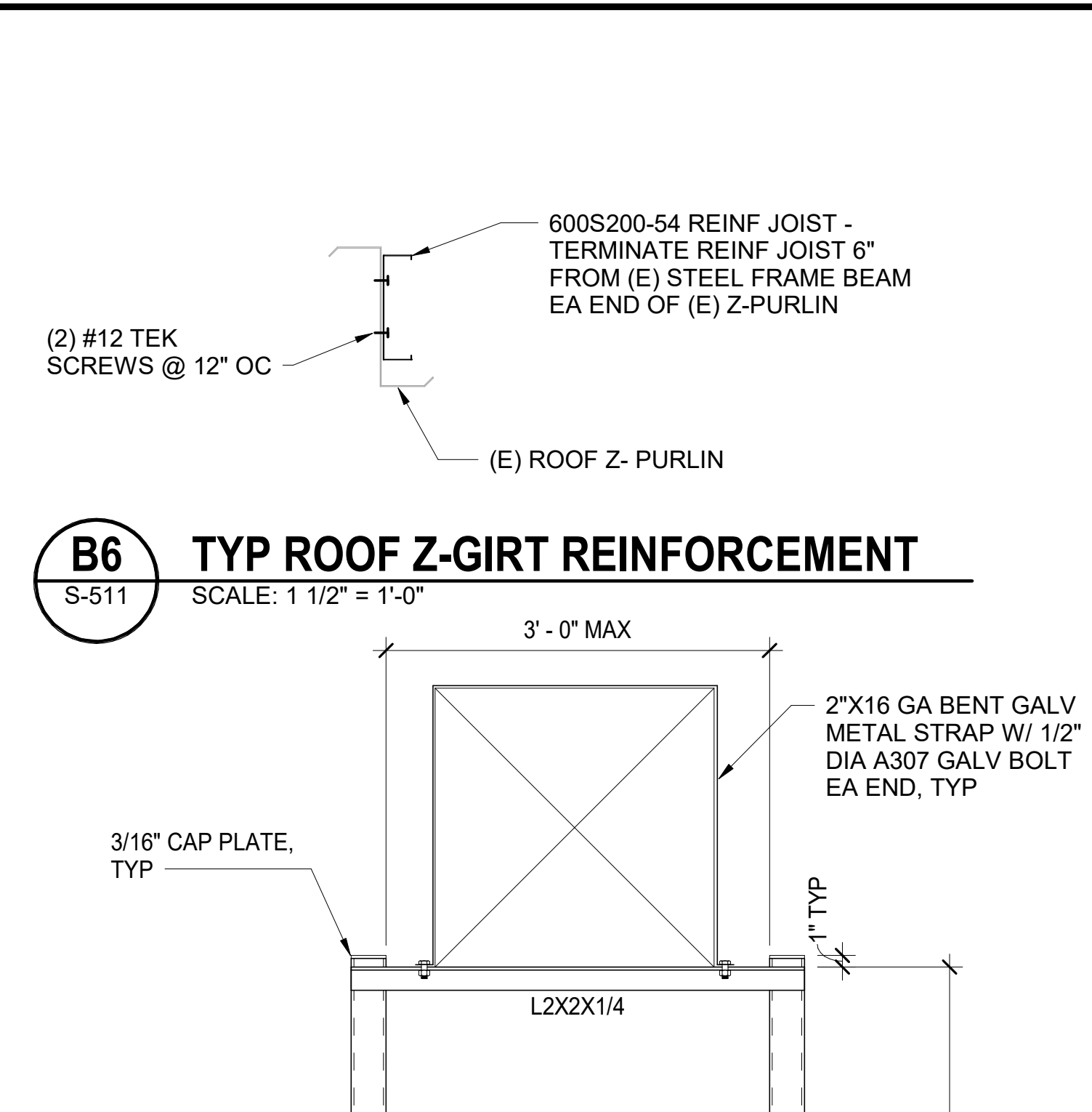
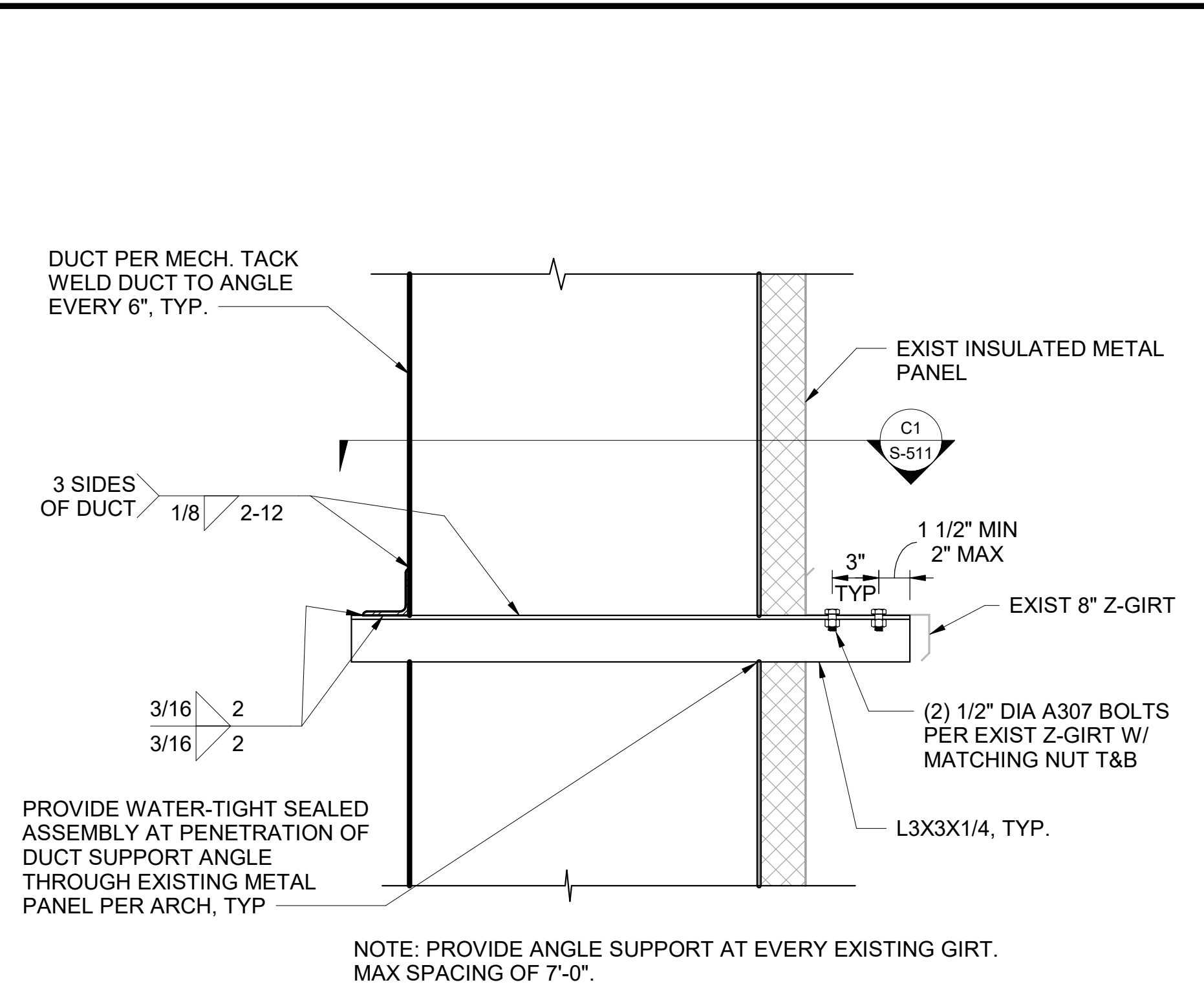
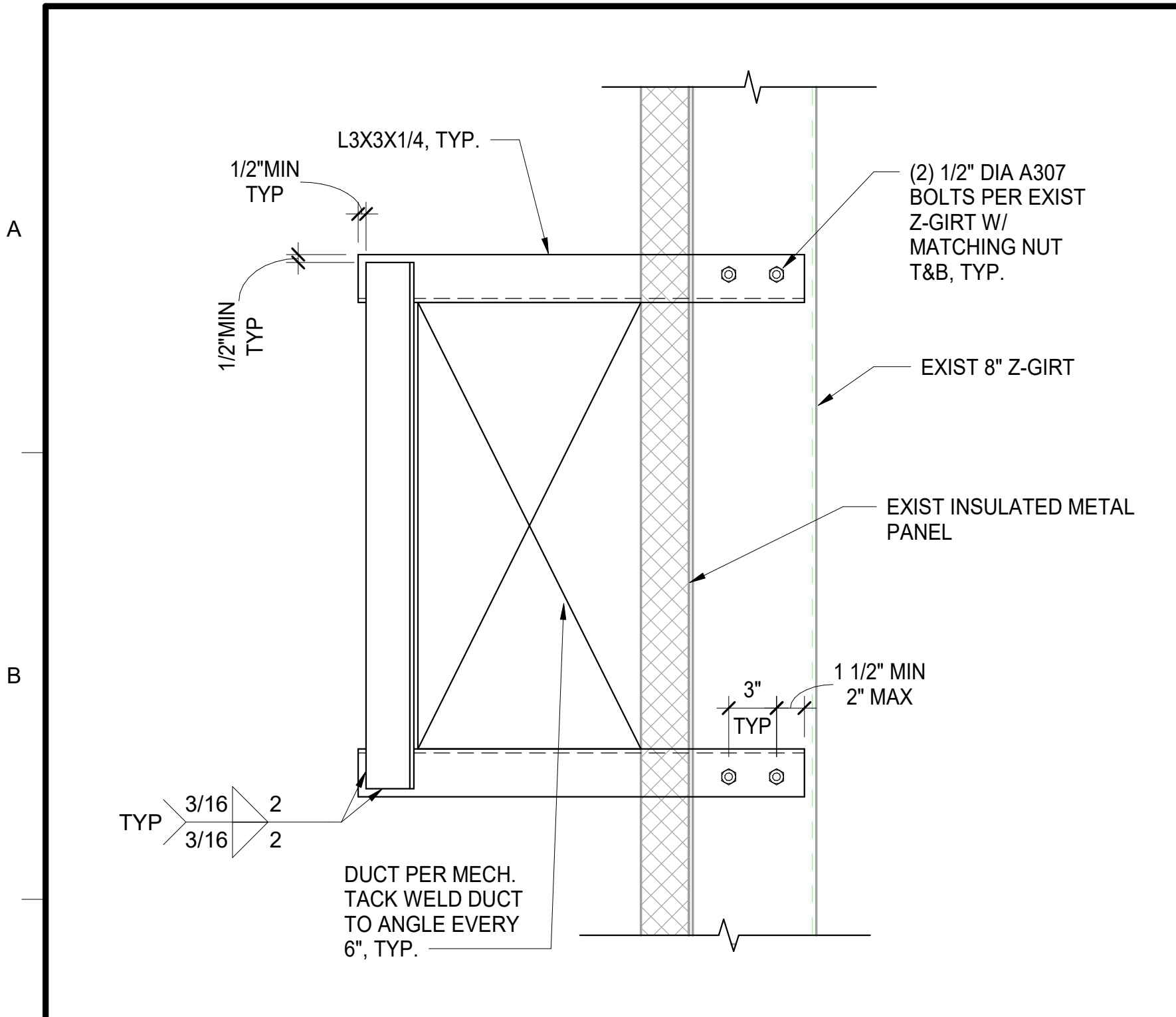
F1
S-502
TYPICAL BEAM CONNECTION-SINGLE PLATE
SCALE: 1" = 1'-0"

F3
S-502
SPECIAL BEAM CONNECTION-SINGLE PLATE
SCALE: 1" = 1'-0"

Sheet Title
STANDARD STEEL DETAILS

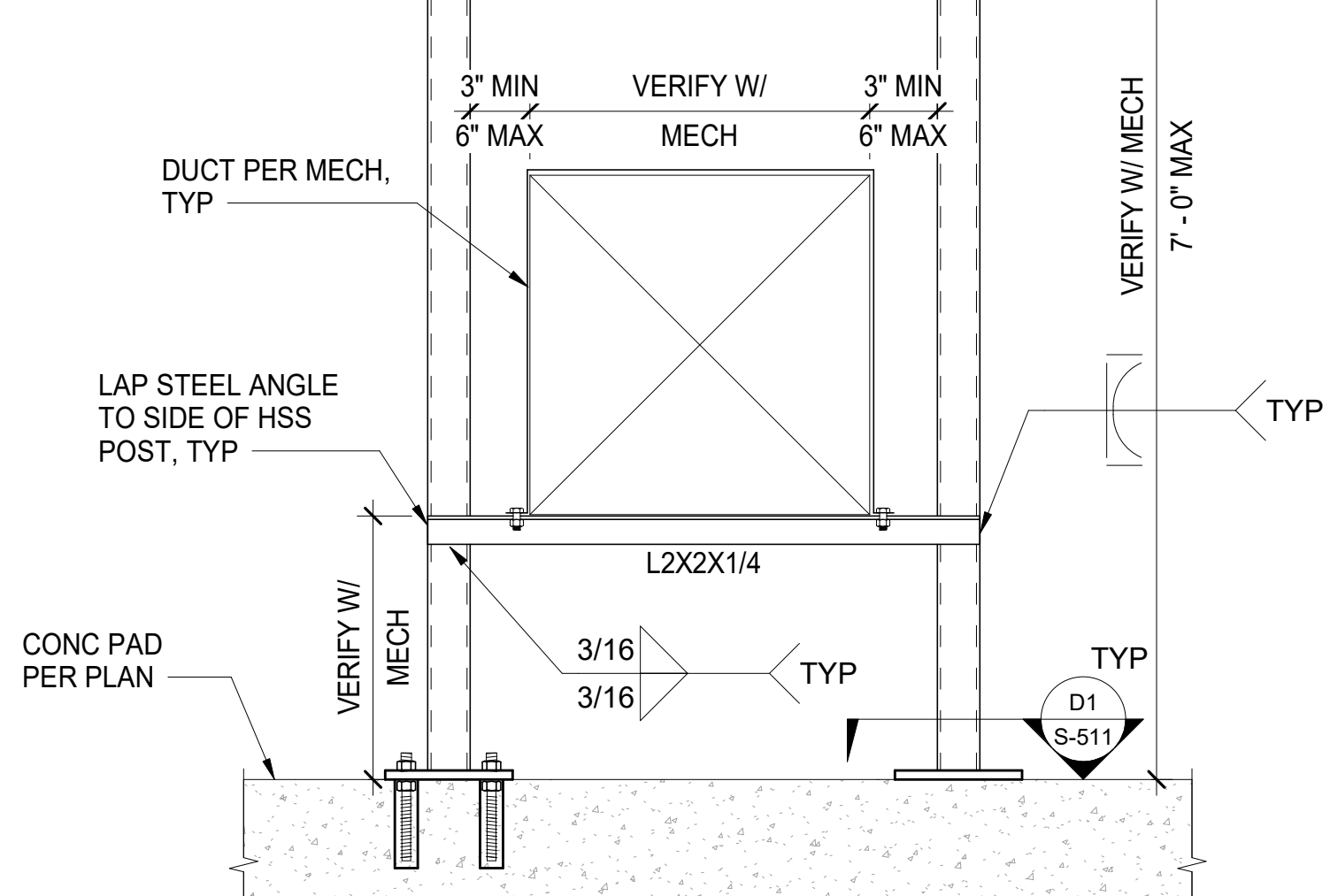
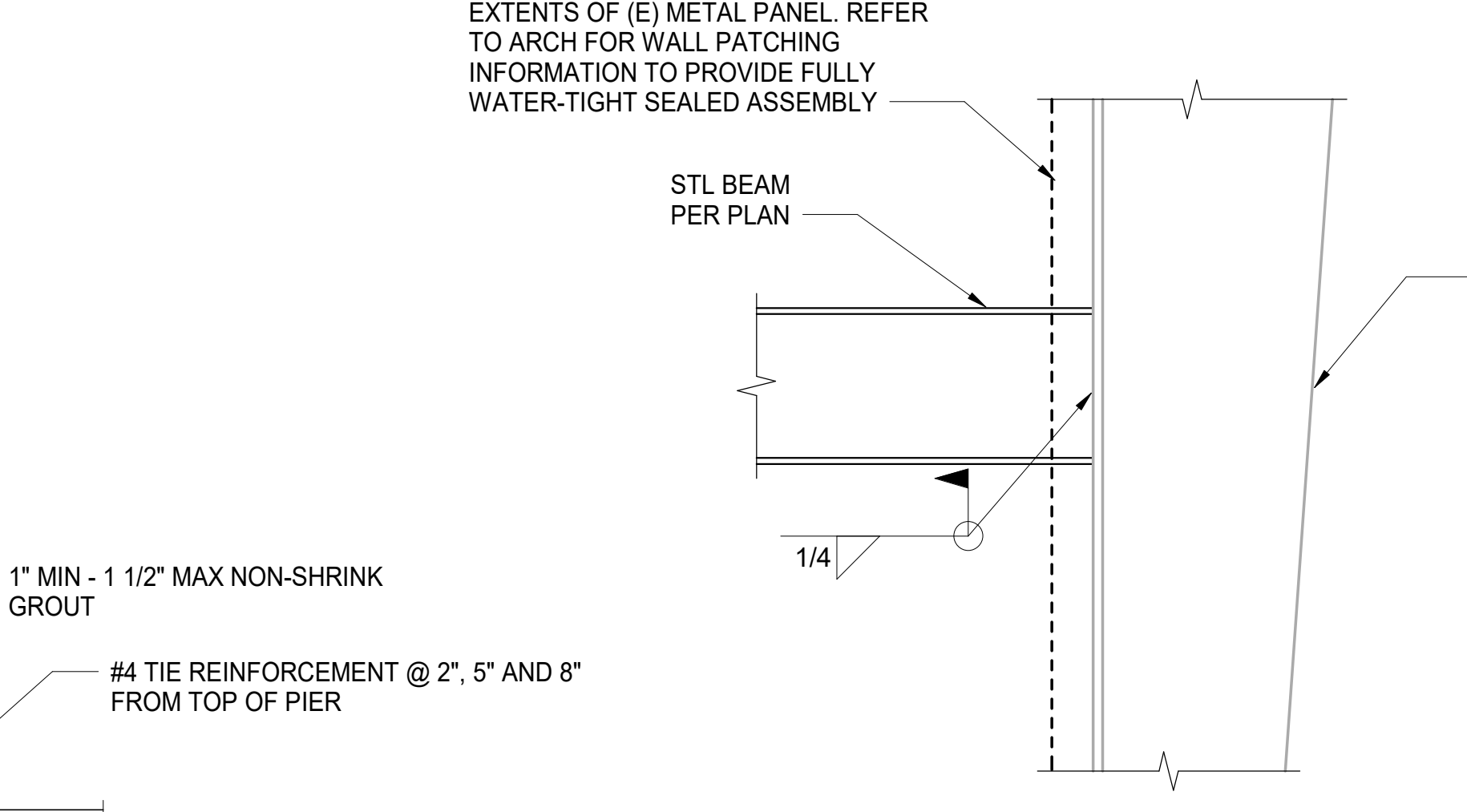
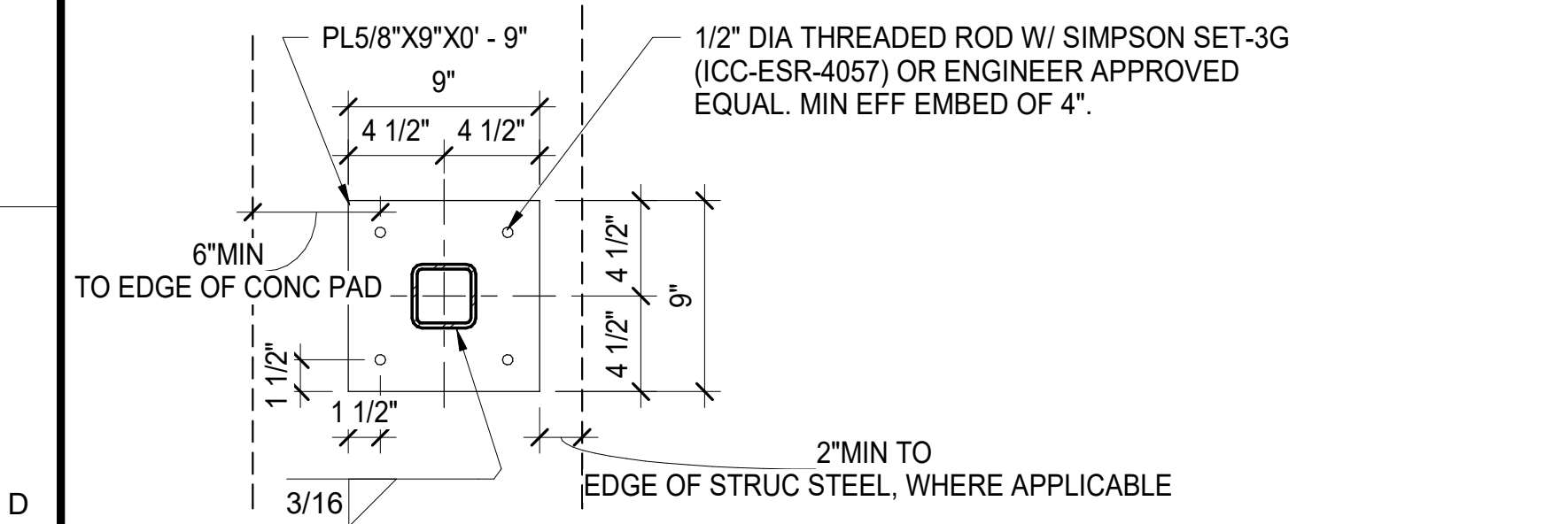
Dwg. No.
PLS2021-0432-0011D

Sht. No.
S-502



C1 DUCT SUPPORT - PLAN VIEW
S-511 SCALE: 1 1/2" = 1'-0"

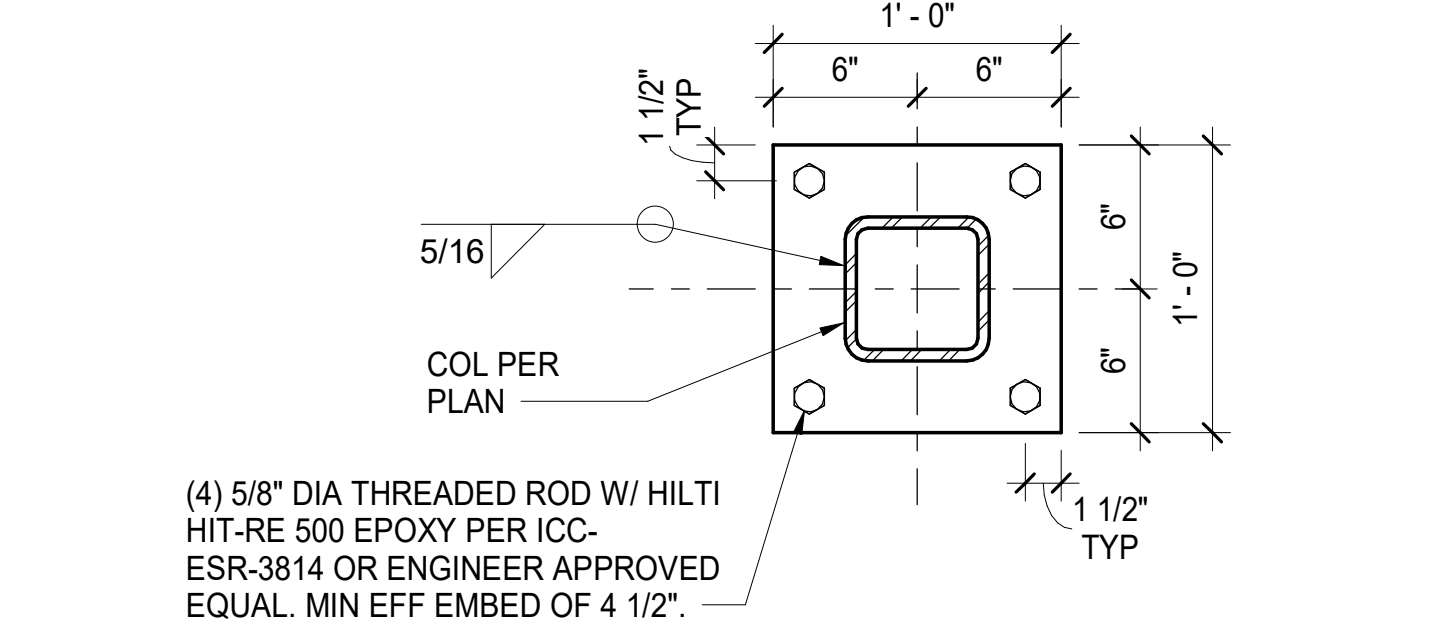
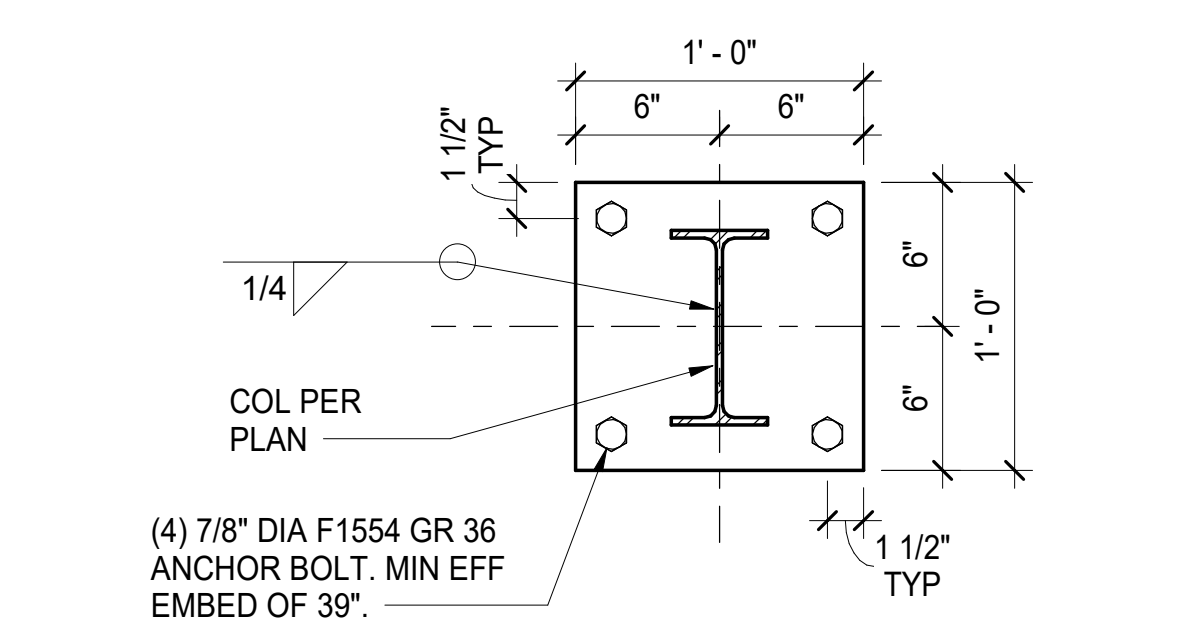
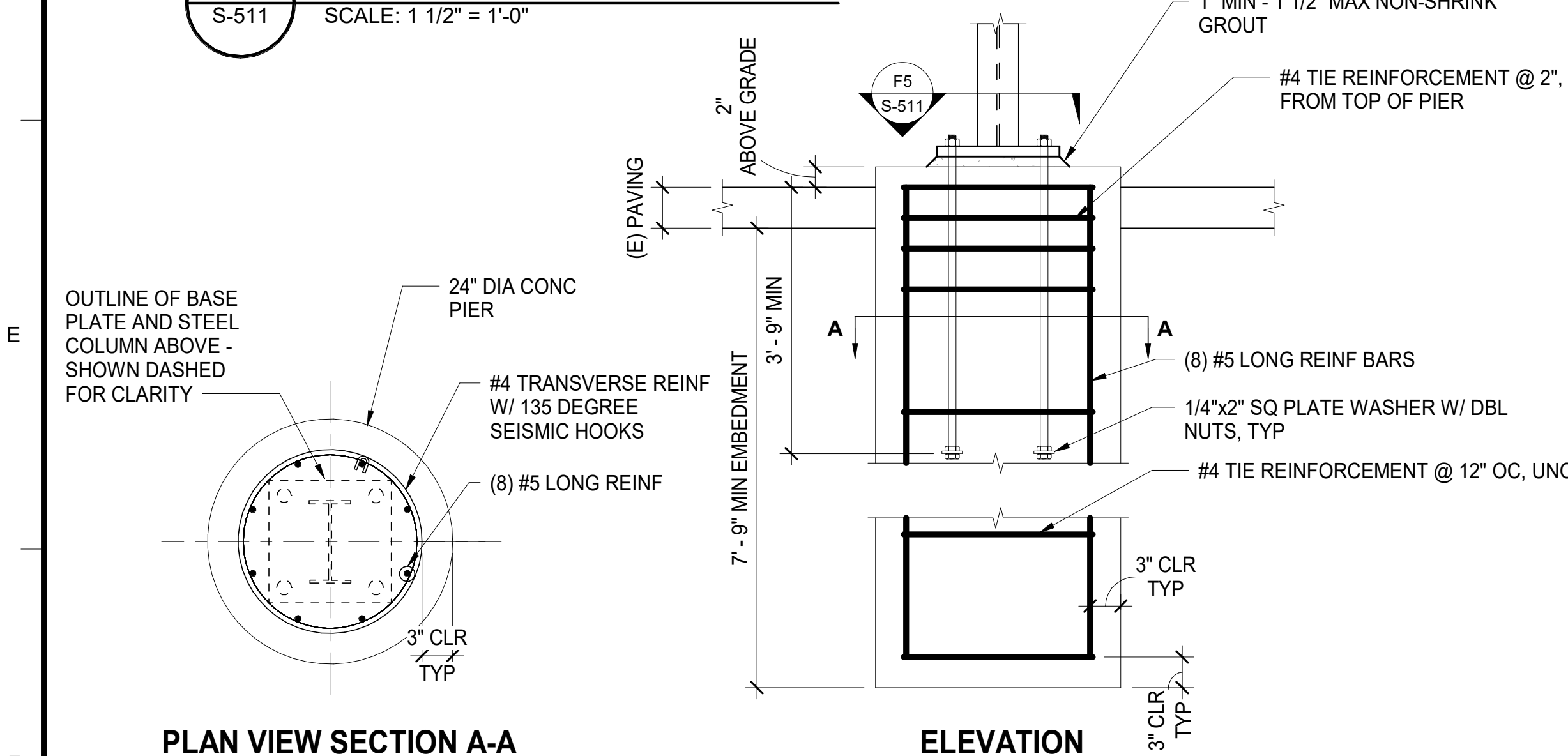
C3 DUCT SUPPORT - SECTION VIEW
S-511 SCALE: 1 1/2" = 1'-0"



D1 TYP HEAT PUMP BASE PLATE
S-511 SCALE: 1 1/2" = 1'-0"

E5 TYP CONNECTION TO (E) PEMB COLUMN
S-511 SCALE: 1 1/2" = 1'-0"

E7 HEAT PUMP DUCT SUPPORT FRAME
S-511 SCALE: 1" = 1'-0"

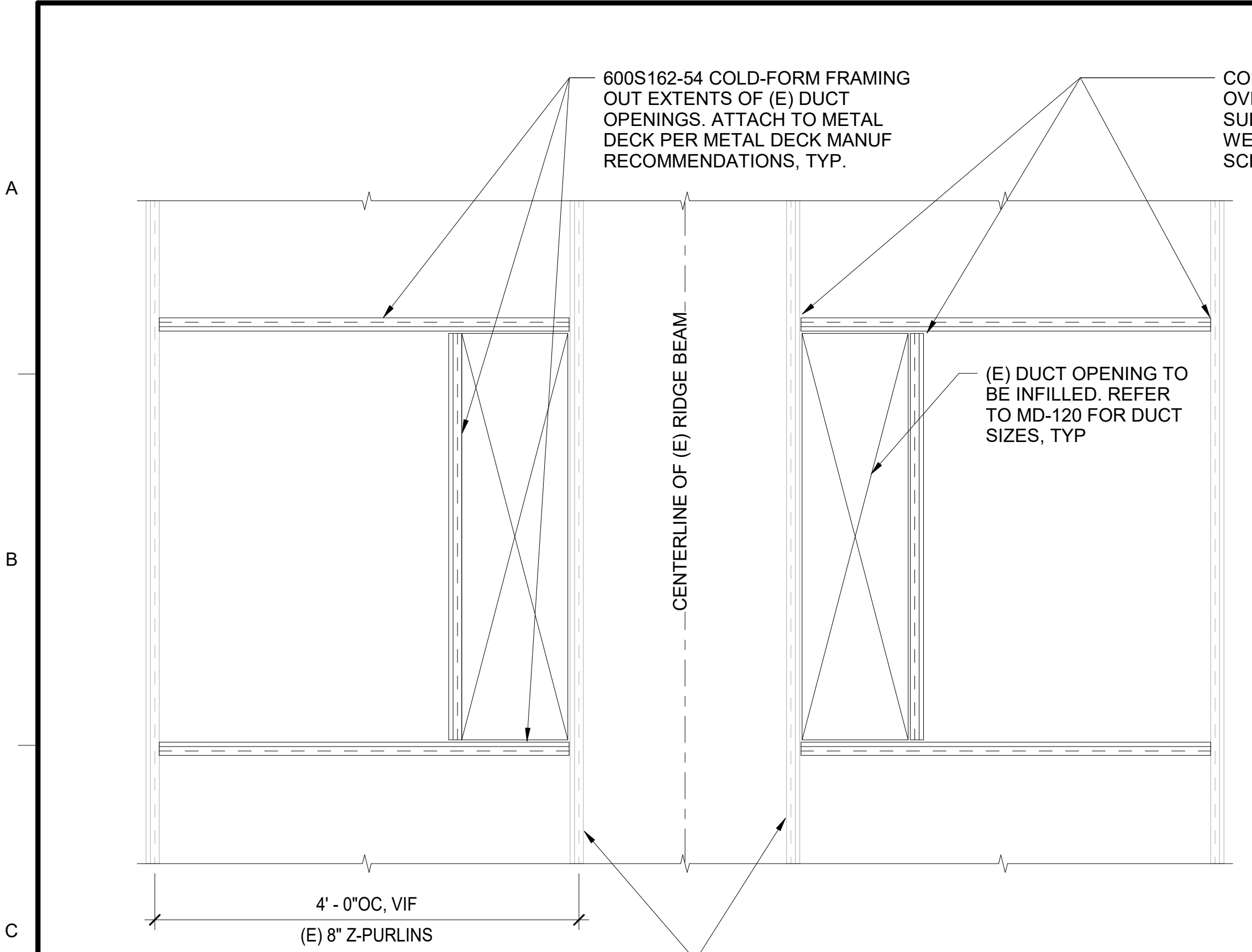


F1 TYP DUCT SUPPORT PIER
S-511 SCALE: 1" = 1'-0"

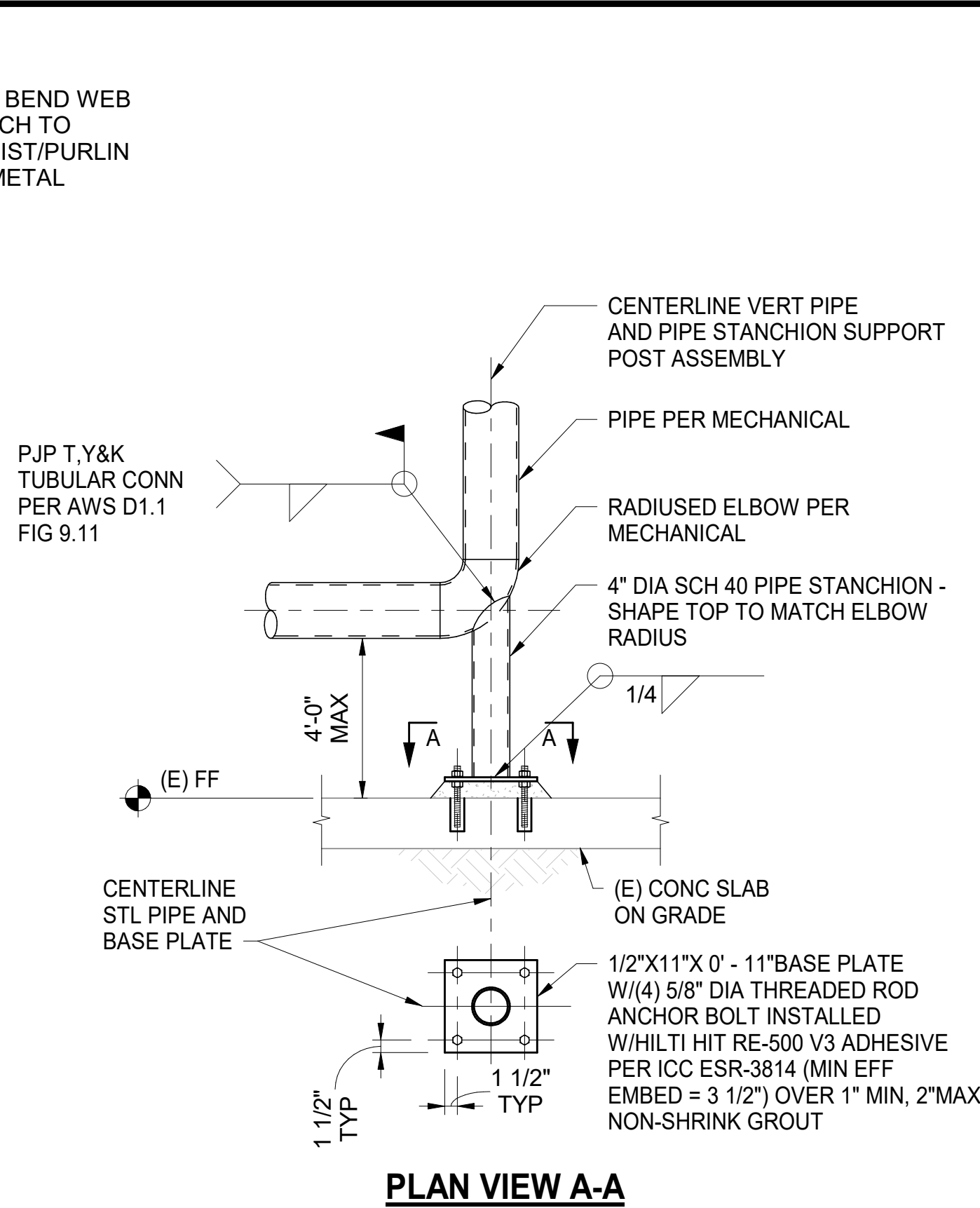
F5 TYP DUCT SUPPORT BASE PLATE
S-511 SCALE: 1 1/2" = 1'-0"

F7 TYP DUCT SUPPORT BASE (OVER ACU)
S-511 SCALE: 1 1/2" = 1'-0"

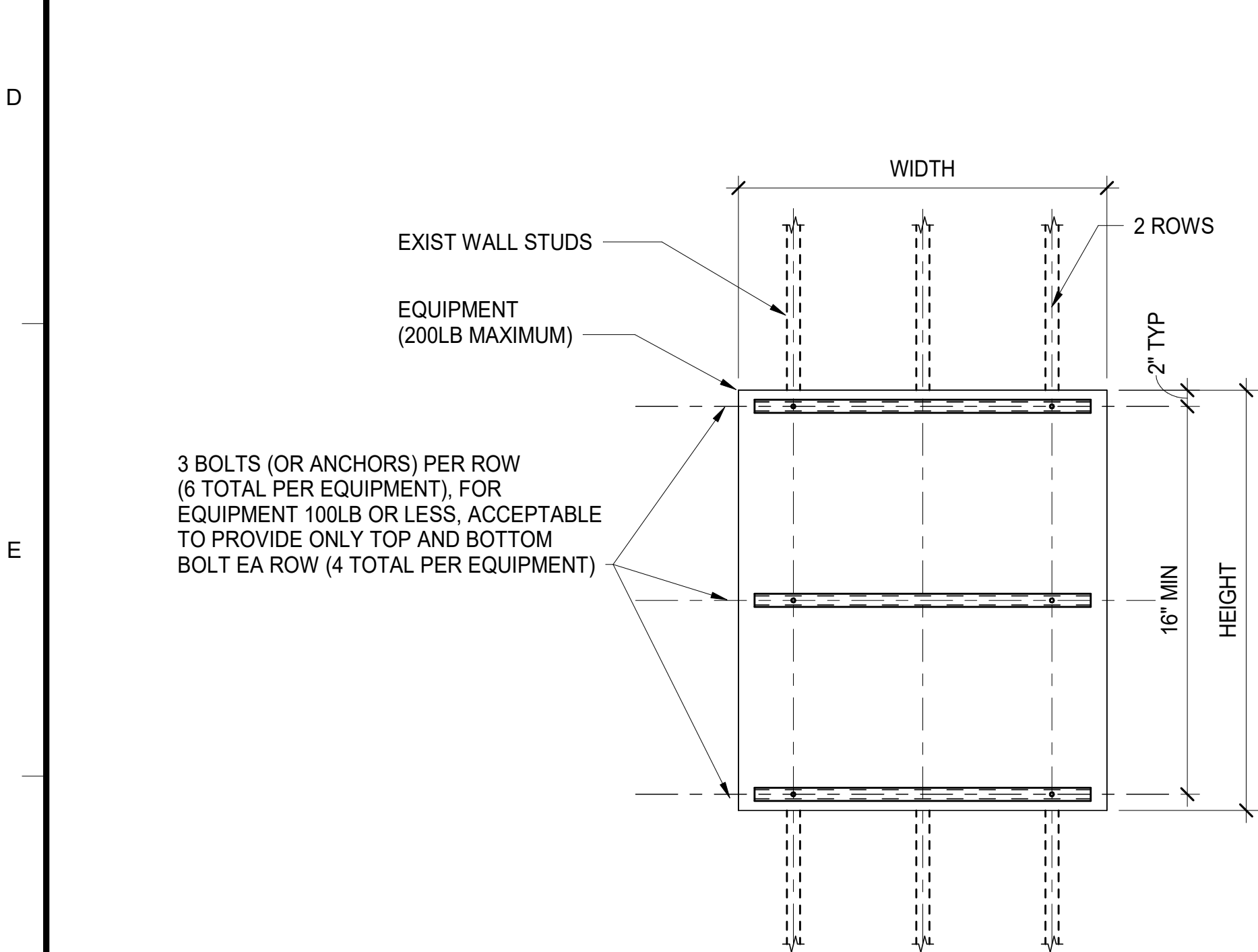
Sheet Title
DUCT SUPPORT DETAILS
Dwg. No.
PLS2021-0432-0012D
Sht. No.
S-511



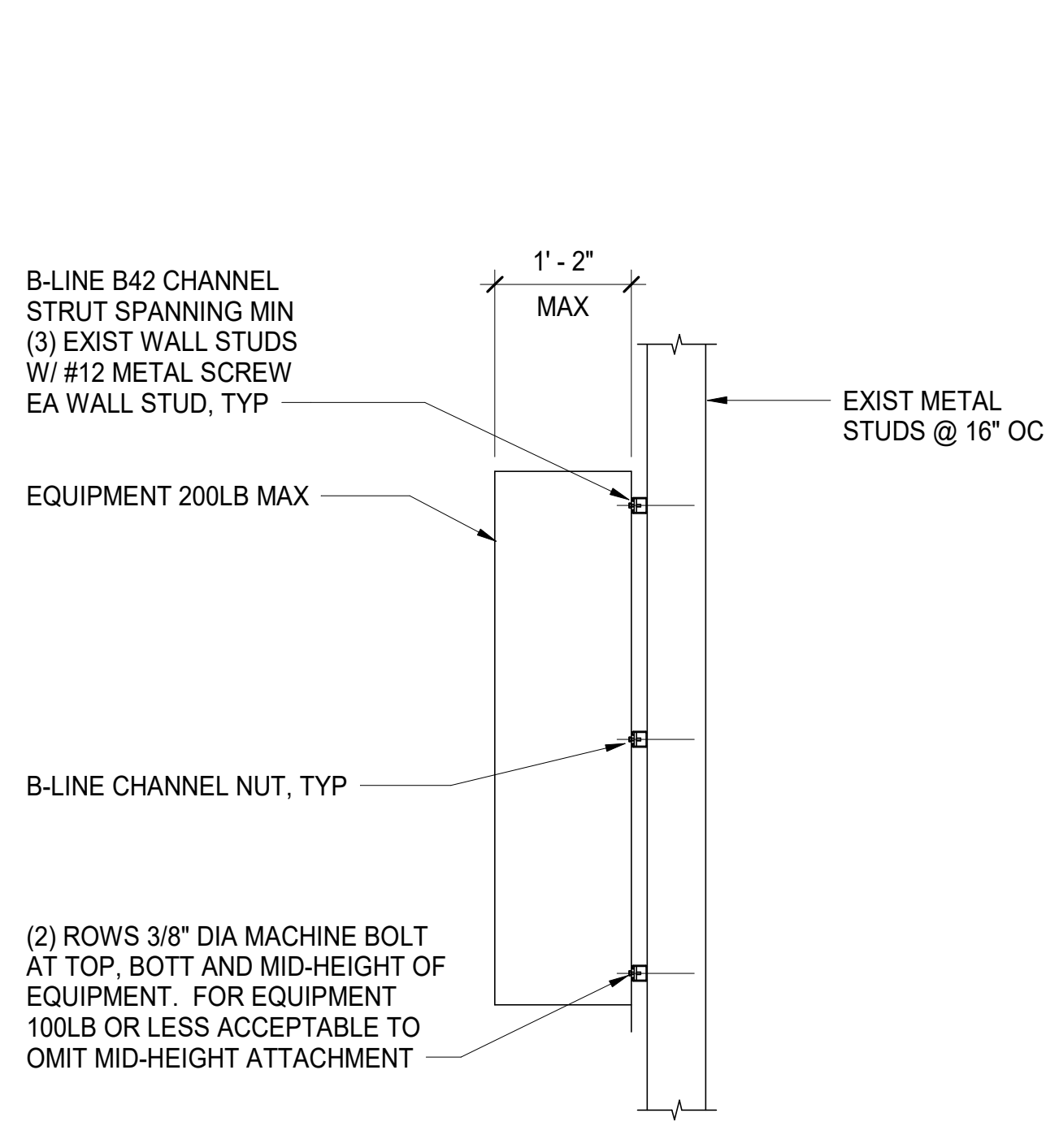
C1 TYP ROOF INFILL
S-512 SCALE: 1" = 1'-0"



C4 TYPICAL PIPE STANCHION SUPPORT AT RADIUS ELBOW
S-512 SCALE: 3/4" = 1'-0"



F1 TYP WALL MOUNTED EQUIPMENT ANCHORAGE
S-512 SCALE: 3/4" = 1'-0"



F6 TYPICAL ELECTRICAL EQUIP STAND AT CONC SLAB
S-512 SCALE: 3/4" = 1'-0"

Sheet Title
MISCELLANEOUS SUPPORT DETAILS

Dwg. No.
PLS2021-0432-0013D

Sht. No.
S-512

EQUIP. ID	EQUIP NAME/DESCRIPTION	WT (LB)	"W" OR DIA (IN)	"L" LENGTH (IN)	"H" HEIGHT (IN)	ANCHORS	REF DETAIL	REMARKS/LOCATIONS
TCHW01-1	BUFFER TANK	11,107	60	-	108	SEE DETAIL	F1/S-513	
PCHW01-1	CHILLED WATER PUMP	425	12	17	38	SEE DETAIL	E6/S-514	
PCHW02-1	CHILLED WATER PUMP	425	12	17	38	SEE DETAIL	E6/S-514	
SDE01-1	AIR SEPERATOR	237	13	-	34	SEE DETAIL	F4/S-513	
TCHWX01-1	BLADDER EXPANSION TANK	270	16	-	48	PER MANUF RECOMMENDATION	N/A	
TCH01-1	CHEMICAL POT FEEDER	40	8	-	20	PER MANUF RECOMMENDATION	N/A	
RGHS01-1	CHILLER	5,800	75	51	71	SEE DETAIL	E1/S-514	
DDCP01	DDC CONTROL PANEL	100	8	30	30	SEE DETAIL	F6/S-512	
ACHPS01-4	AIR CONDITIONING HEAT PUMP	3,331	95	146	68	(4) 5/8" DIA A307 BOLT	B1/S-514	TO STEEL BASE RAIL BEAM
ACHPS02-5	AIR CONDITIONING HEAT PUMP	3,384	95	146	68	(4) 5/8" DIA A307 BOLT	B1/S-514	TO STEEL BASE RAIL BEAM
ACU01-A1	AIR CONDITIONING UNIT	9,685#	100	221	141	5/8" DIA THREADED ROD W/ HILTI HIT-RE 500 V3 W/ 6" EMBED	F1/S-514	
ACU08-2H	AIR CONDITIONING UNIT	70	46	60	43	SEE DETAIL	B4/S-514	
ACU09-3H	AIR CONDITIONING UNIT	60	80	175	59	SEE DETAIL	B4/S-514	
RCUA01-2H	EXTERIOR CONDENSING UNIT	150	37	13	37	(4) 1/2" DIA HILTI KB-TZ2 (ICC-ESR-4266) W/ 3 1/4" EMBED	N/A	
RCUA02-3H	EXTERIOR CONDENSING UNIT	140	37	13	37	(4) 1/2" DIA HILTI KB-TZ2 (ICC-ESR-4266) W/ 3 1/4" EMBED	N/A	

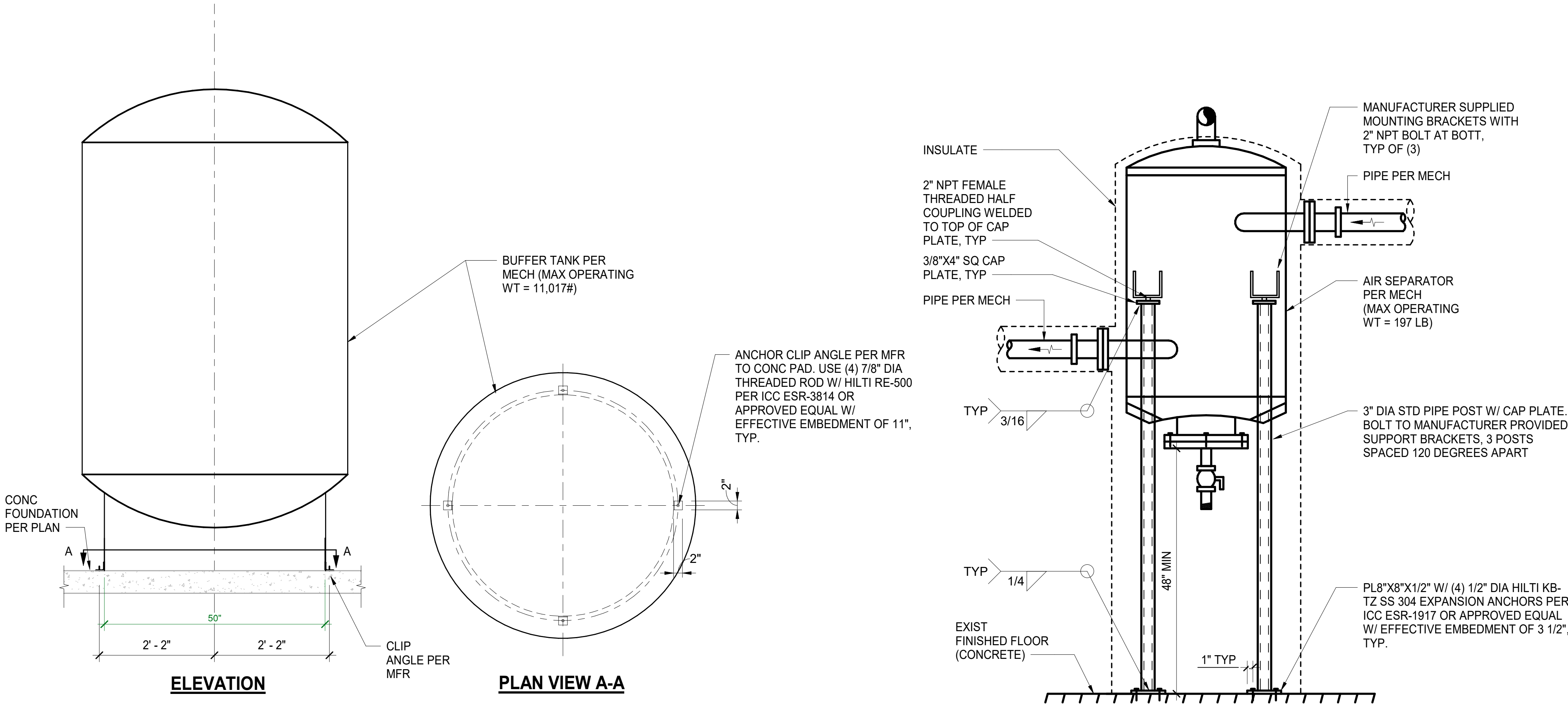
- NOTES:
- ANCHORS SPECIFIED ARE MINIMUM QUANTITIES. ADDITIONAL ANCHORS SHALL BE INSTALLED AS REQUIRED BY MANUFACTURER.
 - SUBCONTRACTOR SHALL SUBMIT ALL EQUIPMENT CUTSHEETS (INDICATING DIMENSIONS & OPERATING WEIGHTS) TO SEOR FOR FINAL REVIEW PRIOR TO INSTALLATION.
 - "DA" INDICATES A DEFERRED APPROVAL.
 - STAINLESS STEEL HARDWARE SHALL HAVE ISOLATION MATERIAL AS REQUIRED AT CARBON STEEL INTERFACES.

B1

S-513

EQUIPMENT ANCHORAGE SCHEDULE

SCALE: 1" = 1'-0"

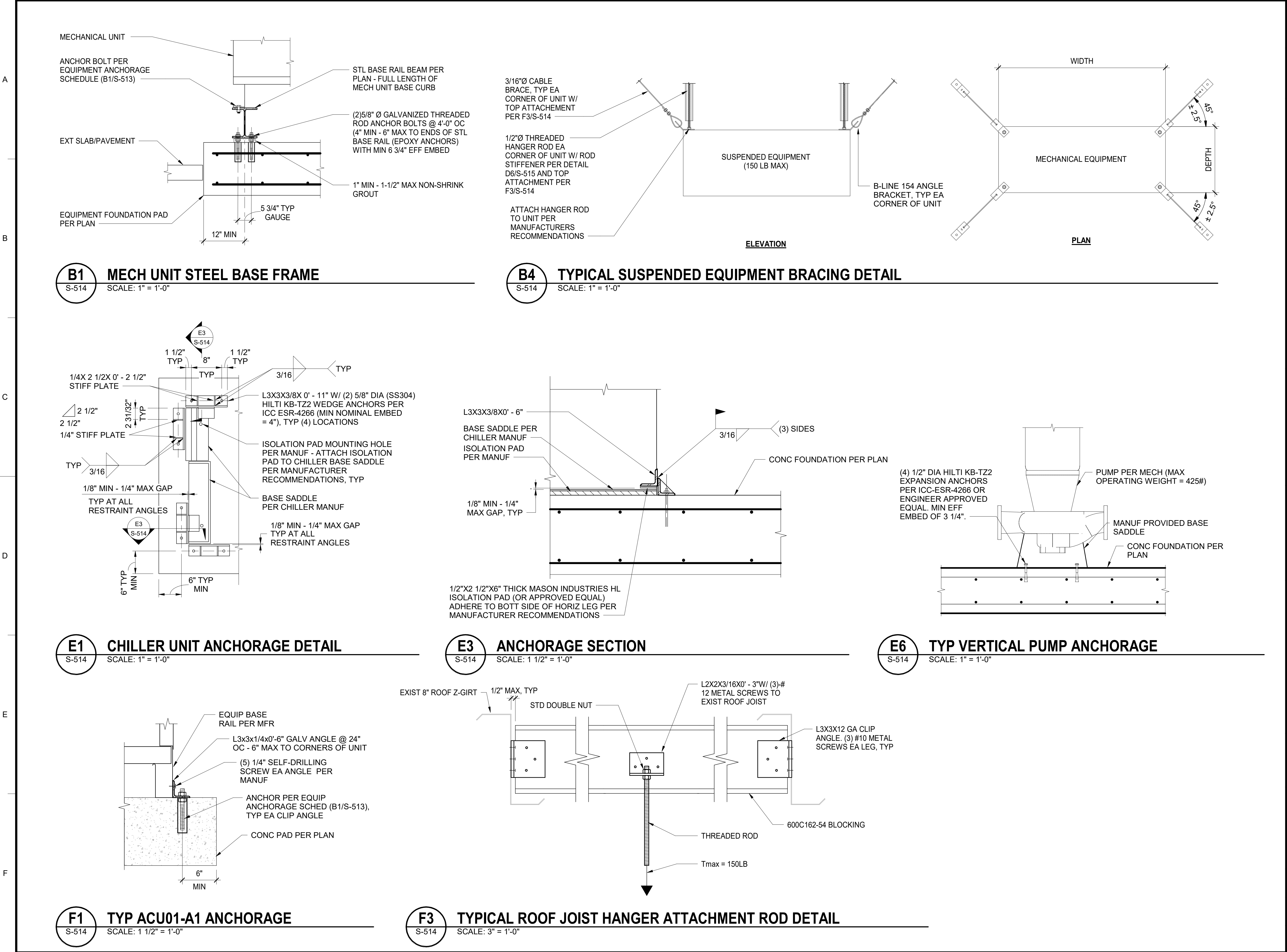


Sheet Title

EQUIPMENT SUPPORT DETAILS

Dwg. No.
PLS2021-0432-0014D

Sht. No.
S-513



Sheet Title

EQUIPMENT SUPPORT
DETAILS

Dwg. No.
PLS2021-0432-0015D

Sht. No.
S-514

A

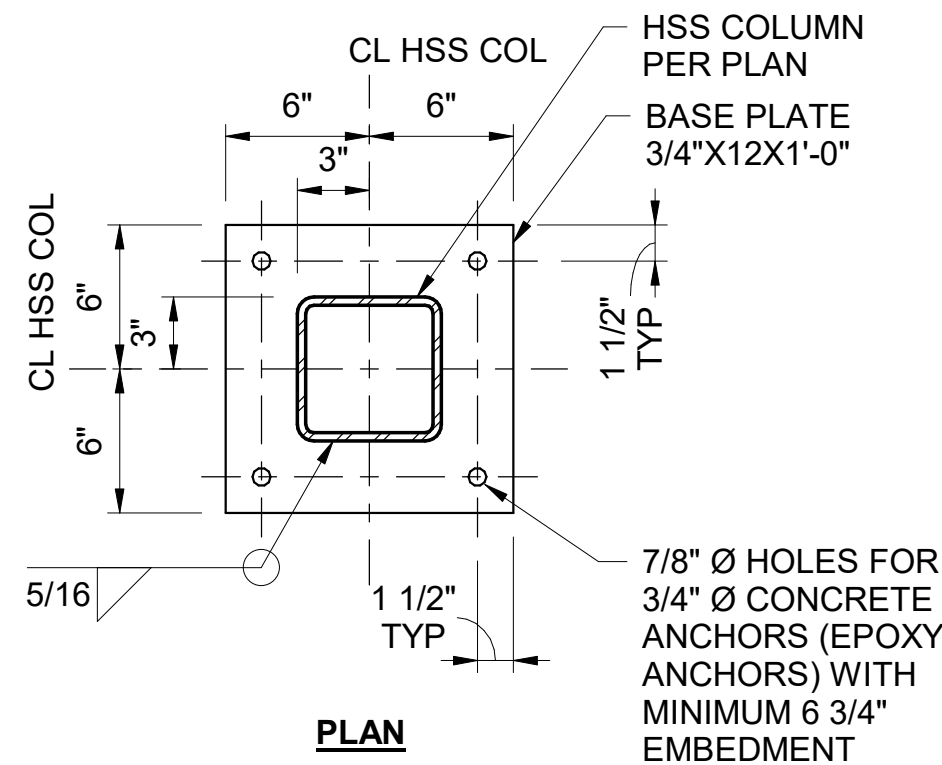
B

C

D

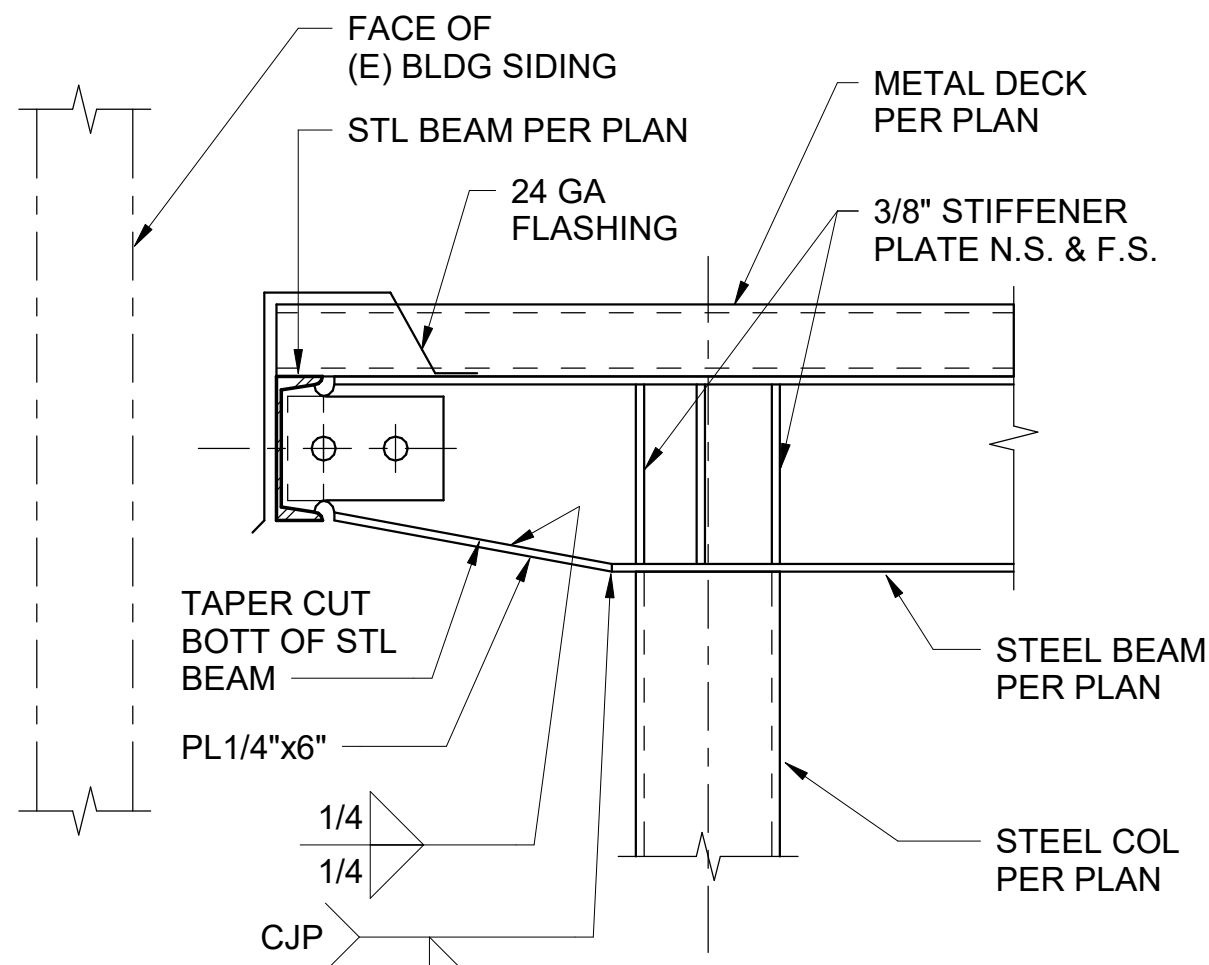
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F



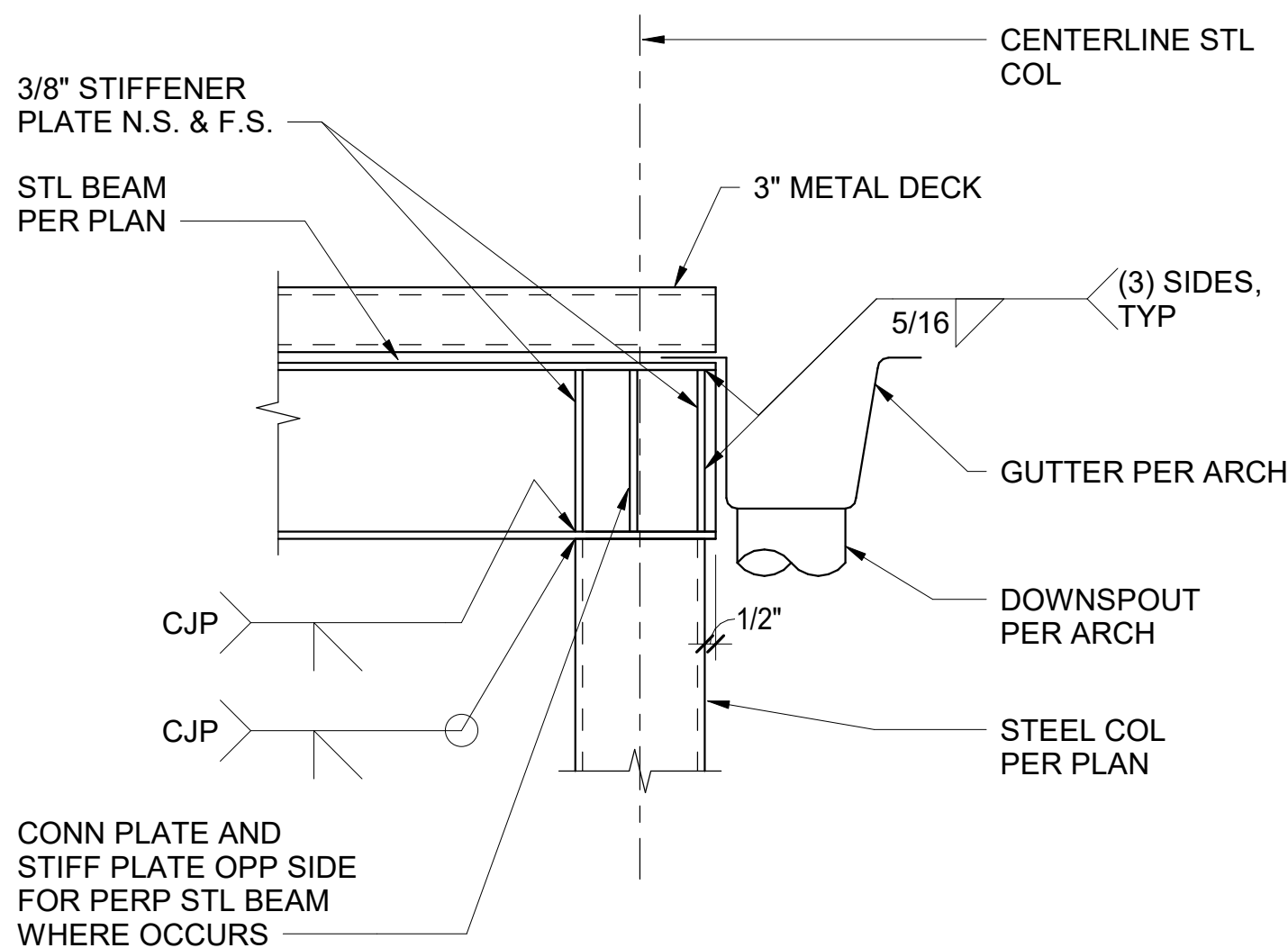
B1 SECTION

S-515 SCALE: 1 1/2" = 1'-0"



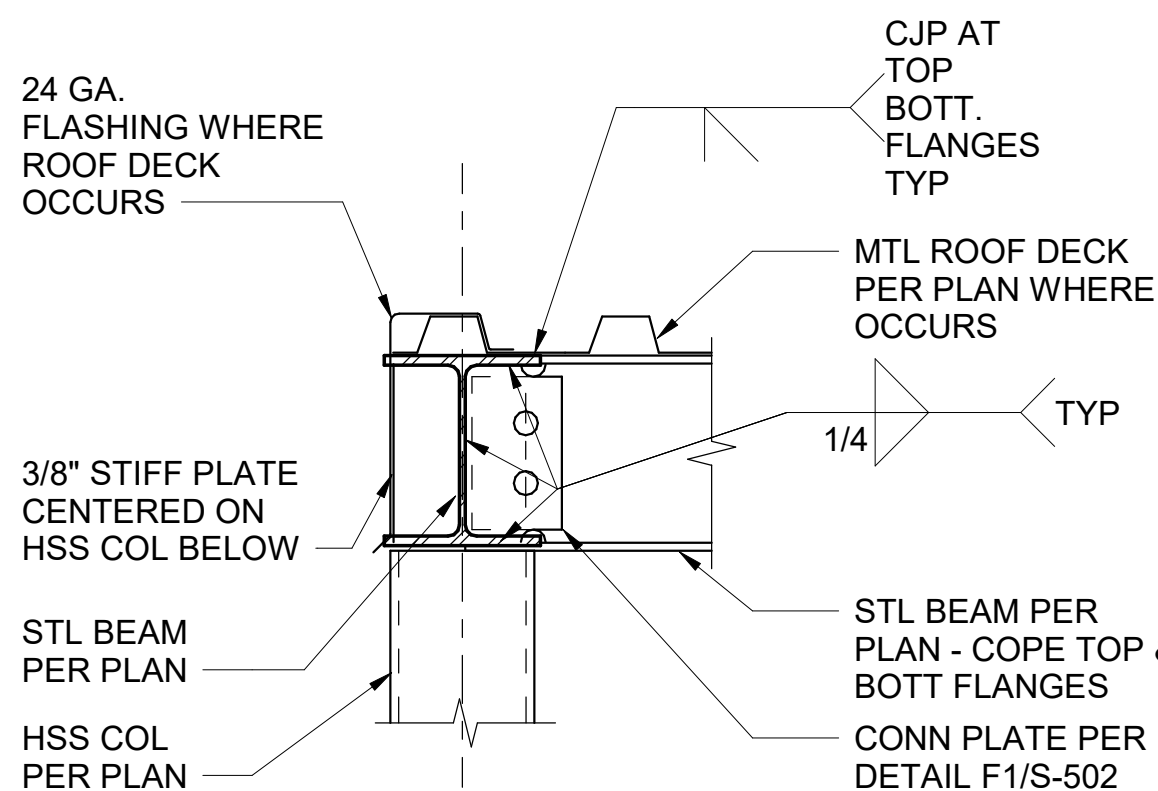
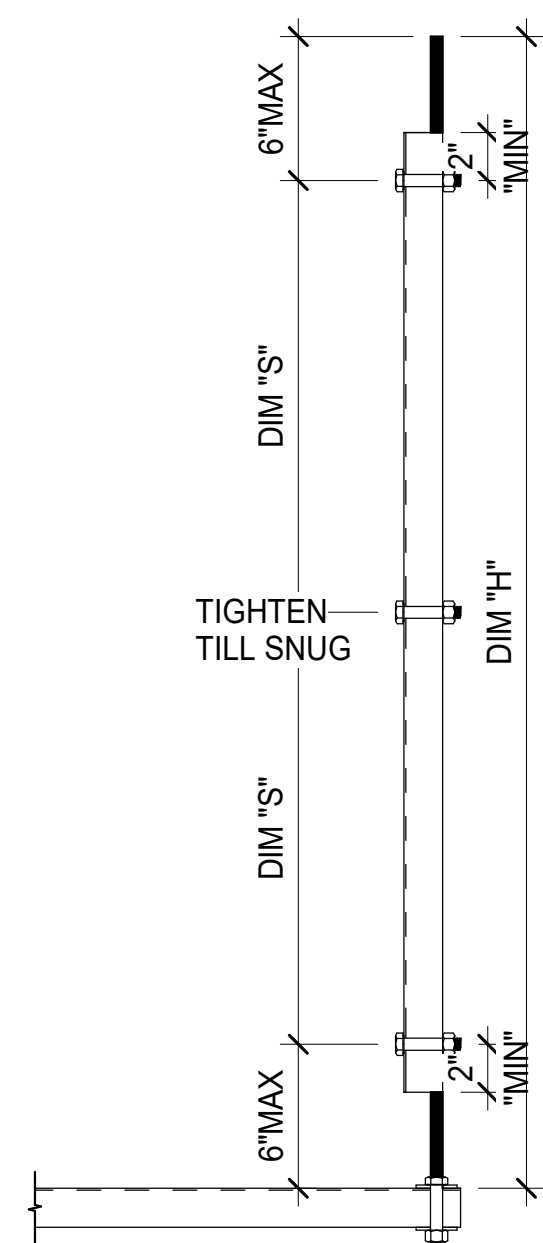
B3 SECTION

S-515 SCALE: 1 1/2" = 1'-0"



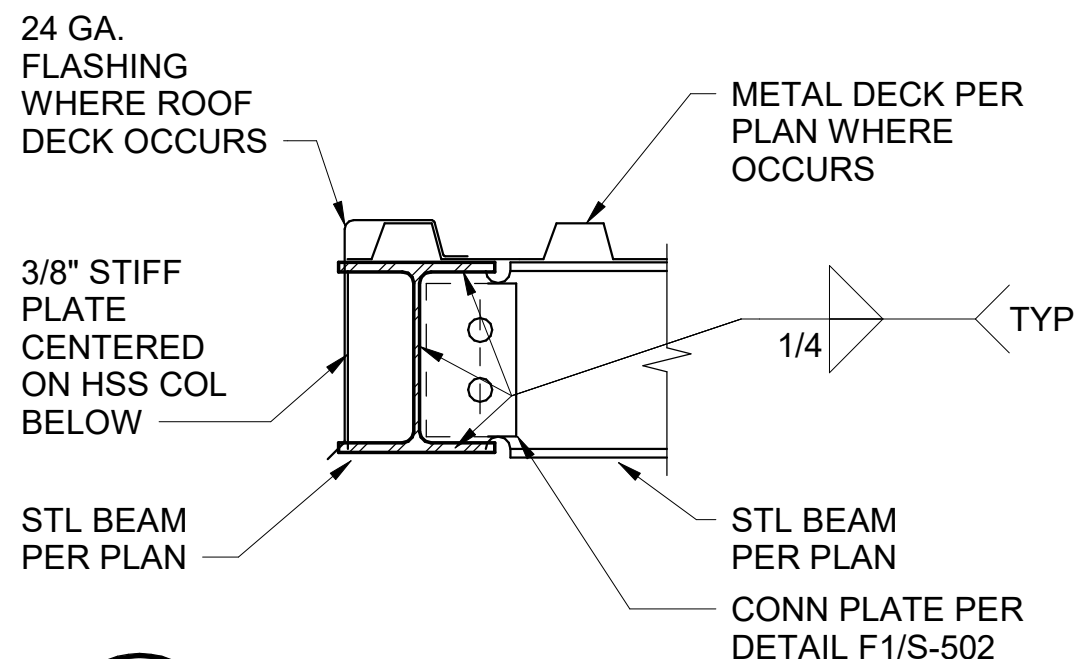
B5 SECTION

S-515 SCALE: 1 1/2" = 1'-0"



D1 SECTION

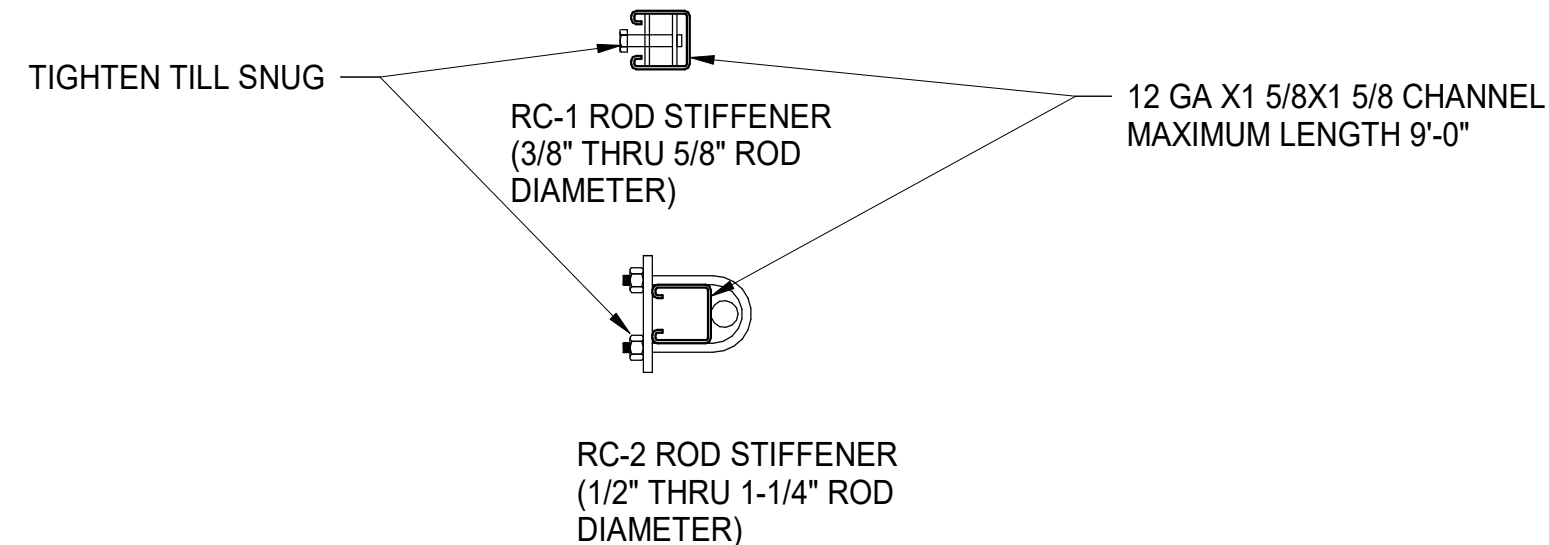
S-515 SCALE: 1 1/2" = 1'-0"



D3 SECTION

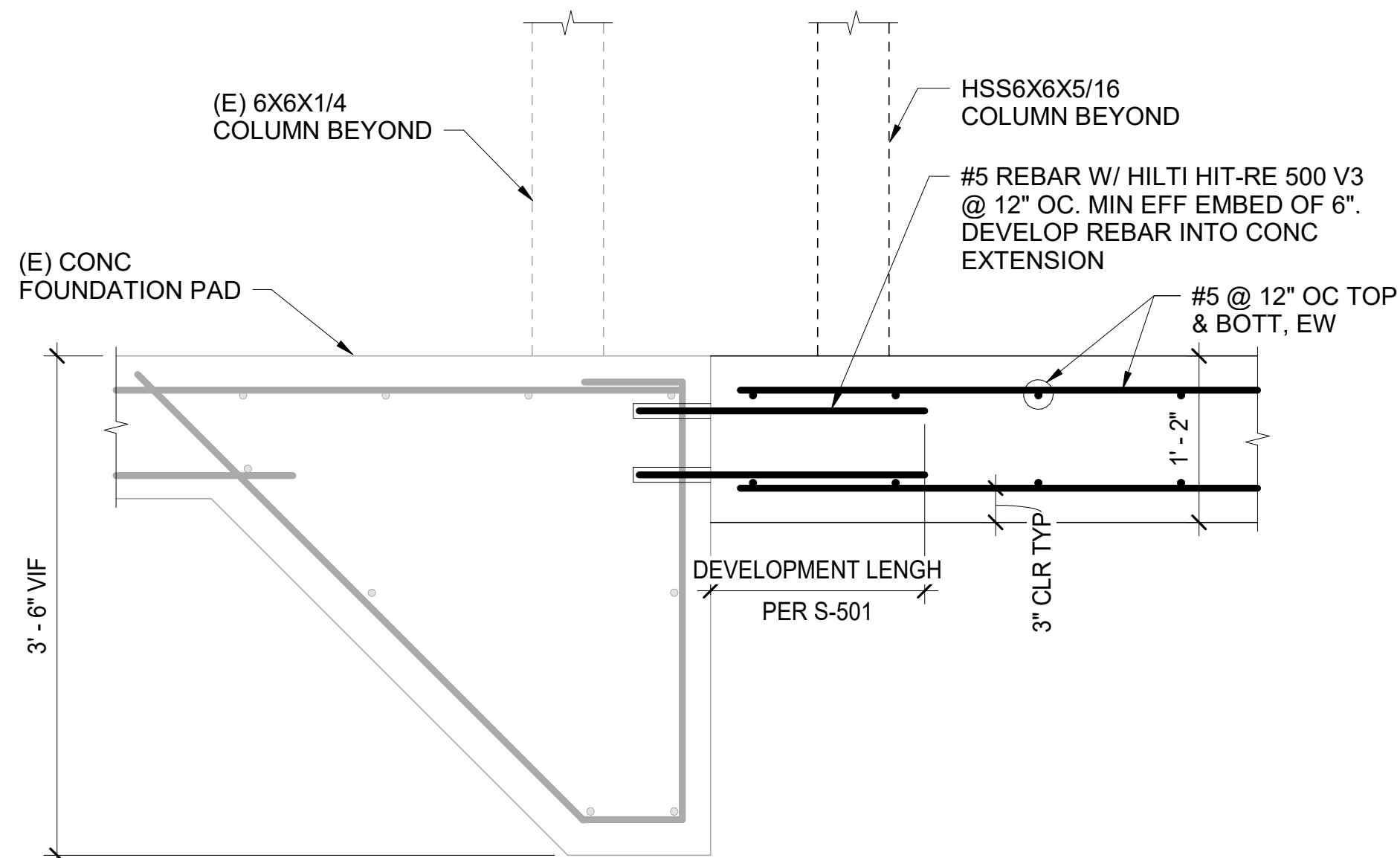
S-515 SCALE: 1 1/2" = 1'-0"

ROD STIFFENER CHART						
THREADED ROD DIAMETER		DIM "H" MAXIMUM ROD LENGTH WITHOUT STIFFENER		DIM "S" MAXIMUM SPACING BETWEEN ROD STIFFENERS		MAX REACTION
IN	(mm)	IN	(mm)	IN	(mm)	LBS
3/8"	(10)	16"	(406)	13"	(330)	790
1/2"	(13)	22"	(559)	18"	(457)	1460



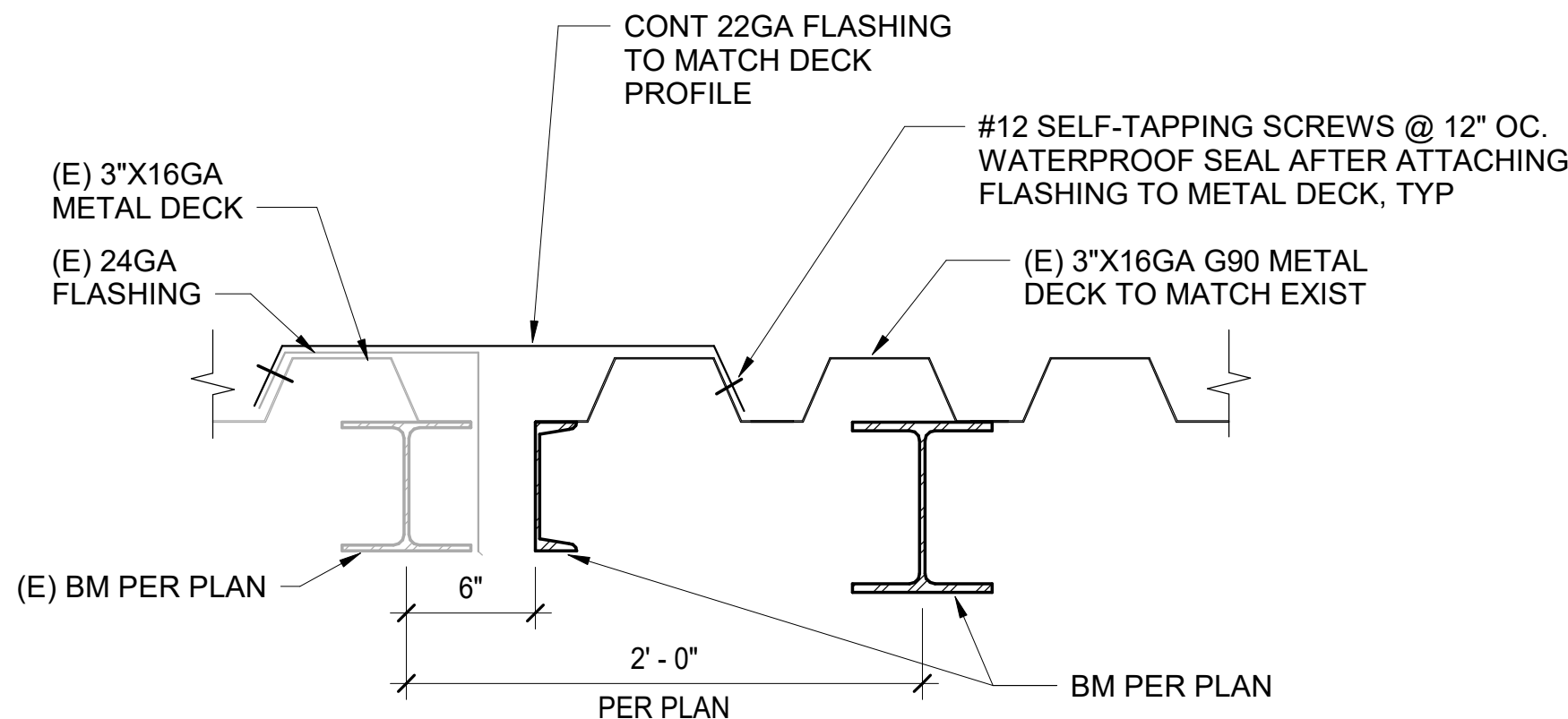
D6 TYPICAL HANGER ROD STIFFENER DETAIL

S-515 SCALE: 1 1/2" = 1'-0"



F1 CONC PAD EXTENSION

S-515 SCALE: 1" = 1'-0"



F4 SECTION

S-515 SCALE: 1 1/2" = 1'-0"

Sheet Title
CANOPY SUPPORT DETAILS

Dwg. No.
PLS2021-0432-0016D

Sht. No.
S-515

FINISH NOTES		GENERAL NOTES		SYMBOL LEGEND		ABBREVIATIONS							
A	GENERAL 1. NOTES PERTAIN TO AREA WITHIN THE SCOPE OF WORK ONLY. 2. IF A COLOR OR FINISH IS NOT SPECIFIED, CONTACT THE CONSTRUCTION MANAGER FOR CLARIFICATION. 3. SUBMIT LABELED SAMPLES FOR EACH MATERIAL SPECIFIED. INCLUDE PRODUCT DATA, SEAMING DIAGRAMS, AND SHOP DRAWINGS PRIOR TO PROCUREMENT OF FABRICATION. 4. PROVIDE REPRESENTATIVE SAMPLES FOR ALL DYE LOTS REQUIRED TO COMPLETE INSTALLATION. 5. PROTECT EXISTING FLOOR, WALL, DOOR AND CEILING FINISHES TO REMAIN AND REPAIR ANY DAMAGE AS A RESULT OF DEMOLITION ON CONSTRUCTION. 6. PROVIDE A SUBMITTAL SCHEDULE FOR ALL LONG LEAD ITEMS TO PREVENT SUBSTITUTIONS.	1. SPECIFICATIONS THE SPECIFICATIONS ARE A PART OF THESE CONTRACT DOCUMENTS. THE SUBCONTRACTOR AND THEIR PERSONNEL MUST BECOME FAMILIAR WITH THE SPECIFICATIONS PRIOR TO BIDDING THE PROJECT AND STARTING ANY CONSTRUCTION.	2. DIMENSIONS DIMENSIONS MUST GOVERN ON DRAWINGS. DO NOT SCALE DRAWINGS.	4. CONFLICT BETWEEN DRAWINGS AND SPECIFICATIONS SHOULD CONFLICTS OCCUR BETWEEN THE DRAWINGS AND SPECIFICATIONS, DRAWINGS MUST GOVERN IN MATTERS OF DIMENSION OR QUANTITY, SPECIFICATIONS SHALL GOVERN IN MATTERS OF MATERIALS OR FINISHES.	5. REVIEW PLANS AND EXISTING CONDITIONS THE SUBCONTRACTOR MUST THOROUGHLY REVIEW PLANS AND EXISTING SITE CONDITIONS AND NOTIFY THE CONSTRUCTION MANAGER OF ANY DISCREPANCIES, ERRORS, OR OMISSIONS PRIOR TO CONSTRUCTION.	6. SUBCONTRACTOR'S RESPONSIBILITIES NEITHER ARCHITECT OF RECORD NOR THE ENGINEER OF RECORD ARE RESPONSIBLE FOR CONSTRUCTION MEANS, METHODS, TECHNIQUES, SEQUENCES OR PROCEDURES OF THE SUBCONTRACTOR, SAFETY PRECAUTIONS AND PROGRAMS OF THE SUBCONTRACTOR, THE ACTS OR OMISSIONS OF THE SUBCONTRACTOR, OR THE FAILURE OF THE SUBCONTRACTOR TO CARRY OUT THE WORK IN ACCORDANCE WITH THE CONTRACT DOCUMENTS.	7. FIELD CONFIRMATION OF DISCREPANCIES FIELD CONFIRMATION OF DISCREPANCIES MUST BE RECORDED AND IMMEDIATELY TRANSMITTED TO THE CONSTRUCTION MANAGER FOR PROJECT RECORD, COORDINATION, AND NECESSARY RESOLUTION PRIOR TO CONTINUING WORK.	8. MATERIAL AND PRODUCT INSTALLATION INSTALL MATERIAL AND PRODUCTS IN STRICT ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS AND APPLICABLE ICC REPORTS.	11. CLEANLINESS UPON COMPLETION OF WORK, JOBSITE MUST BE THOROUGHLY CLEANED PRIOR TO TURN-OVER TO CLIENT.	12. EXISTING CONSTRUCTION AND CONDITIONS: SUBCONTRACTOR MUST FIELD VERIFY THE DIMENSIONS, ELEVATION, MATERIALS, ETC. NECESSARY FOR THE PROPER CONSTRUCTION AND ALIGNMENT OF THE WORK. THE SUBCONTRACTOR MUST MAKE MEASUREMENTS NECESSARY FOR FABRICATION AND ERECTION OF STRUCTURAL MEMBERS.		0 0 GRIDLINE ROOM 1000 ROOM TAG NAME NUMBER 1 1 SECTION SECTION # SHEET # 1 1 ELEVATION(S) ELEVATION # SHEET # 1 1 INTERIOR ELEVATIONS ELEVATION # SHEET # 1 1 DETAIL WORK POINT, CONTROL POINT, DATUM POINT 1 1 WALL TAG 1.1 1.1 WINDOW TAG FLOOR TRANSITION 101 101 DOOR TAG KEYNOTE 1 1 TITLE SCALE: 1/8" = 1'-0" 1 1 TITLE SCALE: NTS.	@ AT A.D. AREA DRAIN ACCESS ACCESSIBLE ACOUS. ACOUSTICAL ADA AMERICANS WITH DISABILITIES ACT ADJ. ADJUSTABLE AFF. ABOVE FINISH FLOOR APPROX APPROXIMATE ARCH ARCHITECTURAL BLDG. BUILDING BLKG. BLOCKING BOT. BOTTOM C.B. CATCH BASIN CL CENTERLINE CLG. CLEAR CMU CONCRETE MASONRY UNITS COL. COLUMN CONC. CONCRETE COORD. COORDINATE D/F DRINKING FOUNTAIN DEMO. DEMOLITION DET. DETAIL DIA DIAMETER DN. DOWN DN. DOWN SPOUT DWG. DRAWING (E) EXISTING EA. EACH EJ. EXPANSION BOLT ELEC. ELECTRICAL EQ. EQUAL EQUIP. EQUIPMENT EXT. EXTERIOR F.A. FIRE ALARM F.D. FLOOR DRAIN F.E. FIRE EXTINGUISHER F.F. FACE OF FIX. FIXED FLR. FLOOR FLUOR. FLUORESCENT FRP FIBERGLASS REINFORCED PLASTIC FTG. FOOTING GA. GALVANIZED GFCI GROUND FAULT CIRCUIT INTERRUPTED GL. GLASS GYP. BD. GYPSUM BOARD H.B. HOSE BIB HE HEAVY EQUIPMENT HORIZ. HORIZONTAL INSUL. INSULATION INT. INTERIOR MAX MAXIMUM MFR. MANUFACTURER MIN. MINIMUM MIR. MIRRORED MTD. MOUNTED (N) NEW NIC. NOT IN SCOPE NTS. NOT TO SCALE O.C. ON CENTER O.F.C.I. OWNER FURNISHED/SUBCONTRACTOR INSTALLED O/ OVER PLYWD PLYWOOD PTD. PAPER TOWEL DISPENSER RCP REFLECTED CEILING PLAN REF. REFERENCE REFRG. REFRIGERATOR REQ. REQUIRED RM. ROOM RWL. RAIN WATER LEADER S.C.D. SEAT COVER DISPENSER S.D. STORM DRAIN S.N.R. SANITARY NAPKIN RECEPTACLE SAFM SELF ADHERING FLASHING SCHED. SCHEDULED SF SQUARE FOOT SIM. SIMILAR SQ. SQUARE SST. STAINLESS STEEL SUSP. SUSPENDED SYM. SYMMETRICAL T&G TONGUE& GROOVE T.P.D. TOILET PAPER DISPENSER THK. THICKNESS TYP. TYPICAL U.L. UNDERWRITERS LABORATORY UON. UNLESS OTHERWISE NOTED VIF. VERIFY IN FIELD W.O. WHERE OCCURS W.S. WOOD SCREW W/ WITH W/O WITHOUT DOWN DOWN SPOUT WITH SPLASH BLOCK
	PATCHING 1. SMALL HOLES < 1" MAY BE PATCHED TO MATCH EXISTING FINISH. PREPARE SURFACES PER MANUFACTURE'S RECOMMENDATIONS. 2. LARGE VOIDS >12" IN ANY DIMENSION SHALL BE PATCHED USING REINFORCING STEEL WELDED TO STEEL FRAMED OPENING OR DOWELED INTO EXISTING CONCRETE OPENING AND A 4000# CONCRETE MIX.												
	FLOOR FINISHES 1. FINISH FLOOR PER MANUFACTURER'S BASE LEVEL SPECIFICATIONS UNLESS OTHERWISE NOTED IN DRAWINGS 2. USE LOW-VOC ADHESIVES.												
B	CEILING FINISHES 1. PAINT EXPOSED COMPONENTS ON OPEN STRUCTURE CEILING - SEE DRAWINGS												
	WALL / DOOR FINISHES 1. PATCH AND PREP EXISTING WALLS THAT HAVE NEW PENETRATIONS OR PENETRATIONS TO BE PATCHED FOLLOWING DEMO. 2. CONFIRM EQUIPMENT MOUNTING HEIGHTS PRIOR TO INSTALLATION OR RE-INSTALLATION. 3. PROVIDE SAMPLES OF PAINT IN SPECIFIED COLOR AND GLOSS. 4. PROVIDE 1 COAT OF PRIMER AND 2 COATS OF FINISH COATS WHERE PAINTING IS CALLED OUT. 5. REMOVE UNUSED UNISTRUT, TAPE, BOLTS, COVER PLATES, AND ANY LEGACY UTILITY LINES WHERE THEY OCCUR. PATCH HOLES TO MATCH EXISTING CONSTRUCTION PRIOR TO FINISHING. 6. REMOVE EXISTING SIGNS, TAGS, ETC PRIOR TO APPLYING FINISHES. RE-INSTALL ONCE FINISHES ARE COMPLETE. PROVIDE NEW IF ORIGINAL LABEL, SIGN, TAG, ETC. IS DAMAGED OR LOST.												
	PAINTING OF MECHANICAL AND ELECTRICAL EQUIPMENT 1. DO NOT PAINT												
C													
D													
E													
F													

Sheet Title

SYMBOLS, ABBREVIATIONS,
GENERAL NOTES

Dwg. No.

PLA2021-0432-0001D

Sht. No.

A-001

A

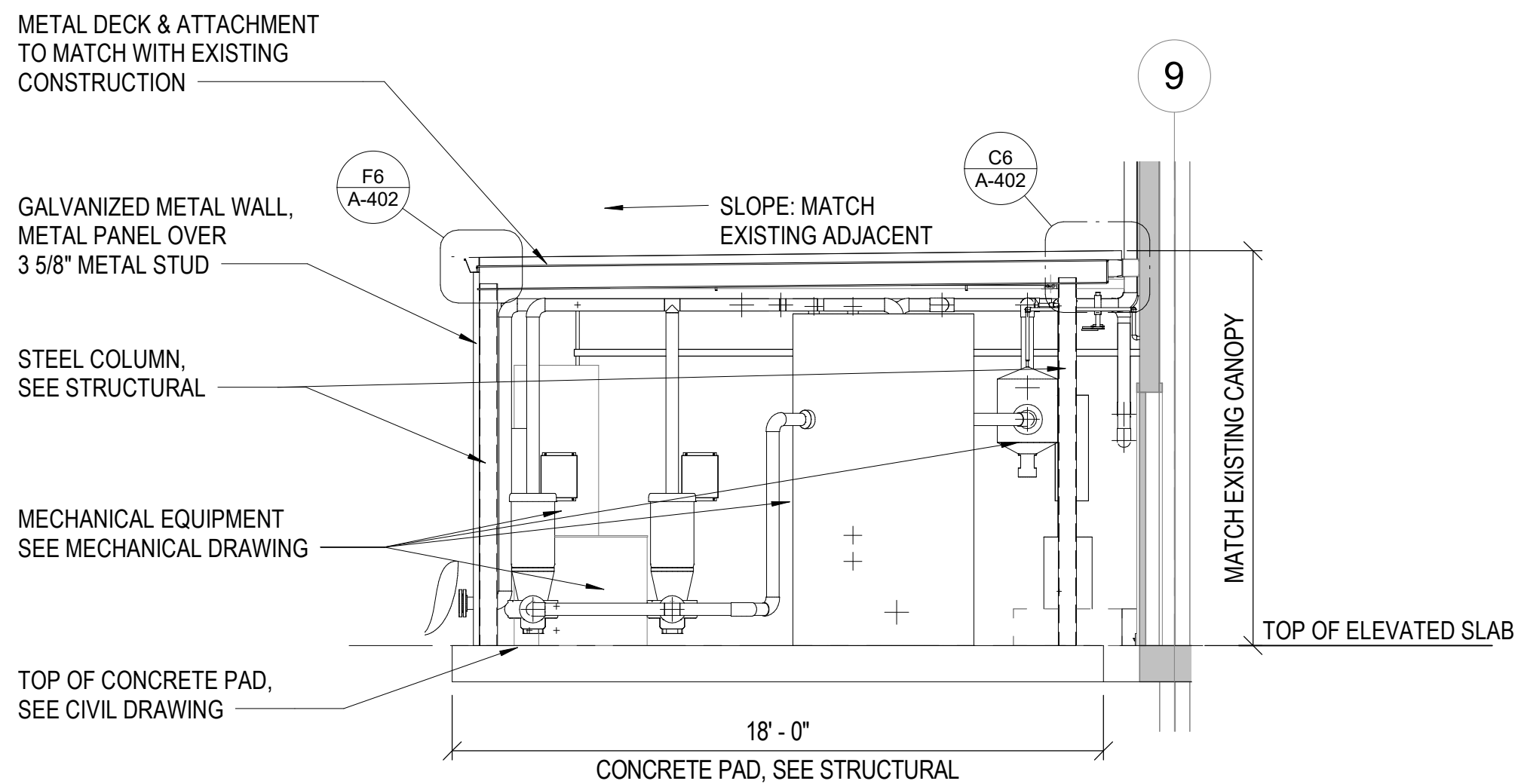
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C

D

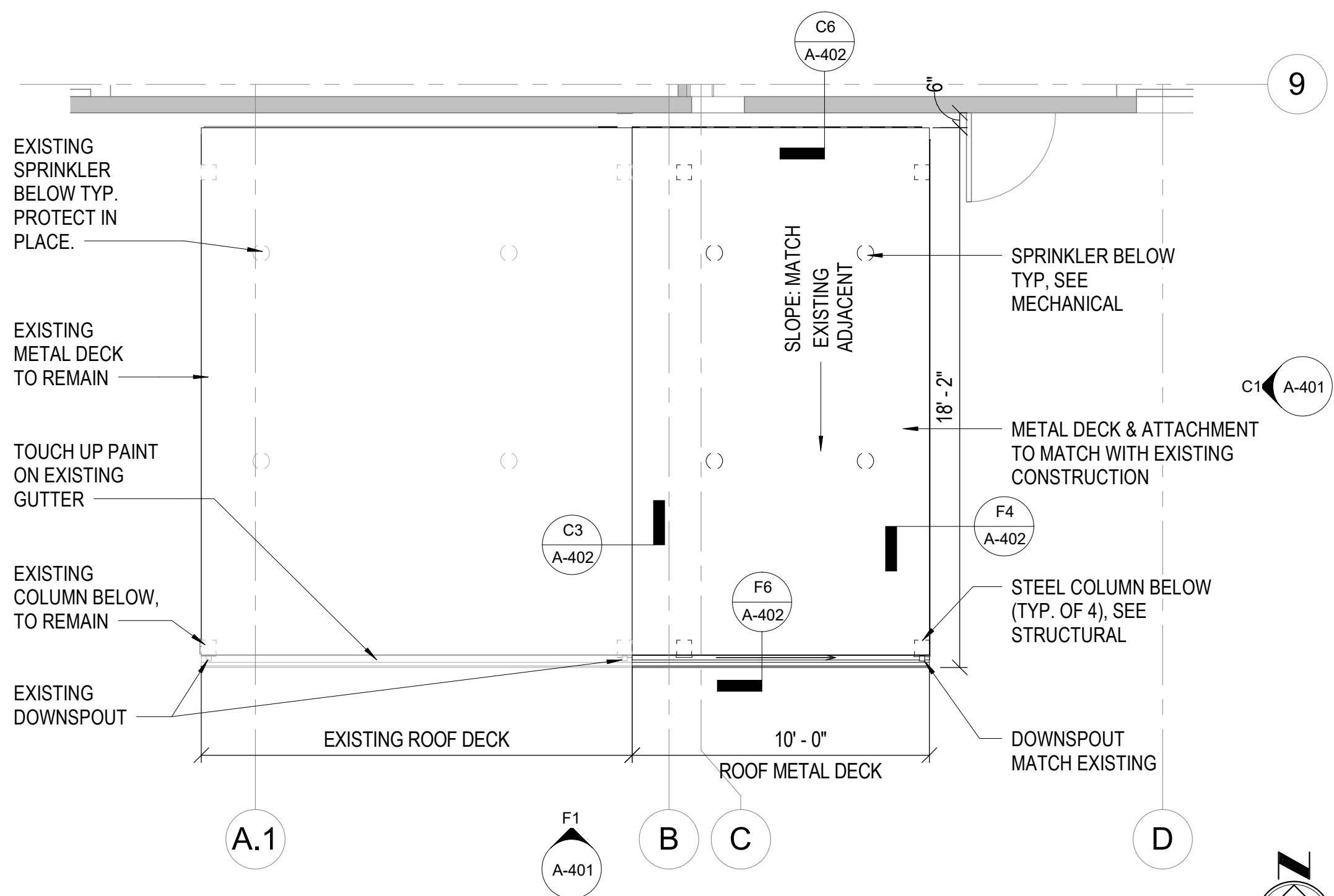
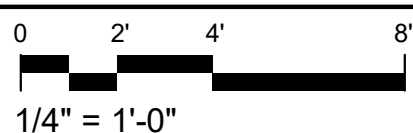
E

F



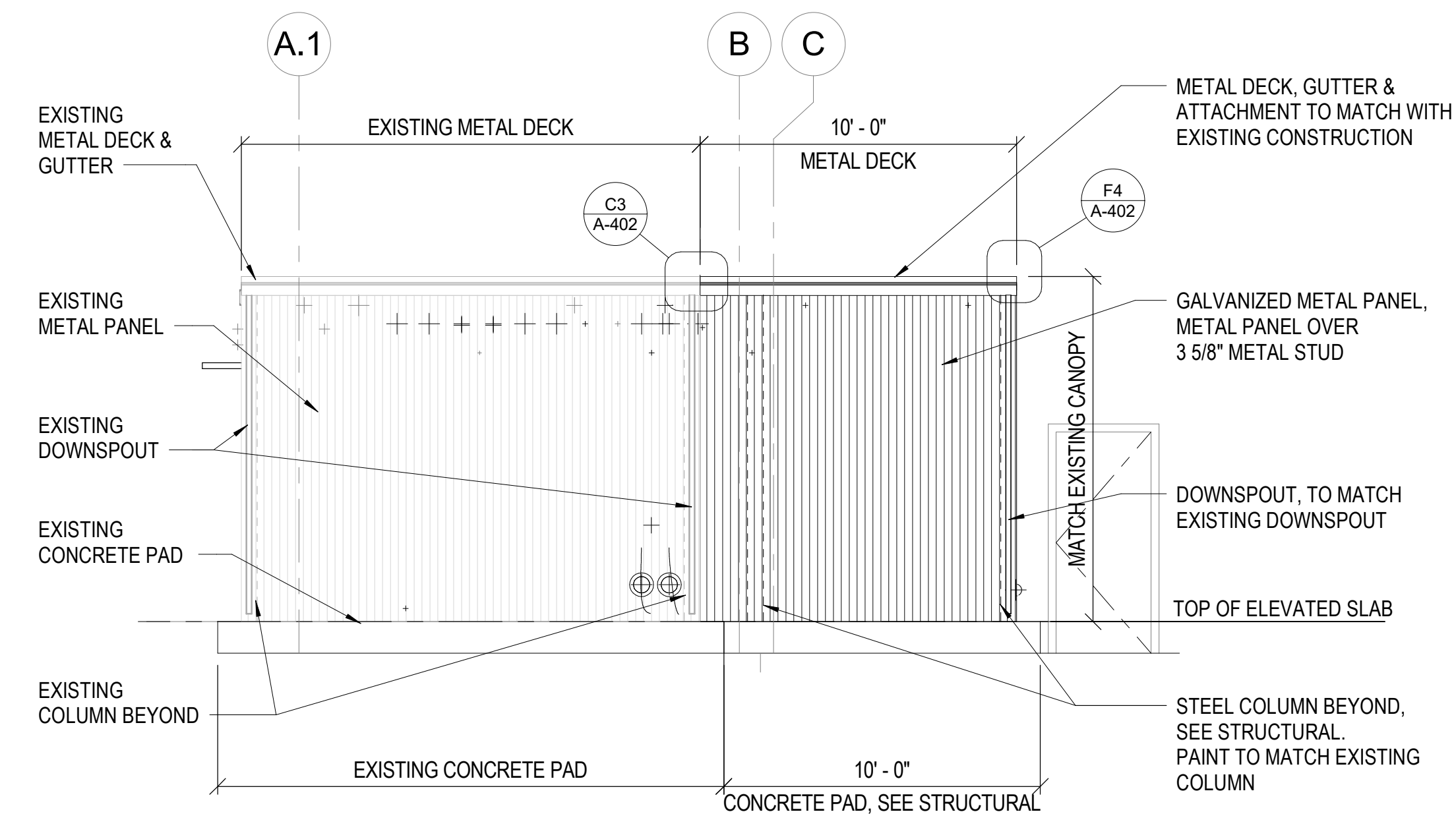
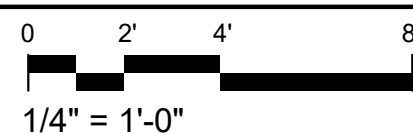
C1 EAST ELEVATION
A-401

SCALE: 1/4" = 1'-0"



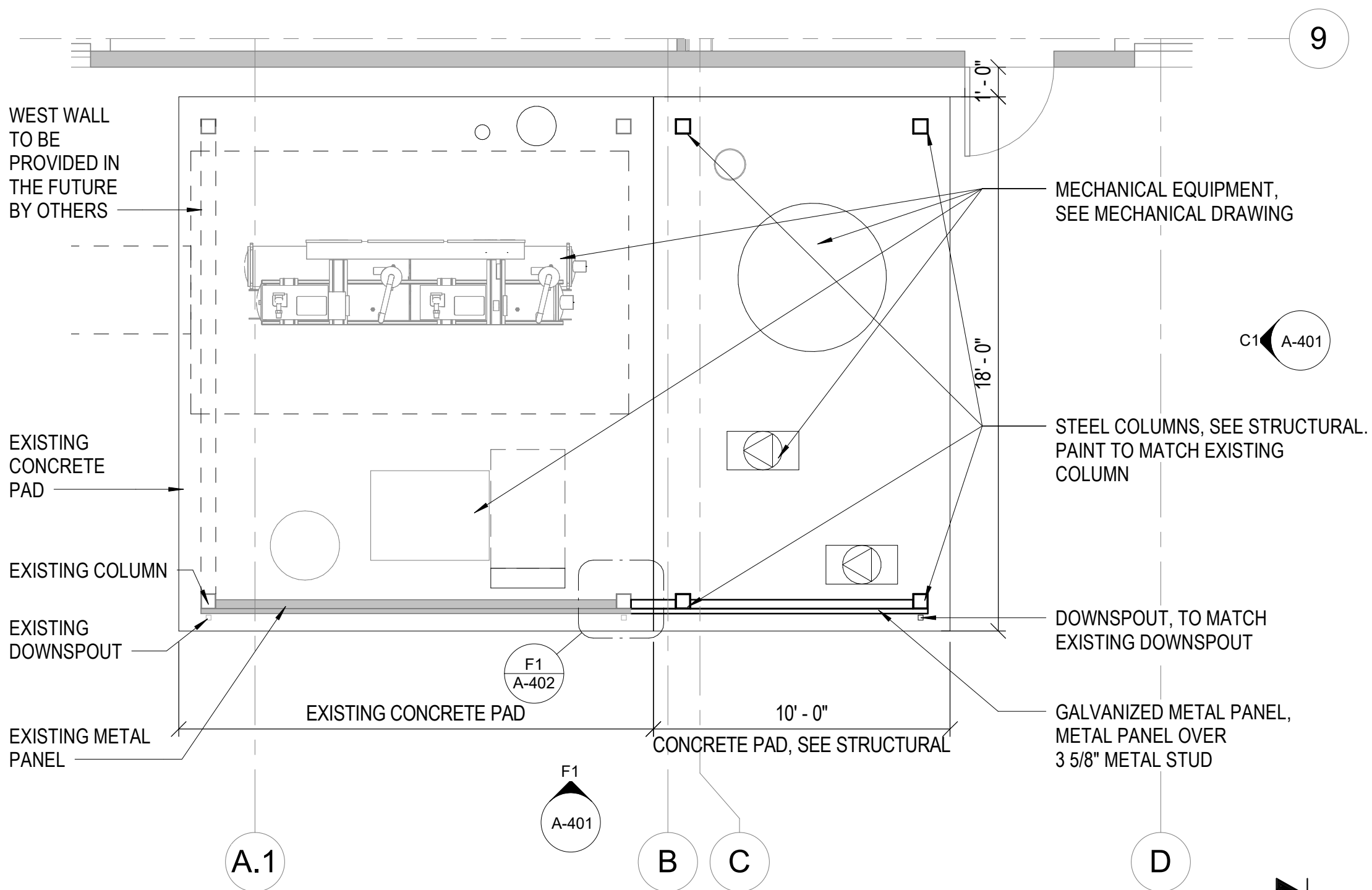
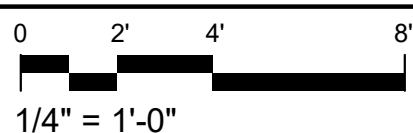
C4 ENLARGED CANOPY PLAN
A-401

SCALE: 1/4" = 1'-0"



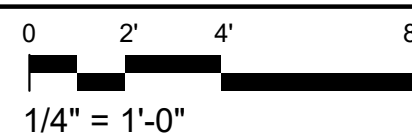
F1 SOUTH ELEVATION
A-401

SCALE: 1/4" = 1'-0"



F4 ENLARGED PLAN
A-401

SCALE: 1/4" = 1'-0"



GENERAL NOTES

1. GENERAL

FOR ARCHITECTURAL GENERAL NOTES, MATERIAL LEGEND, AND ABBREVIATIONS REFER TO SHEET A-001.

2. DIMENSIONS

DIMENSIONS MUST GOVERN ON WORKING DRAWINGS. DO NOT SCALE DRAWINGS. THE CONTRACTOR MUST VERIFY ALL DIMENSIONS, ELEVATIONS AND SITE CONDITIONS BEFORE BEGINNING WORK AND FABRICATION. THE OWNER MUST BE NOTIFIED IN WRITING IMMEDIATELY OF ANY DISCREPANCIES OR INCONSISTENCIES. IN NO CASE MUST DIMENSIONS BE SCALED OFF PLANS, SECTIONS, OR DETAILS FROM THESE DRAWINGS.

3. REVIEW PLANS AND EXISTING CONDITIONS

THE SUBCONTRACTOR MUST THOROUGHLY REVIEW PLANS AND EXISTING SITE CONDITIONS AND NOTIFY THE CONSTRUCTION MANAGER OF ANY DISCREPANCIES, ERRORS, OR OMISSIONS PRIOR TO CONSTRUCTION.

KEY PLAN



KEYNOTES - NEW WORK

NO. DESCRIPTION

LEGEND

	EXISTING WALL
	DEMOLISH WALL
	NEW WALL
	WALL TAG
	DOOR TAG

Sheet Title

CHILLER YARD - PLANS & ELEVATIONS

Dwg. No.

PLA2021-0432-0003D

Sht. No.

A-401

A

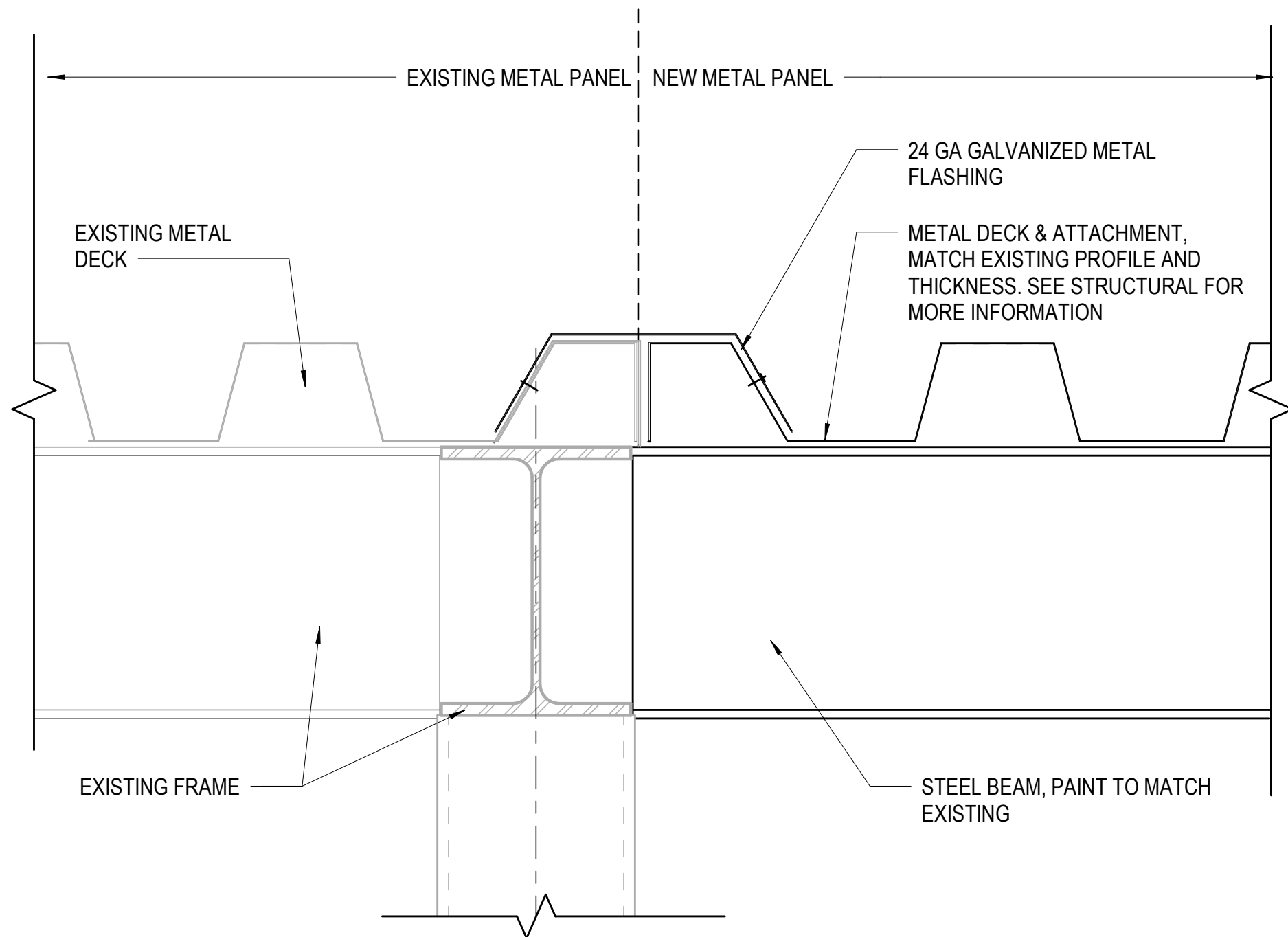
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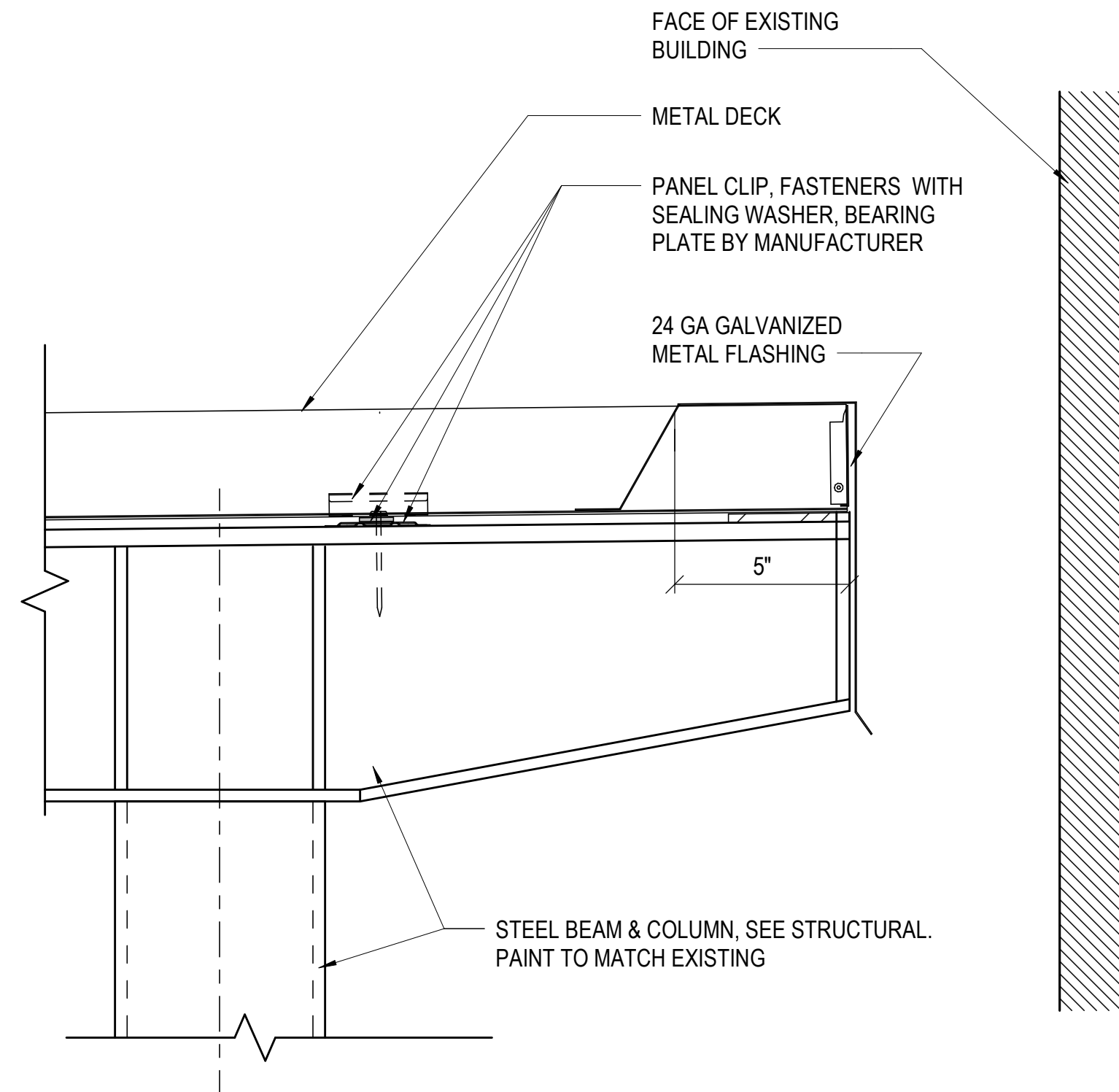
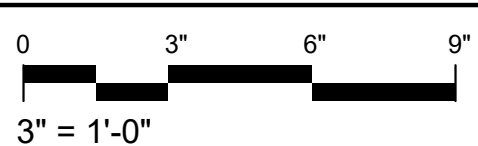
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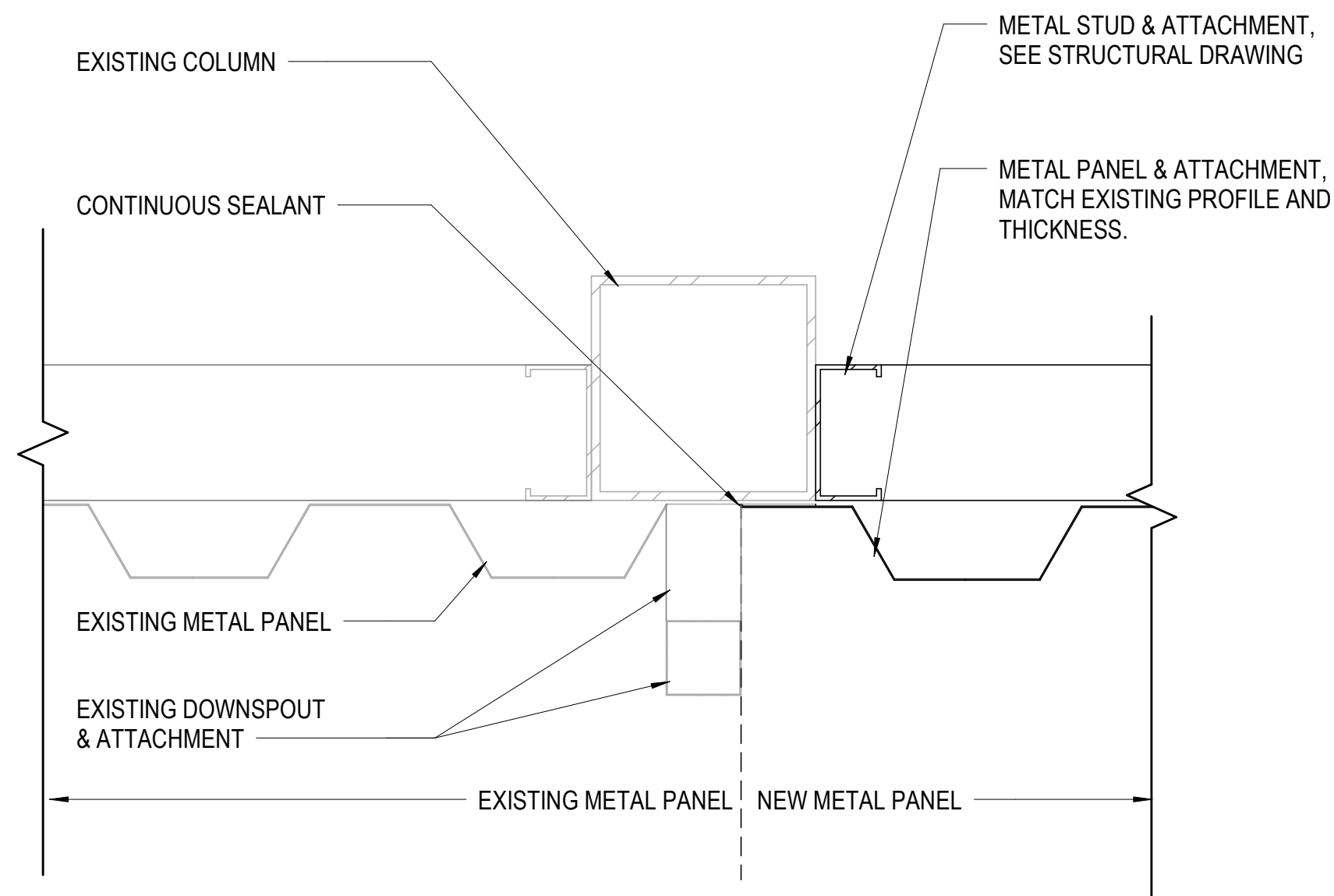
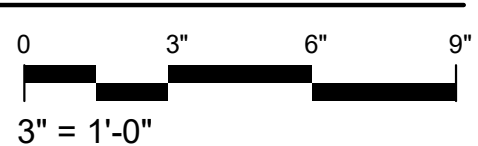
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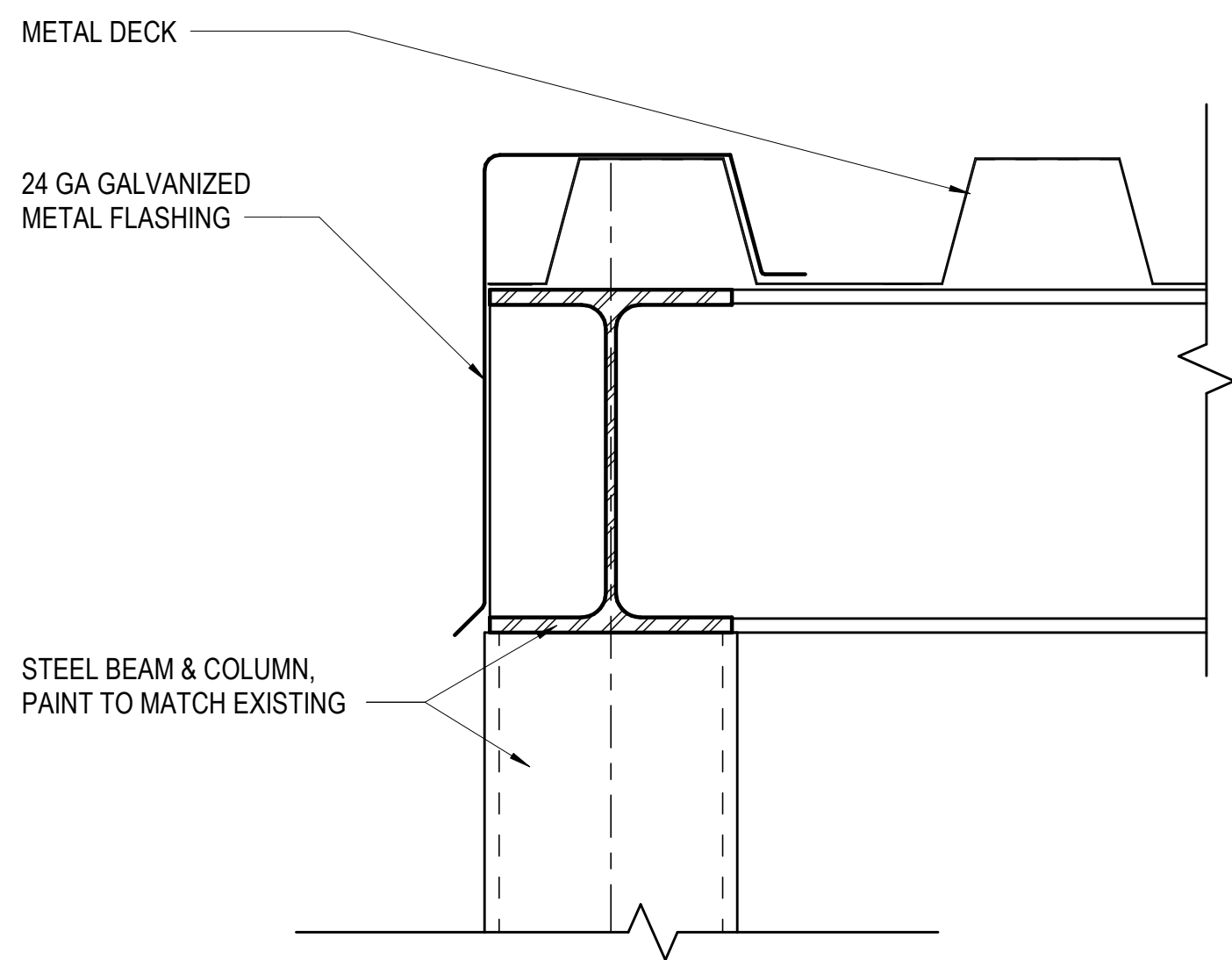
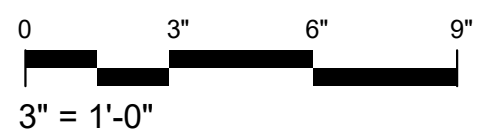
C3 CANOPY DETAIL - NEW & EXISTING
A-402 SCALE: 3" = 1'-0"



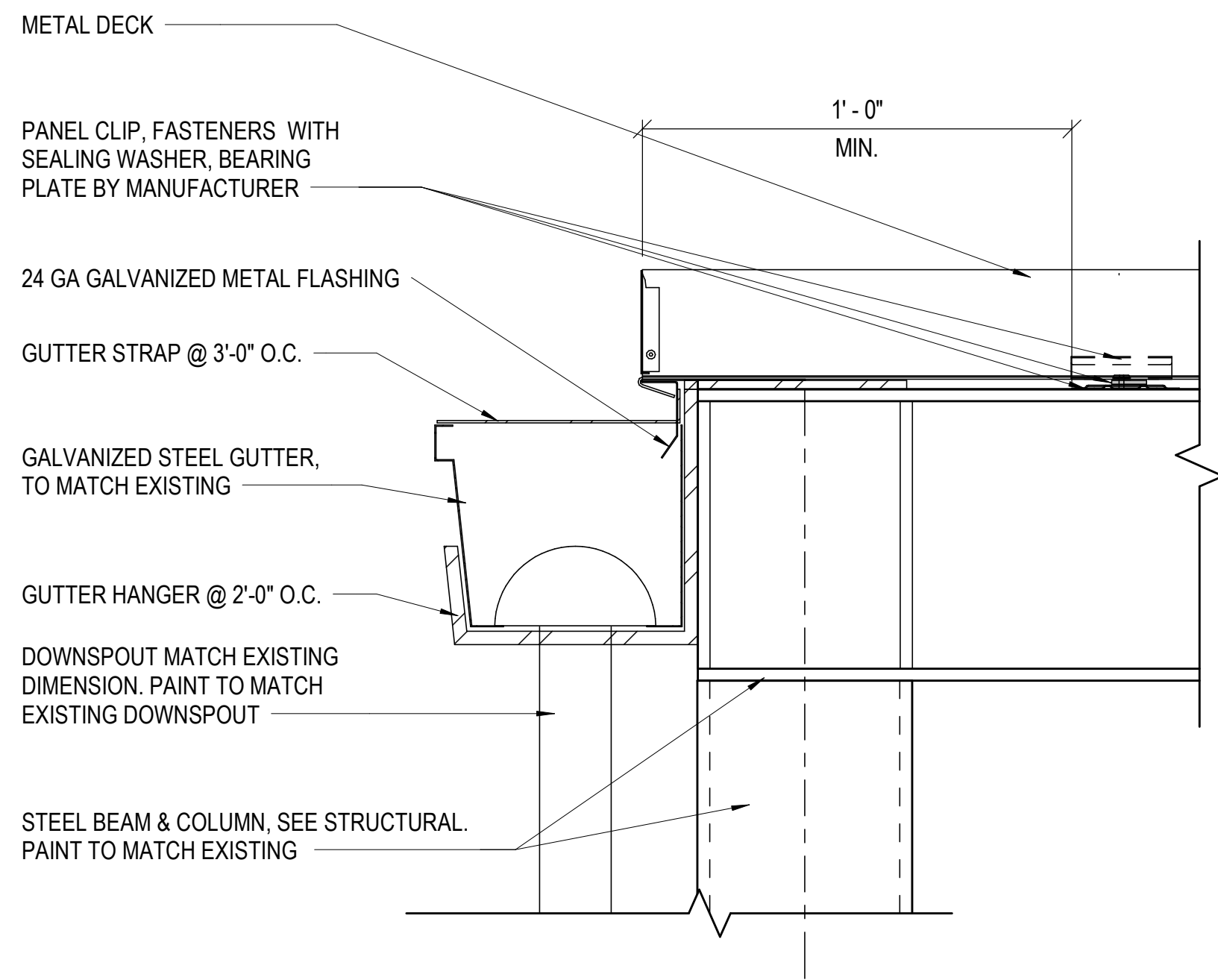
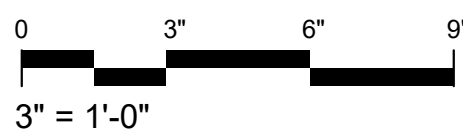
C6 CANOPY DETAIL AT HIGH POINT
A-402 SCALE: 3" = 1'-0"



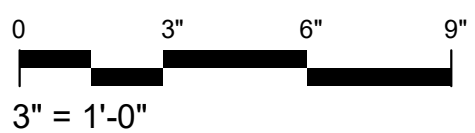
F1 METAL PANEL DETAIL - PLAN
A-402 SCALE: 3" = 1'-0"



F4 CANOPY DETAIL AT EDGE
A-402 SCALE: 3" = 1'-0"



F6 CANOPY DETAIL AT GUTTER
A-402 SCALE: 3" = 1'-0"



KEY PLAN



Sheet Title
**CANOPY AND METAL PANEL
DETAILS - CHILLER YARD**

Dwg. No.
PLA2021-0432-0004D

Sht. No.
A-402

1

2

3

4

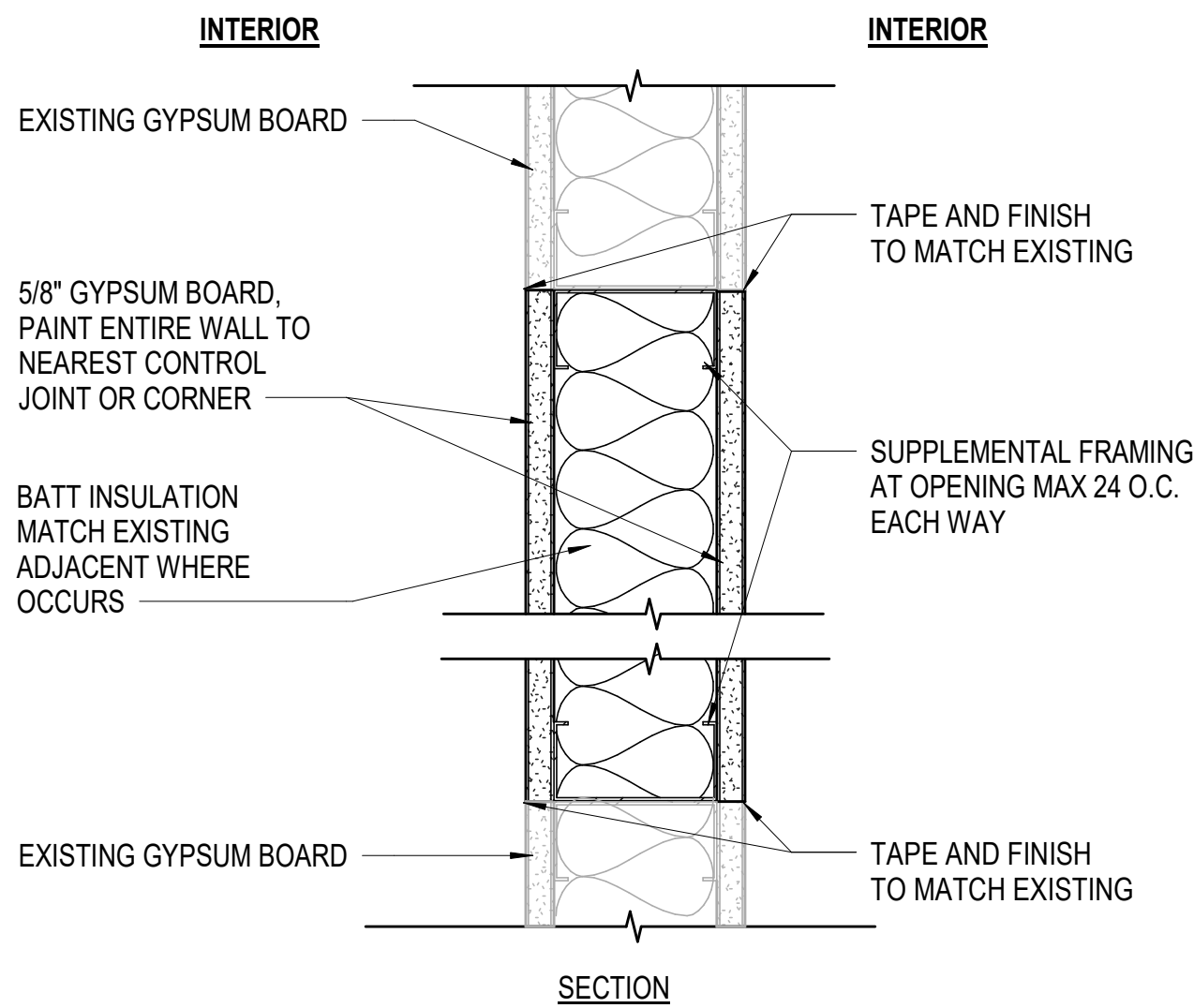
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A
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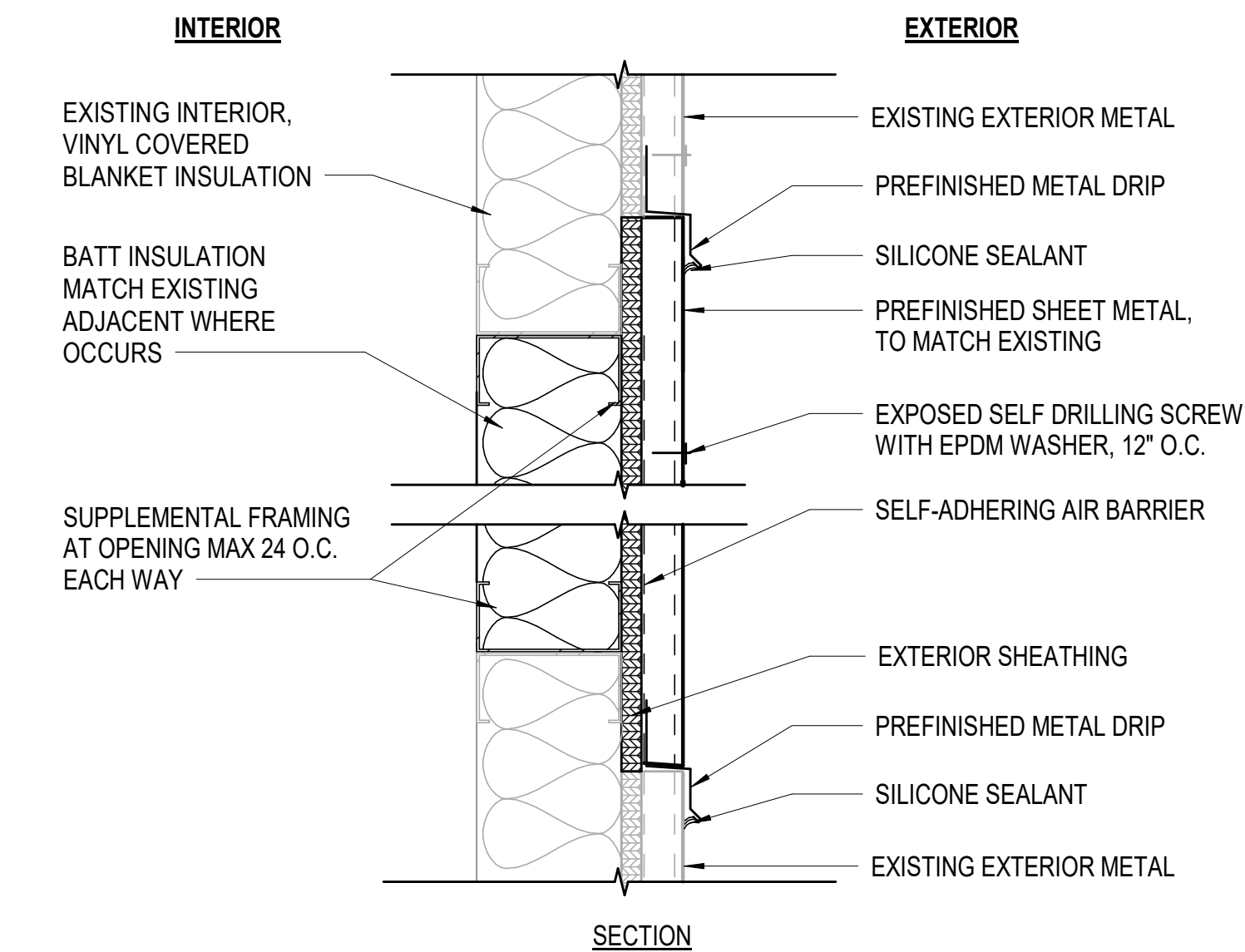


D1 INFILL DETAIL @ INTERIOR GYPSUM BOARD WALL

A-501 SCALE: 3" = 1'-0"

0 3" 6" 9"

3" = 1'-0"

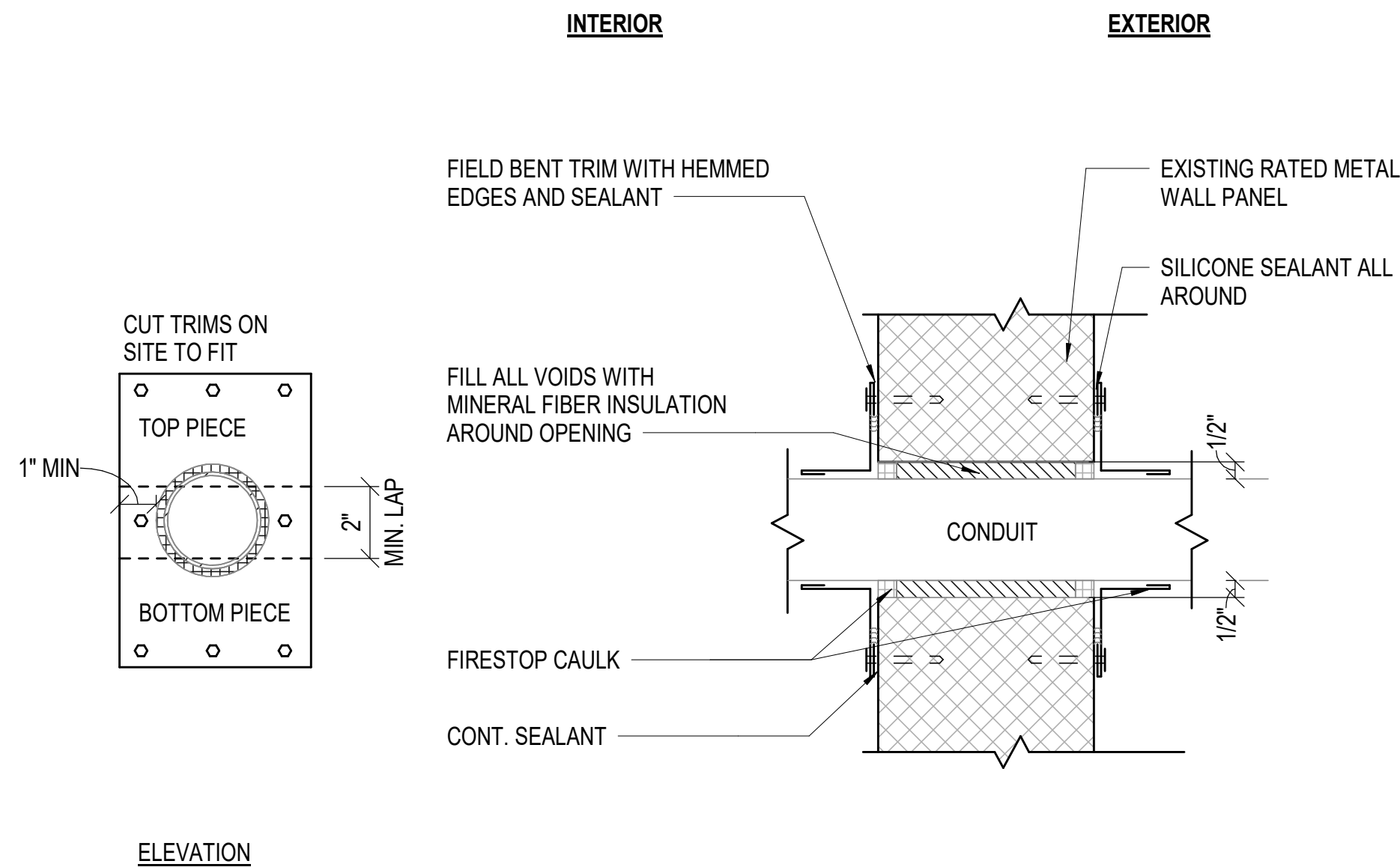


F1 INFILL DETAIL @ EXTERIOR METAL WALL

A-501 SCALE: 3" = 1'-0"

0 3" 6" 9"

3" = 1'-0"



C3 PIPE PENETRATION @ EXTERIOR METAL WALL

A-501 SCALE: 3" = 1'-0"

0 3" 6" 9"

3" = 1'-0"

NEW METAL ROOF TO MATCH EXISTING PROFILE. ATTACH AND OVERLAP WITH EXISTING MIN. 3".

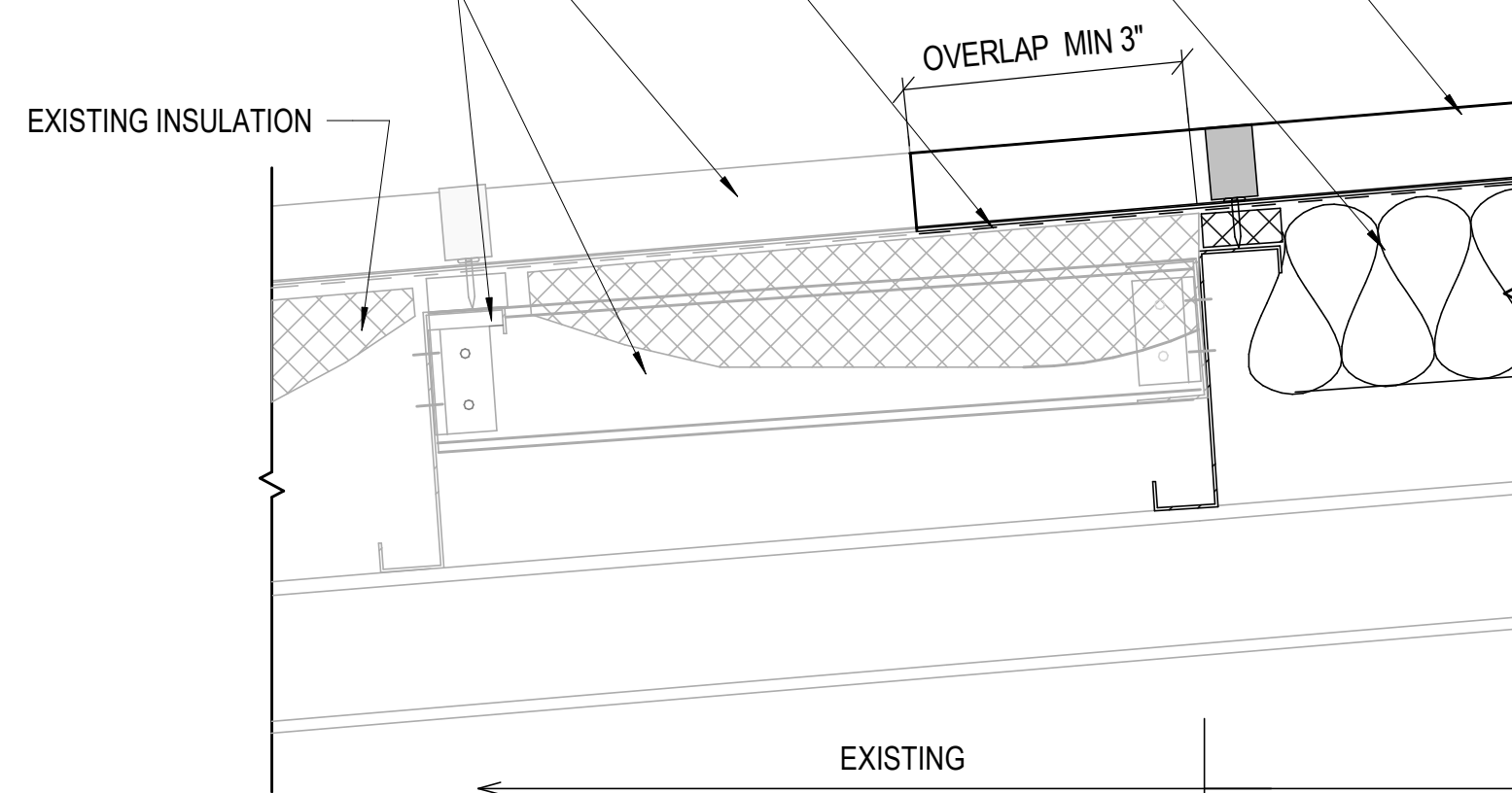
BATT INSULATION MATCH EXISTING ROOF

MECHANICALLY FASTENED WEATHER RESISTANT BARRIER

EXISTING METAL ROOF

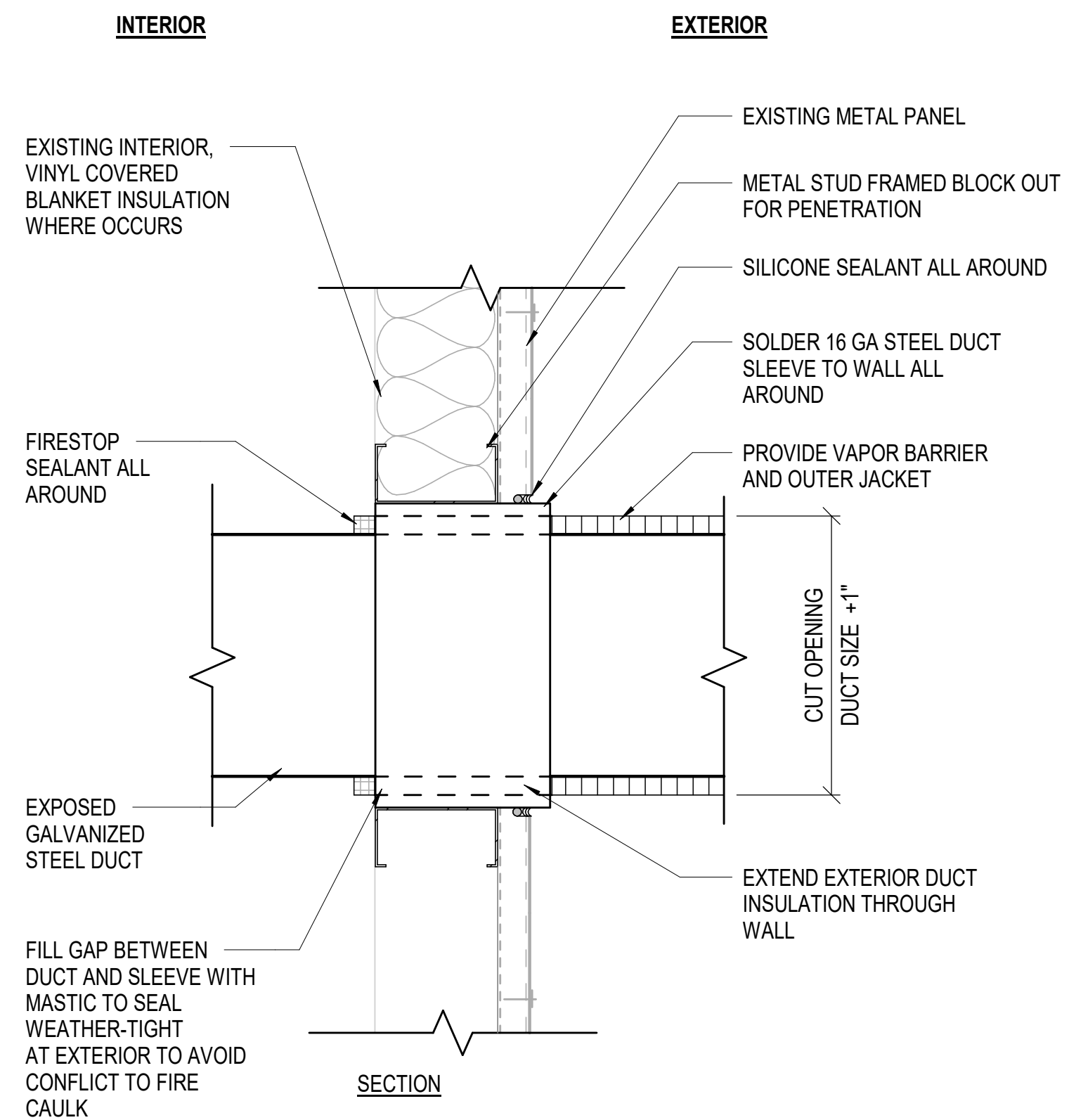
EXISTING ROOF FRAMING

EXISTING INSULATION



F3 ROOF INFILL DETAIL

A-501 SCALE: 3" = 1'-0"

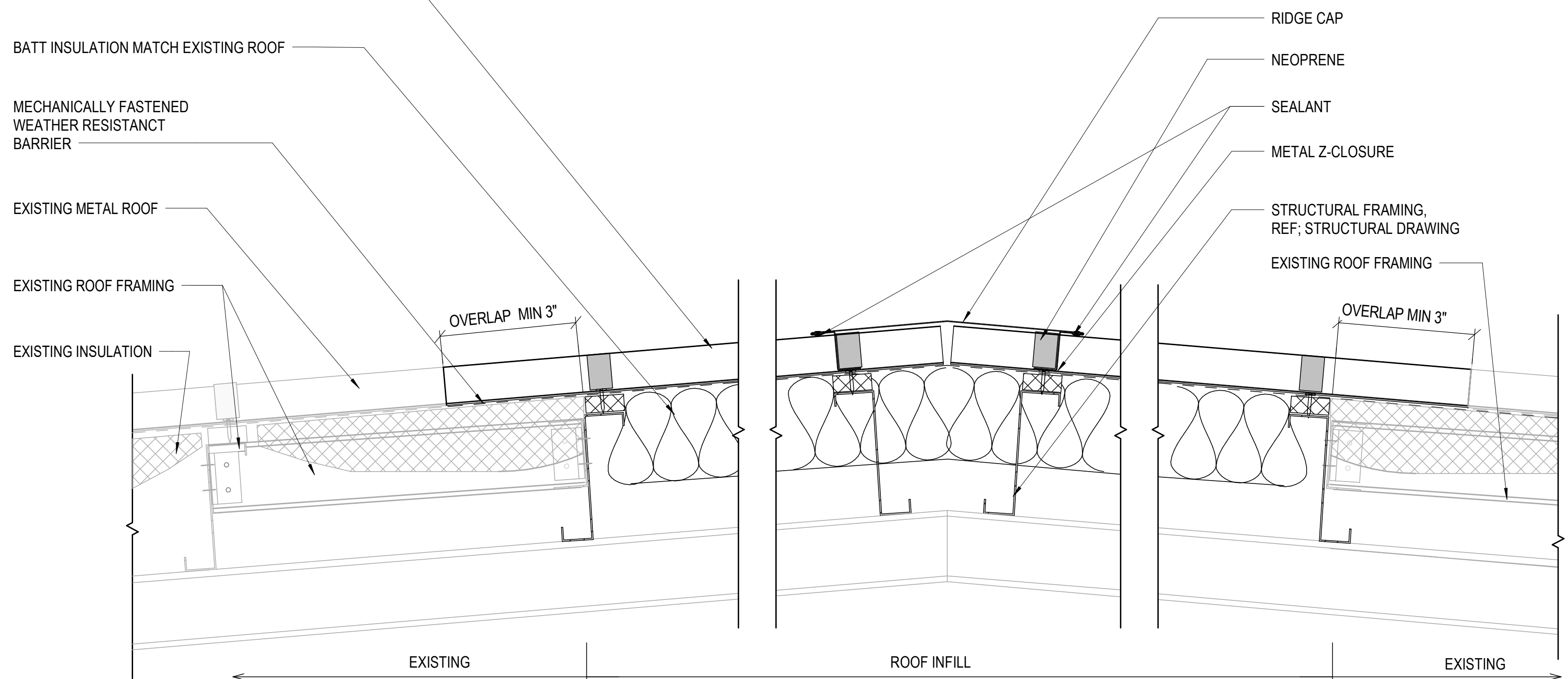


C6 DUCT PENETRATION @ EXTERIOR METAL WALL

A-501 SCALE: 3" = 1'-0"

0 3" 6" 9"

3" = 1'-0"



0 3" 6" 9"

3" = 1'-0"

KEY PLAN



Sheet Title
PENETRATION AND INFILL DETAILS

Dwg. No.
PLA2021-0432-0005D

Sht. No.
A-501

F

Sht. No. A-502

F

DEMOLISH EXISTING BOLLARDS,
SEE CIVIL DRAWINGS FOR
ADDITIONAL SCOPE



AD101

SCALE: NTS.

PREPARE EXISTING METAL DECK FOR ADJACENT NEW METAL DECK



SEE MECHANICAL & ELECTRICAL
DRAWINGS FOR ADDITIONAL
SCOPE

AD101

SCALE: NTS

AD101

SCALE: $1/4" = 1'-0"$

0 2' 4' 8'

1/4" = 1'-0"

8. PRIOR TO COMMENCEMENT WITH ANY DEMOLITION WORK, CONTRACTOR MUST IDENTIFY ALL ELECTRICAL CIRCUITS SERVICING THE AREA INVOLVED WITH THIS DEMOLITION. THOSE CIRCUITS MUST THEN BE LOCKED OUT AND TAGGED OUT IF THEY DO NOT SERVICE ANY OF THE REMAINING BUILDING. THOSE CIRCUITS WHICH ARE IDENTIFIED TO SERVICE BOTH THE AREA TO BE DEMOLISHED AND THE REMAINING BUILDING MUST BE SPLIT SO AS TO KILL ALL ELECTRICAL POWER TO THE AREA TO BE DEMOLISHED WHILE MAINTAINING POWER TO THE REMAINDER OF THE BUILDING.

NO.	DESCRIPTION
-----	-------------

1	EXISTING COLUMN TO REMAIN
2	EXISTING CONCRETE PAD TO REMAIN.
3	EXISTING WALL TO REMAIN
4	DEMOLISH EXISTING BOLLAR
5	EXISTING BUILDING TO REMAIN
6	EXISTING MECHANICAL EQUIPMENT TO REMAIN.
7	DEMOLISH MECHANICAL EQUIPMENT.

DEMOLITION DETAILS - CHILLER YARD

PLA2021-0432-0002D

AD101



KEY PLAN

GENERAL NOTES:

1. SUBCONTRACTOR MUST VERIFY DIMENSIONS AND CONDITIONS PRIOR TO STARTING WORK.
2. SUBCONTRACTOR MUST SUBMIT MATERIAL DATA PER THE SPECIFICATIONS TO CONSTRUCTION MANAGER FOR REVIEW PRIOR TO INSTALLATION
3. THE "REGULATIONS FOR IN-USE OFF-ROAD DIESEL VEHICLES" LIMITS IDLING OF DIESEL-FUELED OFF-ROAD VEHICLES WITH ENGINES GREATER THAN OR EQUAL TO 25 HORSEPOWER THAT ARE NOT DESIGNED FOR USE ON ROADS OR HIGHWAYS (E.G., BACK HOES, ASPHALT COMPACTOR ROLLERS, CRANES, DOZERS, EXCAVATORS, FORKLIFTS, GRADERS, TRACTORS, ETC.). SUBCONTRACTOR MUST RESTRICT IDLING OF ANY DIESEL-FUELED EQUIPMENT WITH AN ENGINE GREATER THAN OR EQUAL TO 25 HORSEPOWER TO NO LONGER THAN FIVE MINUTES. IDLING IS WHEN THE ENGINE IS OPERATING AND THE VEHICLE IS NOT MOVING OR IS NOT PERFORMING WORK. THE ENGINES MUST HAVE A CALIFORNIA AIR CONTROL RESOURCES BOARD (CARB) REGISTRATION AND ASSOCIATED STICKERS.
4. SUBCONTRACTOR MUST BE RESPONSIBLE FOR IMPLEMENTATION OF BEST MANAGEMENT PRACTICES (BMPS) PER THE SWPPP TO LIMIT DISCHARGE OF POLLUTANTS FROM CONSTRUCTION RELATED ACTIVITIES. SEE FACILITY SPECIFICATION 01 35 43 ENVIRONMENTAL PROTECTION.
5. THE PROJECT LAND DISTURBANCE IS OVER ONE ACRE AND IS SUBJECT TO EISA 438 OR CALIFORNIA GENERAL CONSTRUCTION PERMIT REQUIREMENTS.
6. SUBCONTRACTOR MAY REQUEST HORIZONTAL AND VERTICAL CONTROL POINTS, SUBCONTRACTOR MUST COORDINATE WITH CONSTRUCTION MANAGER.
7. CALTRANS SPECIFICATIONS REFERENCES THE 2018 STANDARD SPECIFICATIONS.
8. SUBCONTRACTOR IS REQUIRED TO POTHOLE AND VERIFY TIE-IN UTILITY DEPTH AND MATERIAL TYPE PRIOR TO COMMENCING TRENCHING.
9. THE SUBCONTRACTOR MUST BE SOLELY RESPONSIBLE FOR THE COST OF REPLACING ALL SURVEY WHICH ARE DAMAGED, DESTROYED OR DISTURBED DUE TO THEIR CONSTRUCTION ACTIVITIES. PRIOR TO CONSTRUCTION, SURVEY MONUMENTS THAT ARE LOCATED WITHIN THE CONSTRUCTION WORK AREA MUST BE TIED-OUT AND REFERENCED BY A LAND SURVEYOR LICENSED IN THE STATE OF CALIFORNIA. UPON COMPLETION OF CONSTRUCTION, ALL DAMAGED DESTROYED OR DISTURBED SURVEY MONUMENTS MUST BE REPLACED, AND A CORNER RECORD OF SURVEY MUST BE PREPARED AND FILED WITH THE COUNTY SURVEYOR AS REQUIRED BY THE PROFESSIONAL LAND SURVEYOR'S ACTS, SECTION 8771 OF THE BUSINESS AND PROFESSIONS CODE OF THE STATE OF CALIFORNIA.
10. IF ANY VERTICAL CONTROL IS TO BE DISTURBED OR DESTROYED, MUST BE NOTIFIED IN WRITING, AT LEAST 3 DAYS PRIOR TO THE CONSTRUCTION. THE SUBCONTRACTOR IS SOLELY RESPONSIBLE FOR THE COST ASSOCIATED WITH REPLACING ANY VERTICAL CONTROL BENCHMARKS DESTROYED OR DISTURBED DUE TO THE CONTRACTOR'S CONSTRUCTION ACTIVITIES.
11. SUBCONTRACTOR MUST COORDINATE WITH CM FOR FINAL INSPECTION OF NEW UTILITIES PRIOR TO BACKFILLING OR PRIOR TO PLACING CONCRETE FOR ELECTRICAL AND TELECOM DUCTBANK.
12. PERFORM WORK SAFELY AND PROVIDE A PLACE OF EMPLOYMENT FREE FROM RECOGNIZED HAZARDS THAT MAY CAUSE, OR ARE LIKELY TO CAUSE, DEATH OR SERIOUS PHYSICAL HARM TO EMPLOYEES. THIS INCLUDES PROVIDING PROTECTION FOR THE PUBLIC AND THE ENVIRONMENT. THE SUBCONTRACTOR IS RESPONSIBLE FOR THE SAFE PERFORMANCE OF THE WORK IN ACCORDANCE WITH APPLICABLE FEDERAL, STATE, AND SPECIFIC ES&H REQUIREMENTS. EXERCISE A DEGREE OF CARE COMMENSURATE WITH THE WORK AND THE ASSOCIATED HAZARDS. ENSURE THAT ES&H IS AN INTEGRAL AND TRANSPARENT PART OF THE PLANNING AND EXECUTION OF WORK.
13. STANDARD WORK HOURS ARE MONDAY THROUGH FRIDAY FROM 7:00 AM TO 6:00 PM, EXCEPT HOLIDAYS

a. OPEN TO EARLIER START TIMES DURING SUMMER MONTHS DUE TO SUMMER TEMPERATURES

b. SUBMIT REQUESTS FOR NONSTANDARD WORK HOURS TO THE STR AT LEAST 48 HOURS IN ADVANCE
14. COMPLY WITH, AND ASSIST IN COMPLYING WITH, ES&H REQUIREMENTS OF APPLICABLE LAWS AND REGULATIONS, AND APPLICABLE DIRECTIVES IDENTIFIED IN THIS SUBCONTRACT. COOPERATE WITH FEDERAL, STATE, AND LOCAL AGENCIES HAVING JURISDICTION OVER ES&H MATTERS UNDER THIS SUBCONTRACT.
15. SUBCONTRACTORS ARE REQUIRED TO SUBMIT DOCUMENTATION OF THE WORK TASKS, HAZARDS, CONTROLS, GENERAL SAFETY PRACTICES, AND READINESS TO WORK. THIS INFORMATION IS TERMED "ES&H SUBMITTALS."
16. THE SUBCONTRACTOR IS RESPONSIBLE TO TRAIN ITS EMPLOYEES IN ACCORDANCE WITH LAWS AND STANDARDS, AND INCLUDE ADDITIONAL TRAINING FOR SITE SUPERVISION. CONTINUE TRAINING THROUGH THE TERM OF THE SUBCONTRACT. SUBMIT COPIES OF TRAINING CERTIFICATES FOR EACH EMPLOYEE FOR OPERATIONS THAT REQUIRE SUCH TRAINING PRIOR TO PERFORMING THE WORK
17. CONFINED SPACES. CONDUCT ENTRIES OF PERMIT-REQUIRED CONFINED SPACES IN ACCORDANCE WITH 29 CFR 1910.146, 29 CFR 1926 SUBPART AA, AND ANSI Z88.2. IF THE SUBCONTRACTOR IS THE SOLE ENTRANT, THE ENTRY WILL BE PERFORMED UNDER THE SUBCONTRACTOR'S CONFINED SPACE PROGRAM USING THE SUBCONTRACTOR'S ENTRY PERMIT.

IN ALL CASES, CONDUCT A JOINT PRE-ACTIVITY WALKTHROUGH TO REVIEW CONFINED SPACE HAZARDS AND CONTROLS.
18. WELDING AND BURNING ACTIVITIES. PERFORM WELDING IN ACCORDANCE WITH ANSI Z49.1: SAFETY IN WELDING, CUTTING, AND ALLIED PROCESSES, SECTIONS 4.3 AND E4.3. DO NOT USE THORIATED WELDING RODS WITHOUT THE STR'S APPROVAL IN WRITING. SUBMIT WELDING PROGRAM AS PART OF THE CORPORATE SAFETY PLAN.
19. SILICA DUST. WHEN PERFORMING WORK GENERATING SILICA DUST (E.G., JACKHAMMERING, CORE-DRILLING, OR SAW-CUTTING CONCRETE, REMOVING OR SAWING TILE OR STONE, SAND-BLASTING, REPAVING), PROTECT WORKERS IN ACCORDANCE WITH THE OSHA SILICA STANDARD (29 CFR 1926.1153) USING A COMBINATION OF ADMINISTRATIVE CONTROLS, ENGINEERING CONTROLS, AND PPE TO PREVENT WORKER EXPOSURE TO RESPIRABLE AIRBORNE SILICA FROM EXCEEDING THE ACGIH TLV EXPOSURE LIMITS. AFTER JANUARY 17, 2019, CONSTRUCTION CONTRACTORS MUST OBTAIN EITHER EXPOSURE MONITORING DATA OR USE A COMBINATION OF EXPOSURE MONITORING DATA AND OBJECTIVE DATA TO ACCURATELY CHARACTERIZE WORKER EXPOSURES TO RESPIRABLE SILICA INSTEAD OF RELYING SOLELY ON THE CONTROLS IDENTIFIED 29 CFR 1926.1153(C)(1) TABLE 1. IN THE ABSENCE OF EXPOSURE MONITORING DATA, RESPIRATORY PROTECTION GREATER THAN THAT SPECIFIED IN TABLE 1 MAY BE REQUIRED. ALL MONITORING RESULTS CONDUCTED BY SUBCONTRACTORS ON THEIR EMPLOYEES PERFORMING WORK MUST BE SUBMITTED TO THE STR AND ES&H TEAMS
20. ASBESTOS HAZARD RELATED TO THE CUTTING/REMOVAL OF UNDERGROUND ASBESTOS CEMENT PIPE (TRANSIT PIPE) IS EXPECTED DURING UTILITIES DEMOLITION AND UTILITY TIE-IN WORK. SUCH ACTIVITIES REQUIRE 16 HOURS ASBESTOS TRAINING AND RESPIRATOR PROTECTION PER OSHA ASBESTOS SAFETY CLASS III.
21.
22. TRAFFIC CONTROL PLAN. SUBMIT A TRAFFIC CONTROL PLAN TO PMEC CIVIL ENGINEER AND THE FACILITIES & INFRASTRUCTURE POINT OF CONTACT (FPOC) FOR 02 LAND FOR REVIEW AND APPROVAL PRIOR TO THE START OF CONSTRUCTION. SHOW HOW PEDESTRIAN AND VEHICULAR TRAFFIC WILL BE MAINTAINED DURING CONSTRUCTION ON THE TRAFFIC CONTROL PLAN.
23. DO NOT COMMENCE EARTH-MOVING OPERATIONS UNTIL TEMPORARY SITE FENCING AND EROSION- AND SEDIMENTATION-CONTROL MEASURES SPECIFIED IN SECTION 01 35 43 "ENVIRONMENTAL PROTECTION," 01 50 00 "TEMPORARY FACILITIES AND CONTROLS", AND 31 10 00 "SITE CLEARING" ARE IN PLACE.

PERMITTING GENERAL NOTES:

1. ADVANCE NOTIFICATION REQUIREMENTS:

a. 24 HOURS FOR BURN ACTIVITIES

b. 14 DAYS FOR THE FOLLOWING; SOIL AND EXCAVATION, CONCRETE PENETRATIONS, UTILITY OUTAGE.

c. KEEP THE PERMIT CARD ON THE JOB SITE AT ALL TIMES FOR BUILDING INSPECTOR SIGN-OFF.
- EXCAVATION AND TRENCHING GENERAL NOTES:

1. BEFORE BEGINNING EXCAVATIONS 5 FEET OR MORE IN DEPTH, SUBMIT IN THE CORPORATE SAFETY PLAN THE TRENCHING AND EXCAVATION PROGRAM, AND IN THE JHA OR A SEPARATE DOCUMENT SUBMIT A DETAILED PLAN SHOWING THE DESIGN OF SHORING, BRACING, SLOPING, OR OTHER PROVISIONS TO PROTECT WORKERS FROM THE HAZARD OF CAVING GROUND DURING THE EXCAVATION.

2. SUBCONTRACTOR MUST COORDINATE WITH CONSTRUCTION MANAGER TO OBTAIN SOIL AND EXCAVATION PERMIT AND CONCRETE PENETRATION PERMIT PRIOR TO COMMENCING TRENCHING. TRENCHING MUST CONFORM TO REQUIREMENTS PER SPECIFICATION 31 20 00 EARTH MOVING.

3. PERFORM EXCAVATION UNDER THE SUPERVISION OF A COMPETENT PERSON AS DEFINED BY 29 CFR 1926, SECTIONS 650 AND 651. IF EXCAVATION UNCOVERS AN UNIDENTIFIED UTILITY, STOP EXCAVATION IN THIS AREA AND IMMEDIATELY NOTIFY
- EMERGENCIES:

1. IN AN EMERGENCY AFFECTING THE SAFETY OF PERSONS OR PROPERTY, IMMEDIATELY CALL 911 FROM SYSTEM PHONE OR CELLULAR PHONE, AND TAKE APPROPRIATE ACTION TO PREVENT OR MINIMIZE DAMAGE, INJURY, OR LOSS, AND TO PRESERVE THE INTEGRITY OF THE SCENE FOR FUTURE INVESTIGATION. PROMPTLY NOTIFY THE STR OF THE OCCURRENCE OF SUCH AN EMERGENCY AND ACTIONS TAKEN BY THE SUBCONTRACTOR. THIS NOTICE MAY BE ORAL FOLLOWED BY WRITTEN CONFIRMATION.

CIVIL SHEET INDEX	
SHEET NUMBER	SHEET DESCRIPTION
C-001	GENERAL NOTES AND ABBREVIATIONS
CD-101	DEMOLITION PLAN
C-101	CIVIL SITE PLAN
C-501	CIVIL DETAILS
C-502	CIVIL DETAILS
C-503	CIVIL DETAILS

Existing Condition	Description	Plan
	LIMITS OF CONSTRUCTION	
	GRADING LIMIT LINE	
	EASEMENT LINE	
	CHAIN LINK FENCE	
	HIGH PRESSURE NATURAL GAS	
	ELECTRIC UTILITY/CONDUIT	
	COMMUNICATION LINE	
	STORM DRAINAGE LINE	
	SANITARY SEWER LINE	
	DOMESTIC WATER LINE	
	IRRIGATION LINE	
	MAJOR CONTOURS	
	MINOR CONTOURS	
	MONITORING WELL	
	SEWER STRUCTURE/MANHOLE	
	FIRE DEPARTMENT CONNECTION	
	FIRE HYDRANT	
	POST INDICATOR VALVE	
	TREE OR SHRUB AS INDICATED	
	LIGHT POLE	
	STORM DRAIN STRUCTURE/MANHOLE	
	STORM DRAIN INLET	
	IRRIGATION BOX	
	UTILITY VAULT PER PLAN	
	INLET/OUTLET STRUCTURE	
	WATER VALVE	
	THRUST BLOCK	
	TEE	
	ELBOW	
	CAP OR PLUG	
	SIGN	
	ELEVATION	
	FLOW LINE	

GRID LETTER, GRID NUMBER

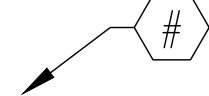
D5

VIEW/SHEET TITLE

SHEET NO

SCALE: 1"= SCALE IN FEET OR NONE

KEYED ITEM/NOTE



GRID LETTER, GRID NUMBER

D7

DETAIL TITLE

SHEET NO

SCALE: 1"=SCALE IN FEET OR NONE

ABBREVIATIONS

ADA	AMERICAN DISABILITIES ACT	LP	LOW POINT
BC	BEGIN CURVE	MH	MANHOLE
BVC	BEGIN VERTICAL CURVE	MOC	MIDDLE OF CURVE
CL	CENTERLINE	O.C.	ON CENTER
CO	CLEANOUT	PB	PULL BOX
DIA	DIAMETER	PCC	PORTLAND CEMENT CONCRETE
DI	DROP INLET	PIV	POST INDICATOR VALVE
EC	END CURVE	SW	SIDEWALK
ECCAB/TRANS	ELECTRICAL CABINET / TRANSFORMER	SDMH	STORM DRAIN MANHOLE
EP	EDGE OF PAVEMENT	TG	TOP OF GRATE
EV	ELECTRIC VEHICLE	TYP	TYPICAL
EVC	END VERTICAL CURVE	UTILS	UTILITIES
EX	EXISTING	WM	WATER METER
FF	FINISH FLOOR	@	AT
FDC	FIRE DEPARTMENT CONNECTION	Ø	DIAMETER/PHASE
FH	FIRE HYDRANT	#	NUMBER
FG	FINISH GROUND	%	PERCENT
FL	FLOW LINE	±	PLUS/MINUS (APPROXIMATE)
FS	FINISH SURFACE		
GB	GRADE BREAK		
HH	HAND HOLE		
HP	HIGH POINT		
INV	INVERT		

NOTE:
SOME ABBREVIATIONS MAY
NOT BE USED ON THIS PLAN

KEY PLAN

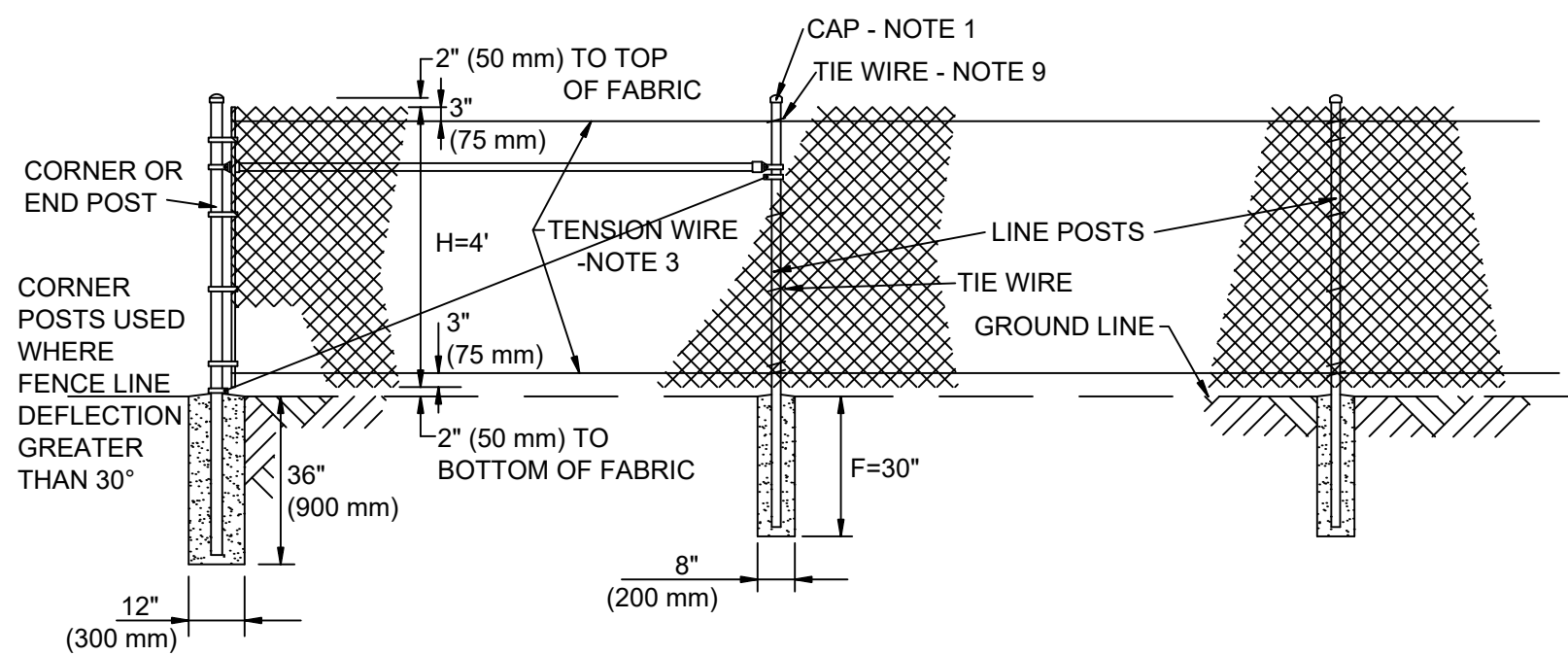
GENERAL NOTES AND ABBREVIATIONS

Dwg. No.
PLC2021-0432-0001D

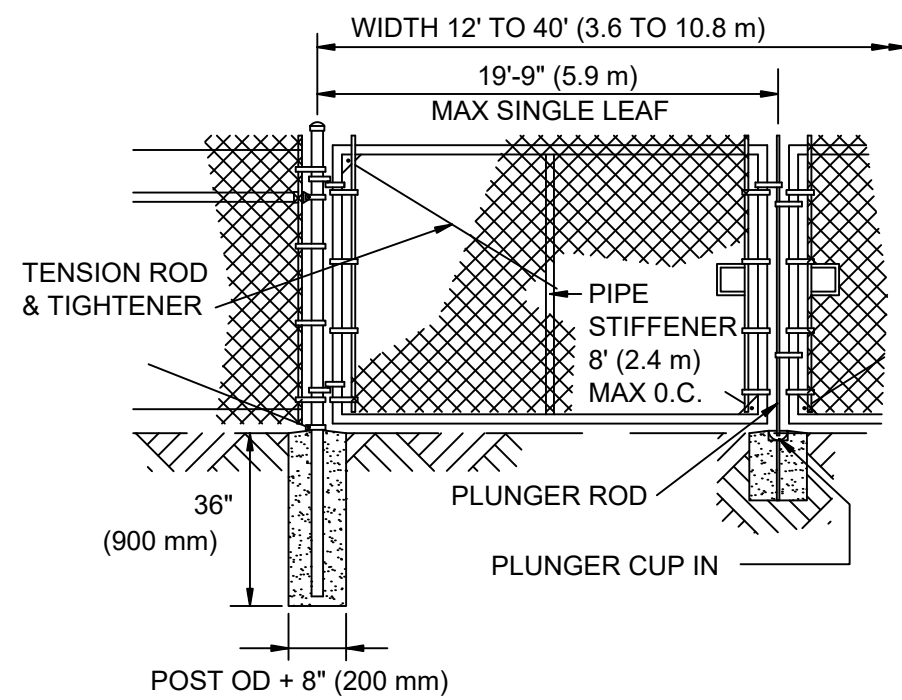
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C-001



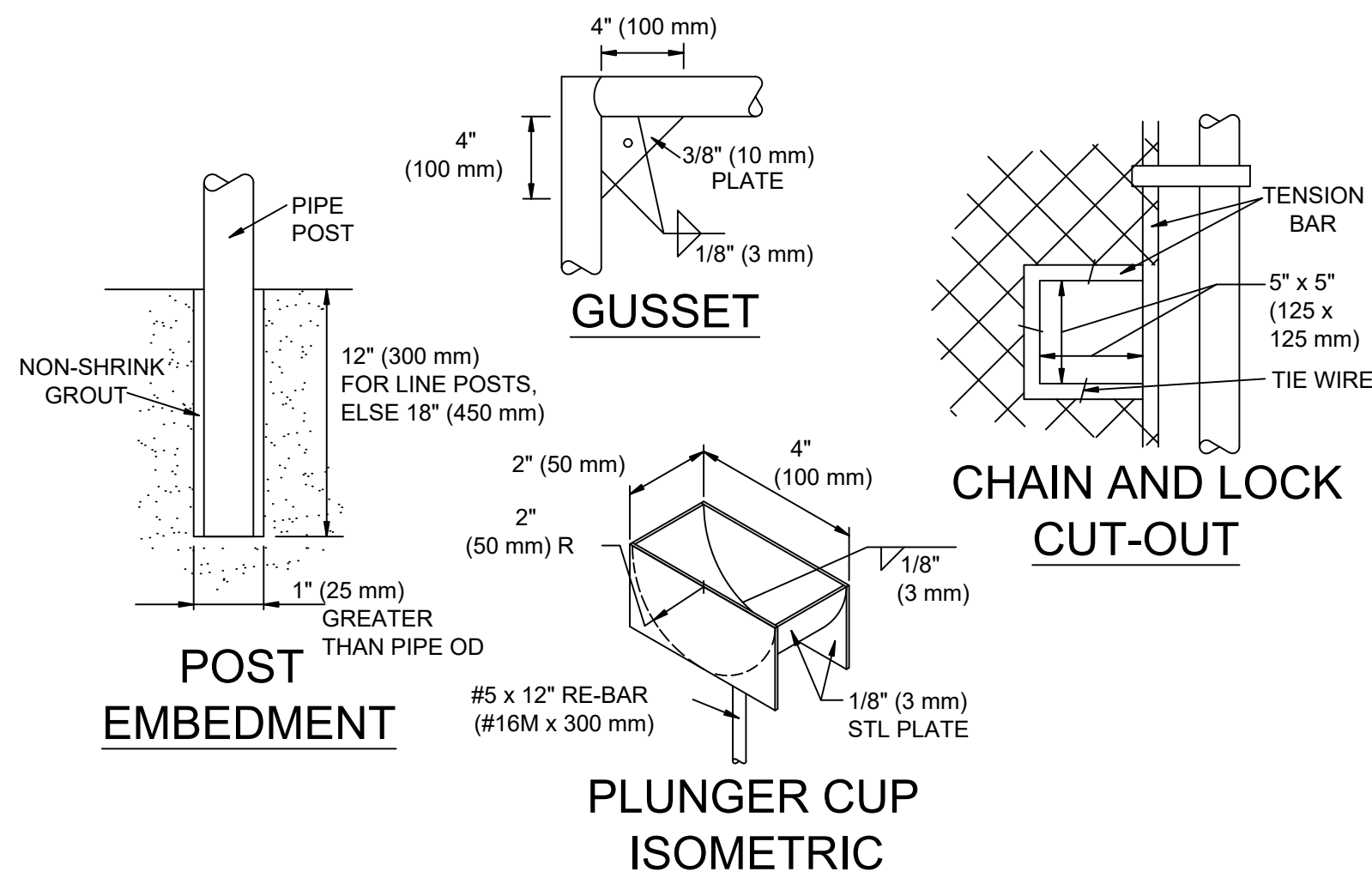
THIS IS PRIVATE PROPERTY AND THE 811 PROCESS DOES NOT PROTECT YOU FROM HITTING PRIVATE UTILITIES. PLEASE USE NON-INVASIVE TECHNIQUES SUCH AS GPR AND ELECTROMAGNETIC TRACING TO LOOK FOR UTILITIES PRIOR TO ANY EXCAVATION.



B1 CHAIN-LINK FENCE W/CORNER POST
C-501 C-101 SCALE: NONE



B4 SWING GATE
C-501 C-101 SCALE: NONE

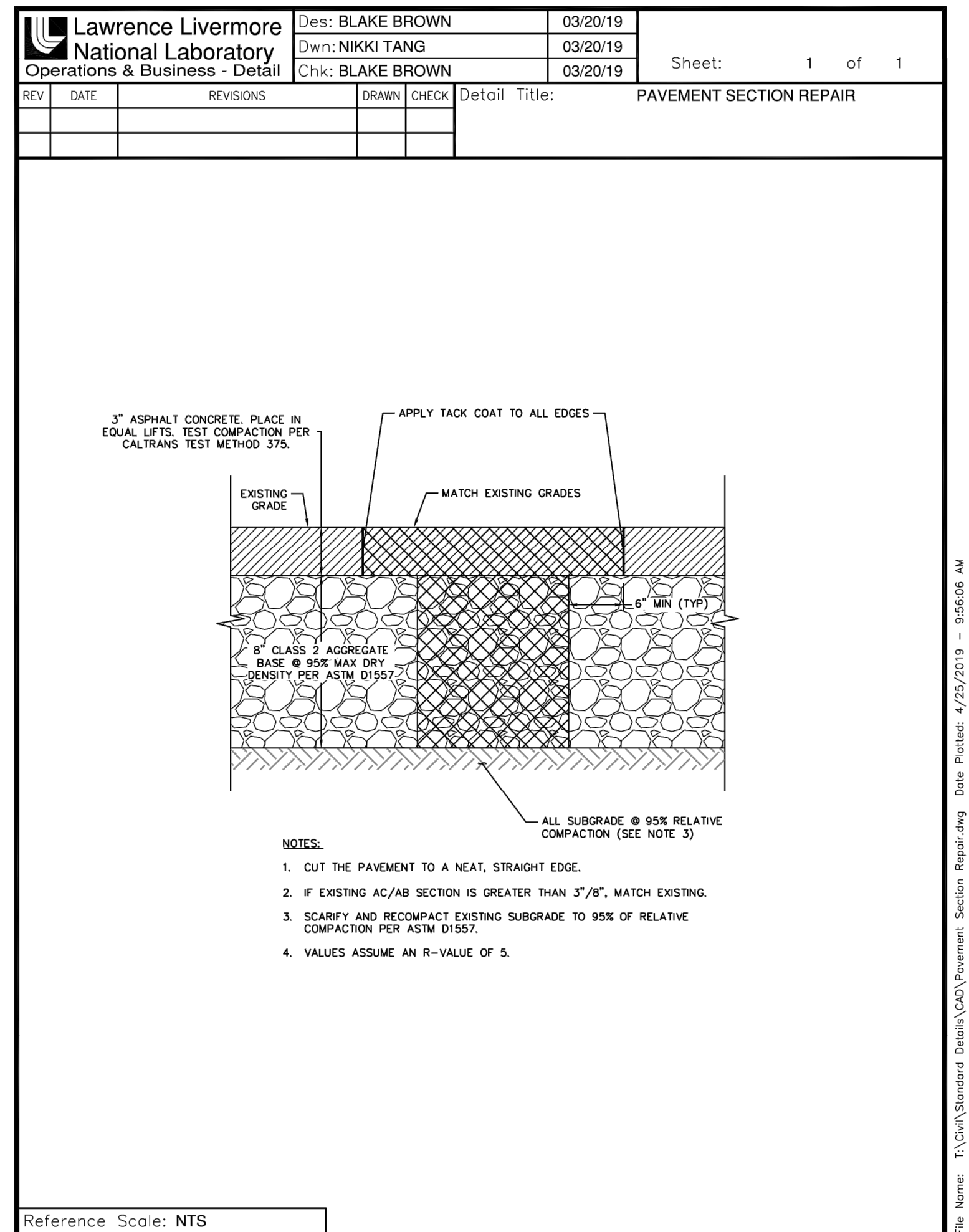


E1 SWING GATE DETAILS
C-501 C-101 SCALE: NONE

NOTES:

1. SECURE DRIVE-FIT GALVANIZED CAP TO POST WITH 1/4" (6 mm) ROUND-HEAD RIVET.
2. H DENOTES FABRIC WIDTH AND NOMINAL FENCE HEIGHT. H = 4' UNLESS OTHERWISE NOTED.
3. IF FENCE WITH TOP RAIL IS SPECIFIED, DELETE STEEL TENSION WIRE AT TOP, AND PIPE RAILS AT INTERMEDIATE, SLOPE, END AND CORNER POSTS. EXTEND TENSION ROD TO TOP RAIL.
4. POST SPACING IS MAXIMUM 10' (3.0 m).
5. FILL CLEAR OPENINGS GREATER THAN 3" (75 mm) WITH FABRIC. FOR OPENINGS LESS THAN 18" (450 mm), TIE FABRIC TO POSTS.
6. USE ONE POST FOR COMBINED SLOPE AND CORNER POST IF TOP OF CHANNEL WALL IS CONSTRUCTED AS SHOWN FOR "ALTERNATE".
7. STEEL BANDS AT TENSION BARS MUST BE 1/8" x 1" (3 x 25 mm), MINIMUM, SPACED AT MAXIMUM 16" (400 mm).
8. SECURE TENSION WIRES TO EACH LINE POST WITH TIE WIRES.

E4 FENCING NOTES
C-501 C-101 SCALE: NONE



E6 PAVEMENT SECTION REPAIR
C-501 C-101 SCALE: NONE

KEY PLAN

Sheet Title

CIVIL DETAILS

Dwg. No.

PLC2021-0432-0004D

Sht. No.

C-501



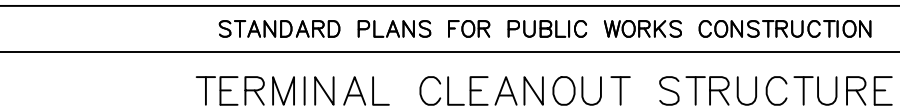
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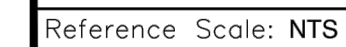
C-502 C-101 SCALE: NONE



C-502 C-101 SCALE: NONE



4



C-502 C-101 SCALE: NONE

Sheet Title

CIVIL DETAILS

Dwg. No.

PLC2021-0432-0005D

Sht. No.

C-502

FOR INFORMATION
NOT SHOWN
OR NOTED, SEE

D4
-

D1 **FIXED BOLLARD**
C-502 C-101 SCALE: NONE

SLOPE CONC.
ASPHALT OR
PAVING

D4 **REMOVABLE BOLLARD**
C-502 C-101 SCALE: NONE

BOLLARD NOTES

- A. STORAGE TANKS AND CONNECTED PIPING, VALVES, AND FITTINGS
B. STORAGE AREAS CONTAINING TANKS OR PORTABLE CONTAINERS EXCEPT WHERE THE EXPOSING VEHICLES ARE LIMITED TO POWERED INDUSTRIAL TRUCKS USED FOR TRANSPORTING THE HAZARDOUS MATERIALS
C. COMPRESSED GAS USE AREAS

LOCATE ALL UNDERGROUND UTILITIES PRIOR TO EXCAVATING BOLLARD FOOTING. CONSULT CIVIL OR STRUCTURAL ENGINEER IF UNDERGROUND UTILITIES ARE WITHIN 18 INCHES OF BOLLARD FOOTING.

PROVIDE 4 FT CLEARANCE AND REMOVABLE BOLLARDS WHERE ACCESS IS REQUIRED FOR SERVICE OR HOT STICK OPERATIONS. SECTIONALIZING SWITCHGEAR TYPICALLY REQUIRES HOT STICK OPERATION SPACES ON TWO SIDES.

HOT DIP GALVANIZE ALL STEEL PARTS AFTER FABRICATION IN ACCORDANCE WITH ASTM A123.

PAINT BOLLARDS WITH TWO COATS OF "SAFETY YELLOW PAINT OVER A SUITABLE PRIMER. SHERMAN WILLIAMS "POLANE" H.S. POLYURETHANE ENAMEL OR BETTER.

YELLOW POLYETHYLENE SLEEVES, SUCH AS THE ARMORCAST "GUARDIAN SLEEVE" MAY BE USED INSTEAD OF PAINTING.

REFLECTIVE TAPE SHALL BE 4" WHITE HIGH INTENSITY PRISMATIC
REFLECTIVE TAPE (ORALITE 5900, ASTM D4956-13 TYPE IV) OR APPROVED
EQUAL.

SUBCONTRACTOR SHALL SUBMIT DIMENSIONED LAYOUT DRAWING FOR APPROVAL PRIOR TO INSTALLATION.

D6 **BOLLARD NOTES**
C-502 C-101 SCALE: NONE

KEY PLAN

Sheet Title

CIVIL DETAILS

Dwg. No.

PLC2021-0432-0006D

Sht. No.

C-503



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PROCESS DOES NOT PROTECT YOU FROM
HITTING PRIVATE UTILITIES. PLEASE USE
NON-INVASIVE TECHNIQUES SUCH AS GPR
AND ELECTROMAGNETIC TRACING TO LOOK
FOR UTILITIES PRIOR TO ANY EXCAVATION.

F

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ELECTRICAL SYMBOL LIST										ABBREVIATIONS	
LAYOUT PLANS											
SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION	A, AMP.	AMPERE
	CONDUIT RUN EXPOSED, CONCEALED IN WALL OR ABOVE CEILING.	GFCI	20A, 125V, 2-POLE 3-WIRE, HEAVY DUTY, SPECIFICATION GRADE, DUPLEX RECEPTACLE, NEMA 5-20R, WHITE. FLUSH MOUNT IN WALL. <u>HUBBELL 5362W UNO</u>		MECHANICAL INTERLOCKED RECEPTACLE		DOWN LIGHT LUMINAIRE		HIGH VOLTAGE TERMINATION	AC	ALTERNATING CURRENT
	CONDUIT, CABLE, OR BARE GROUND OCNDUCTORS, RUN UNDERGROUND, CONCEALED IN OR UNDER FLOOR SLAB OR GRADE.	A/C S R	ABOVE CEILING SURFACE MOUNTED RECESSED	G	GROUNDING OUTLET		DOWN LIGHT LUMINAIRE WITH EMERGENCY BATTERY PACK.		HIGH VOLTAGE SWITCH AND FUSE	AIC	AMPERE INTERRUPTING CAPACITY
	CIRCUIT EXTENDED IN LIQUIDTIGHT FLEXIBLE CONDUIT, AT THE END OF A CONDUIT RUN	GFCI	INDICATES GROUND FAULT CIRCUIT INTERRUPTING TYPE. <u>HUBBELL GFRST20W UNO</u> .	R	RADIO RECEPTACLE		DOWN LIGHT WALL WASH LUMINAIRE		PRIMARY CONNECTION (DELTA)	AF	AMPERE FRAME
	CONDUIT RUN TURNED DOWN	WL,WP	WET LOCATIONS. INDICATES WEATHER-RESISTANT GFCI TYPE RECEPTACLE WITH WEATHERPROOF EXTRA DUTY IN-USE COVER.		SURFACE METAL RACEWAY. <u>LEGRAND WIREMOLD 4000 SERIES UNO</u> .		WALL MOUNTED LUMINAIRE		LIQUID FILLED SUBSTATION TRANSFORMER, TYPE AND RATINGS AS INDICATED ON THE DRAWINGS	AE	ARC-ENERGY REDUCTION
	CONDUIT RUN TURNED UP	RECEPTACLE: COVER:	<u>HUBBELL GFWRST20W UNO</u> . <u>HUBBELL TAYMAC MX320Q UNO</u> .		OVER/IN-FLOOR METAL RACEWAY. <u>CONNECTRACK UNO</u> .		WALL MOUNTED LUMINAIRE WITH EMERGENCY BATTERY PACK.		SECONDARY CONNECTION (GROUNDED WYE)	AFC	ABOVE FINISHED CONCRETE
	CONDUIT CAPPED	DL	DAMP LOCATIONS. INDICATES WEATHER-RESISTANT GFCI TYPE RECEPTACLE WITH FLIP COVER.	S	SPEAKER		STRIP LUMINAIRE		DRAWOUT TYPE POWER BREAKER 1200 AMP FRAME, 1000 AMP TRIP, 3-POLE	AT	AMPERE TRIP
	POWER PANEL NUMBER POWER CIRCUIT NUMBERS	RECEPTACLE: COVER:	<u>HUBBELL GFWRST20W UNO</u> . <u>HUBBELL HBL5206WQ UNO</u> .		2\"/>		STRIP LUMINAIRE WITH A EMERGENCY BATTERY PACK.		L = LONG TIME TRIP S = SHORT TIME TRIP I = INSTANTANEOUS TRIP G = GROUND FAULT TRIP A = ALARM	ATS	AUTOMATIC TRANSFER SWITCH
	CIRCUIT HOMERUN IN MIN. 3/4\"/>		TWO 20A, 125V, 2-POLE 3-WIRE, HEAVY DUTY, SPECIFICATION GRADE DUPLEX RECEPTACLES, NEMA TYPE 5-20R, WHITE. FLUSH MOUNT IN WALL. <u>HUBBELL 5362W UNO</u> .		PRIMARY TELEPHONE/DATA OUTLET: 4-11/16\"/>		1\"/>		MOLDED CASE CIRCUIT BREAKER 600 AMP FRAME, 400 AMP TRIP, 3-POLE SEE DRAWOUT TYPE BREAKER ABBREVIATIONS ABOVE.	AWG	AMERICAN WIRE GAUGE
	208/120V PANELBOARD		DUPLEX FLOOR RECEPTACLE		SECONDARY TELEPHONE/DATA OUTLET: 4-11/16\"/>		1\"/>		CONTINUOUS AMPERE RATING	BAHJ	BUILDING AUTHORITY HAVING JURISDICTION
	480/277V PANELBOARD		DUPLEX CEILING RECEPTACLE		DOUBLE DUPLEX FLOOR RECEPTACLE		2\"/>		MOLDED CASE MOTOR CIRCUIT PROTECTOR TYPE BREAKER	BC	BARE COPPER
	480V PANELBOARD OR MCC		DOUBLE DUPLEX CEILING RECEPTACLE		DOUBLE DUPLEX FLOOR RECEPTACLE		2\"/>		STARTER SIZE	BLDG	BUILDING
	208/120V LOAD CENTER		20A, 125V, 2P, 3W, SINGLE RECEPTACLE (NEMA 5-20R)		DOUBLE DUPLEX CEILING RECEPTACLE		2\"/>		MAGNETIC STARTER	CO	CONDUIT ONLY
	240/120V PANELBOARD		INDICATES HAZARDOUS CLASS II, DIV. 2 RECEPTACLE		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-20R)		2\"/>		PRIMARY CONNECTION (DELTA)	CB	CIRCUIT BREAKER
	EQUIPMENT CONTROL PANEL		15A, 125V DUPLEX RECEPTACLE (NEMA 5-15R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-20R)		2\"/>		TRANSFORMER, DRY-TYPE DISTRIBUTION, RATINGS AS INDICATED ON DRAWINGS	CIR., CKT	CIRCUIT
	TRANSFORMER, SIZE AND RATING AS INDICATED ON DRAWINGS		15A, 125V RECEPTACLE (NEMA L5-15R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		INDICATES WITH ELECTROSTATIC SHIELD (FARADAY)	CTRL	CONTROL
	MULTI-OUTLET BOX, TYPE AND RATINGS AS INDICATED ON THE DRAWINGS AND/OR SPECIFICATIONS		30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		SECONDARY CONNECTION (GROUNDED WYE)	DEMO.	DEMOLISH
	ENCLOSED CIRCUIT BREAKER, TYPE AND RATINGS AS INDICATED ON THE DRAWINGS		15A, 250V RECEPTACLE (NEMA 6-15R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		BUSWAY, CURRENT AND VOLTAGE RATINGS AS INDICATED ON DRAWINGS	DC	DIRECT CURRENT
	PULL BOX		30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		SURGE ARRESTOR	DWG	DRAWING
	JUNCTION BOX, SIZE AS REQUIRED, UNO		30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		GROUND	E	EMERGENCY CIRCUIT
	VARIABLE FREQUENCY DRIVE WITH INTEGRAL BYPASS SWITCH, TYPE AND RATINGS AS INDICATED ON THE DRAWINGS		30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		SEPARABLE CONTACTS	EGC	EQUIPMENT GROUNDING CONDUCTOR
	ELECTRIC MOTOR, HORSEPOWER AND VOLTAGE, AS INDICATED ON DRAWINGS		30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		FUSE, SIZE AND RATINGS AS INDICATED ON THE DRAWINGS	(E)	EXISTING
	SWITCH RATING SIZE OF FUSES (WHEN INDICATED FUSIBLE) 3-POLE, 240VAC-600VAC, HEAVY DUTY, SINGLE THROW NON-FUSIBLE (UNLESS INDICATED OTHERWISE) SAFETY SWITCH, HP RATED IN A NEMA 250, TYPE 1, ENCLOSURE (UNLESS INDICATED OTHERWISE)		30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		FUSED DISCONNECT SWITCH, RATINGS AS INDICATED ON DRAWINGS	EMT	ELECTRICAL METALLIC TUBING
	INDICATES IN A NEMA 250, TYPE 3R, ENCLOSURE (TYP. FOR ALL DEVICES)		30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		DISCONNECT SWITCH, RATINGS AS INDICATED ON DRAWINGS	EOL	END OF LINE DEVICE
	INDICATES FUSIBLE (THREE CLASS RK5 FUSES)		30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		INDUCTION MOTOR, 3Ø, NUMERAL INDICATES HORSEPOWER	EQPM	EQUIPMENT
	MANUAL MOTOR STARTER WITH THERMAL OVERLOAD PROTECTION, TYPE AND RATINGS AS INDICATED ON THE DRAWINGS		30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		INDUCTION MOTOR, 1Ø, NUMERAL INDICATES HORSEPOWER	EV	ELECTRIC VEHICLE
	TOGGLE TYPE MOTOR RATED SWITCHED DISCONNECT AND ENCLOSURE WITH LOCKOUT/TAGOUT		30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		PRIMARY FUSES	EV/A	EVACUATION VOICE/ALARM
	MOTOR STARTER, TYPE AND RATINGS AS INDICATED ON THE DRAWINGS. COORDINATE CONTROL WIRING VOLTAGE WITH MECHANICAL EQUIPMENT.		30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		CONTROL TRANSFORMER, DESCRIPTION AND RATING AS INDICATED ON DRAWINGS	EYS	EXPLOSION PROOF Y SEAL
	COMBINATION MOTOR STARTER, WITH INTEGRAL DISCONNECT SWITCH/MOLDED CASE CIRCUIT BREAKER, HP RATED, TYPE AND RATINGS AS INDICATED ON THE DRAWINGS. COORDINATE CONTROL WIRING VOLTAGE WITH MECHANICAL EQUIPMENT.		30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		GROUND SECONDARY TERMINAL \"X2\"	(F)	FUTURE
	DRIVEN GROUND ROD, 3/4\"/>		30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		SECONDARY FUSE	FA	FIRE ALARM
	DRIVEN GROUND ROD, 3/4\"/>		30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		NORMALLY OPEN CONTACT	FACU	FIRE ALARM CONTROL UNIT
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		NORMALLY CLOSED CONTACT	FFCP	FIREMANS FAN CONTROL PANEL
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>		THERMAL OVERLOAD ELEMENT	FLEX	FLEXIBLE METAL CONDUIT
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			G, GND	GROUND
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			GEC	GROUNDING ELECTRODE CONDUCTOR
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			GFCI	GROUND FAULT CIRCUIT INTERRUPTER
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			GFPE	GROUND FAULT PROTECTION EQUIPMENT
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			H-O-A	HAND-OFF-AUTO
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			HP	HORSEPOWER
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			HV	HIGH VOLTAGE
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			JB, J-BOX	JUNCTION BOX
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			KW	KILOWATT
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			KV	KILOVOLT
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			KVA	KILOVOLT-AMPERE
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			MAX.	MAXIMUM
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			MCB	MAIN CIRCUIT BREAKER
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			MCC	MOTOR CONTROL CENTER
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			MCP	MOTOR CIRCUIT PROTECTOR
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			MFR	MANUFACTURER
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			MIN.	MINIMUM
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			MISC.	MISCELLANEOUS
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			MOB	MULTI-OUTLET BOX
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			MTG HT	MOUNTING HEIGHT
			30A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		20A, 125V, 2P, 3W RECEPTACLE (NEMA L5-30R)		2\"/>			N/A	NOT APPLICABLE

GENERAL

A. PROVIDE NEW ELECTRICAL WORK INDICATED. PERFORM WORK IN ACCORDANCE WITH THE DRAWINGS, SPECIFICATIONS, PROJECT REQUIREMENTS DOCUMENT (PRD), AND OTHER CONSTRUCTION DOCUMENT. IF DISCREPANCIES OR CONFLICTS BETWEEN ANY DOCUMENTS ARE IDENTIFIED, PROMPTLY NOTIFY THE SUBCONTRACT TECHNICAL REPRESENTATIVE (STR) AND DO NOT PROCEED WITH THAT PORTION OF THE WORK UNTIL THE STR PROVIDES DIRECTION FOR CORRECTIVE ACTION.

B. COMPLY WITH THE CURRENT VERSION OF THE NATIONAL ELECTRICAL CODE FOR ELECTRICAL CONNECTIONS AND INSTALLATIONS.

C. THE DRAWINGS AND SPECIFICATIONS DO NOT SPECIFICALLY SHOW OR MENTION EVERY ITEM NECESSARILY REQUIRED. PROVIDE COMPLETE AND OPERABLE SYSTEMS AND EQUIPMENT UNLESS EXPLICITLY STATED OTHERWISE.

D. ELECTRICAL DRAWINGS ARE ESSENTIALLY DIAGRAMMATIC, ALTHOUGH THE LOCATIONS OF EQUIPMENT ARE SHOWN TO SCALE WHEREVER POSSIBLE. VERIFY EQUIPMENT SIZE WITH SHOP DRAWINGS. LAYOUT AND INSTALL WORK TO AVOID INTERFERENCE WITH OTHER TRADES.

E. NOTES ON THIS SHEET ARE GENERAL AND MAY NOT APPLY TO THE PROJECT.

F. ENSURE MATERIALS AND EQUIPMENT ARE INSTALLED CORRECTLY AND MAKE ADJUSTMENTS AS NECESSARY OR REQUIRED TO RESOLVE SPACE PROBLEMS AND PRESERVE SERVICE CLEARANCE. IN THE EVENT OF MAJOR REROUTING OF A SYSTEM APPEARS NECESSARY, PREPARE SHOP DRAWINGS OF THE PROPOSED REARRANGEMENT AND SUBMIT DRAWINGS FOR APPROVAL TO THE STR.

G. SUBMIT FOR APPROVAL EXACT LOCATION OF NEW EQUIPMENT WITH STR.

H. ELECTRICAL MATERIALS MUST BE NEW AND LISTED BY A NATIONALLY RECOGNIZED TESTING LABORATORY (NRTL). ELECTRICAL MATERIALS AND EQUIPMENT MUST MEET THE NRTL REQUIREMENTS AND BEAR THE NRTL LABEL.

I. ENCLOSURE CONSTRUCTION AND METHODS OF ASSEMBLY MUST BE IN ACCORDANCE WITH UL 50 AND NEMA PB 1.

J. NEUTRAL CONDUCTOR FOR ALL NON-LINEAR LOADS (EX. COMPUTER, ELECTRONIC RACKS, ETC.) MUST BE COUNTED AS CURRENT-CARRYING CONDUCTORS. MAXIMUM NUMBER OF CURRENT-CARRYING CONDUCTORS MUST NOT EXCEED 9 IN ANY HOME RUN CONDUIT.

K. CIRCUIT BREAKERS RATED 250 AMPS AND HIGHER MUST BE 100% RATED WITH ADJUSTABLE LONG TIME PICKUP AND DELAY, BANDS, ADJUSTABLE SHORT TIME PICKUP WITH MULTIPLE DELAY BANDS AND ADJUSTABLE INSTANTANEOUS PICKUP FOR OPTIMAL SYSTEM COORDINATION. INCLUDE A 3-PHASE AMMETER IN CIRCUIT BREAKER AND PROVIDE INTERCHANGEABLE/UNIVERSAL RATING PLUGS.

L. PROVIDE FIELD-ADJUSTABLE MAGNETIC INSTANTANEOUS TRIP SETTING FOR THERMAL MAGNETIC CIRCUIT BREAKERS RATED HIGHER THAN 150 AMPS.

M. CIRCUIT BREAKERS RATED, OR CAN BE ADJUSTED TO, 1200 AMPS OR HIGHER MUST HAVE ARC ENERGY REDUCTION MAINTENANCE SWITCH THAT CAN REDUCE CLEARING TIME FOR ARC-ENERGY REDUCTION AND DOCUMENTATION MADE AVAILABLE.

N. SERIES-RATED COMBINATION DEVICES ARE NOT PERMITTED. OVERCURRENT PROTECTION DEVICES MUST HAVE AN INTERRUPTING DUTY THAT EXCEEDS MAXIMUM AVAILABLE FAULT CURRENT.

O. DO NOT MOUNT MAIN CIRCUIT BREAKER IN A BRANCH BREAKER POSITION.

P. INDOOR PANELBOARD MUST BE PROVIDED WITH TWO DOORS. DOOR-IN-DOOR STYLE (INNER AND OUTER) DOORS THAT OPEN TO THE RIGHT. INNER DOOR EXPOSE THE CIRCUIT BREAKERS INSULATED BODY AND HANDLE ONLY WHEN OPEN. OUTER DOOR FULLY EXPOSE GUTTER SPACE, THE WIRING RACEWAY, AND CIRCUIT BREAKER LOAD SIDE TERMINALS WHEN OPEN. EACH DOOR MUST BE HINGED AND HAVE A FLUSH LOCKABLE HANDLE.

Q. OUTDOOR PANELBOARD MUST BE PROVIDED WITH THREE DOORS. NEMA 3R EXTERIOR DOOR WITH HINGED DOOR-IN-DOOR (INNER AND OUTER) STYLE DOORS THAT OPEN TO THE RIGHT. INNER DOOR EXPOSE THE CIRCUIT BREAKERS INSULATED BODY AND HANDLE ONLY WHEN OPEN. OUTER DOOR FULLY EXPOSE GUTTER SPACE, THE WIRING RACEWAY, AND CIRCUIT BREAKER LOAD SIDE TERMINALS WHEN OPEN. EACH DOOR MUST BE HINGED AND HAVE A FLUSH LOCKABLE HANDLE.

R. PANELBOARD DOOR HINGES MUST BE CONCEALED BUTT OR FULL LENGTH PIANO HINGES.

S. PANELBOARDS MUST BE SOURCED FROM INDUSTRIAL ELECTRIC MANUFACTURING (IEM).

T. ELECTRICAL PANELS MUST BE COPPER BUS, FULLY BUSSED WITH MINIMUM 30 TO MAXIMUM 42 CIRCUIT. DEAD-FRONTED SAFETY TYPE NEUTRAL BUS MUST BE FULL-SIZED AND FULLY RATED AND EQUAL TO THE PHASE BUS RATING. CIRCUIT BREAKERS MUST BE MOLDED CASE BOLT-ON TYPE. CIRCUIT BREAKERS MUST BE PROVIDED WITH LOCKING DEVICES SO THAT EACH CIRCUIT BREAKER MAY BE LOCKED IN THE OPEN POSITION. ENSURE THAT DEVICES CANNOT BE MOVED FROM THE FRONT OF PANEL WITH TRIM IN PLACE, AND THAT THE DEVICE IS FIRMLY INSTALLED.

U. SHORT CIRCUIT RATING OF EACH CIRCUIT BREAKER FOR NEW PANELBOARDS MUST BE MINIMUM 14 KAIC RMS SYMMETRICAL FOR 480/277V AND 22 KAIC RMS SYMMETRICAL FOR 240V OR LESS.

W. DRY-TYPE TRANSFORMER WINDINGS MUST BE COPPER, FULLY RATED WITH NO MOVING PARTS, INTERNAL SWITCH OR CIRCUIT BREAKER.

X. ALL CIRCUIT BREAKERS MUST BE MOLDED CASE BOLT-ON TYPE. PLUG-IN CIRCUIT BREAKERS ARE NOT ALLOWED. MULTI-POLE CIRCUIT BREAKERS MUST HAVE A COMMON TRIP MECHANISM. HANDLE TIES ARE NOT ALLOWED.

Y. MOTOR STARTERS MUST BE PROVIDED WITH THE FOLLOWING INDICATING LIGHT COLORS:
- WHITE: INDICATES CONTROL POWER IS ON (LABELED AS "POWER ON")
- GREEN: INDICATES MOTOR IS RUNNING (LABELED AS "RUNNING")
- AMBER: INDICATES OVERLOAD RELAY IS TRIPPED (LABELED AS "OL TRIPPED")
- RED: INDICATES MOTOR STOPPED (LABELED AS "STOPPED")

Z. PROVIDE 600 VOLT CONDUCTORS WITH TYPE THHN/THWN-2 INSULATED RATED FOR 90°C IN BOTH DRY AND WET LOCATIONS. MC CABLE IS PERMITTED TO BE THHN/THWN. ALL CONDUCTORS MUST BE STRANDED FOR ALL SIZES.

AA. PROVIDE COLOR-CODED, INSULATED CONDUCTORS FOR ALL SIZES. WHERE NEUTRAL CONDUCTORS OF DIFFERENT VOLTAGE SYSTEMS ARE MIXED IN THE SAME BOX, RACEWAY, OR ENCLOSURE, THE NEUTRALS MUST BE IDENTIFIED TO DIFFERENTIATE THEM. TRAVELER WIRING MUST BE UNIQUELY IDENTIFIED.

	208Y/120V 3Ø 4W	480Y/277V 3Ø 4W	240/120V 2Ø 3W
A PHASE	BLACK	BROWN	BLACK
B PHASE	RED	YELLOW	BLUE
C PHASE	BLUE	PURPLE	
NEUTRAL	WHITE/GRAY	WHITE/GRAY	WHITE

EXECUTION

A. ELECTRICAL INSTALLATIONS MUST BE INSPECTED BY BUILDING AUTHORITY HAVING JURISDICTION (BAHJ) OR BUILDING AUTHORITY HAVING JURISDICTION FIELD REPRESENTATIVE (BAHJFR). COORDINATE INSPECTION EFFORTS WITH THE STR.

B. FIELD VERIFY EXISTING CONDITIONS, INCLUDING LOCATION OF ELECTRICAL EQUIPMENT.

C. MAINTAIN AT LEAST 6 INCHES CLEARANCE BETWEEN ELECTRICAL CONDUITS AND OTHER SYSTEM PIPING AND TUBING. MAINTAIN 12 INCHES CLEARANCE BETWEEN ELECTRICAL CONDUIT AND STEAM PIPING. MAINTAIN 1 INCH CLEARANCE BETWEEN ELECTRICAL CONDUIT AND DUCTWORK. MAINTAIN 8 INCHES CLEARANCE BETWEEN ELECTRICAL CONDUIT AND ROOF. INSTALL HORIZONTAL RACEWAY RUNS ABOVE WATER AND STEAM PIPING.

D. PROTECT AND REPAIR ADJACENT EXISTING SURFACES AND UTILITIES AND AREAS DAMAGED AS A RESULT OF DEMOLITION OR NEW WORK. PATCH, REPAIR, AND FINISH SURFACES TO MATCH EXISTING ADJACENT SURFACES.

E. PERFORM CUTTING AND DEMOLITION BY METHODS THAT WILL NOT JEOPARDIZE THE STRUCTURAL INTEGRITY OF THE BUILDING AND THAT WILL NOT DAMAGE PORTIONS TO REMAIN.

F. IN ADDITION TO THOSE THAT MAY BE SHOWN ON THE DRAWINGS, PROVIDE PULL BOXES AND JUNCTION BOXES AS NEEDED TO FACILITATE THE INSTALLATION OF WIRING. EXTENSION BOXES ARE NOT ALLOWED ON NEW CONSTRUCTION.

G. PROVIDE DEDICATED NEUTRAL CONDUCTOR FOR EACH BRANCH CIRCUIT. ALL NEUTRALS FOR BRANCH CIRCUITS MUST BE CONSIDERED CURRENT-CARRYING CONDUCTORS.

H. PROVIDE GALVANIZED RIGID METAL CONDUIT AND FITTINGS IF EXPOSED AND WHEN INSTALLED IN AREAS SUBJECT TO PHYSICAL DAMAGE.

I. PROVIDE ELECTRICAL POWER WIRING FOR THE HVAC SYSTEM AND OTHER MECHANICAL, PLUMBING, AND FIRE PROTECTION EQUIPMENT, INCLUDING WIRING THROUGH LINE VOLTAGE CONTROL DEVICES. COORDINATE WITH DRAWINGS AND PROVIDE NECESSARY CONTROL WIRING.

J. COORDINATE EXACT LOCATIONS OF HVAC UNITS, DISCONNECTS, AND DEVICES IN FIELD. VERIFY AND CONFIRM THE ACTUAL MOUNTING LOCATION OF THE HVAC UNIT FOR PLACEMENT OF THE DISCONNECT.

K. DISCONNECT SWITCH MUST BE MOUNTED ON WALL OR STRUT, INDEPENDENT OF THE EQUIPMENT AND ANY OTHER STRUCTURES (FREE STANDING). LOCATION MUST COMPLY WITH NEC DEPTH AND HEIGHT OF WORKING SPACE.

L. EXISTING PANEL SCHEDULES FOR PANELS AFFECTED BY THIS PROJECT ARE SHOWN ON THE DRAWINGS. UPDATE EXISTING PANEL SCHEDULES FOR PANELS AFFECTED BY THIS PROJECT. PROVIDE NEW PANEL SCHEDULES FOR PANELS INCLUDED IN THIS PROJECT.

M. CIRCUIT NUMBERS TO EXISTING PANELS AS SHOWN ARE FOR REFERENCE. FIELD TRACE DOWN AND VERIFY CIRCUITS BEFORE WORK STARTS. FIELD RED LINE FINDINGS ON DRAWING FOR RECORD. PANEL SCHEDULE CHANGES ARE RED FOR DELETE AND GREEN FOR ADDITION.

N. MOUNT OUTLET BOXES FOR NOTED DEVICES AS FOLLOWS UNO. MEASUREMENTS ARE TO TOP OF BOX UNO.:

- 120V POWER RECEPTACLES AT 15" AFF TO BOTTOM OF BOX
- DATA OUTLETS AT 36" AFF
- TELEPHONE OUTLETS AT 48" AFF
- LIGHT SWITCHES AT 48" AFF
- COUNTERTOP POWER RECEPTACLES AND TELE/DATA OUTLETS AT 8" ABOVE COUNTER OR BACKSPLASH IF SPECIFIED.

O. DO NOT MOUNT OUTLETS OR BOXES BACK-TO-BACK. MOUNT OUTLETS AND BOXES MINIMUM HORIZONTAL DISTANCE OF ONE STUD BAY OR 16" APART, OR 24" APART IN FIRE-RATED WALLS.

P. ELECTRICAL COMPONENTS INSTALLED BEHIND LADDERS: DO NOT CREATE OBSTRUCTIONS THAT VIOLATE 29 CFR 1910.23(D)(2). PROVIDE AT LEAST 7 INCHES OF PERPENDICULAR CLEAR SPACE FROM THE FACE OF NEW WORK TO A PLANE RUNNING THROUGH THE LONGITUDINAL CENTERLINES OF THE LADDER RUNGS.)

Q. UNLESS NOTED OTHERWISE PERFORM THE FOLLOWING WORK FOR ANY DEVICE OR EQUIPMENT IDENTIFIED TO BE DEMOLISHED AS A PART OF THIS PROJECT SCOPE:

- REMOVE ALL EXPOSED CONDUIT AND ALL WIRING BACK TO SOURCE PANELBOARD OR NEXT ACTIVE DEVICE TO REMAIN IN SERVICE.
- CONCEALED OR UNDERGROUND CONDUIT MUST BE ABANDONED IN PLACE. CUT CONCEALED CONDUIT FLUSH WITH SURFACE. CUT AND CAP UNDERGROUND CONDUIT 6" ABOVE FLOOR.

R. EXISTING LIGHTNING CIRCUITS MAY HAVE SHARED NEUTRALS. FIELD VERIFY ALL EXISTING LIGHTING CIRCUITS WITHIN PANELBOARD BEFORE DEMOLITION AND SAFE END SHARED NEUTRALS.

S. NO UNTERMINATED WIRING IS ALLOWED, AND WIRING ASSOCIATED WITH THE DEMOLITION SCOPE OF THIS PROJECT MUST NOT BE ABANDONED.

T. METAL-CLAD CABLE (MC) IS PERMITTED ONLY FOR CONCEALED APPLICATIONS IN DRY LOCATIONS SERVING BRANCH WIRING DEVICES AND LIGHTING WITH THE FOLLOWING RESTRICTIONS:

- DO NOT USE FOR CIRCUITS OVER 250V TO GROUND.
- DO NOT USE FOR HOMERUNS.
- BRANCH CIRCUIT RUNS MUST BE LIMITED TO 75 FEET.
- DO NOT USE AROUND HORIZONTAL CORNERS WHEN CONCEALED IN WALLS.
- DO NOT USE WHERE EXPOSED TO PHYSICAL DAMAGE.
- DO NOT USE IN CORROSIVE LOCATIONS.
- PROVIDE WITH ADDITIONAL ISOLATED INSULATED GROUNDING CONDUCTOR FOR ISOLATED GROUND CIRCUITS.

W. MAINTAIN CONTINUITY AND WIRING FOR EXISTING CIRCUITS AND SYSTEMS REMAINING IN SERVICE. IF CIRCUITS ARE INTERRUPTED, PROVIDE NEW WIRING TO MAINTAIN SYSTEM OPERATION.

X. PROVIDE GFCI PROTECTION FOR RECEPTACLES PER NEC.

Y. ALL NONHAZARDOUS ELECTRICAL EQUIPMENT INSTALLED INDOORS WITHIN 6 FEET FROM ANY WATER SOURCE (SINK, EYEWASH, SHOWER, ETC.) MUST BE RATED FOR INGRESS OF WATER (DRIPPING AND LIGHT SPLASHING).

Z. DO NOT ATTACH ELECTRICAL EQUIPMENT, CONDUIT OR DEVICES TO SEISMIC BRACING.

AA. DO NOT USE EQUIPMENT TO SUPPORT CONDUIT AND VICE VERSA.

AB. FOR UNDERGROUND GROUNDING ELECTRODE SYSTEM CONNECTIONS, USE IRREVERSIBLE CONNECTIONS. MECHANICAL CONNECTORS ENCASED IN CONCRETE ARE CONSIDERED IRREVERSIBLE. USE REMOVABLE COMPRESSION-TYPE CONNECTIONS AT THE GROUND RODS UNLESS BURIED.

AC. IN UNDERGROUND INSTALLATIONS, INSTALL AN ELECTRICALLY DETECTABLE WARNING TAPE ABOVE THE ENTIRE LENGTH OF THE DUCT BANK OR DIRECT BURIED CONDUIT.

AD. PROVIDE SCHEDULE 40 RIGID PVC CONDUIT, CONCRETE ENCASED, FOR UNDERGROUND INSTALLATIONS THAT ARE SUBJECT TO TRAFFIC BOTH DURING AND AFTER CONSTRUCTION.

AE. PROVIDE SCHEDULE 80 RIGID PVC CONDUIT FOR UNDERGROUND INSTALLATIONS THAT ARE PERMANENTLY IN NON-TRAFFIC AREAS.

AF. MAINTAIN AT LEAST 6 INCHES BETWEEN ELECTRICAL AND TELECOMMUNICATION UNDERGROUND CONDUITS, AND AT LEAST 12 INCHES BETWEEN ELECTRICAL OR TELECOMMUNICATION UNDERGROUND CONDUITS AND OTHER SYSTEM PIPING.

AG. VERTICAL STUB-UPS, HORIZONTAL BENDS AND OFF-SETS GREATER THAN 22 DEGREES IN THE DUCT BANK RUN MUST USE RIGID STEEL ELBOWS AND RISERS, WRAPPED WITH TWO LAYERS OF HALF-LAPPED, 10MIL THICK, BLACK VIRGIN POLYETHYLENE TAPE, OR FACTORY APPLIED EPOXY PVC COATING.

AH. JOINT UTILITY TRENCHES MUST ONLY CONTAIN ELECTRICAL AND TELECOMMUNICATION CONDUITS. NO OTHER SYSTEMS SHOULD BE LOCATED IN THE SAME TRENCH AS ELECTRICAL OR TELECOM.

AJ. OCCUPANCY SENSORS MUST NOT BE INSTALLED WITHIN 4 FEET OF ANY MECHANICAL DIFFUSERS.

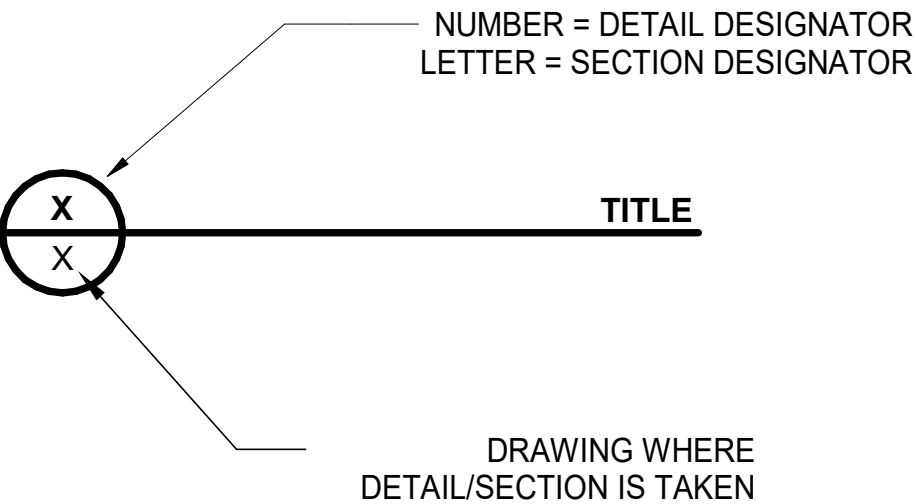
AK. PHOTOCELLS MUST NOT BE INSTALLED FACING INTO ELECTRICAL LIGHT THAT IS BEING CONTROLLED.

AL. DO NOT SPLICE FEEDER CONDUCTORS.

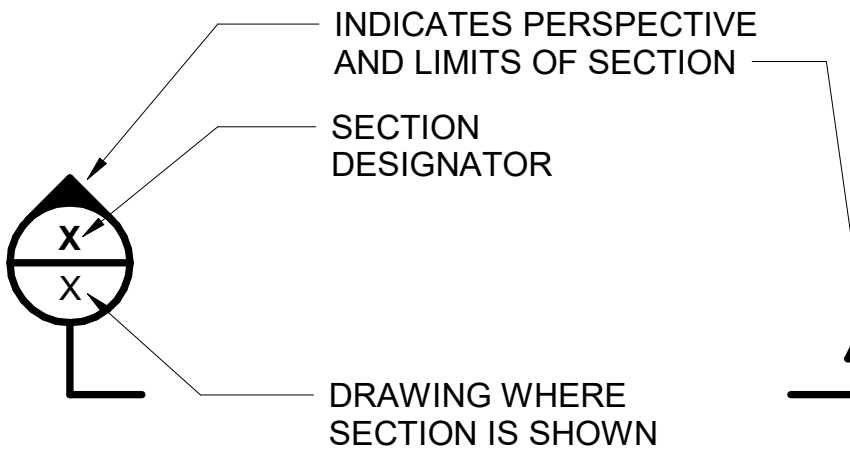
AM. DO NOT SPLICE CONDUCTORS IN PANELBOARDS, MOTOR CONTROL CENTERS, DISTRIBUTION PANELS, SWITCHBOARD, OR SWITCHGEAR.

AN. DO NOT USE GUTTERS FOR BRANCH CIRCUIT COLLECTION AT PANELBOARDS.

ELECTRICAL LEGENDS



SECTION CUT SYMBOL

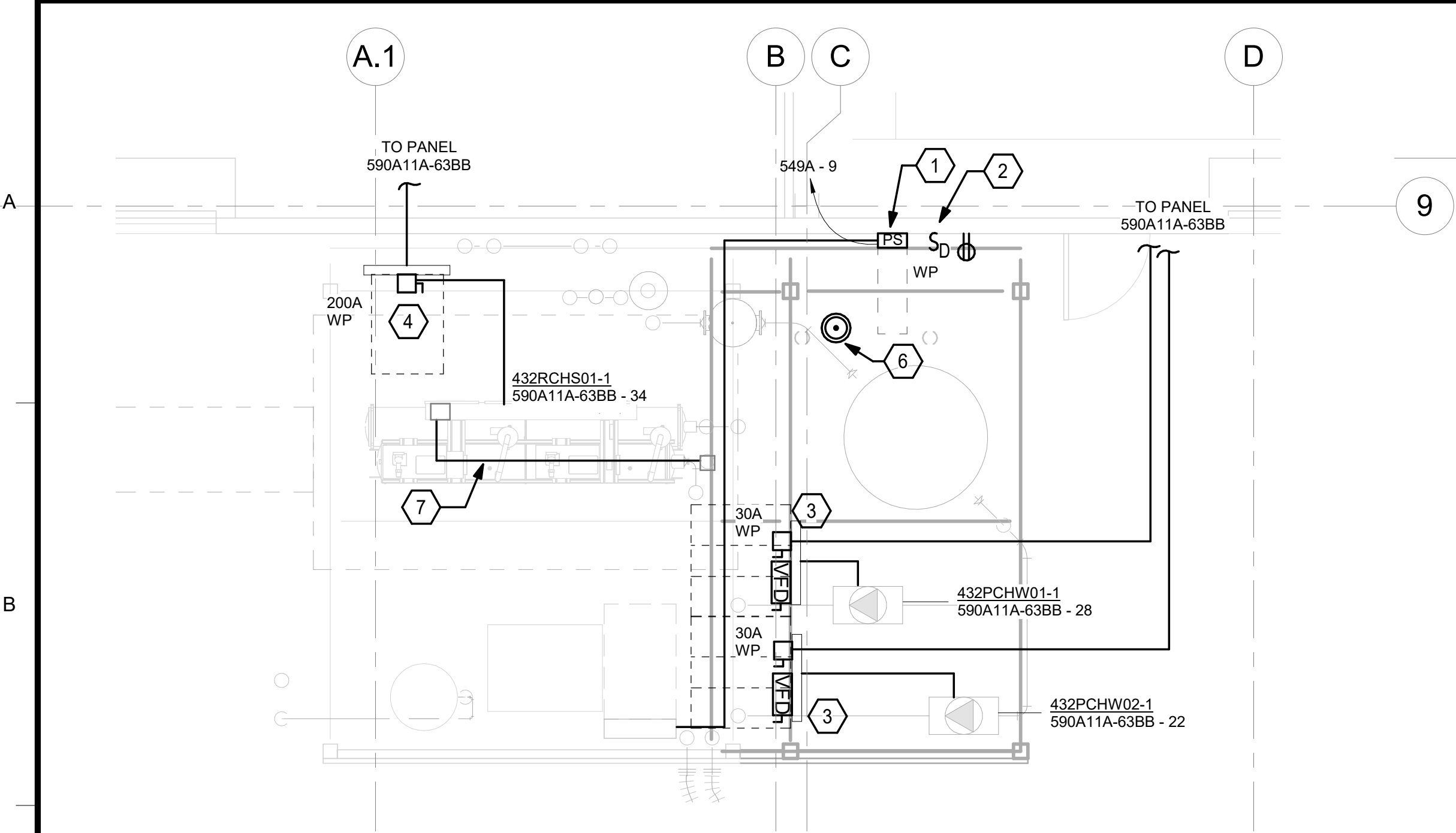


Sheet Title
ELECTRICAL NOTES

Dwg. No.
PLE2021-0432-0002D

Sht. No.
E-002

A	GENERAL	TESTING	COMMISSIONING	CLOSEOUT DOCUMENTATION
	A. BASIS OF ACCEPTANCE 1. BASIS FOR ACCEPTANCE IS COMPLIANCE WITH REQUIREMENTS SET FORTH IN SUBCONTRACT, AND OTHER TERMS AND CONDITIONS OF SUBCONTRACT. CORRECT REJECTED DELIVERABLE ITEMS IN ACCORDANCE WITH APPLICABLE CLAUSES.	A. SCOPE OF TESTING 1. PROVIDE THIRD PARTY ENGINEERING TECHNICIAN, CERTIFIED BY A NATIONAL ORGANIZATION AND WITH RELEVANT EXPERIENCE, FOR ELECTRICAL ACCEPTANCE AND FUNCTION TESTS TO ENSURE TOTAL SYSTEM OPERATION OF PROVIDED ELECTRICAL EQUIPMENT IS OPERATIONAL WITHIN INDUSTRY AND MANUFACTURERS' TOLERANCES, AND AS SPECIFIED.	A. SCOPE OF COMMISSIONING 1. ENSURE THAT SYSTEMS ARE COMPLETE, FUNCTIONING IN ACCORDANCE WITH THE DESIGN INTENT. 2. PROVIDE DOCUMENTED CONFIRMATION THAT THE FACILITY FULFILLS THE FUNCTIONAL AND PERFORMANCE REQUIREMENTS. ESTABLISH DESIGN INTENT/CRITERIA; VERIFY AND DOCUMENT COMPLIANCE WITH THESE CRITERIA THROUGHOUT CONSTRUCTION, START-UP, AND THE INITIAL PERIOD OF OPERATION. COMPLETE OPERATION AND MAINTENANCE MANUALS. 3. DEVELOP FUNCTIONAL TEST PROCEDURES FOR COMMISSIONED EQUIPMENT AND ASSEMBLIES, AND DEVELOP FUNCTIONAL TEST REPORT FOR REVIEW AND APPROVAL. 4. ENSURE THAT COMMISSIONING ACTIVITIES ARE INCLUDED IN THE PROJECT SCHEDULES, KEPT UP TO DATE, AND COMMUNICATED TO THE STR. 5. DEVELOP A START-UP AND INITIAL SYSTEMS CHECK PLAN FOR SELECTED EQUIPMENT. 6. DOCUMENT SYSTEMS START-UP; REVIEW START-UP REPORTS. 7. COORDINATE FUNCTIONAL TESTING FOR COMMISSIONED SYSTEMS AND ASSEMBLIES. 8. AFTER MANUAL TESTING AND INITIAL TROUBLE-SHOOTING IS COMPLETE, MONITOR SYSTEM OPERATION AND PERFORMANCE FOR SELECTED DATA POINTS FOR UP TO ONE WEEK BY REQUESTING TREND LOGS. ANALYZE MONITORED DATA TO VERIFY OPERATION AND PERFORMANCE AND ISSUE A REPORT (INCLUDE IN FINAL COMMISSIONING REPORT). 9. REVISE CONSTRUCTION PHASE COMMISSIONING PLANS AS NECESSARY. 10. COMPILE A COMMISSIONING RECORD. 11. EQUIPMENT MANUFACTURER, IF APPLICABLE, MUST BE ENGAGED DURING COMMISSIONING AND START-UP. B. COMMISSIONING REQUIREMENTS 1. IMPLEMENT FINAL ADJUSTMENTS ON PROTECTIVE DEVICE SETTINGS PER COORDINATION STUDY AND REPORT. 2. LIGHTING SYSTEM: PERFORM THE FOLLOWING FUNCTION TESTS FOR APPLICABLE LIGHTING CONTROL DEVICES: a. OCCUPANCY SENSORS: SIMULATE OCCUPIED AND UNOCCUPIED STATES. VERIFY OPERATION MATCHES DESIGN DRAWING INTENT. LIGHTS MUST TURN ON, OFF, OR DIM WITHOUT FLICKER. b. DIMMABLE WALL SWITCHES: TEST DIMMABLE STATES FOR FULL RANGE OF DIMMING SWITCH CAPABILITIES. LIGHTS MUST NOT FLICKER AT ANY POINT. c. INDOOR PHOTO SENSORS: PERFORM FULL, PARTIAL, AND NO DAYLIGHT TESTS TO ENSURE PROPER OPERATION. DAYLIGHT SENSORS MUST BE COORDINATED WITH SHADE OPERATION IF APPLICABLE. d. OUTDOOR PHOTO SENSORS: VERIFY LIGHTS TURN ON WHEN NO DAYLIGHT IS AVAILABLE. 3. MEASURE AND VERIFY LIGHTING LEVEL PER DESIGN DRAWINGS, AND SET ALL ADJUSTMENTS ON LIGHTING DEVICES AND LIGHTING CONTROL PANELS PER DESIGN DRAWINGS OR MANUFACTURER REQUIREMENTS. PROVIDE THE FOLLOWING DATA: AVERAGE LIGHT LEVEL FOR EACH SPACE TYPE FROM A MINIMUM OF NINE POINTS SPREAD OUT UNIFORMLY IN THE SPACE. 4. AFTER INSTALLATION AND INSPECTION APPROVAL, BUT BEFORE APPLYING CIRCUIT LABELS, VERIFY BRANCH CIRCUIT RECEPTACLES ARE CONNECTED TO APPROPRIATE CIRCUIT BY OPENING CORRESPONDING BRANCH CIRCUIT BREAKER AND TESTING FOR VOLTAGE. RECORD CONNECTED BRANCH CIRCUIT IF DIFFERENT TO WHAT IS ON PLAN. BRANCH CIRCUITING MUST FOLLOW DESIGN INTENT, LATEST BULLETIN, AND UPDATED PANEL SCHEDULES.	A. AT COMPLETION OF PROJECT, THE FOLLOWING DOCUMENTS MUST BE PROVIDED: 1. ALL CONSTRUCTION DOCUMENT MARKUPS MUST BE PROVIDED TO THE DESIGNER OF RECORD, EVALUATED, AND INCORPORATED INTO THE FINAL RECORD DOCUMENTS. 2. ALL FINAL SHOP DRAWINGS 3. ALL WARRANTY DOCUMENTATION FOR EQUIPMENT 4. ALL UPDATED PANEL SCHEDULES B. PROVIDE TRAINING FOR NEW SYSTEMS TO RELEVANT PERSONNEL
B				
C		B. ELECTRICAL TESTS 1. ONLY THE FOLLOWING TESTS ARE REQUIRED. SECTIONS NOTED ARE FROM NETA ATS: a. LOW-VOLTAGE CIRCUIT BREAKERS: VISUAL AND MECHANICAL INSPECTIONS AND TESTS PER SECTION 7.6.1.1(A) & (C) ONLY. b. LOW-VOLTAGE, 250 AMP FRAME AND HIGHER CIRCUIT BREAKERS: REMOVE AND DELIVER TO SITE 200 CIRCUIT BREAKER TESTING FACILITY. c. GROUNDING SYSTEM PER SECTION 7.13. d. LOW-VOLTAGE CABLES, 600 VOLTS MAXIMUM: VISUAL AND MECHANICAL INSPECTIONS AND TESTS PER SECTION 7.3.2 FOR ALL CONDUCTORS. ELECTRICAL INSPECTIONS AND TESTS PER SECTION 7.3.2 FOR CONDUCTORS #1 AWG AND LARGER. FIELD MEGGER AND TORQUE TESTS WITH REPORTS. USE BOTH LOW-RESISTANCE OHMMETER AND TORQUE-WRENCH METHOD WHEN INSPECTING BOLTED ELECTRICAL CONNECTIONS FOR HIGH RESISTANCE PER SECTION 7.3.2.A.3. DO NOT USE THERMOGRAPHIC SURVEY. e. BOLT-TORQUE VALUES FOR ELECTRICAL CONNECTIONS MUST BE IN ACCORDANCE WITH MANUFACTURERS PUBLISHED DATA. IN ABSENCE OF MANUFACTURERS PUBLISHED DATA, USE TABLE 100.12.1 IN NETA ATS. f. LICENSED ELECTRICIAN MUST TORQUE ALL ELECTRICAL CONNECTIONS AND BEAHJ OR DELEGATED AUTHORITY MUST WITNESS. TORQUE TEST VERIFICATION MUST BE PERFORMED BY BEAHJ OR DELEGATED AUTHORITY. g. SWITCHGEAR AND SWITCHBOARD ASSEMBLIES PER SECTION 7.1. h. TRANSFORMERS PER SECTION 7.2. i. MOTOR CONTROL CENTERS AND MOTOR STARTERS PER SECTION 7.16. j. ADJUSTABLE SPEED DRIVE SYSTEMS PER SECTION 7.17. k. SURGE ARRESTERS PER SECTION 7.19. l. OUTDOOR BUS STRUCTURES PER SECTION 7.21. m. EMERGENCY SYSTEMS: ENGINE GENERATOR, UNINTERRUPTIBLE POWER SYSTEMS, AND AUTOMATIC TRANSFER SWITCHES PER SECTION 7.22.		
D				
E				
F				



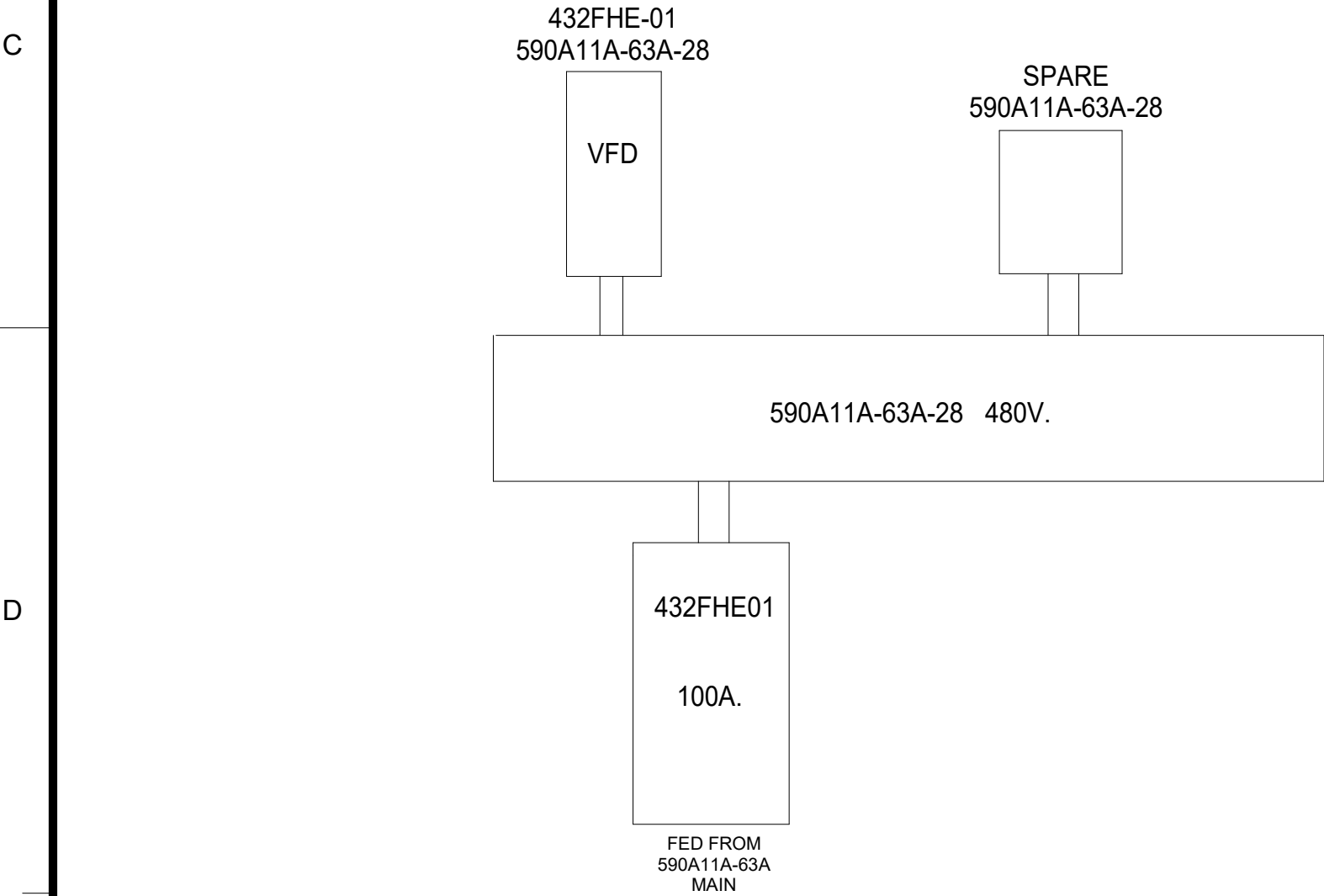
C1
E-401

ENLARGED NEW WORK PLAN - SOUTH BUILDING EXTERIOR

SCALE: 1/4" = 1'-0"

0 2' 4' 8'

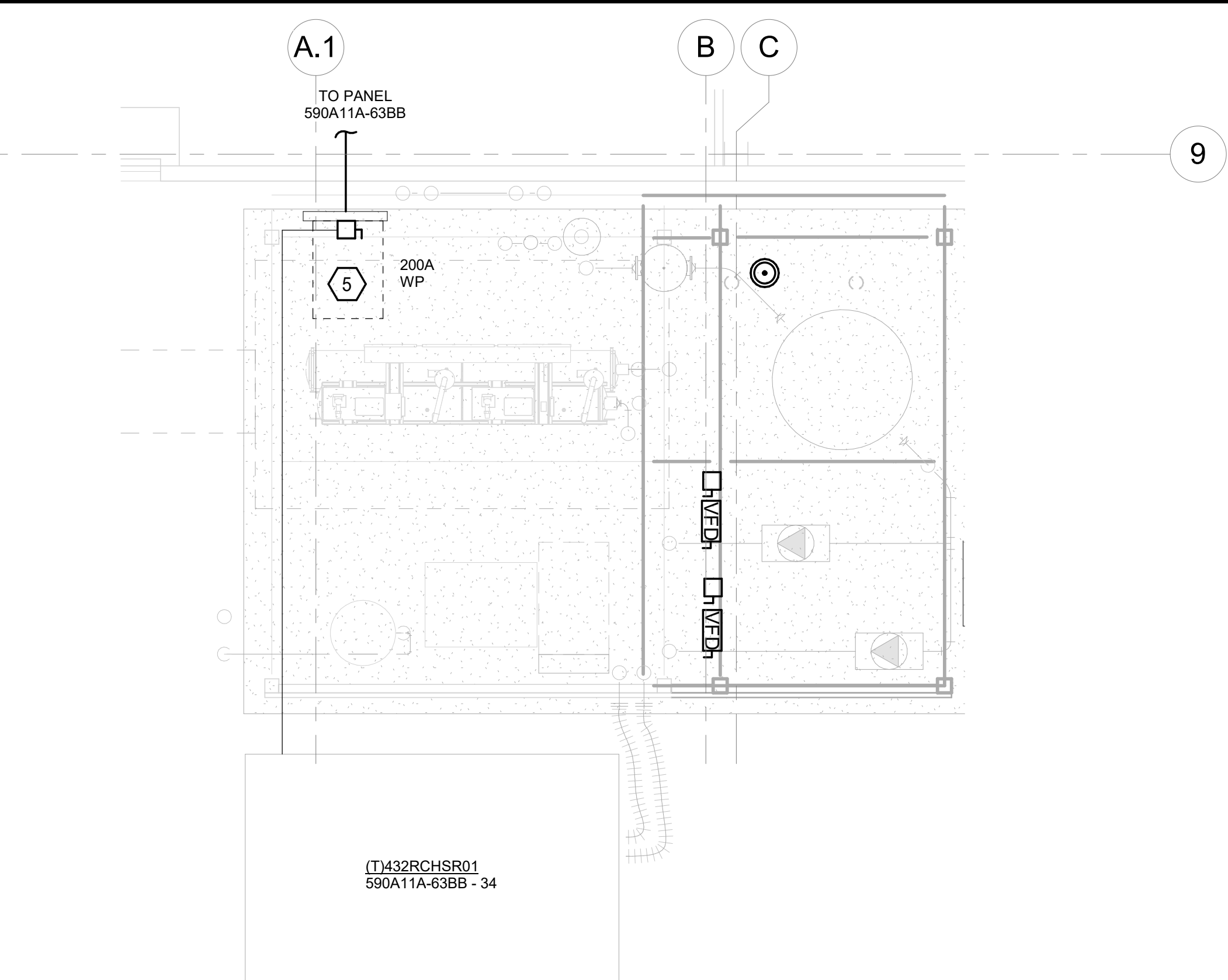
SCALE IN FEET



E1
E-401

590A11A-63A-28 NEW WORK ELEVATION

SCALE: NO SCALE SHOWN



C4
E-401

ENLARGED NEW WORK PLAN - TEMPORARY CHILLER

SCALE: 1/4" = 1'-0"

0 2' 4' 8'

SCALE IN FEET

GENERAL NOTES

1. FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS, REFER TO SHEET E-001.
2. UNLESS OTHERWISE NOTED, LIGHT LINEWEIGHT INDICATES EXISTING EQUIPMENT TO REMAIN AND HEAVY LINEWEIGHT INDICATES NEW WORK.
3. SUBCONTRACTORS MUST FIELD VERIFY LOCATION OF EXISTING EQUIPMENT AND EQUIPMENT TO BE DISCONNECTED AND REMOVED.
4. SUBCONTRACTOR MUST FIELD VERIFY CIRCUITS INDICATED AND LOADS SERVED BY THEM.
5. SUBCONTRACTORS MUST WORK CLOSELY WITH MECHANICAL SUBCONTRACTOR TO VERIFY PHASING OF DEMOLITION. THIS IS TO AVOID ANY UNNECESSARY OR UNPLANNED OUTAGES FOR EQUIPMENT OR THE FACILITY.
6. SUBCONTRACTOR MUST REFER TO DIVISION 1 SPECIFICATIONS FOR CONSTRUCTION SEQUENCING AND PHASING.
7. REFER TO SHEETS E-801 AND E-802 FOR ADDITIONAL WORK AT ELECTRICAL PANELS.
8. IT HAS BEEN FIELD VERIFIED THAT THERE ARE EXISTING MAINTENANCE RECEPTACLES WITHIN 25' OF MECHANICAL EQUIPMENT.
9. SUBCONTRACTOR MUST VERIFY THAT NEW ELECTRICAL EQUIPMENT HAVE PROPER WORKING CLEARANCE PRIOR TO INSTALLATION.
10. ALL NEW CONDUITS MUST BE PAINTED TO MATCH EXISTING IN AREA.
11. ALL CONDUCTORS FROM VFD TO MOTOR MUST BE SYMMETRICAL SHIELDED VFD CABLE TYPE INSTALLED IN CONDUIT.
12. LFMC CONNECTIONS CAN ONLY BE 6' MAX IN LENGTH AND MUST BE SECURED AND SUPPORTED WITHIN 12" OF TERMINATIONS AND NO MORE THAN 4'-6" OTHERWISE.
13. VERIFY EXISTING CONDITIONS MEET NEC STANDARDS.

KEYED NOTES

1. 120V/24V TRANSFORMER FOR MECHANICAL CONTROLS. PROVIDE FUNCTIONAL DEVICES #PSH500AB10-LVC OR APPROVED EQUAL. PROVIDE ADDITIONAL NEMA 3R ENCLOSURE TO SURROUND THE NEMA 1 ENCLOSURE.
2. PROVIDE A LOCKABLE TOGGLE SWITCH FOR 120V/24V TRANSFORMER.
3. INSTALL DISCONNECT AND VFD ON UNISTRUT STAND. PROVIDE AUXILIARY CONTACTS IN VFD DISCONNECTS. REFER TO STRUCTURAL DETAIL SHEETS FOR ADDITIONAL INFORMATION.
4. INSTALL DISCONNECT ON NEW UNISTRUT STAND. REFER TO STRUCTURAL SHEETS FOR ADDITIONAL INFORMATION.
5. DISCONNECT MUST FEED TEMPORARY CHILLER DURING CONSTRUCTION. REUSE DISCONNECT TO FEED PERMANENT CHILLER UPON REMOVAL OF TEMPORARY CHILLER. EXTEND CONDUCTORS TO NEW CHILLER LOCATON.
6. EXISTING GROUND WELL TO BE RAISED TO BE ACCESSABLE FROM NEW PAD. VERIFY REMOVABLE CLAMPS WILL REMAIN ACCESSIBLE AND REMOVABLE.
7. PROVIDE CONDUIT AND CABLING FOR 120V CONNECTION FROM CHILLER CONTROL POWER TRANSFORMER TO FIELD INSTALLED CONDENSER CONTROL VALVE.

ELECTRICAL MECHANICAL NEW WORK SCHEDULE											
EQUIPMENT NAME	LOCATION	Phase	VOLTAGE	MCA / LOAD	OCPD	PANEL	CIRCUIT	FEEDER	DISCONNECT SIZE	DISCONNECT RATING	REMARKS
(T)432RCHSR01	S. BUILDING EXTERIOR	3	480 V	179 A	200	590A11A-63BB	34	(3) #250, #6G, 2 1/2"C	200A NONFUSIBLE	NEMA 3R	PROVIDE PADLOCKABLE, HEAVY DUTY DISCONNECT SWITCH.
432ACHPS01-4	E. BUILDING EXTERIOR	3	480 V	54 A	60	(N) 590A11B	4	(3) #6, #10G, 1"C	60A NONFUSIBLE	NEMA 3R	PROVIDE PADLOCKABLE, HEAVY DUTY DISCONNECT SWITCH.
432ACHPS02-5	E. BUILDING EXTERIOR	3	480 V	52 A	60	(N) 590A11B	3	(3) #6, #10G, 1"C	60A NONFUSIBLE	NEMA 3R	PROVIDE PADLOCKABLE, HEAVY DUTY DISCONNECT SWITCH.
432ACU01-A1 CIRCUIT 1	W. BUILDING EXTERIOR	3	480 V	20 A	25	590A11A-63A-45	4	(3) #10, #10G, 3/4"C	30A NONFUSIBLE	NEMA 3R	PROVIDE PADLOCKABLE, HEAVY DUTY DISCONNECT SWITCH.
432ACU01-A1 CIRCUIT 2	W. BUILDING EXTERIOR	3	480 V	60 A	70	(N) 590A11B	21	(3) #3, #6G, 1"	100A NONFUSIBLE	NEMA 3R	PROVIDE PADLOCKABLE, HEAVY DUTY DISCONNECT SWITCH.
432ACU08-2H	RM 2007	1	208 V	1 A	20	POWERED BY 432RCU01-2H	(2) #12, #12G, 3/4"C		30A NONFUSIBLE	NEMA 1	SINGLE POINT POWER CONNECTION WITH 432RCUA02-3H. PROVIDE PADLOCKABLE, HEAVY DUTY DISCONNECT SWITCH.
432ACU9-3H	RM 2005	1	208 V	1 A	20	POWERED BY 432RCU02-3H	(2) #12, #12G, 3/4"C		30A NONFUSIBLE	NEMA 1	SINGLE POINT POWER CONNECTION WITH 432RCUA02-3H. PROVIDE PADLOCKABLE, HEAVY DUTY DISCONNECT SWITCH.
432HEA01-A1	RM 1200	3	480 V	25 A	35	590A11A-63A-39	15	(3) #10, #10G, 3/4"	60A NONFUSIBLE	NEMA 1	PROVIDE PADLOCKABLE, HEAVY DUTY DISCONNECT SWITCH.
432HEA02-4	E. BUILDING EXTERIOR	3	480 V	13 A	30	590A11B-16	28	(3) #12, #12G, 3/4"	30A NONFUSIBLE	NEMA 3R	PROVIDE PADLOCKABLE, HEAVY DUTY DISCONNECT SWITCH.
432HEA03-5	E. BUILDING EXTERIOR	3	480 V	13 A	30	590A11A-63A-39	28	(3) #12, #12G, 3/4"	30A NONFUSIBLE	NEMA 3R	PROVIDE PADLOCKABLE, HEAVY DUTY DISCONNECT SWITCH.
432PCHW01-1	S. BUILDING EXTERIOR	3	480 V	21 A	30	590A11A-63BB	28	(3) #10, #10G, 1"C	30A NONFUSIBLE	NEMA 3R	VFD SHALL BE ABB ACH550 WITH ELECTRONIC BYPASS. PROVIDE PADLOCKABLE, HEAVY DUTY DISCONNECT SWITCH.
432PCHW02-1	S. BUILDING EXTERIOR	3	480 V	21 A	30	590A11A-63BB	22	(3) #10, #10G, 1"C	30A NONFUSIBLE	NEMA 3R	VFD SHALL BE ABB ACH550 WITH ELECTRONIC BYPASS. PROVIDE PADLOCKABLE, HEAVY DUTY DISCONNECT SWITCH.
432RCHS01-1	S. BUILDING EXTERIOR	3	480 V	108 A	150	590A11A-63BB	34	(3) #2/0, #6G, 2"C	200A NONFUSIBLE	NEMA 3R	PROVIDE PADLOCKABLE, HEAVY DUTY DISCONNECT SWITCH.
432RCUA01-2H	E. BUILDING EXTERIOR	1	208 V	19 A	20	799A47	19	(2) #10, #10G, 3/4"C	30A NONFUSIBLE	NEMA 3R	PROVIDE 2-POLE PADLOCKABLE, NEMA 3R, MOTOR_RATED TOGGLE SWITCH. HUBBELL CIRCUIT-LOCK OR APPROVED EQUAL.
432RCUA02-3H	E. BUILDING EXTERIOR	1	208 V	11 A	20	799A47	28	(2) #10, #10G, 3/4"C	30A NONFUSIBLE	NEMA 3R	PROVIDE 2-POLE PADLOCKABLE, NEMA 3R, MOTOR_RATED TOGGLE SWITCH. HUBBELL CIRCUIT-LOCK OR APPROVED EQUAL.

LOAD AND MOCF PER MECHANICAL EQUIPMENT MANUFACTURER'S SPECIFICATIONS

Sheet Title

**ELECTRICAL ENLARGED
PLANS - NEW WORK**

Dwg. No.

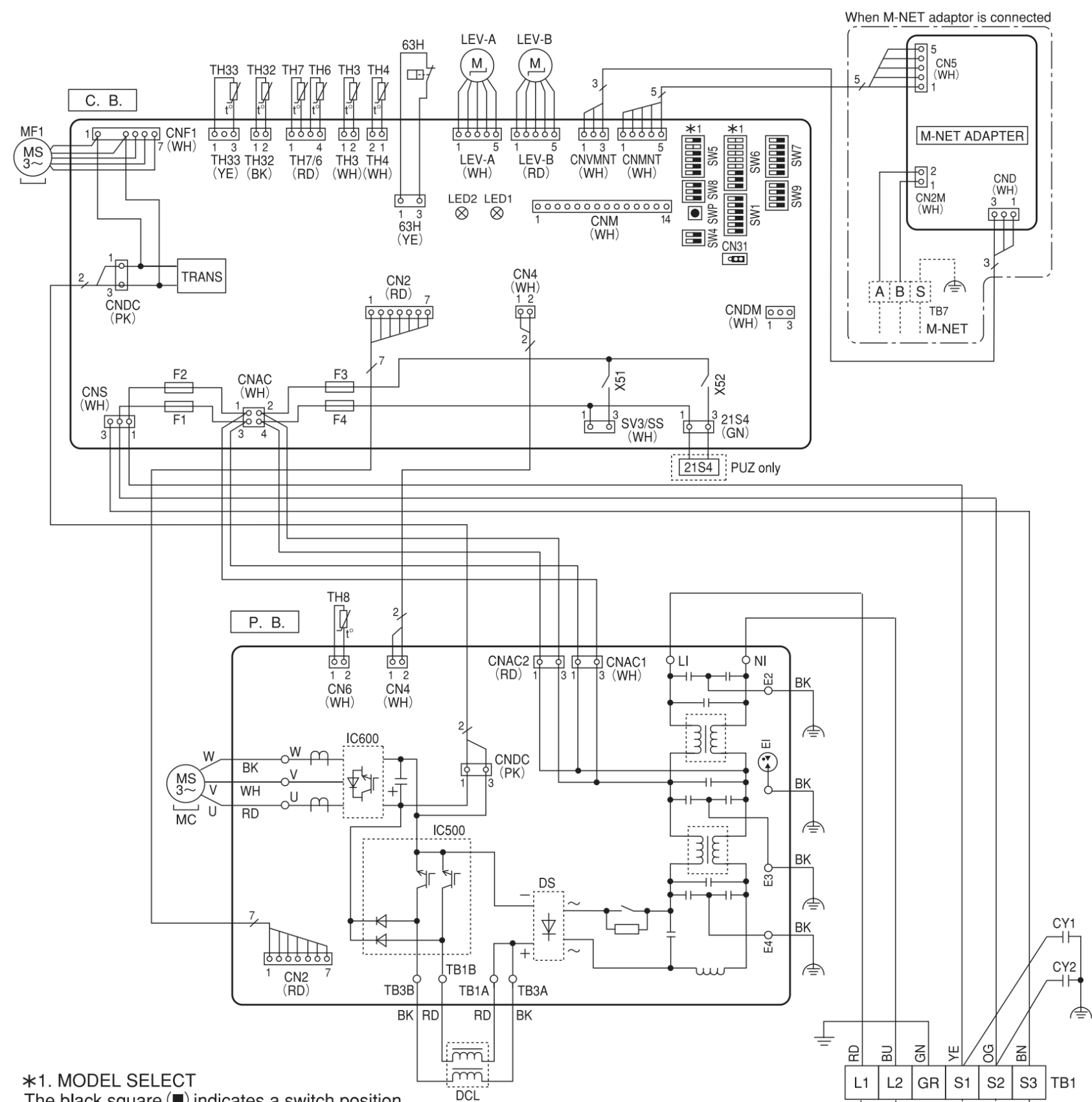
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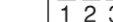
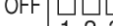
Sht. No.

E-401

PUY-A30NHA7
PUY-A30NHA7-BS

SYMBOL	NAME	SYMBOL	NAME	SYMBOL	NAME
TB1	Terminal Block (Power Supply, Indoor/Outdoor)	TH33	Thermistor (Comp. Surface)	SW5	Switch (Function Switch, Model Select)
MC	Motor for Compressor	LEV-A, LEV-B	Linear Expansion Valve	SW6	Switch (Model Select)
MF1	Fan Motor	DLA	Reactor	SW7	Switch (Function Switch)
2154	Solenoid Valve (4-Way Valve)	CY1, CY2	Capacitor	SW8	Switch (Function Switch)
63H	High Pressure Switch	P. B.	Power Circuit Board	SW9	Switch (Function Switch)
TH3	Thermistor (Liquid)	C. B.	Controller Circuit Board	SWP	Switch (Pump Down)
TH4	Thermistor (Discharge)	F1, F2	Fuse (1TADL250V)	CNM	Connector (Connection for Operation)
TH6	Thermistor (2-Phase Pipe)	F3, F4	Fuse (16.3AL250V)	CN3	Connector (Emergency Operation)
TH7	Thermistor (Ambient)	SW1	Switch (Manual Defrost, Defect History Record Reset, Refrigerant Address)	CNM1	Connector (Connection for Operation)
TH8	Thermistor (Heat Sink)			SV/SS3	Connector (Connection for Operation)
TH32	Thermistor (Suction)	SW4	Switch (Function Switch)		



MODEL	SW6-4, 5, 6, 7, 8 *2	SW5-5 *2
PUZ-A24	ON OFF  1 2 3 4 5 6 7 8	ON OFF  1 2 3 4 5 6
PUZ-A30	ON OFF  1 2 3 4 5 6 7 8	ON OFF  1 2 3 4 5 6
PUY-A24	ON OFF  1 2 3 4 5 6 7 8	ON OFF  1 2 3 4 5 6
PUY-A30	ON OFF  1 2 3 4 5 6 7 8	ON OFF  1 2 3 4 5 6

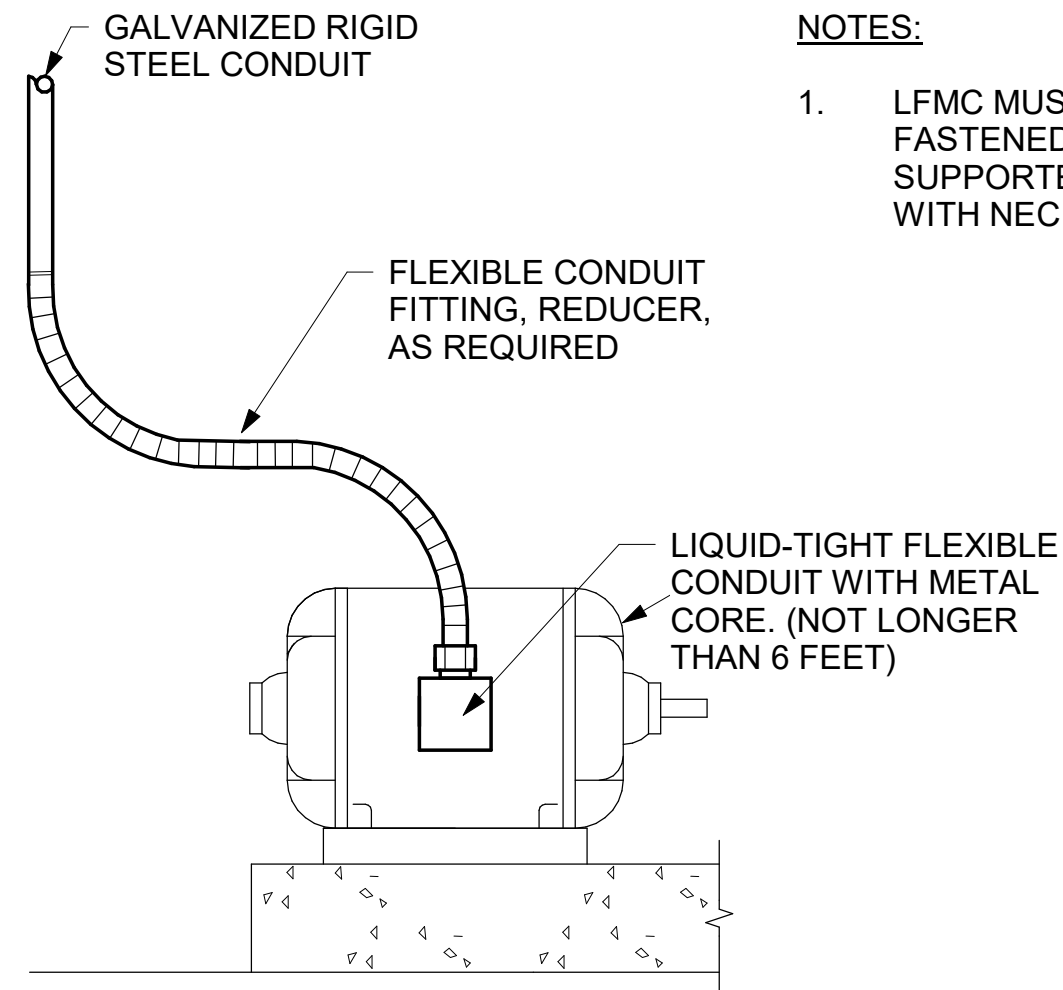
SYMBOL	NAME
TB7	Terminal Block (M-NET connection)
CN5	Connector (Transmission)
CND	Connector (Power Supply)
CN2M	Connector (M-NET communication)

POWER SUPPLY UNIT
208/230 V AC 60 Hz
*Use copper supply wires.
*Utilisez des d' simentation en cuivre.

18

E-501

SCALE:NOT TO SCALE



1. LPMC MUST BE SECURELY FASTENED IN PLACE AND SUPPORTED IN ACCORDANCE WITH NEC 350.30(A) AND (B).

E-501

TYP. OVERHEAD CONDUIT SERVICE FOR SMALL AND MEDIUM SIZE MOTORS

SCALE: NO SCALE SHOWN

1. FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS, REFER TO SHEET E-001.
2. REFER TO STRUCTURAL SHEET S-501 FOR ADDITIONAL REQUIREMENTS.

ELECTRICAL DETAILS

PLE2021-0432-0012D

E-501

432		PANEL 590A11A-63A (EXISTING)		BLDG. 432		LOCATION RM. 1111 E. WALL	
MFR G.E.	PANEL TYPE CCB	BREAKER TYPE THEF, T/FK	VOLTS 480	AMPS 600			
SUPPLY DISCONNECT CKT 590A11A-B3	LOCATION BLDG. 436 N.E. WALL		SUBSTATION 590				
BREAKER KAIC (MIN) 25K							

(OLD 590A11-25A)		PAGE 1 OF 2	
DATE	DESCRIPTION	WIRE SIZE	DESCRIPTION
	T0548 150KVA RM. 2022 W. WALL	4/0 200 TFK 4/0	PNL 590A11A-63A-4 RM. 1000 W. WALL
	432ACU02-2 RM. 2100 MEZZ	12 15 THED 12	DISCONNECT SW. RM. 1100
	432PCHW01-2 OUTSIDE NORTH	12 15 12	WIRES SAFE END IN J-BOX RM. 1300 S. WALL
	SPARE	50 21 22 70 THED	480V. 3-PH. RECEPT. RM. 1113 S.E. WALL
	480V. 3-PH. RECEPT. RM. 1000 W. WALL	1 70 THED 1	2 432FHE01 ROOF OVER RM. 1200; RM. 1200 2 432ACU01-A1 CEILING MTD. N. END
	L-650 MOORE LATHE RM. 1000	12 20 12 4/0	L-658 MOORE LATHE RM. 1000
	PNL 590A11A-63A-39 RM. 1204 N. WALL	4/0 225 4/0	432RCHSR01-2 OUTSIDE NORTH

BUS ENTRY B		CONNECTED LOAD		ARC FLASH CATEGORY		FEEDER SIZE		REVISION SUBMITTED BY:	
FDR BREAKER 600	PHASE "A" kW			2		WIRE 2 - #500MCM/MPH THW		NFA70E	
DATE INST 1970	PHASE "B" kW					NEUT NONE		J.O./W.T. 03/81/PW 03S 5200.51	
LAST PM DATE: 2015	PHASE "C" kW					GND. 1 - #410 B.C.		CAD REV BY ML	
						CONDUIT 4" C.		CAD REV DATE 9/10/15	

[illegible][illegible][illegible]

432 (EXISTING)

SCHEDULE

PANEL	828A13A GUTTER	432	LOCATION	RM. 1400 N. WALL
MFR.		PANEL TYPE	BREAKER TYPE	VOLTS .480 AMPS
SUPPLY DISCONNECT OKT.	590A11B MAIN	LOCATION	BLDG. 436 O/S N.E.	SUBSTATION 590
BREAKER KAC (MIN.)	35K			

DATE: _____

MARKED UP BY: _____

ACC. NO.: _____

828A13A 480V. RM. 1400

828A13A-8 60A. 3#4

432ACHPS01-X 828A13A-2 70A. 3#4

432ACHPS02 828A13A-3 40A. 3#8

432CRB02 RM. 1300 30A. 3#2

MAIN DISC. 590A11B 480/277V. 400A. 3#4/0

T1225 75KVA 828A13A-4 100A. 3#4/0

T1236 75KVA 828A13A-6 200A. 3#4/0

T1224 75KVA 828A13A-5 125A. 3#4/0

828A13A MAIN 600A. 828A13A-1 FED FROM 590A11B MAIN

PANEL 828A13A-7 200A. 3#4/0

FROM BLDG. 436 OUTSIDE N. WALL RM. 1477 FEEDS 828A13A MAIN

NOTE: NEUT. WIRE ENDED IN THIS MAIN SW. BOX

BUS ENTRY	CONNECTED LOAD	ARC FLASH CATEGORY	FEEDER SIZE	REVISION SUBMITTED BY:
FDR. BREAKER 400	PHASE "A" kW	4	WIRE 2 - #500MCM/PH.	NFPA70E
DATE INST.	PHASE "B" kW		NEUT. _____	J.O./W.T. 32887/PH/BS/STO/JS
LAST PM DATE: 6/1/15	PHASE "C" kW		GND. _____	CAD REV. BY ML
			CONDUIT _____	CAD REV. DATE 6/1/15

432		PANEL BOARD SCHEDULE	
PANEL 590A11B (MODIFIED)		BLDG. 432	LOCATION RM. 1400 N. WALL
MFR. G.E.	PANEL TYPE CCB	BREAKER TYPE THEF. TFK. TJK	VOLTS 480 AMPS 400
SUPPLY DISCONNECT OKT. 590A11B MAIN		LOCATION BLDG. 432 O/S N.E.	SUBSTATION 590
BREAKER KAIC (MIN.) 35K			

DATE:	DESCRIPTION	WIRE SIZE	Circuit Breaker				WIRE SIZE	DESCRIPTION
			A	B	C	D		
	432ACHPS02-5	6					6	432ACHPS01-4
	OUTSIDE EAST	6	60	3	4	60	6	OUTSIDE EAST
		6					6	
	T1225 75KVA	2					4/0	
		2	100	9	10	200	4/0	T1236 75KVA
		2					4/0	
		6					4/0	
	METER POWER	6	60	15	16	200	4/0	PNL 590A11B-16
		6					4/0	RM. 1400
		4					1	
	432ACU01-A1 CIRCUIT 2	4	70	21	22	125	1	T1236 75KVA
	W. BUILDING EXTERIOR	4					1	
		10						
	432CRB02	10	30	27	28	20		SPARE
	RM. 1300	10						
	SPACE			33	34			SPACE
	MAIN DISC	500						SPACE
	(ENCLOSED CIRCUIT BREAKER ADJACENT TO PANEL)	500	400	39	40			
		500						

BUS ENTRY FOR BREAKER _____ DATE INST. _____ LAST PM DATE: _____	CONNECTED LOAD PHASE "A" _____ kW PHASE "B" _____ kW PHASE "C" _____ kW	ARC FLASH CATEGORY _____	FEEDER SIZE WIRE _____ NEUT. _____ GND. _____ CONDUIT _____	REVISION SUBMITTED BY: J.O./W.T. _____ CAD REV. BY _____ CAD REV. DATE _____
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- ## GENERAL NOTES

1. FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS, REFER TO SHEET E-001.
2. UNLESS OTHERWISE NOTED, LIGHT LINEWEIGHT INDICATES EXISTING EQUIPMENT TO REMAIN AND HEAVY LINEWEIGHT INDICATES EQUIPMENT TO BE MODIFIED.
3. SUBCONTRACTORS MUST FIELD VERIFY LOCATION OF EXISTING EQUIPMENT AND EQUIPMENT TO BE INSTALLED.
4. SUBCONTRACTOR MUST FIELD VERIFY CIRCUITS INDICATED AND LOADS SERVED BY THEM.
5. SUBCONTRACTOR MUST UPDATE ALL MODIFIED PANELBOARD SCHEDULES.
6. PANEL SCHEDULES FOR AFFECTED PANELS ARE SHOWN. COORDINATE EQUIPMENT SHUTDOWN WITH CONSTRUCTION MANAGER / SUBCONTRACT TECHNICAL REPRESENTATIVE DURING CONSTRUCTION.
7. CONDUCTOR SPLICES ARE NOT ALLOWED WITHOUT LEAD ELECTRICAL ENGINEER APPROVAL

KEYED NOTES

1. DISCONNECT AND REMOVE EXISTING LOAD IN EXISTING PANEL. SPLICE CONDUCTORS AND REFEED TO SERVE EXISTING 432FH01 LOAD. PROVIDE NEW CONDUCTORS TO FEED NEW 432ACU01 LOAD FROM NEW PANEL 590A11B. SPLICES MUST BE APPROVED
2. DEMOLISH EXISTING GUTTER AND REPLACE WITH 480V-3PH-3W PANELBOARD. PROVIDE NEW CONDUCTORS TO RE-FEED EXISTING EQUIPMENT. EXISTING CONDUIT MUST REMAIN AND TO BE REUSE FOR NEW WORK. SPLICE EXISTING 432CRB02 LOAD AND RECONNECT TO NEW PANELBOARD. SPLICE MUST OCCUR IN SPLICE BOX. SPLICES MUST BE APPROVED

Sheet Title

ELECTRICAL PANEL SCHEDULES

Dwg. No.

PLE2021-0432-0014D

Sht. No.

E-801

[illegible][illegible][illegible][illegible][illegible]

GENERAL NOTES

1. FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS, REFER TO SHEET E-001.
2. UNLESS OTHERWISE NOTED, LIGHT LINEWEIGHT INDICATES EXISTING EQUIPMENT TO REMAIN AND HEAVY LINEWEIGHT INDICATES EQUIPMENT TO BE MODIFIED.
3. SUBCONTRACTORS MUST FIELD VERIFY LOCATION OF EXISTING EQUIPMENT AND EQUIPMENT TO BE INSTALLED.
4. SUBCONTRACTOR MUST FIELD VERIFY CIRCUITS INDICATED AND LOADS SERVED BY THEM.
5. SUBCONTRACTOR MUST UPDATE MODIFIED PANELBOARD SCHEDULES.
6. PANEL SCHEDULES FOR AFFECTED PANELS ARE SHOWN. COORDINATE EQUIPMENT SHUTDOWN WITH CONSTRUCTION MANAGER / SUBCONTRACT TECHNICAL REPRESENTATIVE DURING CONSTRUCTION.
7. CONDUCTOR SPLICES ARE NOT ALLOWED WITHOUT LEAD ELECTRICAL ENGINEER APPROVAL

KEYED NOTES

1. EXISTING LOAD TO BE DISCONNECTED AND REPURPOSED FOR NEW LOAD. EXISTING CONDUCTORS MUST BE EXTENDED TO NEW EQUIPMENT LOCATION.
2. DISCONNECT AND REMOVE EXISTING CIRCUIT BREAKER IN EXISTING PANEL. REPLACE WITH NEW 3P-480V-30A CIRCUIT BREAKER IN EXISTING PANEL TO SERVE NEW LOAD.
3. UTILIZE EXISTING SPARE 3P-480V-20A CIRCUIT BREAKER IN EXISTING PANEL TO SERVE NEW LOAD.
4. UTILIZE EXISTING SPARE 1P-120V-20A CIRCUIT BREAKER IN EXISTING PANEL TO SERVE NEW LOAD.
5. UTILIZE EXISTING BREAKER TO SERVE NEW LOAD.
6. EXISTING LOAD TO BE DISCONNECTED AND REMOVED BACK TO SOURCE.
7. DISCONNECT AND REMOVE EXISTING CIRCUIT BREAKER IN EXISTING PANEL.
8. CIRCUIT MUST FEED TEMPORARY CHILLER DURING CONSTRUCTION. INSTALL NEW 3P-480V-200A CIRCUIT BREAKER TO POWER TEMPORARY LOAD. TEMPORARY CIRCUIT BREAKER CHARACTERISTICS MUST MATCH EXISTING.
9. UPON REMOVAL OF THE TEMPORARY CHILLER DISCONNECT AND REMOVE TEMPORARY 3P-480V-200A CIRCUIT BREAKER. REPLACE WITH NEW 3P-480V-150A CIRCUIT BREAKER TO POWER NEW PERMANENT CHILLER LOAD. NEW CIRCUIT BREAKER CHARACTERISTICS MUST MATCH EXISTING.

Sheet Title

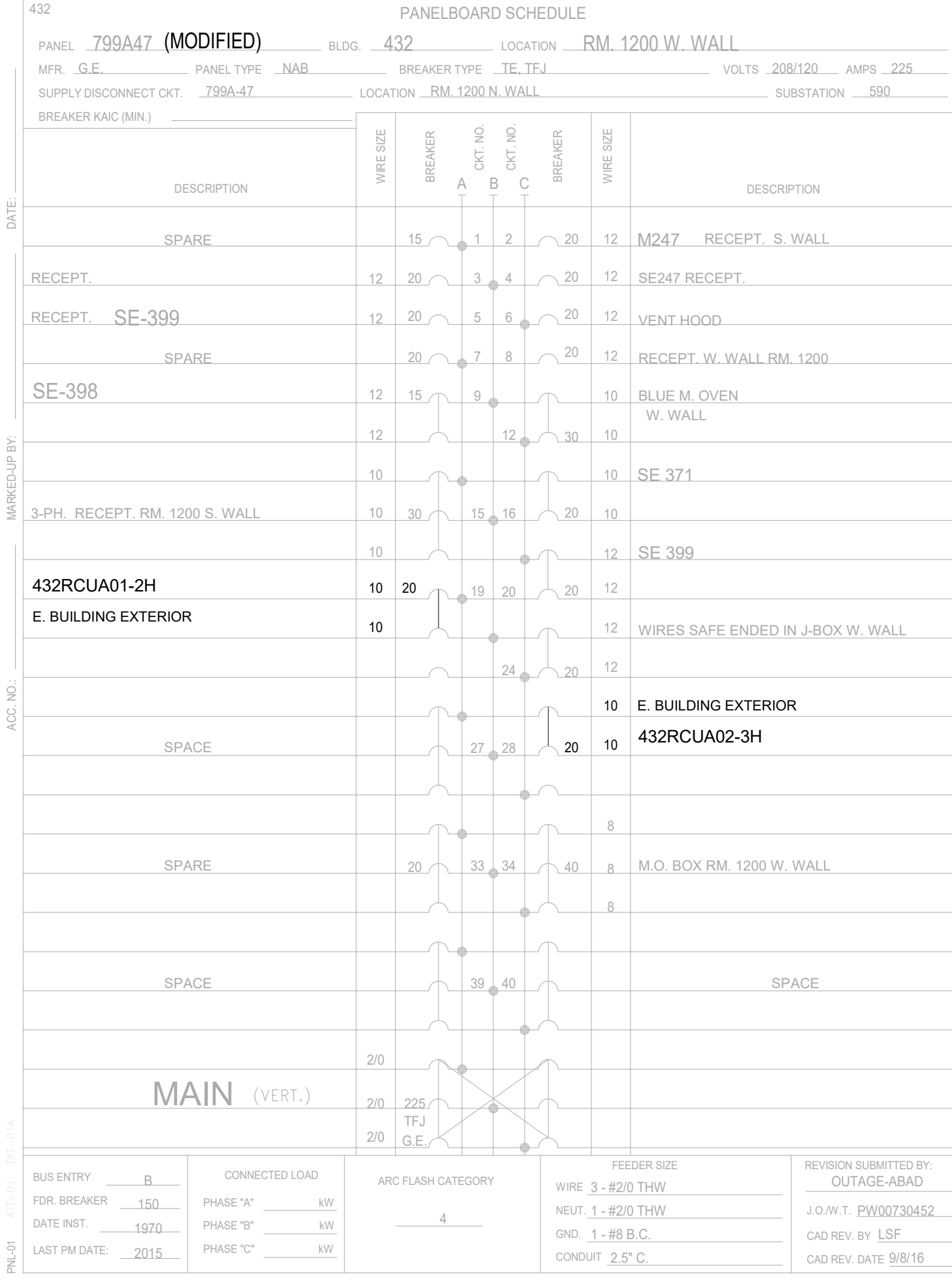
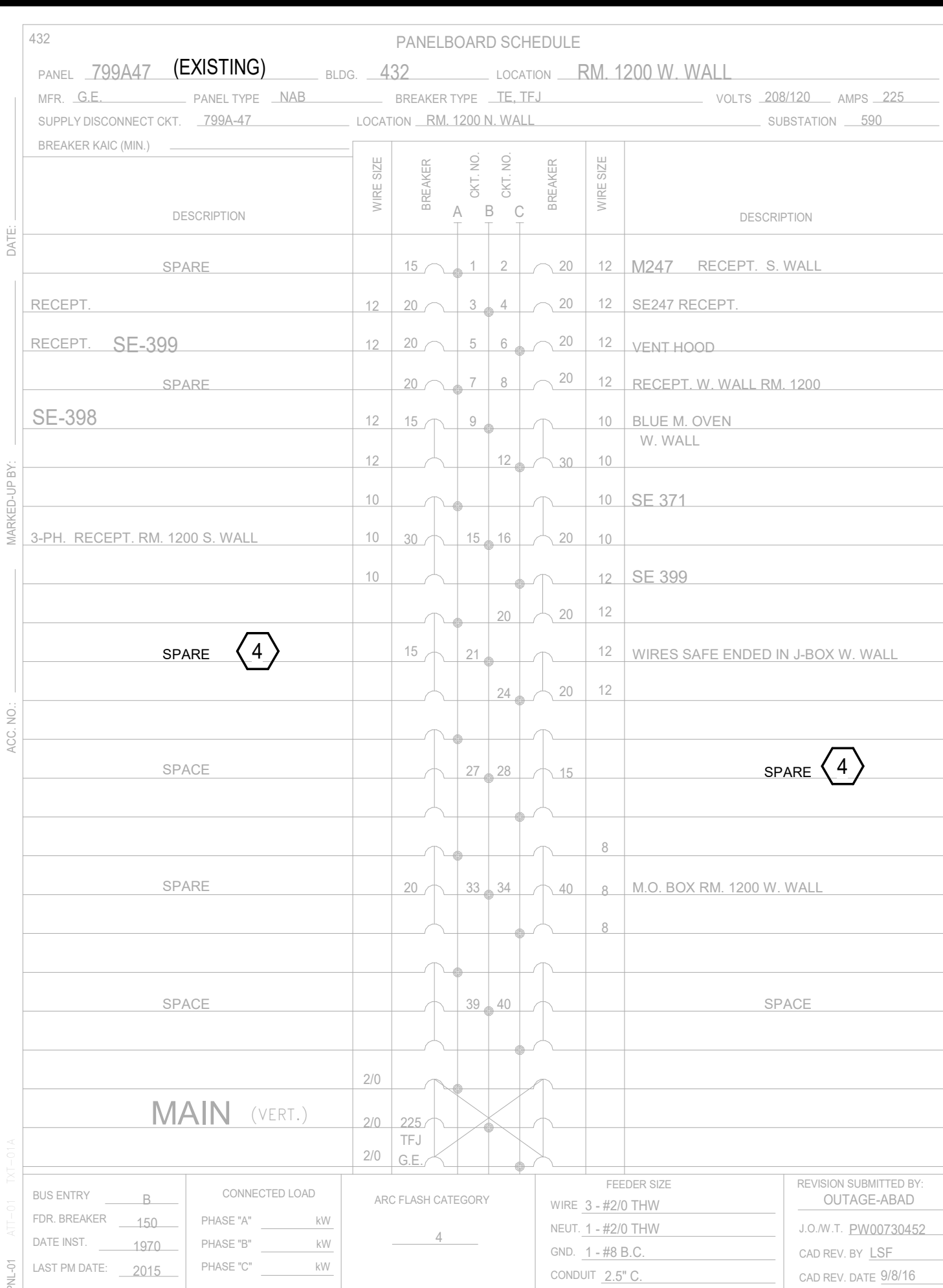
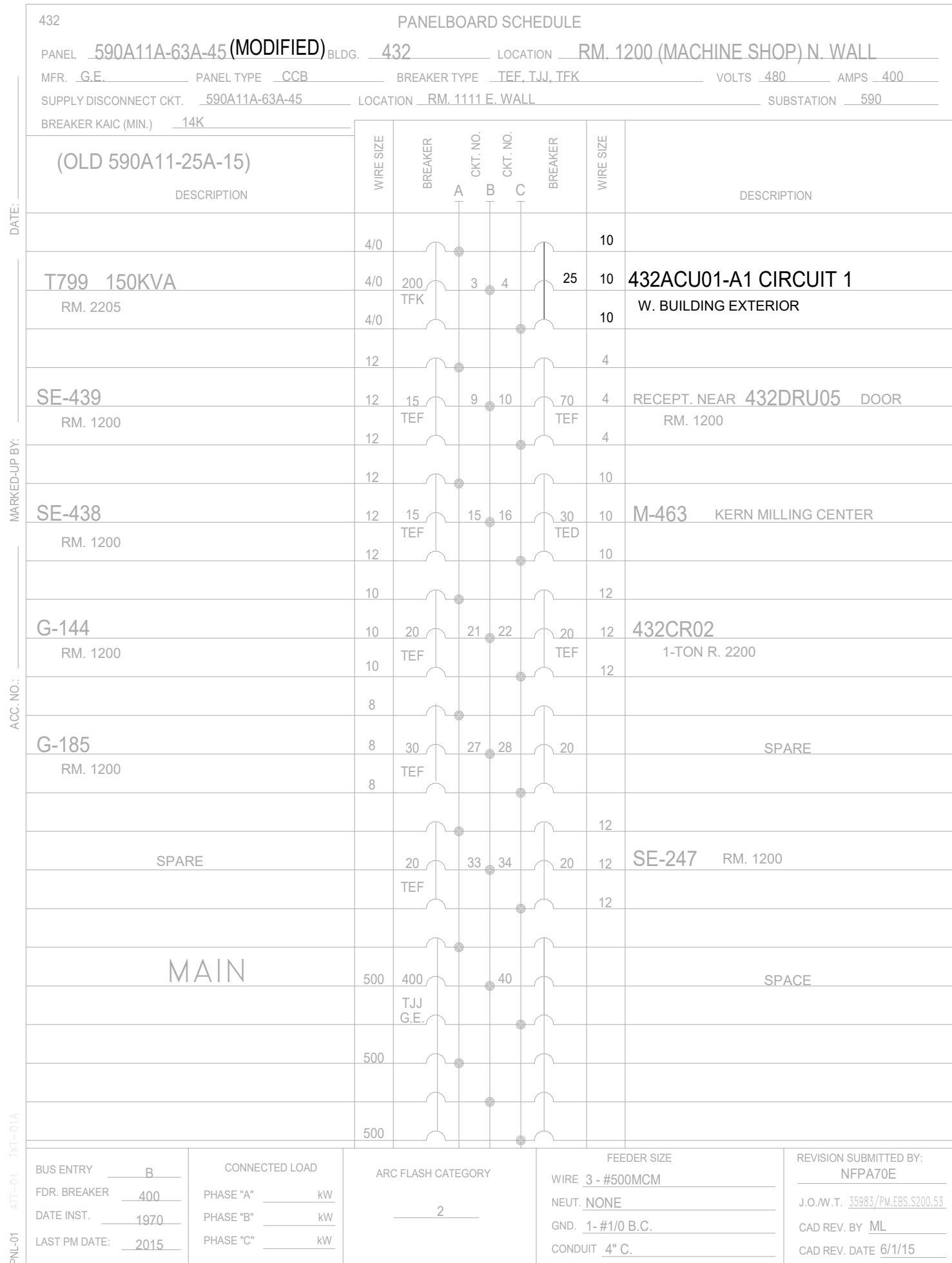
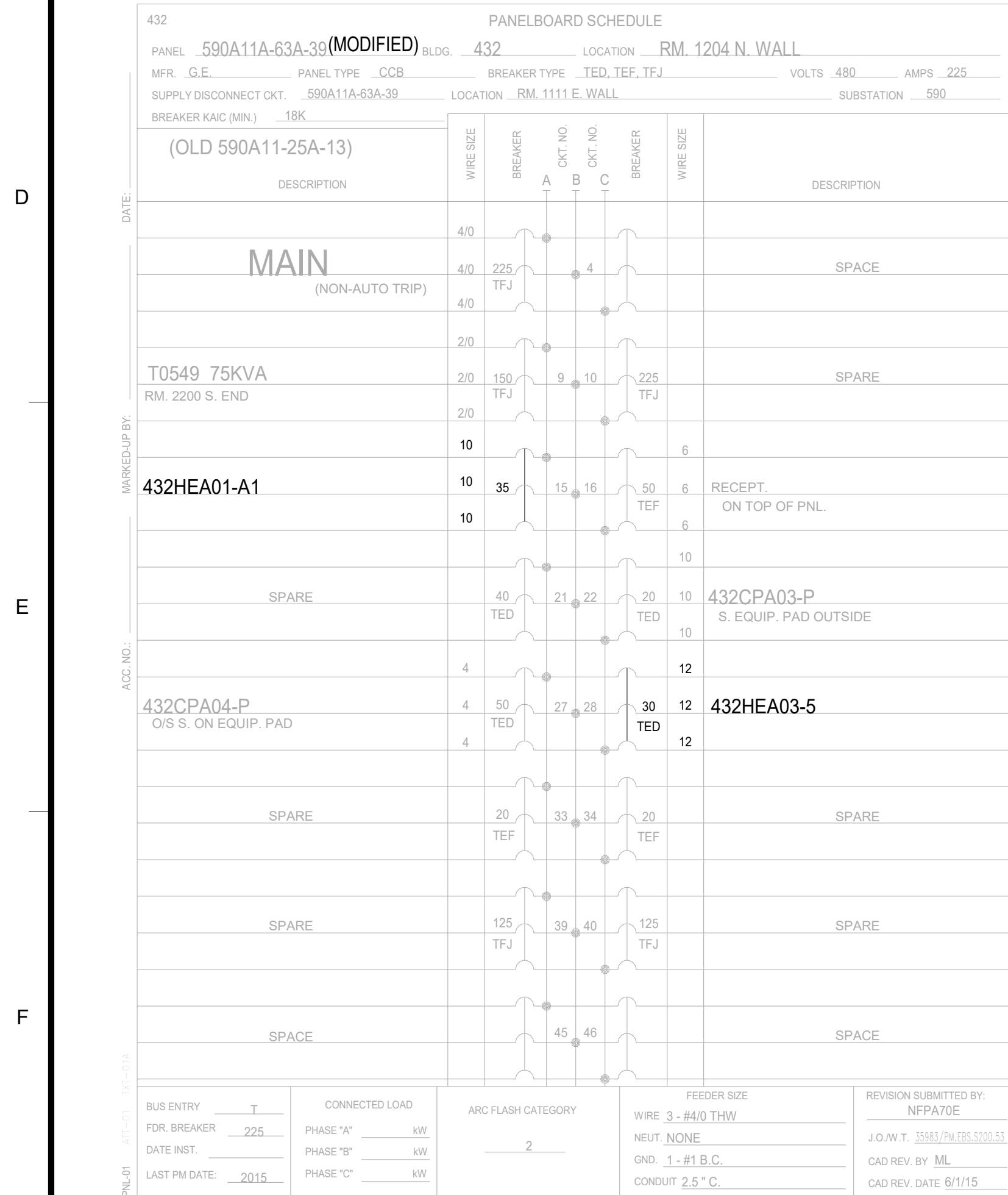
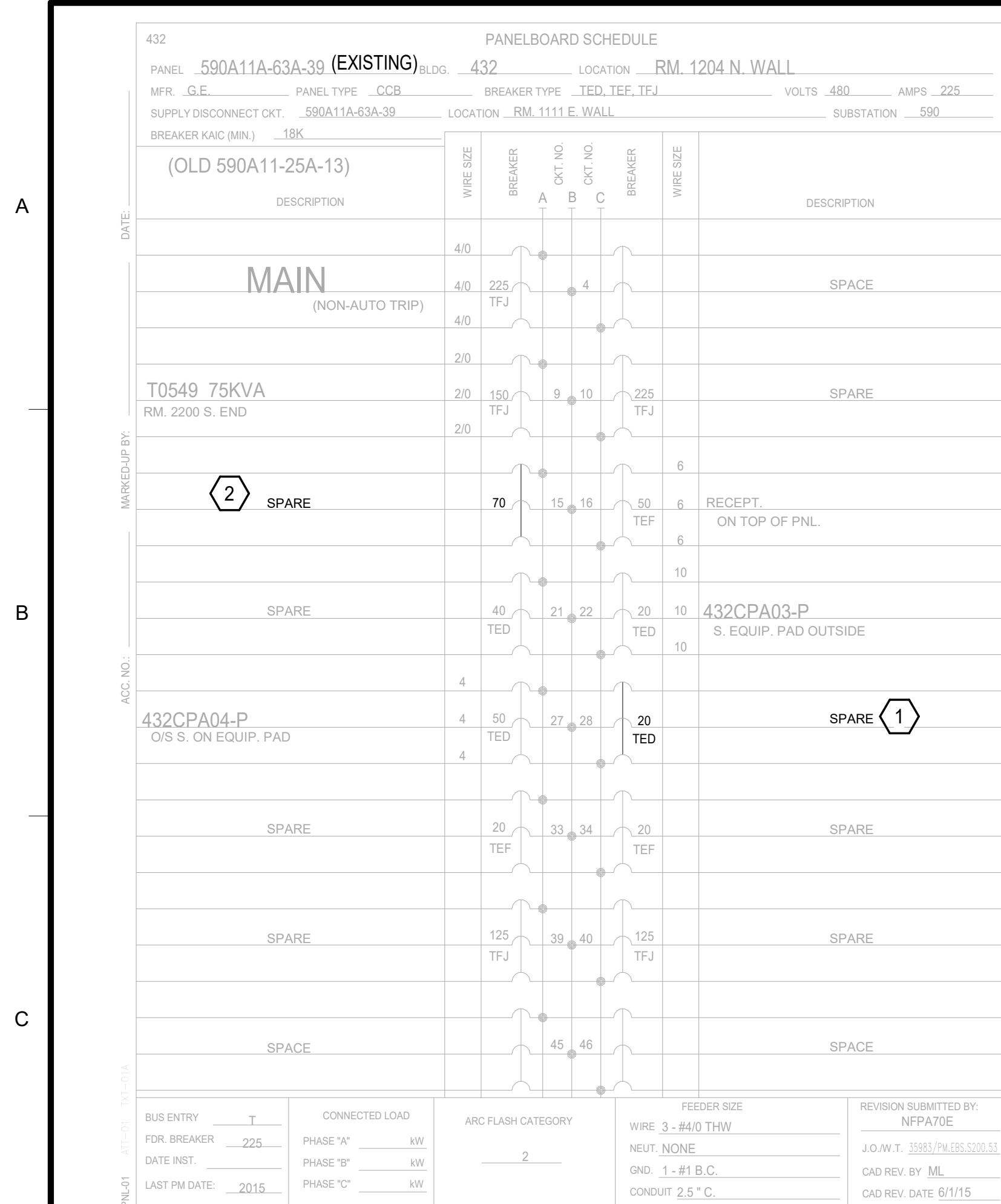
ELECTRICAL PANEL SCHEDULES

Dwg. No.

PLE2021-0432-0015D

Sht. No.

E-802



- ## GENERAL NOTES

1. FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS, REFER TO SHEET E-001.
2. UNLESS OTHERWISE NOTED, LIGHT LINEWEIGHT INDICATES EXISTING EQUIPMENT TO REMAIN AND HEAVY LINEWEIGHT INDICATES EQUIPMENT TO BE MODIFIED.
3. SUBCONTRACTORS MUST FIELD VERIFY LOCATION OF EXISTING EQUIPMENT AND EQUIPMENT TO BE INSTALLED.
4. SUBCONTRACTOR MUST FIELD VERIFY CIRCUITS INDICATED AND LOADS SERVED BY THEM.
5. SUBCONTRACTOR MUST UPDATE MODIFIED PANELBOARD SCHEDULES.
6. PANEL SCHEDULES FOR AFFECTED PANELS ARE SHOWN. COORDINATE EQUIPMENT SHUTDOWN WITH CONSTRUCTION MANAGER / SUBCONTRACT TECHNICAL REPRESENTATIVE DURING CONSTRUCTION.
7. CONDUCTOR SPLICES ARE NOT ALLOWED WITHOUT LEAD ELECTRICAL ENGINEER APPROVAL.

KEYED NOTES

1. DISCONNECT AND REMOVE EXISTING CIRCUIT BREAKER IN EXISTING PANEL. REPLACE WITH NEW 2P-208V-30A CIRCUIT BREAKER IN EXISTING PANEL TO SERVE NEW LOAD.
2. DISCONNECT AND REMOVE EXISTING CIRCUIT BREAKER IN EXISTING PANEL. REPLACE WITH NEW 3P-480V-80A CIRCUIT BREAKER TO SERVE NEW LOAD. NEW CIRCUIT BREAKER CHARACTERISTICS MUST MATCH EXISTING.
3. DISCONNECT AND REMOVE EXISTING CIRCUIT BREAKER IN EXISTING PANEL. REPLACE WITH NEW 3P-480V-110A CIRCUIT BREAKER TO SERVE NEW LOAD. NEW CIRCUIT BREAKER CHARACTERISTICS MUST MATCH EXISTING.
4. DISCONNECT AND REMOVE EXISTING CIRCUIT BREAKER IN EXISTING PANEL. REPLACE WITH NEW 2P-208V-20A CIRCUIT BREAKER IN EXISTING PANEL TO SERVE NEW LOAD.

Sheet Title

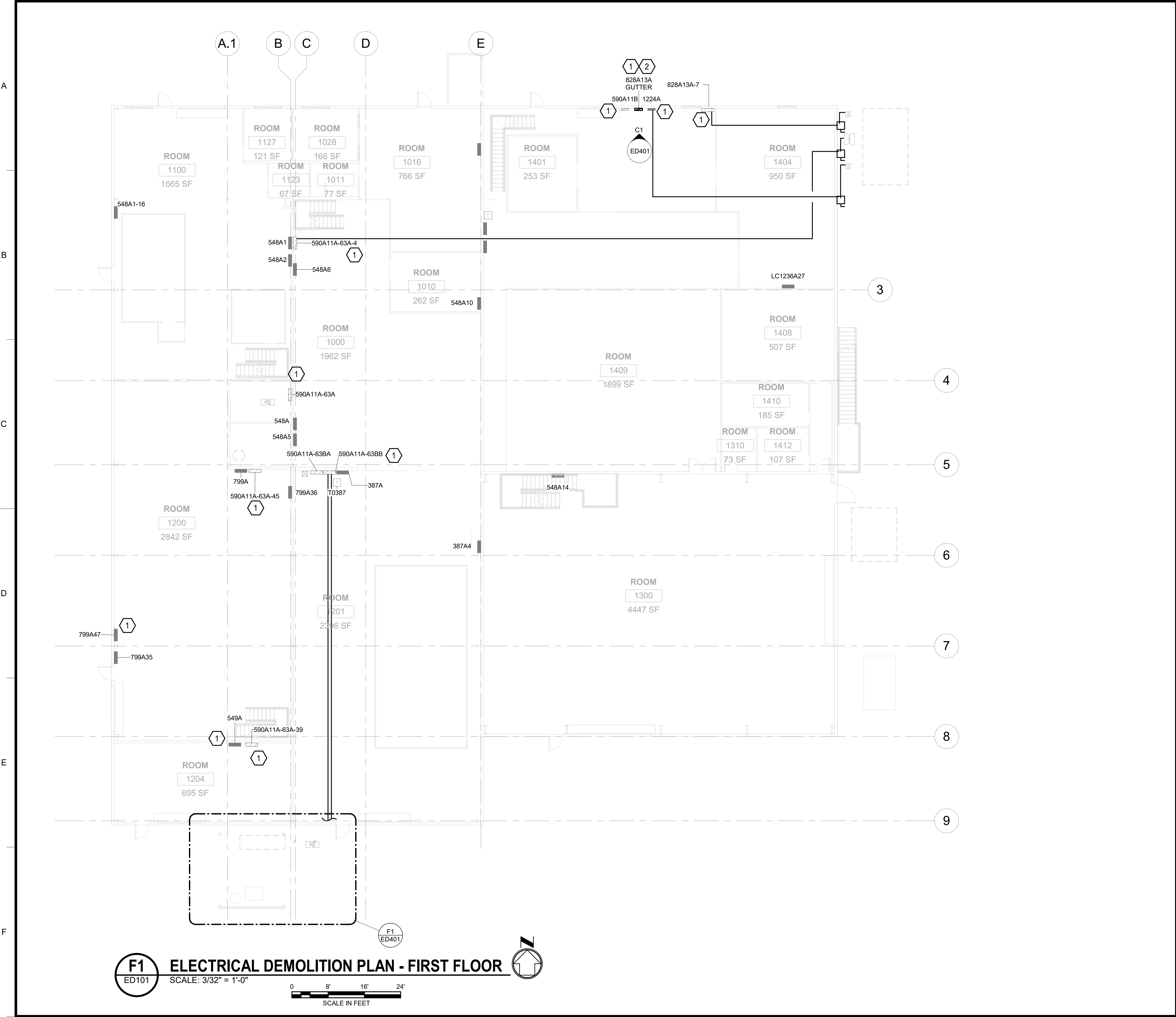
ELECTRICAL PANEL SCHEDULES

Dwg. No.

PLE2021-0432-0017D

Sht. No.

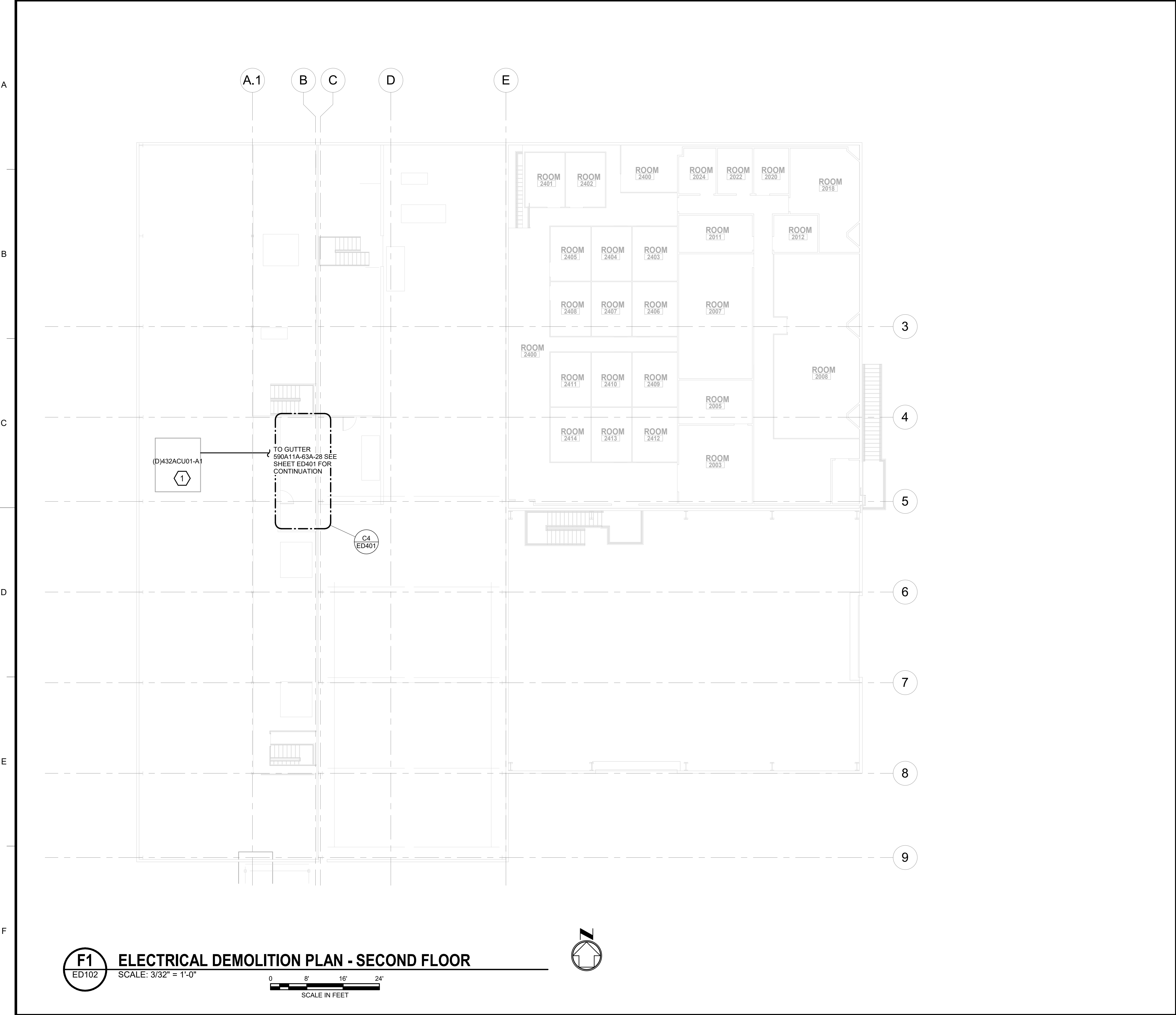
E-804



- ### GENERAL NOTES
1. FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS, REFER TO SHEET E-001.
 2. UNLESS OTHERWISE NOTED, LIGHT LINEWEIGHT INDICATES EXISTING EQUIPMENT TO REMAIN AND HEAVY LINEWEIGHT INDICATES DEMOLITION WORK.
 3. SUBCONTRACTORS MUST FIELD VERIFY LOCATION OF EXISTING EQUIPMENT AND EQUIPMENT TO BE DISCONNECTED AND REMOVED.
 4. SUBCONTRACTOR MUST FIELD VERIFY CIRCUITS INDICATED AND LOADS SERVED BY THEM.
 5. SUBCONTRACTORS MUST WORK CLOSELY WITH MECHANICAL SUBCONTRACTOR TO VERIFY PHASING OF DEMOLITION. THIS IS TO AVOID ANY UNNECESSARY OR UNPLANNED OUTAGES FOR EQUIPMENT OR THE FACILITY.
 6. SUBCONTRACTOR MUST REFER TO DIVISION 1 SPECIFICATIONS FOR CONSTRUCTION SEQUENCING AND PHASING.
 7. REFER TO SHEETS E-801, E-802 AND E-803 FOR ADDITIONAL WORK AT ELECTRICAL PANELS.
 8. CONDUIT ROUTING IS FOR DIAGRAMMATIC PURPOSE ONLY. SUBCONTRACTOR MUST VERIFY EXACT CONDUIT ROUTES
 9. DEMOLITION WORK MUST COMPLY WITH NEC ARTICLE 725.25 AND 800.25.

- ### # KEYED NOTES
1. EXISTING PANEL TO BE AFFECTED BY THIS PROJECT. COORDINATE SHUTDOWN WITH SUBCONTRACT TECHNICAL REPRESENTATIVE DURING CONSTRUCTION. SEE SHEET E-801, E-802 AND E-803 FOR ADDITIONAL WORK.
 2. DISCONNECT AND REMOVE EXISTING GUTTER, CONDUCTORS, DISCONNECT SWITCH, VFDS AND PLATFORM SERVING EXISTING EQUIPMENT BACK TO SOURCE PANEL. EXISTING CONDUIT MUST REMAIN AND TO BE REUSED FOR NEW WORK.

Sheet Title	
ELECTRICAL DEMOLITION PLAN - FIRST FLOOR	
Dwg. No.	PLE2021-0432-0004D
Sht. No.	ED101



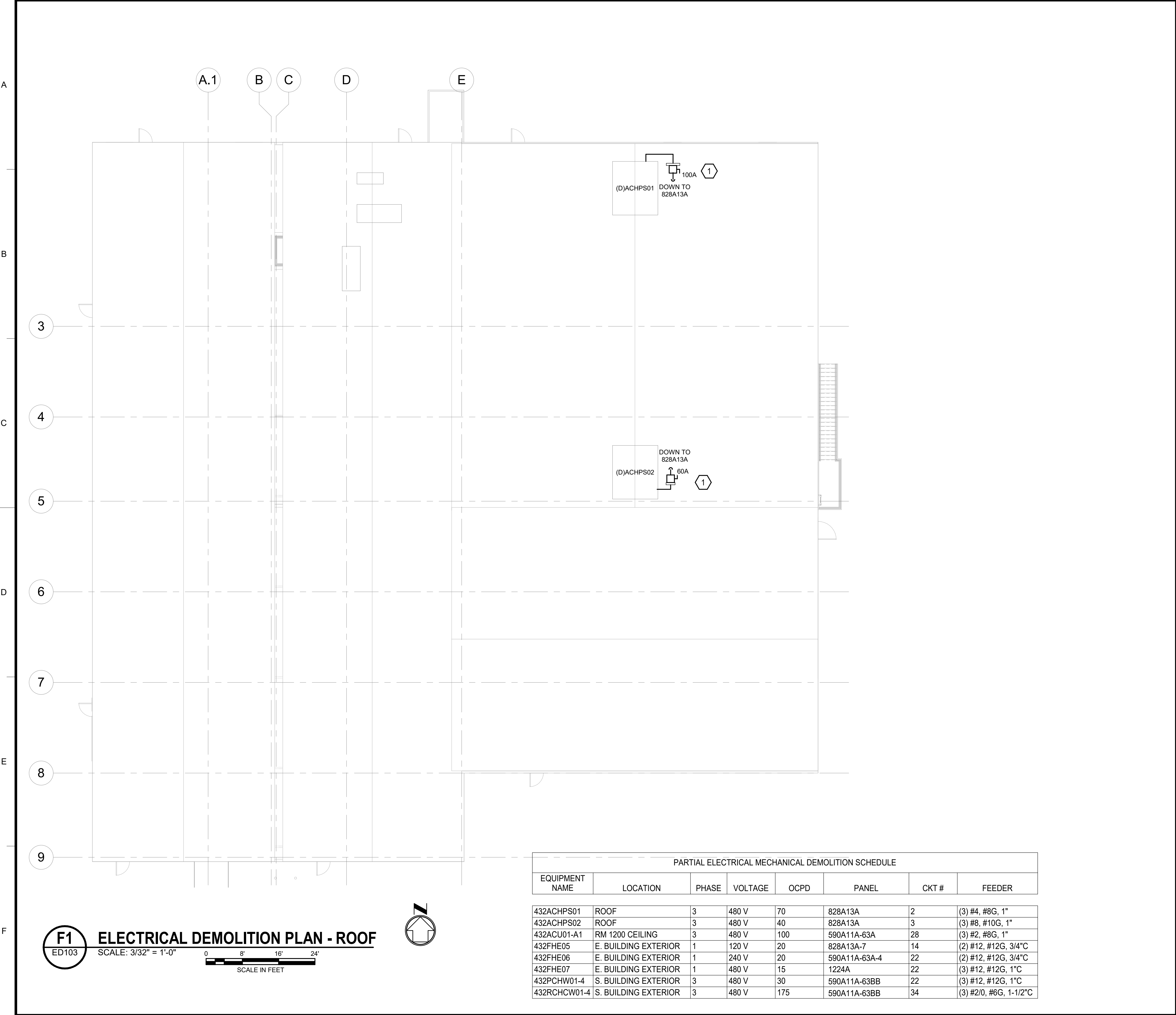
GENERAL NOTES

1. FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS, REFER TO SHEET E-001.
2. UNLESS OTHERWISE NOTED, LIGHT LINEWEIGHT INDICATES EXISTING EQUIPMENT TO REMAIN AND HEAVY LINEWEIGHT INDICATES DEMOLITION WORK.
3. SUBCONTRACTORS MUST FIELD VERIFY LOCATION OF EXISTING EQUIPMENT AND EQUIPMENT TO BE DISCONNECTED AND REMOVED.
4. SUBCONTRACTOR MUST FIELD VERIFY CIRCUITS INDICATED AND LOADS SERVED BY THEM.
5. SUBCONTRACTORS MUST WORK CLOSELY WITH MECHANICAL SUBCONTRACTOR TO VERIFY PHASING OF DEMOLITION. THIS IS TO AVOID ANY UNNECESSARY OR UNPLANNED OUTAGES FOR EQUIPMENT OR THE FACILITY.
6. SUBCONTRACTOR MUST REFER TO DIVISION 1 SPECIFICATIONS FOR CONSTRUCTION SEQUENCING AND PHASING.
7. REFER TO SHEETS E-801, E-802 AND E-803 FOR ADDITIONAL WORK AT ELECTRICAL PANELS.
8. CONDUIT ROUTING IS FOR DIAGRAMMATIC PURPOSE ONLY. SUBCONTRACTOR MUST VERIFY EXACT CONDUIT ROUTES
9. DEMOLITION WORK MUST COMPLY WITH NEC ARTICLE 725.25 AND 800.25.

KEYED NOTES

1. DEMOLISH EXISTING EQUIPMENT AND ALL ASSOCIATED DISCONNECTS, CONDUCTORS AND CONDUITS.

Sheet Title	
ELECTRICAL DEMOLITION PLAN - SECOND FLOOR	
Dwg. No.	PLE2021-0432-0005D
Sht. No.	ED102



GENERAL NOTES

1. FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS, REFER TO SHEET E-001.
2. UNLESS OTHERWISE NOTED, LIGHT LINEWEIGHT INDICATES EXISTING EQUIPMENT TO REMAIN AND HEAVY LINEWEIGHT INDICATES DEMOLITION WORK. .
3. SUBCONTRACTORS MUST FIELD VERIFY LOCATION OF EXISTING EQUIPMENT AND EQUIPMENT TO BE DISCONNECTED AND REMOVED.
4. SUBCONTRACTOR MUST FIELD VERIFY CIRCUITS INDICATED AND LOADS SERVED BY THEM.
5. SUBCONTRACTORS MUST WORK CLOSELY WITH MECHANICAL SUBCONTRACTOR TO VERIFY PHASING OF DEMOLITION. THIS IS TO AVOID ANY UNNECESSARY OR UNPLANNED OUTAGES FOR EQUIPMENT OR THE FACILITY.
6. SUBCONTRACTOR MUST REFER TO DIVISION 1 SPECIFICATIONS FOR CONSTRUCTION SEQUENCING AND PHASING.
7. REFER TO SHEETS E-801, E-802 AND E-803 FOR ADDITIONAL WORK AT ELECTRICAL PANELS.
8. CONDUIT ROUTING IS FOR DIAGRAMMATIC PURPOSE ONLY. SUBCONTRACTOR MUST VERIFY EXACT CONDUIT ROUTES
9. DEMOLITION WORK MUST COMPLY WITH NEC ARTICLE 725.25 AND 800.25.

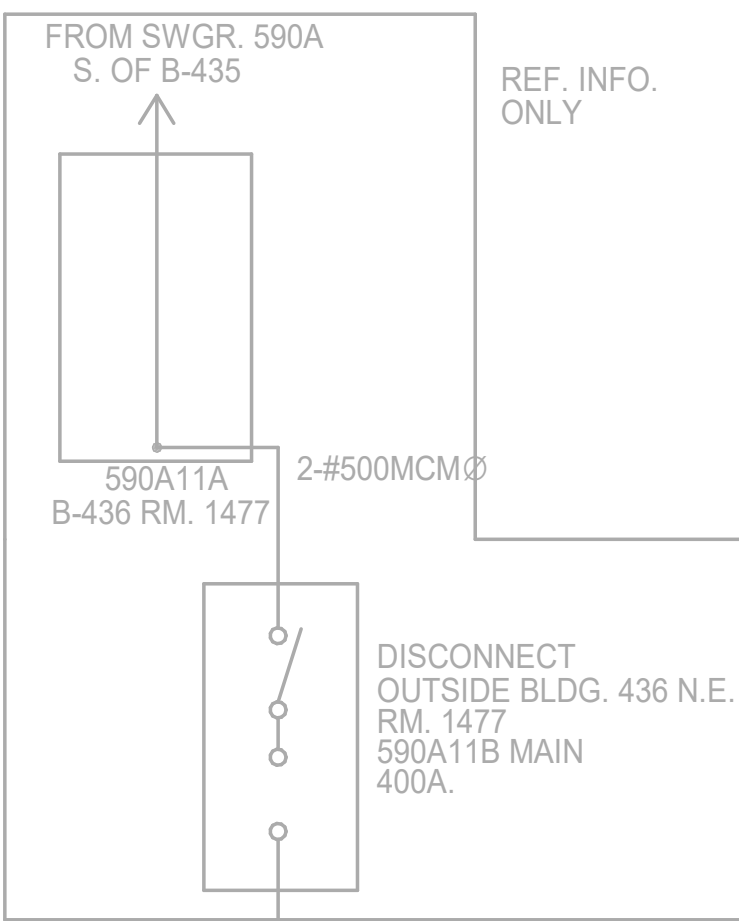
KEYED NOTES

1. DISCONNECT AND REMOVE EXISTING CONDUIT, CONDUCTORS AND DISCONNECT SWITCH SERVING EXISTING EQUIPMENT. PROVIDING SEALING FOR CONDUIT PENETRATIONS UPON EQUIPMENT REMOVAL.

Sheet Title
ELECTRICAL DEMOLITION PLAN - ROOF
Dwg. No. PLE2021-0432-0006D
Sht. No. ED103



ED401



Sht. No. ED701

F1
ED701

SCALE: NO SCALE SHOWN